

spintent

SPANISH PARSE INTENTS

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CTAN: <https://www.ctan.org/pkg/spintent>

 <https://github.com/pablgonz/spintent>

Resumen

El paquete **spintent** proporciona una serie de utilidades para docentes de educación primaria y secundaria que necesiten crear documentos PDF *accesibles* (*tagged* PDF) en español usando Lua[†] $\TeX$.

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Motivación y agradecimientos

Desde que desarrollé `scontents`[7] en 2019 y, más tarde, `enumext`[8] en 2024, me he ido interesando cada vez más por la accesibilidad en los documentos PDF. Al principio era un tema prácticamente desconocido para mí, pero a medida que fui aprendiendo sobre él comprendí la importancia que tiene, especialmente en el ámbito educativo.

El equipo del proyecto de $\TeX$ [19] ha realizado un trabajo extraordinario en este campo. Hoy es posible generar documentos etiquetados, validarlos con herramientas como `veraPDF`[20] o inspeccionar su estructura con `ngpdf`[21]. Sin embargo, siempre me quedó una duda: cuando un estudiante utiliza NVDA[11] junto con MathCAT[12], ¿realmente escucha lo que el o la docente quiso expresar?

Basta con generar un documento accesible siguiendo las recomendaciones de *The LaTeX Tagged PDF Project* y escucharlo con NVDA/MathCAT para darse cuenta de que, en muchos casos, la *verbalización* no coincide con lo que uno espera. Esto no resulta extraño: las reglas del español, y en particular el lenguaje matemático, dependen en gran medida del contexto en que se utilizan.

El objetivo de **spintent** nace precisamente de esta necesidad: proporcionar herramientas para que el o la docente pueda indicar explícitamente el contexto de determinadas expresiones matemáticas desde el propio código fuente de $\TeX$, de modo que el documento PDF resultante no solo sea *accesible*, sino que también produzca una *verbalización más natural* y adecuada para el estudiante al utilizar NVDA/MathCAT.

El nuevo estándar PDF 2.0[14, 18, 16], las especificaciones de WTPDF[15] y la guía más reciente de la PDF Association, «Best Practice Guide: Math in PDF»[17], coinciden en que el camino hacia una accesibilidad matemática de calidad pasa por el uso de MathML. En ese contexto, Lua[†] $\TeX$ ofrece una plataforma especialmente adecuada para incorporar esta tecnología.

*Este archivo describe la documentación para la versión v0.99 revisada por última vez el 2026-09-05.

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Licencia y requerimientos

El paquete `spintent` carga y requiere el paquete `unicode-math`[1] y que se ejecute bajo Lua $\TeX$ por lo que es necesario contar con una distribución moderna de $\TeX$ como $\TeX$ Live 2026. Ha sido probado con las clases estándar proporcionadas por $\LaTeX$: `book`, `report`, `article` y `letter` en tamaños `10pt`, `11pt` y `12pt`.

Se otorga permiso para copiar, distribuir y/o modificar este paquete bajo los términos de $\LaTeX$ Project Public License (LPPL), versión 1.3 o posterior (<https://www.latex-project.org/lppl.txt>). El paquete tiene la condición de mantenido al estilo «A Work in Progress»¹.

- El requerimiento mínimo es $\LaTeX$ release 2026-11-01.

Este paquete solo se puede utilizar solo con Lua $\TeX$ y *tagged* PDF activo. Está presente en $\TeX$ Live y MiK $\TeX$, para instalarlo, utilice el gestor de paquetes de su distribución. Si desea una instalación manual, descargue `spintent.zip` y descomprímalo, luego ejecute `luatex spintent.ins`, mueva todos los archivos a las ubicaciones correspondientes y luego ejecute `mktexlsr`. Para generar la documentación, ejecute `arara spintent.dtx`.

```
spintent.sty  => TDS:tex/lualatex/spintent/
spintent.lua  => TDS:tex/lualatex/spintent/
spintent.pdf  => TDS:doc/lualatex/spintent/
README.md    => TDS:doc/lualatex/spintent/
spintent.dtx => TDS:source/lualatex/spintent/
spintent.ins => TDS:source/lualatex/spintent/
```

1 Descripción

El nombre `spintent` puede entenderse como «*Spanish Parse Intent*» o también como «*School Parse Intent*». Ambas interpretaciones reflejan el objetivo del paquete y el público para el que fue diseñado: docentes de educación primaria y secundaria.

Está pensado para crear «*apuntes*», «*hojas de ejercicios*», «*clases escritas*» enfocadas en el estudiante. Solo un docente sabe cual es el nivel de aprendizaje del estudiante. El paquete intenta sobrescribir muchas de las reglas dictadas por la heurística de NVDA/MathCAT teniendo esto presente.

- El paquete `spintent` usa y abusa del atributo `:intent` de MathML por medio del comando `\MathMLintent` definido en `latex-lab-mathintent`[30], un encapsulado de `\luamml_attribute:een` proveída por `luamml`[2]. ¡Gracias Marcel!

1.1 Carga del paquete

El paquete `spintent` trabaja solo con Lua $\TeX$ y *tagged* PDF activo usando `tagging=on` dentro del comando `\DocumentMetadata` al inicio del archivo. Las llaves (*keys*) aceptadas por `\DocumentMetadata` y en general para el código de *tagged* PDF están documentadas en los paquetes `tagpdf`[29] `documentmetadata-support`[30], es deber del usuario revisar la documentación asociada a ellos.

```
\DocumentMetadata
{
  lang=es-CL,
  pdfversion=2.0,
  pdfstandard=ua-2,
  tagging=on,
  tagging-setup={math/setup={mathml-SE}},
}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage{spintent}
```

1.2 El paquete babel

El paquete `babel-spanish`[10] está marcado como «parcialmente compatible» para *tagged* PDF, pero, es absolutamente necesario para la creación de documentos en español. En caso de encontrar problemas de compatibilidad con otros paquetes se puede desactivar usando:

```
\usepackage[spanish,shorthands=off]{babel}
```

Muchas de las «shorthands» proveídas por `babel-spanish` hoy no son necesarias, más allá de `\sptext` para las abreviaturas y ordinales el resto son solo atajos de teclas para escritura que hoy en día no se ocupan. El paquete `spintent` provee el comando `\spshort` (§3.4) que cubre estos casos, de esta manera podemos cargar `babel`[9] con todas sus funcionalidades sin problemas.

¹A *Work in Progress* es un álbum en vivo de Neil Peart (* 1952–† 2020), baterista y principal letrista de Rush. El nombre no pretende formar parte de la terminología de la LPPL; simplemente describe el estado permanente de este paquete mientras sigo aprendiendo sobre *accesibilidad* en PDF.

1.3 Fuentes tipográficas

Como `spintent` es un paquete Lua $\TeX$ debemos distinguir entre las fuentes tipográficas OpenType/TrueType versus las fuentes de 8bit estándar usadas por $\TeX$ para el *modo texto* y el *modo matemático*. Para evitar problemas con *tagged* PDF preferiremos no combinar fuentes OpenType/TrueType con fuentes 8bit y trabajaremos solo con fuentes OpenType/TrueType que tengan soporte para matemáticas.

La carga de las fuentes tipográficas para Lua $\TeX$ se maneja por medio del paquete `fontspec`[3], si no se especifica ninguna se usarán las fuentes Latin Modern para el *modo texto* y Latin Modern Math para el *modo matemático*.

El tipo de fuente tipográfica escogida es preferencia de cada usuario, en lo personal utilizo las fuente Libertinus para el *modo texto* por medio del paquete `libertinus-otf`[4] y la fuente Erewhon-Math para el modo matemático cargada por el paquete `fourier-otf`[5]. Adicionalmente cargo la fuente Source Code Pro para el texto tipo «máquina de escribir» por medio del paquete `sourcecodepro`[6].

```
% \DocumentMetadata{...}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\usepackage{spintent}
```

1.4 El paquete hyperref

El paquete `hyperref`[23] es «casi obligatorio» cuando se crean documentos *tagged* PDF, se encarga de las referencias cruzadas `\label` y `\ref`, las notas al pie `\footnote`, los hipervínculos `\href` y muchos otros elementos como marcadores, índices y metadatos.

```
\DocumentMetadata
{
  lang=es-CL,
  pdfversion=2.0,
  pdfstandard=ua-2,
  tagging=on,
  tagging-setup={math/setup={mathml-SE}},
}
\documentclass{article}
\usepackage[spanish]{babel}
\usepackage[osf,nomath,mono=false]{libertinus-otf}
\usepackage[osf,semibold]{sourcecodepro}
\usepackage[no-text]{fourier-otf}
\setmathfont{Erewhon-Math}[range={cal,bfcal},StylisticSet=1]
\usepackage{spintent, hyperref}
\hypersetup
{
  colorlinks      = true,
  pdfstartview    = FitH,
  pdfdisplaydoctitle = true,
  pdftitle        = {Test para el paquete spintent},
  pdfinfo         =
  {
    Title   = {Test},
    Subject = {spintent},
  },
}
```

2 El comando \setspintent

`\setspintent` `\setspintent{⟨comando⟩}{⟨key-uno = valor, key-dos = valor, key-tres = valor, ...⟩}`

El comando `\setspintent` establece las llaves (*⟨keys⟩*) de forma global para los comandos provistos por `spintent`. Puede utilizarse tanto en el preámbulo como en el cuerpo del documento tantas veces como se desee. Las llaves (*⟨keys⟩*) definidas en el *argumento opcional* [*⟨key = val⟩*] de los comandos tienen prioridad sobre las establecidas por este, anulando las configuraciones globales de `\setspintent`.

Muchas de las utilidades que provee el paquete `spintent` implementan el sistema [*⟨key = val⟩*] y están pensadas para ser utilizadas en la generación de «*hojas de ejercicios*» o «*clases escritas*». Se recomienda usar `\setspintent` y establecerlas de manera global para cada documento, para mantener la coherencia en la verbalización de NVDA/MathCAT a lo largo del mismo.

El sistema [*⟨key = val⟩*] utilizado por `spintent` se implementa usando el módulo `l3keys` de `expl3`[24]. Por diseño, las llaves que reciben un *valor* después del signo '=' NO aceptan *valores vacíos* y es obligatorio pasar uno válido; las llaves documentadas como *⟨valor prohibido⟩* no reciben *ningun valor*. En caso de omitir un *valor* válido o pasar un *valor vacío* a una llave que acepte *valor* o pasar un *valor* a una llave del tipo *⟨valor prohibido⟩* se devolverá un "error" y se detendrá la compilación del documento.

3 The big four

El paquete `spintent` provee cuatro grandes comandos («*the big four*²»): `\spnum` para el manejo de números, `\spunit` para el manejo de unidades, `\spmoney` para el manejo de dinero y `\spshort` para el manejo de abreviaturas, acrónimos y números ordinales.

- También se incluyen las utilidades `\spnewunit` y `\spunitalias` asociadas a al comando `\spunit`.

3.1 El comando \spnum

`\spnum` `\spnum{⟨número⟩}` `\spnum[⟨key = val⟩]{⟨número,número;número⟩}`
`\spnum[⟨key = val⟩]{⟨+número⟩}` `\spnum[⟨key = val⟩]{⟨número.número;número⟩}`
`\spnum[⟨key = val⟩]{⟨-número⟩}` `\spnum[⟨key = val⟩]{⟨número;número⟩}`
`\spnum[⟨key = val⟩]{⟨número,número⟩}` `\spnum[⟨key = val⟩]{⟨número e exponente⟩}`
`\spnum[⟨key = val⟩]{⟨número.número⟩}` `\spnum[⟨key = val⟩]{⟨número E exponente⟩}`

El comando `\spnum` trabaja en *modo matemático* y recibe el *argumento obligatorio* {*⟨número⟩*}, donde la «*parte entera*» puede iniciar con los signos '+' o '-', la «*parte decimal finita*» **debe** iniciar con una coma ',' o con un punto '.', y la «*parte decimal infinita (periódica)*» **debe** iniciar con un punto y coma '; '.

Si no se proporciona el *argumento opcional* [*⟨key = val⟩*] y se ejecuta `\spnum{+23}` se imprimirá '+23' en la salida PDF y NVDA/MathCAT verbalizará «*más veintitrés*», si se ejecuta `\spnum{3,14}` se imprimirá '3,14' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce*», si se ejecuta `\spnum{3,1;4}` se imprimirá '3,14' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma uno con cuatro periódico*».

Además, el *argumento obligatorio* {*⟨número⟩*} admite al final de este un sufijo de «*notación científica*». Este se indica mediante las letras 'e' o 'E', acompañadas de los signos opcionales '+' o '-' seguido por los dígitos del exponente. Por ejemplo `\spnum{3,14e3}` imprimirá '3,14 · 10³' en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*».

- La coma ',' o el punto '.' son *separadores decimales* y NO de millares, por lo tanto solo pueden aparecer una «única vez» dentro del {*⟨argumento⟩*}. El {*⟨argumento⟩*} puede contener espacios matemáticos válidos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo. Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

3.1.1 Llaves para \spnum

`cifras-sep = {⟨true | false⟩}`

default: *true*

Establece el *espaciado* utilizado para agrupar los dígitos en bloques de tres, tanto en la parte entera como en la parte decimal finita. Si se establece en *true* se siguen las normas de la RAE y se utiliza un *espacio fino* '\,' entre las cifras; si se establece en *false*, se respetan los espacios matemáticos introducidos en el argumento. Por ejemplo `\spnum{100000}` imprimirá '100 000' en la salida PDF y `\spnum[cifras-sep=false]{10\,00\,00}` imprimirá '10 00 00' en la salida PDF.

`read-sign = {⟨terse | clear⟩}`

default: *terse*

Establece la *forma* en la que NVDA/MathCAT verbalizará los signos '+' o '-'. Si se establece en *terse*, NVDA/MathCAT verbalizará «*más*» o «*menos*» **antes** de leer el número; si se establece en *clear*, NVDA/MathCAT verbalizará «*positivo*» o «*negativo*» **después** de leer el número. Por ejemplo `\spnum{-23}` imprimirá '-23' en la salida PDF y NVDA/MathCAT verbalizará «*menos veintitrés*» y `\spnum[read-sign=clear]{-23}` imprimirá '-23' en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés negativo*».

`read-period-sep = {⟨coma | punto⟩}`

default: *coma*

²Los «*cuatro grandes*» del Thrash Metal: Metallica, Megadeth, Slayer y Anthrax, que me acompañan mientras desarrollo `spintent`.

Establece la *forma* en la que NVDA/MathCAT verbalizará e imprimirá el signo de separación ‘;’ de la «*parte decimal infinita (periódica)*» cuando no se tiene la «*parte decimal finita*». Si se establece en *coma*, se imprimirá ‘,’ en la salida PDF y NVDA/MathCAT verbalizará «*coma*»; si se establece en *punto*, se imprimirá ‘.’ en la salida PDF y NVDA/MathCAT verbalizará «*punto*». Por ejemplo `\spnum{3;4}` imprimirá ‘3,4’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma cuatro periódico*» y `\spnum[read-period-sep=punto]{3;4}` imprimirá ‘3.4’ en la salida PDF y NVDA/MathCAT verbalizará «*tres punto cuatro periódico*»

- Esta llave es necesaria SOLO cuando se utilizan «*decimales periódicos puros*», es decir, cuando no se ha incluido en el `{(argumento)}` una *coma* ‘,’ o un *punto* ‘.’ antes del signo de separación ‘;’ periódica.

`notation-sym = {(times | cdot)}`

default: *cdot*

Establece el *símbolo* que se utilizará para imprimir la «*notación científica*» al usar ‘e’ o ‘E’. Si se establece en *times* imprimirá ‘×’ en la salida PDF y NVDA/MathCAT verbalizará «*por*»; si se establece en *cdot* imprimirá ‘·’ en la salida PDF y NVDA/MathCAT verbalizará «*por*». Por ejemplo `\spnum{3,14e3}` imprimirá ‘3,14 · 10³’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*» y `\spnum[notation-sym=times]{3,14e3}` imprimirá ‘3,14 × 10³’ en la salida PDF y NVDA/MathCAT verbalizará «*tres coma catorce por diez al cubo*».

- Decir «*notación científica*» aquí es un poco exagerado, en realidad es «*multiplicación por potencias de 10*», no se verifica si la entrada cumple o no con la condición de estar escrita en «*notación científica*».

3.2 El comando \spunit

<code>\spunit{(unidad)}</code>	<code>\spunit[key = val]{(número unidad*unidad)}</code>
<code>\spunit[key = val]{(unidad)}</code>	<code>\spunit[key = val]{(número unidad*unidad/unidad)}</code>
<code>\spunit[key = val]{(número unidad)}</code>	<code>\spunit[key = val]{(número unidad*unidad/unidad*unidad)}</code>

El comando `\spunit` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{(unidad)}`, el cual puede contener una «*parte numérica*» opcional la cual se procesa internamente por el comando `\spnum` al inicio del `{(argumento)}` seguido de una `{(unidad)}` o un `{(alias)}` valido descritos en §3.2.2 o una `{(unidad)}` creada por el comando `\spnewunit` (§3.2.3) o un `{(alias)}` creado por el comando `\spunitalias` (§3.2.4).

Si no se proporciona el *argumento opcional* `[key = val]` y se ejecuta `\spunit{23 m}` se imprimirá ‘23 m’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros*».

La multiplicación entre las `{(unidades)}` se realiza mediante un *asterisco* ‘*’ y las potencias de estas se indican con un *circunflejo seguido del exponente entre llaves* ‘^`{(exponente)}`’. Por ejemplo si se ejecuta `\spunit{23 m*s^2}` se imprimirá ‘23 m s²’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros segundos al cuadrado*».

Se pueden usar *fracciones lineales* con las `{(unidades)}` utilizando una «*única barra* ‘/’ para la separación entre las `{(unidades)}` del numerador y el denominador, teniendo presente que el denominador NO puede contener números de ningún tipo. Por ejemplo si se ejecuta `\spunit{23 m/s^2}` se imprimirá ‘23 m/s²’ en la salida PDF y NVDA/MathCAT verbalizará «*veintitrés metros por segundos al cuadrado*».

- Solo imprimiremos *fracciones lineales*, si se requiere otra forma fraccionaria para explicaciones y otros usos se debe usar el comando por separado. El `{(argumento)}` puede contener espacios matemáticos validos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{#1}` o `\mathbf{#1}` por ejemplo, NO puede contener más de una «*única barra* ‘/’ y NO puede contener números en el denominador. Cualquier entrada fuera de esta sintaxis devolverá un “error” deteniendo la compilación del documento.

3.2.1 Llaves para \spunit

El comando `\spunit` posee las *meta-keys* `cifras-sep`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1).

`print-cdot = {true | false}`

default: *false*

Establece si se *imprime* `\cdot` ‘·’ o un *espacio fino* ‘\,’ para la multiplicación de `{(unidades)}`. Si se establece en *true*, se imprimirá ‘·’ entre las `{(unidades)}` en la salida PDF; si se establece en *false* se imprimirá un *espacio fino* ‘\,’ entre las `{(unidades)}` en la salida PDF.

`read-cdot = {true | false}`

default: *false*

Establece la *forma* en la que NVDA/MathCAT verbalizará la multiplicación de `{(unidades)}`. Si se establece en *true*, NVDA/MathCAT verbalizará «*por*» entre las `{(unidades)}`; si se establece en *false*, NVDA/MathCAT no verbalizará nada entre las `{(unidades)}`.

`read-slash = {por | partido-por}`

default: *por*

Establece la *forma* en la que NVDA/MathCAT verbalizará la «*barra* ‘/’ de separación entre el numerador y el denominador. Si se establece en *por*, NVDA/MathCAT verbalizará «*por*»; si se establece en *partido-por*, NVDA/MathCAT verbalizará «*partido por*».

3.2.2 Argumentos para \spunit

Resumen de `{(unidades)}` y `{(alias)}` soportados por `\spunit`.

Entrada	Alias soportados	Nombre unidad	Salida	NVDA/MathCAT
m	metro, metros	Metro	m	«metro»
g	gramo, gramos	Gramo	g	«gramo»
s	segundo, segundos	Segundo (Tiempo)	s	«segundo (Tiempo)»
l	litro, l-litro	Litro (minúscula)	l	«litro (minúscula)»
h	hora, horas	Hora	h	«hora»
d	dia, dias	Día	d	«día»
A	amperio, ampere	Amperio	A	«amperio»
V	voltio, volt	Voltio	V	«voltio»
W	vatio, watt	Vatio	W	«vatio»
J	julio, joule	Julio	J	«julio»
N	newton	Newton	N	«newton»
Pa	pascal	Pascal	Pa	«pascal»
Hz	hercio, hertz	Hercio	Hz	«hercio»
Ω	ohmio, ohm, Omega	Ohmio	Ω	«Ohmio»
F	faradio	Faradio	F	«Faradio»
C	culombio	Culombio	C	«Culombio»
K	kelvin, Kelvin	Kelvin	K	«Kelvin»
°C	celsius, degC, °C	Grados Celsius	°C	«Grados Celsius»
°F	fahrenheit, degF, °F	Grados Fahrenheit	°F	«Grados Fahrenheit»
cal	caloria, calorías	Caloría	cal	«Caloría»
b	bit	Bit	b	«Bit»
B	byte, bytes	Byte	B	«Byte»
in	pulgada, pulgadas	Pulgada	in	«Pulgada»
ft	pie, pies	Pie	ft	«Pie»
yd	yarda, yardas	Yarda	yd	«Yarda»
mi	milla, millas	Milla	mi	«Milla»
lb	libra, libras	Libra	lb	«Libra»
oz	onza, onzas	Onza	oz	«Onza»
gal	galon, galones	Galón	gal	«Galón»
Å	angstrom	Angstrom	Å	«Angstrom»
μ	micro	Micro (Prefijo)	μ	«Micro (Prefijo)»
Ω	mho	Mho / Siemens	Ω	«Mho / Siemens»
'	minmin	Minuto (Arco)	'	«Minuto (Arco)»
"	segseg	Segundo (Arco)	"	«Segundo (Arco)»
u	u-masa	Unidad masa atómica	u	«Unidad masa atómica»

Unidades directas (sin alias):

L	-	Litro (Mayúscula)	L	«Litro (Mayúscula)»
ℓ	-	Litro (Cursiva)	ℓ	«Litro (Cursiva)»
w	-	Vatio (Minúscula)	w	«Vatio (Minúscula)»
°K	-	Grado Kelvin	°K	«Grado Kelvin»
Bq	-	Becquerel	Bq	«Becquerel»
Da	-	Dalton	Da	«Dalton»
Gy	-	Gray	Gy	«Gray»
S	-	Siemens	S	«Siemens»
Sv	-	Sievert	Sv	«Sievert»
T	-	Tesla	T	«Tesla»
Wb	-	Weber	Wb	«Weber»
atm	-	Atmósfera	atm	«Atmósfera»
au	-	Unidad astronómica	au	«Unidad astronómica»
bar	-	Bar	bar	«Bar»
cd	-	Candela	cd	«Candela»
dB	-	Decibelio	dB	«Decibelio»
dyn	-	Dina	dyn	«Dina»
eV	-	Electronvoltio	eV	«Electronvoltio»
erg	-	Ergio	erg	«Ergio»
ha	-	Hectárea	ha	«Hectárea»
kat	-	Katal	kat	«Katal»
kg	-	Kilogramo	kg	«Kilogramo»
km	-	Kilómetro	km	«Kilómetro»
lm	-	Lumen	lm	«Lumen»

Continúa en la siguiente página...

Continuación de la tabla 1				
Entrada	Alias soportados	Nombre unidad	Salida	MathCAT
lx	-	Lux	lx	«Lux»
cm	-	Centímetro	cm	«Centímetro»
min	-	Minuto (Tiempo)	min	«Minuto (Tiempo)»
mm	-	Milímetro	mm	«Milímetro»
mol	-	Mol	mol	«Mol»
pc	-	Pársec	pc	«Pársec»
rad	-	Radián	rad	«Radián»
rpm	-	Rev. por minuto	rpm	«Rev. por minuto»
sr	-	Estereorradián	sr	«Estereorradián»
t	-	Tonelada	t	«Tonelada»
a	-	Año / Área (are)	a	«Año / Área (are)»
y	-	Año (Year)	y	«Año (Year)»
°	-	Grado (Arco)	°	«Grado (Arco)»

3.2.3 El comando \spnewunit

\spnewunit

```
\spnewunit{<nueva unidad>}{<verbalización NVDA>}
```

El comando `\spnewunit` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: `{<nueva unidad>}` que establece lo que se imprimirá en la salida PDF y `{<verbalización NVDA>}` que establece como NVDA/MathCAT verbalizará la `{<nueva unidad>}`.

La `{<nueva unidad>}` no debe estar registrada como `{<unidad>}` o `{<alias>}` para `\spunit` descritos en §3.2.2. Si la `{<nueva unidad>}` ya está registrada se devolverá un “error” y se detendrá la compilación del documento.

La declaración de la `{<nueva unidad>}` es «global» y se almacena en el módulo `spintent.lua` como una `{<unidad>}` valida para pasar como `{<argumento>}` a `\spunit`. Por ejemplo: `\spnewunit{px}{píxeles}` creará la `{<nueva unidad>}` ‘px’ y cuando se utilice `\spunit{100 px}` se imprimirá ‘100 px’ en la salida PDF y NVDA/MathCAT verbalizará «cien píxeles».

3.2.4 El comando \spunitalias

\spunitalias

```
\spunitalias{<nuevo alias>}{<expresión de unidades>}
```

El comando `\spunitalias` trabaja en *modo texto* y recibe dos *argumentos obligatorios*: `{<nuevo alias>}` que establece es el nuevo `{<alias>}` de `{<argumento>}` para `\spunit` y `{<expresión de unidades>}` que puede contener multiplicación o división de `{<unidades>}` ya registradas que se imprimirán en la salida PDF.

El `{<nuevo alias>}` no debe estar registrado como `{<unidad>}` o `{<alias>}` para `\spunit` descritos en §3.2.2 o estar definido como `{<unidad>}` por medio del comando `\spnewunit`. Si el `{<nuevo alias>}` ya está registrado se devolverá un “error” y se detendrá la compilación del documento.

La declaración de un `{<nuevo alias>}` es «global» y se almacena en el módulo `spintent.lua` como una `{<alias>}` valido para pasar como `{<argumento>}` a `\spunit`. Por ejemplo: `\spunitalias{caudal}{m^3/s}` creará el `{<alias>}` ‘caudal’ y cuando se utilice `$\spunit{50 caudal}$` se imprimirá ‘50 m³/s’ en la salida PDF y NVDA/MathCAT verbalizará «cincuenta metros al cubo por segundos».

3.3 El comando \spmoney

\spmoney

```
\spmoney{<cifra>}
\spmoney{<cifra moneda>}
\spmoney[<key = val>]{<cifra moneda>}
```

El comando `\spmoney` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<cifra>}` que contiene la «cifra numérica» positiva o negativa la se cual procesa internamente por el comando `\spnum` al inicio del `{<argumento>}` acompañada de un «símbolo de moneda» o «alias de moneda» opcional descritos en §3.3.2.

Si no se proporciona el *argumento opcional* `[<key = val>]` y se ejecuta `$\spmoney{500}$` se imprimirá ‘\$500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos».

El comportamiento de `\spmoney` depende de como se pase el *argumento obligatorio* que tiene prioridad por sobre los valores por defecto. Por ejemplo: `$\spmoney{500 dolares}$` imprimirá ‘\$500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos dolares» aunque la moneda por defecto sean pesos.

Si se pasa un «alias de moneda» del tipo ISO (código de divisas) como CLP o USD (mayúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país y se ajusta la salida PDF al estandar ISO. Por ejemplo: `$\spmoney{500 CLP}$` imprimirá ‘CLP500’ en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos chilenos».

Si se pasa un «alias de moneda» del tipo ISO (código de divisas) como clp o usd (minúsculas) se modifica la verbalización en NVDA/MathCAT agregando el país, pero, NO se modifica la salida PDF imprimiendo el

«*símbolo de moneda*» correspondiente. Por ejemplo: `\$ \spmoney{500 c\p}` imprimirá '\$500' en la salida PDF y NVDA/MathCAT verbalizará «quinientos pesos chilenos»

Pasar de manera directa en el *argumento obligatorio* un «*símbolo de moneda*» como '\$', '€', '£' o un «*alias de moneda*» como peso, euro o libra produce la salida PDF y verbalización estándar esperada para NVDA/MathCAT.

- La posición del «*símbolo de moneda*» a la izquierda o a la derecha de la «*cifra numérica*» y la verbalización para NVDA/MathCAT se manejan desde el módulo Lua `spintent.lua` y no se proveen llaves para modificar este comportamiento.
- El «*símbolo de moneda*» es tomado de la fuente tipográfica definida para el *modo matemático* al momento de ejecutar el comando, se debe tener presente en caso de querer usar un «*símbolo de moneda*» poco habitual que esté soportada por `spintent`. El `{\argumento}` puede contener espacios matemáticos válidos, pero, NO puede contener comandos con o sin argumentos como `\textstyle`, `\mathrm{\#1}` o `\mathbf{\#1}` por ejemplo. La coma ',' o el punto '.' como *separador decimal* solo pueden aparecer una «única vez». Cualquier entrada fuera de esta sintaxis devolverá un "error" deteniendo la compilación del documento.

3.3.1 Llaves para \spmoney

El comando `\spmoney` posee la *meta-key* `cifras-sep` pasada al comando `\spnum` (§3.1.1).

`currency = {\moneda | alias}` default: peso

Establece el *símbolo de moneda* por defecto que se imprimirá en la salida PDF y cómo lo verbalizará NVDA/MathCAT cuando SOLO se pasa una «*cifra numérica*» como `{\argumento}` al comando `\spmoney`. El «*símbolo de moneda*» o el «*alias de moneda*» pasado a esta llave debe ser alguno de los descritos en §3.3.2.

`sub-units = {\true | false}` default: false

Establece si se usarán *subunidades* de la moneda en curso para la verbalización en NVDA/MathCAT. La moneda en curso debe aceptar *subunidades*. Por ejemplo: `\$ \spmoney[sub-units=true]{100,5 dolares}` imprimirá '\$100,5' en la salida PDF y NVDA/MathCAT verbalizará «*cient dólares con cincuenta centavos*».

`dolar-math = {\true | false}` default: true

Establece si el *símbolo* '\$' se imprimirá en *modo texto* o en *modo matemático*. Si se establece en `true`, se usará la fuente tipográfica definida para el *modo matemático* en curso para imprimir '\$' en la salida PDF; si se establece en `false`, se usará la fuente tipográfica definida para el *modo texto* para imprimir '\$' en la salida PDF.

3.3.2 Monedas y alias para \spmoney

Resumen de los `{\símbolos de moneda}` y `{\alias de moneda}` soportados como `{\argumento}` para `\spmoney`.

3.4 El comando \spshort

`\spshort` `\spshort{\abreviaturas | acrónimos | números ordinales}`
`\spshort[key = val]{\abreviaturas | acrónimos | números ordinales}`
`\spshort[read-arg={\verbalización del argumento para NVDA}]{\comandos LATEX}`

El comando `\spshort` trabaja *solo en modo texto* y recibe un *argumento obligatorio* del tipo `{\abreviaturas}`, `{\acrónimos}` o `{\números ordinales}` descritos en §3.4.2.

Si no se proporciona el *argumento opcional* `[key = val]` y se ejecuta `\spshort{dr. a}` se imprimirá 'Dr.^a' en la salida PDF y NVDA/MathCAT verbalizará «*doctora*»; si se ejecuta `\spshort{onu}` se imprimirá 'ONU' en la salida PDF y NVDA/MathCAT verbalizará «*organización de las naciones unidas*» y si se ejecuta `\spshort{mineduc}` se imprimirá 'Mineduc' en la salida PDF y NVDA/MathCAT verbalizará «*ministerio de educación*».

Para la escritura de `{\números ordinales}` se puede ejecutar `\spshort{1. a}` que imprimirá '1.^a' en la salida PDF y NVDA/MathCAT verbalizará «*primera*»; si se ejecuta `\spshort{1. o}` imprimirá '1.^o' en la salida PDF y NVDA/MathCAT verbalizará «*primero*».

Para crear una nueva `{\abreviatura}`, `{\acrónimo}` o `{\número ordinal}` que no se encuentre registrada por `spintent` dentro del módulo Lua `spintent.lua` descritas en §3.4.2 se debe utilizar la llave `read-arg` y definir esta dentro del *argumento obligatorio* que puede contener `{\comandos LATEX}` en *modo texto* los cuales se imprimirán en salida PDF.

- Para la escritura de `{\números ordinales}` se debe tener cuidado y no confundir la «*letra o voladita*» 'o' con el símbolo de grado '°'. Si la fuente tipográfica del *modo texto* está utilizando `Numbers=OldStyle` los números pueden quedar muy separados de la «*letra voladita*», la solución para esto es agregar `\addfontfeatures{Numbers=Lining}` dentro de un grupo junto al comando: `{\addfontfeatures{Numbers=Lining}\spshort{1. o}}`.
- El comando `\spshort` está pensado como un reemplazo para *shorthands* y el comando `\sptext` provistos por el paquete `babel` en español compatible con *tagged* PDF. Es de las pocas utilidades que provee el paquete `spintent` que trabaja *solo en modo texto*. Internamente utiliza la llave `E` (*expansion text*) proveída por el paquete `tagpdf`[29].

Entrada	Alias	Nombre	Salida	NVDA/MathCAT
\$	peso, pesos	peso	\$	«peso / pesos»
€	euro, euros	euro	€	«euro / euros»
£	libra, libras	libra	£	«libra / libras»
¥	yen, yenes	yen	¥	«yen / yenes»
¥	yuan, yuanes	yuan	¥	«yuan / yuanes»
₩	won, wons	won	₩	«won / wons»
₪	shekel, shekels, sequel, sequeles	séquel	₪	«séquel / séqueles»
₹	rupia, rupias	rupia	₹	«rupia / rupias»
₽	rublo, rublos	rublo	₽	«rublo / rublos»
₺	lira, liras	lira	₺	«lira / liras»
₾	grivna, grivnas	grivna	₾	«grivna / grivnas»
¢	centavo, centavos	centavo	¢	«centavo / centavos»

Divisas en formato ISO (Mayúsculas)

CLP	—	peso chileno	CLP	«peso chileno / pesos chilenos»
MXN	—	peso mexicano	MXN	«peso mexicano / pesos mexicanos»
USD	—	dólar	USD	«dólar estadounidense / dólares estadounidenses»
EUR	—	euro	EUR	«euro / euros»
GBP	—	libra	GBP	«libra / libras»
JPY	—	yen	JPY	«yen / yenes»
CNY	—	yuan	CNY	«yuan / yuanes»
KRW	—	won	KRW	«won / wons»
ILS	—	séquel	ILS	«séquel / séqueles»
INR	—	rupia	INR	«rupia / rupias»
RUB	—	rublo	RUB	«rublo / rublos»
TRY	—	lira	TRY	«lira / liras»
UAH	—	grivna	UAH	«grivna / grivnas»

Códigos ISO en minúsculas

clp	—	peso chileno	\$	«peso chileno / pesos chilenos»
mxn	—	peso mexicano	\$	«peso mexicano / pesos mexicanos»
usd	dolar, dolares	dólar	\$	«dólar estadounidense / dólares estadounidenses»
eur	—	euro	€	«euro / euros»
gbp	—	libra	£	«libra / libras»
jpy	—	yen	¥	«yen / yenes»
cny	—	yuan	¥	«yuan / yuanes»
krw	—	won	₩	«won / wons»
ils	—	séquel	₪	«séquel / séqueles»
inr	—	rupia	₹	«rupia / rupias»
rub	—	rublo	₽	«rublo / rublos»
try	—	lira	₺	«lira / liras»
uah	—	grivna	₾	«grivna / grivnas»

3.4.1 Llaves para \spshort

`read-arg = {⟨verbalización NVDA⟩}` default: no usado

Establece la verbalización del argumento {⟨comandos $\LaTeX$ ⟩} de la nueva {⟨abreviatura⟩}, {⟨acrónimo⟩} o {⟨número ordinal⟩} para NVDA. El valor de esta llave siempre debe pasarse entre '{ }'. Por ejemplo: `\spshort[read-arg={octavo}]{8.\textsuperscript{\emph{avo}}}` creará una nueva {⟨abreviatura⟩} que imprimirá '8.^{avo}' en la salida PDF y NVDA verbalizará «octavo».

`small-caps = {⟨true | false⟩}` default: false

Establece si se usarán versalitas para imprimir {⟨acrónimos⟩} y para la «letra volada» en {⟨abreviaturas⟩} y {⟨números ordinales⟩} respetando las reglas de la RAE para el uso de estas. Si se establece en `true`, las abreviaturas con mayúscula como EE. UU. y los acrónimos de cinco o más letras como Unicef no se verán afectados; si se establece en `false`, no se aplicarán versalitas. Por ejemplo: `\spshort[small-caps=true]{dr.a}` imprimirá 'Dr.^a' en la salida PDF y NVDA verbalizará «doctora».

3.4.2 Argumentos para \spshort

El cuadro 2 resume las abreviaturas, expresiones latinas y títulos registrados en el módulo Lua `spintent.lua`.

Cuadro 2: Abreviaturas soportadas

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
Abreviaturas con letras voladas			
dr.a	dr. ^a	Dr. ^a	«doctora»
dr.as	—	Dr. ^{as}	«doctoras»
prof.a	—	Prof. ^a	«profesora»
d.a	d. ^a	D. ^a	«doña»
m.a	—	M. ^a	«maría»
n.o	n. ^o	N. ^o	«número»
n.os	—	N. ^{os}	«números»
c.ia	—	C. ^{ia}	«compañía»
Abreviaturas comunes y expresiones latinas			
pág.	pag.	pág.	«página»
vol.	—	vol.	«volumen»
etc.	—	etc.	«etcétera»
et al.	et. al.	et al.	«y otros»
ibíd.	ibid.	ibíd.	«ibídem»
op. cit.	op.cit.	op. cit.	«obra citada»
loc. cit.	loc.cit.	loc. cit.	«lugar citado»
v. gr.	v.gr.	v. gr.	«verbigracia»
i. e.	i.e.	i. e.	«esto es»
e. g.	e.g.	e. g.	«por ejemplo»
p. ej.	p.ej.	p. ej.	«por ejemplo»
Títulos, profesiones y cargos			
sr.	—	Sr.	«señor»
sra.	—	Sra.	«señora»
srta.	—	Srta.	«señorita»
dr.	—	Dr.	«doctor»
dra.	—	Dra.	«doctora»
dres.	—	Dres.	«doctores»
dras.	—	Dras.	«doctoras»
prof.	—	Prof.	«profesor»
profa.	—	Profa.	«profesora»
profs.	—	Profs.	«profesores»
ing.	—	Ing.	«ingeniero»
ings.	—	Ings.	«ingenieros»
lic.	—	Lic.	«licenciado»
v. b.	v.b.	V. B.	«visto bueno»
Abreviaturas compuestas e institucionales			
s. a.	s.a.	S. A.	«sociedad anónima»
ee. uu.	ee.uu.	EE. UU.	«estados unidos»
dd. hh.	dd.hh.	DD. HH.	«derechos humanos»
jj. oo.	jj.oo.	JJ. OO.	«juegos olímpicos»

Continúa en la siguiente página...

Cuadro 2: (Continuación - Abreviaturas soportadas)

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
ff. aa.	ff.aa.	FF. AA.	«fuerzas armadas»
rr. ee.	rr.ee.	RR. EE.	«relaciones exteriores»
cc. aa.	cc.aa.	CC. AA.	«comunidades autónomas»
p. d.	p.d.	P. D.	«posdata»
a. c.	a.c.	a. C.	«antes de Cristo»
d. c.	d.c.	d. C.	«después de Cristo»
d. n. i.	d.n.i.	D. N. I.	«documento nacional de identidad»
r. u. t.	r.u.t.	R. U. T.	«rol único tributario»
r. u. n.	r.u.n.	R. U. N.	«rol único nacional»

El cuadro 3 detalla las siglas y acrónimos registrados en el módulo Lua `spintent.lua`.

Cuadro 3: Siglas y acrónimos soportados

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
Siglas (Deletreo o lectura expandida)			
dni	—	DNI	«documento nacional de identidad»
rut	—	RUT	«rol único tributario»
run	—	RUN	«rol único nacional»
onu	—	ONU	«organización de las naciones unidas»
rae	—	RAE	«real academia española»
ong	ongs	ONG	«organización no gubernamental»
urss	—	URSS	«unión de repúblicas socialistas soviéticas»
oea	—	OEA	«organización de los estados americanos»
oms	—	OMS	«organización mundial de la salud»
fmi	—	FMI	«fondo monetario internacional»
bid	—	BID	«banco interamericano de desarrollo»
otan	—	OTAN	«organización del tratado del atlántico norte»
ue	—	UE	«unión europea»
ru	—	RU	«reino unido»
Acrónimos silábicos			
unicef	—	Unicef	«fondo de las naciones unidas para la infancia»
unesco	—	Unesco	«organización de las naciones unidas para la educación, la ciencia y la cultura»
mercosur	—	Mercosur	«mercado común del sur»
cepal	—	Cepal	«comisión económica para américa latina»
mineduc	—	Mineduc	«ministerio de educación»
conicet	—	Conicet	«consejo nacional de investigaciones»

El cuadro 4 muestra ejemplos de números ordinales registrados en el módulo Lua `spintent.lua`.

Cuadro 4: Números ordinales soportados

Entrada	Alias / Variantes	Salida PDF	Lectura NVDA
Sufijos para ordinales (Ejemplos)			
1.a	1. ^a	1. ^a	«primera»
2.o	2. ^o	2. ^o	«segundo»
3.er	—	3. ^{er}	«tercer»
4.os	—	4. ^{os}	«cuartos»
5.as	—	5. ^{as}	«quintas»

4 Utilidades generales

Además de los «cuatro grandes» y sus asociados, el paquete `spintent` provee tres utilidades de uso general: `\spsiglo` para la escritura y lectura de siglos, `\sptime` para la escritura y lectura de tiempo y `\spdate` para la escritura y lectura de fechas.

El comando `\spsiglo`

```
\spsiglo \spsiglo{<siglo>}
\spsiglo[<key = val>]{<siglo>}
```

El comando `\spsiglo` trabaja en *modo texto* y en *modo matemático*, recibe el *argumento obligatorio* `{<siglo>}`, el cual puede ser un *número arábigo* ‘21’ o un *número romano* ‘XXI’ mayor que ‘0’ y menor que ‘4000’.

Si no se proporciona el *argumento opcional* `[<key = val>]` y se ejecuta `\spsiglo{21}` o `\spsiglo{XXI}` se imprimirá ‘xxi’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno».

- El *argumento obligatorio* `{<siglo>}` «siempre» es convertido a *número romano en versalitas* y es verbalizado por NVDA/MathCAT como *número arábigo*. Esta es la única utilidad que provee el paquete `spintent` que trabaja en *modo texto* y en *modo matemático*. Internamente utiliza la llave `alt` proveída por el paquete `tagpdf` en *modo texto* y usa `\MathMLintent` para el *modo matemático*.

Llaves para `\spsiglo`

```
print-era = {<ac | dc | none>} default: none
```

Establece si se *imprime* o no la abreviatura después de imprimir el *argumento* `{<siglo>}`. Si se establece en `ac`, `\spsiglo[print-era=ac]{XXI}` imprimirá ‘xi a. C.’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno antes de cristo»; si se establece en `dc`, `\spsiglo[print-era=dc]{XXI}` imprimirá ‘xi d. C.’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno después de cristo»; si se establece en `none`, `\spsiglo[print-era=none]{XXI}` imprimirá ‘xi’ en la salida PDF y NVDA/MathCAT verbalizará «veintiuno».

El comando `\sptime`

```
\sptime \sptime{<horas : minutos>}
\sptime{<horas : minutos : segundos>}
```

El comando `\sptime` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<horas : minutos>}` o `{<horas : minutos : segundos>}`. Por ejemplo: `\sptime{23:45}` imprimirá ‘23:45’ en *modo texto* la salida PDF y NVDA/MathCAT verbalizará «las veintitrés y cuarenta y cinco minutos».

El comando `\spdate`

```
\spdate \spdate{<DD/ MM/YYYY>} \spdate{<DD- MM-YYYY>}
\spdate{<YYYY/ MM/DD>} \spdate{<YYYY- MM-DD>}
```

El comando `\spdate` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<DD/MM/YYYY>}`, `{<DD-MM-YYYY>}` o `{<YYYY-MM-DD>}`. Por ejemplo: `\spdate{1981-01-02}` y `\spdate{02/01/1981}` imprimirán en *modo texto* la salida PDF ‘1981-01-02’ y ‘02/01/1981’ respectivamente y en ambos casos, NVDA/MathCAT verbalizará «dos de enero de mil novecientos ochenta y uno».

- Si se pasa el *argumento* `{<YYYY/MM/DD>}`, internamente se convertirá en `{<DD/MM/YYYY>}` y se imprimirá de esa forma por compatibilidad con NVDA/MathCAT.

5 Utilidades para símbolos

El paquete `spintent` provee tres comandos `\spread`, `\spdsym` y `\spmsym` para trabajar con «símbolos matemáticos» descritos en esta sección. Están pensando desde el punto de vista docente, para situaciones donde la lectura de un «símbolo» (acorde al contexto) es relevante.

- A diferencia de las otras utilidades implementados por `spintent` (que analizan y dan formato a la salida), estos no procesan los «símbolos matemáticos», solo afectan la *forma* (semántica) en que serán verbalizados por NVDA/MathCAT.

El comando `\spread`

```
\spread \spread{<verbalización NVDA>}{<símbolo matemático>}
```

El comando `\spread` trabaja en *modo matemático* y recibe dos *argumentos obligatorios*: El primer *argumento* `{<verbalización NVDA>}` establece la «forma» en la cual NVDA/MathCAT verbalizará al segundo *argumento* `{<símbolo matemático>}`. Por ejemplo: `\spread{es-paralela}{\parallel}` imprimirá ‘||’ en la salida PDF y NVDA/MathCAT verbalizará «es paralela»; `\spread{es-paralelo}{\parallel}` imprimirá ‘||’ en la salida PDF y NVDA/MathCAT verbalizará «es paralelo».

El comando `\spusep`

```
\spusep <separador>
\spusep[<key = val>]{<separador>}
```

El comando `\spusep` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<separador>}`, el cual debe ser uno de los «*delimitadores*» soportados: ‘(’, ‘)’, ‘[’, ‘]’, ‘{’, ‘}’ o ‘\’. Imprime el *delimitador* indicado con el tamaño establecido por la llave `size` y controla si NVDA/MathCAT lo verbaliza o no mediante la llave `silent`. Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spusep{< >}` imprimirá ‘(’ en la salida PDF y NVDA/MathCAT verbalizará «*paréntesis izquierdo*».

- El motivo de existencia de `\spusep` es que los comandos `\left`, `\right`, `\Uleft` y `\Uright` provistos por LuaTeX crean un *grupo* implícito que dificulta el código para *tagged* PDF cuando se necesita interrumpir algo entre ellos, por ejemplo al construir expresiones complejas con `\spnZ`, `\spmQ` o `\spdiv`. El comando `\spusep` imprime el *delimitador* sin crear ningún grupo.

Llaves para `\spusep`

```
size = {<none | big | Big | bigg | Bigg>} default: none
```

Establece el *tamaño* del `{<delimitador>}` que se imprimirá en la salida PDF. Los valores corresponden a los cuatro tamaños fijos provistos por L^AT_EX: si se establece en `none`, el `{<delimitador>}` se imprime en el tamaño por defecto del modo matemático en curso; si se establece en `big`, se usa `\bigl/\bigr`; si se establece en `Big`, se usa `\Bigl/\Bigr`; si se establece en `bigg`, se usa `\biggl/\biggr`; y si se establece en `Bigg`, se usa `\Biggl/\Biggr`. El tamaño se aplica automáticamente en el lado correcto (izquierdo o derecho) según el `{<delimitador>}` pasado.

```
silent = {<true | false>} default: false
```

Establece si NVDA/MathCAT verbalizará el `{<delimitador>}` impreso por `\spusep`. Si se establece en `true`, el `{<delimitador>}` se marcará con el *intent* `:silent` y NVDA/MathCAT no lo verbalizará; si se establece en `false`, NVDA/MathCAT verbalizará el `{<delimitador>}` de forma normal.

El comando `\spdsym`

```
\spdsym
\spdsym[<key = val>]
```

El comando `\spdsym` trabaja en *modo matemático* y recibe el *argumento opcional* `[<key = val>]` mediante el cual se establece la «*forma*» en la que se imprimirá el *símbolo de división* y como se verbalizará en NVDA/MathCAT. Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spdsym` imprimirá ‘:’ en la salida PDF y NVDA/MathCAT verbalizará «*dividido*».

Llaves para `\spdsym`

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «*antes*» de verbalizar el `{<valor>}` establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

```
suffix = {<sufijo>} default: no usado
```

Establece el *sufijo* que será verbalizado por NVDA/MathCAT «*después*» de verbalizar el `{<valor>}` establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

```
set-sym = {<old-style | two-dots | slash>} default: two-dots
```

Establece el *símbolo de división* que se imprimirá en la salida PDF. Si se establece en `old-style`, imprimirá ‘÷’; si se establece en `two-dots`, imprimirá ‘:’ y si se establece `slash`, imprimirá ‘/’.

```
read-sym = {<dividido | dividido-por | dividido-entre>} default: dividido
```

Establece la *forma* con la cual NVDA/MathCAT verbalizará el *símbolo de división*. Si se establece en `dividido`, NVDA/MathCAT verbalizará «*dividido*»; si se establece en `dividido-por`, NVDA/MathCAT verbalizará «*dividido por*» y si se establece en `dividido-entre`, NVDA/MathCAT verbalizará «*dividido entre*».

El comando `\spmsym`

```
\spmsym
\spmsym[<key = val>]
```

El comando `\spmsym` trabaja en *modo matemático* y recibe el *argumento opcional* `[<key = val>]` mediante el cual se establece la «*forma*» en la que se imprimirá el *símbolo de multiplicación* y como se verbalizará en NVDA/MathCAT.

Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spmsym` imprimirá ‘·’ en la salida PDF y NVDA/MathCAT verbalizará «*por*».

Llaves para `\spmsym`

`prefix = {\langle prefijo \rangle}` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar el `\{ \langle valor \rangle \}` establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

`suffix = {\langle sufijo \rangle}` default: no usado

Establece el *sufijo* que será verbalizado por NVDA/MathCAT «después» de verbalizar el `\{ \langle valor \rangle \}` establecido por la llave `read-sym`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

`set-sym = {\langle cross | dot | asterisk \rangle}` default: dot

Establece el *símbolo de multiplicación* que se imprimirá en la salida PDF. Si se establece en `cross`, imprimirá `×`; si se establece en `dot`, imprimirá `·` y si se establece en `asterisk`, imprimirá `*`.

`read-sym = {\langle por | multiplicado-por | veces \rangle}` default: por

Establece la *forma* con la cual NVDA/MathCAT verbalizará el *símbolo de multiplicación*. Si se establece en `por`, NVDA/MathCAT verbalizará «por»; si se establece en `multiplicado-por`, NVDA/MathCAT verbalizará «multiplicado por» y si se establece en `veces`, NVDA/MathCAT verbalizará «veces».

`not-print = {\langle true | false \rangle}` default: false

Deshabilita la *impresión* del *símbolo de multiplicación*. Cuando se establece esta llave, NVDA/MathCAT verbalizará el *símbolo de multiplicación* acorde al valor establecido por la llave `read-sym`, pero, no se imprimirá este en la salida PDF.

6 Utilidades para números

El paquete `spintent` provee utilidades para trabajar con números descritas en esta sección.

El comando `\spdiv`

`\spdiv` `\spdiv{\langle dividendo \rangle}{\langle divisor \rangle}`
`\spdiv[\langle key = val \rangle]{\langle dividendo \rangle}{\langle divisor \rangle}`

El comando `\spdiv` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* `\{ \langle dividendo \rangle \}` y `\{ \langle divisor \rangle \}`.

Si no se proporciona el *argumento opcional* `[\langle key = val \rangle]`, `\$ \spdiv{20}{26} \$` imprimirá `20 : 26` en la salida PDF y NVDA/MathCAT verbalizará «veinte dividido veinte y seis»

Llaves para `\spdiv`

El comando `\spdiv` posee las *meta-keys* `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las *meta-keys* `set-sym` y `read-sym` pasadas al comando `\spmsym` (pág. 12).

`wrap-parens = {\langle true | false \rangle}` default: false

Establece si se *imprimen* paréntesis redondos `(...)` alrededor de los *argumentos* `\{ \langle dividendo \rangle \}` y `\{ \langle divisor \rangle \}` pasados a `\spdiv`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens = {\langle true | false \rangle}` default: true

Establece si NVDA/MathCAT verbalizará los paréntesis redondos `(...)` alrededor de los *argumentos* `\{ \langle dividendo \rangle \}` y `\{ \langle divisor \rangle \}` pasados a `\spdiv`. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en `false`, NVDA/MathCAT no los verbalizará.

El comando `\spmul`

`\spmul` `\spmul{\langle factor \rangle}{\langle factor \rangle}`
`\spmul[\langle key = val \rangle]{\langle factor \rangle}{\langle factor \rangle}`

El comando `\spmul` trabaja en *modo matemático* y recibe dos *argumentos obligatorios* `\{ \langle factor \rangle \}` y `\{ \langle factor \rangle \}`.

Si no se proporciona el *argumento opcional* `[\langle key = val \rangle]`, `\$ \spmul{20}{26} \$` imprimirá `20 · 26` en la salida PDF y NVDA/MathCAT verbalizará «veinte por veinte y seis».

Llaves para `\spmul`

El comando `\spmul` posee las *meta-keys* `cifras-sep`, `read-sign`, `read-period-sep` y `notation-sym` pasadas al comando `\spnum` (§3.1.1) junto a las *meta-keys* `set-sym` y `read-sym` pasadas al comando `\spmsym` (pág. 13).

`wrap-parens = {\langle true | false \rangle}` default: false

Establece si se *imprimen* paréntesis redondos `(...)` alrededor de los *argumentos* `\{ \langle factor \rangle \}` y `\{ \langle factor \rangle \}` pasados a `\spmul`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor de ambos *argumentos*; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens = {\langle true | false \rangle}` default: true

Establece si NVDA/MathCAT verbalizará los paréntesis redondos ‘(…)’ alrededor de los *argumentos* $\{ \langle \textit{factor} \rangle \}$ y $\{ \langle \textit{factor} \rangle \}$ pasados a `\spmul`. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en `false`, NVDA/MathCAT no los verbalizará.

El comando `\spnD`

```
\spnD \spnD
\spnD(\<número natural>)
\spnD[key = val](\<número natural>)
```

El comando `\spnD` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* $(\langle \textit{número natural} \rangle)$, por ejemplo: ‘(6)’. Si se usa sin $(\langle \textit{argumento} \rangle)$, `\spnD` imprimirá ‘D(*n*)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de *n*».

Si se usa con $(\langle \textit{argumento} \rangle)$, por ejemplo, `\spnD(6)`, imprimirá ‘D(6)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de seis».

Llaves para `\spnD`

`prefix = { \<prefijo> }` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar `\spnD(\<argumento>)`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result = { \<true | false> }` default: false

Establece si se *imprime* el resultado de calcular los *divisores* del $(\langle \textit{número natural} \rangle)$ pasado a `\spnD`. Si se establece en `true`, `\spnD[result=true](6)` imprimirá ‘D(6) = {1, 2, 3, 6}’ en la salida PDF y NVDA/MathCAT verbalizará «los divisores de seis son uno, dos, tres y seis»; si se establece en `false`, no se imprimirá en la salida PDF el resultado de calcular los *divisores*.

`result-num = { \<número natural> }` default: 5

Establece la *cantidad* de números que se mostrarán en la salida PDF al activar `result=true`. Si se establece en `2`, solo se imprimirán dos valores en la salida PDF; si no se establece, se imprimirán como máximo `5` valores y luego se añadirá ‘...’ en la salida PDF.

`sample-arg = { \<true | false> }` default: false

Desactiva la *validación* interna del $(\langle \textit{argumento} \rangle)$ pasado a `\spnD`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en `true`, `\spnD[sample-arg=true](m)` imprimirá ‘D(*m*)’ en la salida PDF y NVDA/MathCAT verbalizará «divisores de *m*»; si se establece en `false`, el $(\langle \textit{argumento} \rangle)$ debe ser un número natural, en caso contrario se devolverá un “error”.

`silent-cmd = { \<true | false> }` default: false

Establece si se *silencia* la verbalización para NVDA/MathCAT de `\spnD(\<argumento>)` cuando se establece `sample-arg=true` o cuando `\spnD` se usa sin $(\langle \textit{argumento} \rangle)$. Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spnM`

```
\spnM \spnM
\spnM(\<número natural>)
\spnM[key = val](\<número natural>)
```

El comando `\spnM` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* $(\langle \textit{número natural} \rangle)$, por ejemplo: ‘(6)’. Si se usa sin $(\langle \textit{argumento} \rangle)$, `\spnM` imprimirá ‘M(*n*)’ en la salida PDF y NVDA/MathCAT verbalizará «múltiplos de *n*».

Si se usa con $(\langle \textit{argumento} \rangle)$, por ejemplo, `\spnM(6)`, imprimirá ‘M(6)’ en la salida PDF y NVDA/MathCAT verbalizará «múltiplos de seis».

Llaves para `\spnM`

`prefix = { \<prefijo> }` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar `\spnM(\<argumento>)`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result = { \<true | false> }` default: false

Establece si se *imprime* el resultado de calcular los *múltiplos* del $(\langle \textit{número natural} \rangle)$ pasado a `\spnM`. Si se establece en `true`, `\spnM[result=true](6)` imprimirá ‘M(6) = {6, 12, 18, 24, 30, ...}’ en la salida PDF y NVDA/MathCAT verbalizará «los múltiplos de seis son seis, doce, dieciocho, veinticuatro, treinta puntos suspensivos»; si se establece en `false`, no se imprimirá en la salida PDF el resultado de calcular los *múltiplos*.

`result-num = { \<número natural> }` default: 5

Establece la *cantidad* de números que se mostrarán en la salida PDF al activar `result=true`. Si se establece en `2`, solo se imprimirán dos valores en la salida PDF seguidos de ‘...’; si no se establece, se imprimirán por defecto los `5` primeros valores y luego se añadirá ‘...’ en la salida PDF.

`sample-arg = { \<true | false> }` default: false

Desactiva la *validación* interna del (*argumento*) pasado a `\spnM`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en `true`, `\spnM[sample-arg=true](m)` imprimirá ‘M(*m*)’ en la salida PDF y NVDA/MathCAT verbalizará «múltiplos de *m*»; si se establece en `false`, el (*argumento*) debe ser un número natural, en caso contrario se devolverá un “error”.

`silent-cmd = { (true | false) }` default: false

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spnM`(*argumento*) cuando se establece `sample-arg=true` o cuando `\spnM` se usa sin (*argumento*). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT lo verbalizará.

El comando `\spmc`

```
\spmc \spmc
\spmc( (números separados por coma) )
\spmc[ (key = val) ] ( (números separados por coma) )
```

El comando `\spmc` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (*números separados por coma*), el cual debe ser una lista de *números naturales*, por ejemplo: ‘(2, 3, 6)’. Si se usa sin (*argumento*) `\spmc` imprimirá ‘mcd’ en la salida PDF, y NVDA/MathCAT verbalizará «máximo común divisor» o «*m c d*», acorde al {*valor*} establecido por la llave `read-cmd`.

Si se usa con (*argumento*) `\spmc(2, 3, 6)` imprimirá ‘mcd(2, 3, 6)’ en la salida PDF, y NVDA/MathCAT verbalizará «máximo común divisor entre dos, tres y seis» o «*m c d* entre dos, tres y seis» acorde al {*valor*} establecido por la llave `read-cmd`.

Llaves para `\spmc`

`upper-case = { (true | false) }` default: false

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmc`. Si se establece en `true`, se imprimirá ‘MCD’ en la salida PDF; si se establece en `false`, se imprimirá ‘mcd’ en la salida PDF.

`read-cmd = { (terse | clear) }` default: clear

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmc`. Si se establece en `terse`, NVDA/MathCAT verbalizará «*m c d*»; si se establece en `clear`, NVDA/MathCAT verbalizará «máximo común divisor».

`prefix = { (prefijo) }` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «antes» de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result = { (true | false) }` default: false

Establece si se *imprime* el resultado de calcular el «máximo común divisor» del (*argumento*) pasado a `\spmc`. Si se establece en `true`: `\spmc[result=true](2, 3, 6)` imprimirá ‘mcd(2, 3, 6) = 1’ en la salida PDF y NVDA/MathCAT verbalizará «máximo común divisor entre dos, tres y seis es igual a uno»; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el «máximo común divisor».

`sample-arg = { (true | false) }` default: false

Desactiva la *validación* interna del (*argumento*) pasado a `\spmc`, permitiendo usar ejemplos «no numéricos» como entrada. Si se establece en `true`: `\spmc[sample-arg=true](m, n)`, imprimirá ‘mcd(*m*, *n*)’ en la salida PDF y NVDA/MathCAT verbalizará «máximo común divisor entre *m* y *n*»; si se establece en `false` el (*argumento*) debe ser una lista de *números naturales*, en caso contrario se devolverá un “error”.

`silent-cmd = { (true | false) }` default: false

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spmc`(*argumento*) cuando se establece `sample-arg=true` o cuando `\spmc` se usa sin (*argumento*). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spmc`

```
\spmc \spmc
\spmc( (números separados por coma) )
\spmc[ (key = val) ] ( (números separados por coma) )
```

El comando `\spmc` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis* (*números separados por coma*), el cual debe ser una lista de *números naturales*, por ejemplo: ‘(2, 3, 6)’. Si se usa sin (*argumento*) `\spmc` imprimirá ‘mcm’ en la salida PDF, y NVDA/MathCAT verbalizará «mínimo común múltiplo» o «*m c m*», acorde al {*valor*} establecido por la llave `read-cmd`.

Si se usa con (*argumento*) `\spmc(2, 3, 6)` imprimirá ‘mcm(2, 3, 6)’ en la salida PDF, y NVDA/MathCAT verbalizará «mínimo común múltiplo entre dos, tres y seis» o «*m c m* entre dos, tres y seis» acorde al {*valor*} establecido por la llave `read-cmd`.

Llaves para `\spmc`

`upper-case` = {`<true>` | `<false>`} default: `false`

Establece si se usarán *mayúsculas* o *minúsculas* en la salida PDF para el comando `\spmc`. Si se establece en `true` se imprimirá ‘MCM’ en la salida PDF; si se establece en `false` se imprimirá ‘mcm’ en la salida PDF.

`read-cmd` = {`<terse>` | `<clear>`} default: `clear`

Establece la *forma* en la que NVDA/MathCAT verbalizará el comando `\spmc`. Si se establece en `terse`, NVDA/MathCAT verbalizará «*m c m*»; si se establece en `clear`, NVDA/MathCAT verbalizará «*mínimo común múltiplo*».

`prefix` = {`<prefijo>`} default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT «*antes*» de verbalizar el valor establecido por la llave `read-cmd`. El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`result` = {`<true>` | `<false>`} default: `false`

Establece si se *imprime* el resultado de calcular el «*mínimo común múltiplo*» del (`<argumento>`) pasado a `\spmc`. Si se establece en `true`: `\spmc[result=true](2, 3, 6)` imprimirá ‘mcm(2, 3, 6) = 6’ en la salida PDF y NVDA/MathCAT verbalizará «*mínimo común múltiplo entre dos, tres y seis es igual a seis*»; si se establece en `false` no se imprimirá en la salida PDF el resultado de calcular el «*mínimo común múltiplo*».

`sample-arg` = {`<true>` | `<false>`} default: `false`

Desactiva la *validación* interna del (`<argumento>`) pasado a `\spmc`, permitiendo usar ejemplos «*no numéricos*» como entrada. Si se establece en `true`: `\spmc[sample-arg=true](m, n)`, imprimirá ‘mcm(*m*, *n*)’ en la salida PDF y NVDA/MathCAT verbalizará «*mínimo común múltiplo entre m y n*»; si se establece en `false` el (`<argumento>`) debe ser una lista de *números naturales*, en caso contrario se devolverá un “error”.

`silent-cmd` = {`<true>` | `<false>`} default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT de `\spmc(<argumento>)` cuando se establece `sample-arg=true` o cuando `\spmc` se usa sin (`<argumento>`). Si se establece en `true` NVDA/MathCAT NO lo verbalizará; si se establece en `false` NVDA/MathCAT SÍ lo verbalizará.

El comando `\spnZ`

`\spnZ` `\spnZ{<número entero>}`
`\spnZ[<key = val>]{<número entero>}`

El comando `\spnZ` trabaja en *modo matemático* y recibe el *argumento* obligatorio {`<número entero>`} el cual se procesa internamente bajo el comando `\spnum`. Este comando está dedicado exclusivamente al manejo de «*números enteros*», y añade algunas utilidades prácticas para el demarcado e interacción con NVDA/MathCAT.

Llaves para `\spnZ`

El comando `\spnZ` posee las *meta-keys* `cifras-sep` y `read-sign` pasadas al comando `\spnum` (§3.1.1).

`wrap-parens` = {`<true>` | `<false>`} default: `false`

Establece si se *imprimen* paréntesis redondos ‘(…)’ alrededor del argumento {`<número entero>`} pasado a `\spnZ`. Si se establece en `true`, se imprimirán paréntesis redondos alrededor del {`<número entero>`}; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens` = {`<true>` | `<false>`} default: `false`

Establece si NVDA/MathCAT *verbalizará* los paréntesis redondos ‘(…)’ alrededor del *argumento* {`<número entero>`} pasado a `\spnZ`. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura del paréntesis ‘(…)»; si se establece en `false`, NVDA/MathCAT no los verbalizará.

El comando `\spmQ`

`\spmQ` `\spmQ{<número entero>}{<numerador>}{<denominador>}`
`\spmQ[<key = val>]{<número entero>}{<numerador>}{<denominador>}`

El comando `\spmQ` trabaja en *modo matemático* y recibe *tres argumentos obligatorios*: {`<número entero>`}, {`<numerador>`} y {`<denominador>`}, los procesa internamente y devuelve un «*número mixto*».

Si no se proporciona el *argumento opcional* [`<key = val>`], `\spmQ{2}{5}{7}` imprimirá ‘ $2\frac{5}{7}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «*dos y cinco séptimos*».

- En el caso de las fracciones, y en especial con los «*números mixtos*», no se pueden manejar muchas cosas desde el lado de **spintent**. Por ejemplo, la entrada `\frac{5}{7}` con *tagged* PDF se verbaliza por NVDA/MathCAT como «*tres más cinco séptimos*», lo cual no está mal, pero pedagógicamente hablando no es del todo correcto.

Llaves para `\spmQ`

`join-word = {⟨entero | enteros | y⟩}`

default: `y`

Establece la *forma* en la que NVDA/MathCAT verbaliza la conexión entre el entero y la fracción que forman el *número mixto*. Si se establece en `entero`, al usar `\spmQ[join-word=entero]{1}{5}{7}` imprimirá $1\frac{5}{7}$ en la salida PDF, y NVDA/MathCAT verbalizará «un entero cinco séptimos»; si se establece en `enteros`, al usar `\spmQ[join-word=enteros]{2}{5}{7}` imprimirá $2\frac{5}{7}$ en la salida PDF, y NVDA/MathCAT verbalizará «dos enteros cinco séptimos»; si se establece en `y`, al usar `\spmQ[join-word=y]{2}{5}{7}` imprimirá $2\frac{5}{7}$ en la salida PDF, y NVDA/MathCAT verbalizará «dos y cinco séptimos».

`use-spnZ = {⟨true | false⟩}`

default: `true`

Establece si el `{⟨numerador⟩}` y el `{⟨denominador⟩}` que forman la fracción del *número mixto* se procesan internamente bajo el comando `\spnZ`, añadiendo así *semántica de entero* en la estructura MathML de la salida PDF. Si se establece en `true`, `{⟨numerador⟩}` y `{⟨denominador⟩}` se pasarán a `\spnZ` y NVDA/MathCAT los verbalizará correctamente como *números ordinales*; si se establece en `false`, los argumentos se tipografían directamente sin procesamiento semántico adicional.

`print-dfrac = {⟨true | false⟩}`

default: `false`

Establece si la fracción del *número mixto* se imprime usando `\dfrac` en lugar de `\frac`. Si se establece en `true`, al usar `\spmQ[print-dfrac=true]{2}{5}{7}` imprimirá la fracción en tamaño *display* $2\frac{5}{7}$ en la salida PDF; si se establece en `false`, la fracción se imprime con `\frac` siguiendo el tamaño matemático del contexto en curso.

`wrap-parens = {⟨true | false⟩}`

default: `false`

Establece si se *imprimen* paréntesis redondos ‘(...)’ alrededor del *número mixto* completo devuelto por `\spmQ`. Si se establece en `true`, se imprimirán paréntesis redondos cuyo tamaño se ajusta automáticamente al estilo matemático en curso; si se establece en `false`, no se imprimirán los paréntesis.

`read-parens = {⟨true | false⟩}`

default: `true`

Establece si NVDA/MathCAT verbalizará los paréntesis redondos ‘(...)’ alrededor del *número mixto* cuando la llave `wrap-parens` está activa. Si se establece en `true`, NVDA/MathCAT verbalizará la lectura de los paréntesis; si se establece en `false`, NVDA/MathCAT no los verbalizará.

7 Utilidades para geometría plana

El paquete `spintent` provee comandos para trabajar con los objetos y medidas de la *geometría plana* descritos en esta sección. Todos los comandos trabajan en *modo matemático* y comparten el mismo módulo `spintent.lua` para construir la verbalización correcta.

- Todos los comandos de esta sección aceptan como puntos solo *letras mayúsculas* como base, con modificadores opcionales de subíndice ‘`_{\dots}`’ o prima ‘`’`’. La única excepción es `\sprecta` cuando el *argumento* es de la forma `{⟨letra⟩}`, que acepta *letras minúsculas*. Los subíndices siempre deben escribirse de la forma correcta usando `_{\dots}` como `A_{\dots}` y nunca como `A_n` sin llaves que funciona solo por coincidencia.

El comando `\spPoint`

`\spPoint {⟨punto⟩}`

`\spPoint [⟨key = val⟩] {⟨punto⟩}`

El comando `\spPoint` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{⟨punto⟩}` el cual es una *única letra mayúscula* con un modificador opcional de subíndice ‘`_{\dots}`’ o prima ‘`’`’.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\spPoint{A}` imprimirá ‘A’ en la salida PDF y NVDA/MathCAT verbalizará «punto A»; `\spPoint{A_{1}}` imprimirá ‘A₁’ y verbalizará «punto A sub uno».

- El comando `\spPoint` solo acepta *un punto*. Para pares o listas de puntos (rectas, rayos, lados, ángulos, arcos, polígonos), véanse los comandos correspondientes de esta sección.

Llaves para `\spPoint`

`read-cmd = {⟨school | mathcat⟩}`

default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{⟨punto⟩}`. Si se establece en `school`, NVDA/MathCAT verbalizará el argumento `{⟨punto⟩}` sin el prefijo «mayúscula», `\spPoint{A}` se verbaliza «punto A»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}`

default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «punto». El valor de esta llave *siempre* debe pasarse entre llaves ‘`{ }`’.

`silent-cmd = {⟨true | false⟩}`

default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «punto». Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\sprecta`

```
\sprecta <punto, punto> <letra>
\sprecta[key = val]{<punto, punto> <letra>}
```

El comando `\sprecta` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{<letra>}` o `{<punto, punto>}`. Si el *argumento* es de la forma `{<punto, punto>}` se aplica `\overleftrightharpoonrightarrow` en la impresión; si el *argumento* no contiene coma y es de la forma `{<letra>}` (mayúscula o minúscula), se imprimirá solo la `{<letra>}` sin aplicar `\overleftrightharpoonrightarrow`.

Si no se proporciona el *argumento opcional* `[key = val]`, `\sprecta{A, B}` imprimirá $\overrightarrow{AB}$ en la salida PDF y NVDA/MathCAT verbalizará «recta A B»; `\sprecta{m}` imprimirá m en la salida PDF y NVDA/MathCAT verbalizará «recta m».

Llaves para `\sprecta`

```
read-cmd = {<school | mathcat>} default: school
```

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{<punto, punto>}` y el argumento `{<letra>}` cuando está en mayúscula. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sprecta{A, B}` se verbaliza «recta A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «recta». El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

```
silent-cmd = {<true | false>} default: false
```

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «recta». Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\sprayo`

```
\sprayo <punto, punto>
\sprayo[key = val]{<punto, punto>}
```

El comando `\sprayo` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<punto, punto>}`. Representa un *rayo* o *semirrecta* con origen en el primer punto y dirección hacia el segundo, aplicando `\overrightarrow` en la impresión.

Si no se proporciona el *argumento opcional* `[key = val]`, `\sprayo{A, B}` imprimirá $\overrightarrow{AB}$ en la salida PDF y NVDA/MathCAT verbalizará «rayo A B».

Llaves para `\sprayo`

```
read-cmd = {<school | mathcat>} default: school
```

Establece la *forma* en la que NVDA/MathCAT verbalizará el argumento `{<punto, punto>}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sprayo{A, B}` se verbaliza «rayo A B»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves `{ }`.

```
silent-cmd = {<true | false>} default: false
```

Establece si se *silencia* la verbalización en NVDA/MathCAT del término establecido por la llave `read-sty`. Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

```
read-sty = {<rayo | semirrecta>} default: rayo
```

Establece el *término* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se utiliza `\sprayo[read-sty=rayo]{A, B}`, NVDA/MathCAT verbalizará «rayo A B»; si se utiliza `\sprayo[read-sty=semirrecta]{A, B}`, NVDA/MathCAT verbalizará «semirrecta A B».

- La impresión $\overrightarrow{AB}$ en la salida PDF es idéntica en ambos casos. La elección depende de la terminología del libro de texto o del currículo nacional: en Chile y España se prefiere «semirrecta»; en México y otros países de América Latina se usa «rayo».

El comando `\spside`

```
\spside <punto, punto>
\spside[key = val]{<punto, punto>}
```

El comando `\spside` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<punto, punto>}`. Representa un *lado*, *segmento* o *trazo* determinado por dos puntos, aplicando `\overline` en la impresión.

Si no se proporciona el *argumento opcional* `[key = val]`, `\spside{A, B}` imprimirá $\overline{AB}$ en la salida PDF y NVDA/MathCAT verbalizará «lado A B».

Llaves para `\spside`

`read-cmd` = { `<school | mathcat>` }

default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento { `<punto, punto>` }. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$(\spside{A, B})$` se verbaliza «*lado A B*»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = { `<prefijo>` }

default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`silent-cmd` = { `<true | false>` }

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término establecido por la llave `read-sty`. Si se establece en *true*, NVDA/MathCAT NO verbalizará el término; si se establece en *false*, NVDA/MathCAT SÍ lo verbalizará.

`read-sty` = { `<lado | segmento | trazo>` }

default: *lado*

Establece el *término* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se utiliza `$(\spside[read-sty=lado]{A, B})$`, NVDA/MathCAT verbalizará «*lado A B*»; si se utiliza `$(\spside[read-sty=segmento]{A, B})$`, NVDA/MathCAT verbalizará «*segmento A B*» y si se utiliza `$(\spside[read-sty=trazo]{A, B})$`, NVDA/MathCAT verbalizará «*trazo A B*».

`read-arg` = { `<verbalización NVDA>` }

default: *no usado*

Sobrescribe el *término* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'. Por ejemplo: `$(\spside[read-arg={hipotenusa}]{A, B})$` imprimirá ' $\overline{AB}$ ' en la salida PDF y NVDA/MathCAT verbalizará «*hipotenusa A B*».

El comando `\spmside`

`\spmside` `\spmside{<punto, punto>}`

`\spmside[<key = val>]{<punto, punto>}`

El comando `\spmside` trabaja en *modo matemático* y recibe el *argumento obligatorio* { `<punto, punto>` }. Representa la *medida* del *lado*, *segmento* o *trazo* determinado por dos puntos. La distinción entre `\spside` y `\spmside` es semántica: `\spside` representa el *objeto geométrico* ' $\overline{AB}$ '; `\spmside` representa su *medida* (un número real positivo).

Si no se proporciona el *argumento opcional* [`<key = val>`], `$(\spmside{A, B})$` imprimirá ' $\overline{AB}$ ' en la salida PDF y NVDA/MathCAT verbalizará «*medida del lado A B*».

Llaves para `\spmside`

`read-cmd` = { `<school | mathcat>` }

default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento { `<punto, punto>` }. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `$(\spmside{A, B})$` se verbaliza «*medida del lado A B*»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix` = { `<prefijo>` }

default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar «*medida del*» seguido por el término establecido por la llave `read-sty`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`silent-cmd` = { `<true | false>` }

default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT de «*medida del*» seguido por el término establecido por la llave `read-sty`. Si se establece en *true*, NVDA/MathCAT NO verbalizará «*medida del*» seguido por el término; si se establece en *false*, NVDA/MathCAT SÍ lo verbalizará.

`read-sty` = { `<lado | segmento | trazo>` }

default: *lado*

Establece el *término* que verbalizará NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. Si se utiliza `$(\spmside[read-sty=lado]{A, B})$`, NVDA/MathCAT verbalizará «*medida del lado A B*»; si se utiliza `$(\spmside[read-sty=segmento]{A, B})$`, NVDA/MathCAT verbalizará «*medida del segmento A B*» y si se utiliza `$(\spmside[read-sty=trazo]{A, B})$`, NVDA/MathCAT verbalizará «*medida del trazo A B*».

`read-arg` = { `<verbalización NVDA>` }

default: *no usado*

Sobrescribe el *término* establecido por la llave `read-sty` que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'. Por ejemplo: `$(\spmside[read-arg={medida cateto}]{A, B})$` imprimirá ' $\overline{AB}$ ' en la salida PDF y NVDA/MathCAT verbalizará «*medida cateto A B*».

`print-m` = { `<true | false>` }

default: *false*

Establece si se *imprime* la notación visual 'm' antes del argumento. Si se establece en *true*, `$(\spmside[print-m=true]{A, B})$` imprimirá ' $m\overline{AB}$ ' en la salida PDF, si se establece en *false* imprimirá ' $\overline{AB}$ ' en la salida PDF. En ambos casos NVDA/MathCAT verbalizará «*medida del lado A B*».

El comando `\spangle`

<code>\spangle</code>	<code>\spangle{⟨punto letra griega número⟩}</code>	<code>\spangle{⟨punto, punto, punto⟩}</code>
	<code>\spangle[⟨key = val⟩]{⟨punto letra griega número⟩}</code>	<code>\spangle[⟨key = val⟩]{⟨punto, punto, punto⟩}</code>

El comando `\spangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{⟨punto | letra griega | número⟩}` o `{⟨punto, punto, punto⟩}` aplicando `\angle` o `\rightangle` en la impresión.

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\spangle{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo en A»; `\spangle{\alpha}` imprimirá ‘ $\angle \alpha$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo alfa» y `\spangle{A, O, B}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo A O B».

Llaves para `\spangle`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{⟨punto⟩}` o `{⟨punto, punto, punto⟩}`. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spangle{A, O, B}` se verbaliza «ángulo A O B»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «ángulo». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «ángulo». Si se establece en *true*, NVDA/MathCAT NO verbalizará el término; si se establece en *false*, NVDA/MathCAT SÍ lo verbalizará.

`recto = {⟨true | false⟩}` default: *false*

Establece si se usará `\rightangle` en la impresión añadiendo el término «recto» a la verbalización en NVDA/MathCAT. Si se establece en *true*, `\spangle[recto=true]{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo recto en A», `\spangle[recto=true]{A, O, B}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «ángulo recto A O B»; si se establece en *false*, imprimirá usando `\angle` en la salida PDF y NVDA/MathCAT omitirá el término «recto».

El comando `\spmangle`

<code>\spmangle</code>	<code>\spmangle{⟨punto letra griega número⟩}</code>	<code>\spmangle{⟨punto, punto, punto⟩}</code>
	<code>\spmangle[⟨key = val⟩]{⟨punto letra griega número⟩}</code>	<code>\spmangle[⟨key = val⟩]{⟨punto, punto, punto⟩}</code>

El comando `\spmangle` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{⟨punto | letra griega | número⟩}` o `{⟨punto, punto, punto⟩}` aplicando `\measuredangle` o `\measuredrightangle` en la impresión. La distinción entre `\spangle` y `\spmangle` es semántica: `\spangle` representa el *objeto geométrico* ‘ $\angle AOB$ ’; `\spmangle` representa su *medida* (un valor en grados).

Si no se proporciona el *argumento opcional* `[⟨key = val⟩]`, `\spmangle{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo en A»; `\spmangle{\alpha}` imprimirá ‘ $\angle \alpha$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo alfa» y `\spmangle{A, O, B}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo A O B».

Llaves para `\spmangle`

`read-cmd = {⟨school | mathcat⟩}` default: *school*

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{⟨punto⟩}` o `{⟨punto, punto, punto⟩}`. Si se establece en *school*, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spmangle{A, O, B}` se verbaliza «medida del ángulo A O B»; si se establece en *mathcat*, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {⟨prefijo⟩}` default: *no usado*

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «medida del ángulo». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`silent-cmd = {⟨true | false⟩}` default: *false*

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del ángulo». Si se establece en *true*, NVDA/MathCAT NO verbalizará el término; si se establece en *false*, NVDA/MathCAT SÍ lo verbalizará.

`recto = {⟨true | false⟩}` default: *false*

Establece si se usará `\measuredrightangle` en la impresión añadiendo el término «recto» a la verbalización en NVDA/MathCAT. Si se establece en *true*, `\spmangle[recto=true]{A}` imprimirá ‘ $\angle A$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo recto en A», `\spmangle[recto=true]{A, O, B}` imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo recto A O B».

imprimirá ‘ $\angle AOB$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del ángulo recto $A O B$ »; si se establece en `false`, imprimirá usando `\measuredangle` en la salida PDF y NVDA/MathCAT omitirá el término «recto».

`print-m = {\true | false}` default: `false`

Establece si se *imprime* la notación visual ‘m’ antes del argumento. Si se establece en `true`, `\spmangle[print-m=true]{A}`, imprimirá ‘ $m\angle AOB$ ’ en la salida PDF, si se establece en `false` imprimirá ‘ $\angle AOB$ ’ en la salida PDF. En ambos casos NVDA/MathCAT verbalizará «medida del ángulo $A O B$ ».

El comando `\spang`

```
\spang \spang{\langle grados \rangle}
\spang \spang{\langle grados : minutos \rangle}
\spang \spang{\langle grados : minutos : segundos \rangle}
```

El comando `\spang` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{\langle grados \rangle}`, `{\langle grados : minutos \rangle}` o `{\langle grados : minutos : segundos \rangle}`. Los procesa internamente bajo los comandos `\spnum` y `\spunit` para imprimir y verbalizar el «ángulo sexagesimal». Por ejemplo, `\spang{45:30:15}` imprimirá ‘ $45^{\circ}30'15''$ ’ en la salida PDF y NVDA/MathCAT verbalizará «cuarenta y cinco grados treinta minutos quince segundos».

El comando `\sparc`

```
\sparc \sparc{\langle punto, punto \rangle} \sparc{\langle punto, punto, punto \rangle}
\sparc \sparc[\langle key = val \rangle]{\langle punto, punto \rangle} \sparc[\langle key = val \rangle]{\langle punto, punto, punto \rangle}
```

El comando `\sparc` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{\langle punto, punto \rangle}` o `{\langle punto, punto, punto \rangle}`. Representa un arco determinado por dos o tres puntos aplicando `\widearc` en la impresión.

Si no se proporciona el *argumento opcional* `[\langle key = val \rangle]`, `\sparc{A, B}` imprimirá ‘ $\widehat{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «arco $A B$ ».

Llaves para `\sparc`

`read-cmd = {\langle school | mathcat \rangle}` default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{\langle punto, punto \rangle}` o `{\langle punto, punto, punto \rangle}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\sparc{A, B}` se verbaliza «arco $A B$ »; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {\langle prefijo \rangle}` default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «arco». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`silent-cmd = {\true | false}` default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «arco». Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spmarc`

```
\spmarc \spmarc{\langle punto, punto \rangle} \spmarc{\langle punto, punto, punto \rangle}
\spmarc \spmarc[\langle key = val \rangle]{\langle punto, punto \rangle} \spmarc[\langle key = val \rangle]{\langle punto, punto, punto \rangle}
```

El comando `\spmarc` trabaja en *modo matemático* y recibe un *argumento obligatorio* el cual puede ser de la forma `{\langle punto, punto \rangle}` o `{\langle punto, punto, punto \rangle}`. Representa la *medida* del arco determinado por dos o tres puntos. La distinción entre `\sparc` y `\spmarc` es semántica: `\sparc` representa el *objeto geométrico* ‘ $\widehat{AB}$ ’; `\spmarc` representa su *medida* (un valor en grados).

Si no se proporciona el *argumento opcional* `[\langle key = val \rangle]`, `\spmarc{A, B}` imprimirá ‘ $m\widehat{AB}$ ’ en la salida PDF y NVDA/MathCAT verbalizará «medida del arco $A B$ ».

Llaves para `\spmarc`

`read-cmd = {\langle school | mathcat \rangle}` default: `school`

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{\langle punto, punto \rangle}` o `{\langle punto, punto, punto \rangle}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spmarc{A, B}` se verbaliza «medida del arco $A B$ »; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {\langle prefijo \rangle}` default: `no usado`

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «medida del arco». El valor de esta llave *siempre* debe pasarse entre llaves ‘{ }’.

`silent-cmd = {\true | false}` default: `false`

Establece si se *silencia* la verbalización en NVDA/MathCAT del término «medida del arco». Si se establece en `true`, NVDA/MathCAT NO verbalizará el término; si se establece en `false`, NVDA/MathCAT SÍ lo verbalizará.

El comando `\spPoly`

```
\spPoly {<punto, punto, punto, ...>}
\spPoly[<key = val>]{<punto, punto, punto, ...>}
```

El comando `\spPoly` trabaja en *modo matemático* y recibe el *argumento obligatorio* `{<punto, punto, punto, ...>}`, una lista de al menos tres `{<puntos>}` separados por comas. El tipo de *símbolo* en la impresión y la verbalización se determinan acorde al número de puntos en el argumento: Si son tres puntos se aplica `\triangle` y añade el término «triángulo» a la verbalización; si son cuatro puntos se aplica `\mdlgwhsquare` y añade el término «cuadrado» a la verbalización; si son cinco o más puntos no se aplica ningún símbolo ni se añade verbalización.

Si no se proporciona el *argumento opcional* `[<key = val>]`, `\spPoly{A, B, C}` imprimirá ' $\triangle ABC$ ' en la salida PDF y NVDA/MathCAT verbalizará «triángulo A B C»; `\spPoly{A, B, C, D}` imprimirá ' $\square ABCD$ ' y verbalizará «cuadrado A B C D»; `\spPoly{A, B, C, D, E}` imprimirá ' $ABCDE$ ' y verbalizará «A B C D E».

Llaves para `\spPoly`

```
read-cmd = {<school | mathcat>} default: school
```

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento `{<punto, punto, punto, ...>}`. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spPoly{A, B, C}` se verbaliza «triángulo A B C»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el argumento o los términos «triángulo» y «cuadrado» establecido por la llave `print-sym`. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

```
print-sym = {<true | false>} default: true
```

Establece si se usará `\triangle` o `\mdlgwhsquare` en la impresión añadiendo el término correspondiente a la verbalización en NVDA/MathCAT. Si se establece en `true`, `\spPoly[print-sym=true]{A, B, C}` imprimirá ' $\triangle ABC$ ' en la salida PDF y NVDA/MathCAT verbalizará «triángulo A B C»; si se establece en `false`, imprimirá ' ABC ' en la salida PDF y NVDA/MathCAT omitirá el término «triángulo», verbalizando solo «A B C».

```
silent-cmd = {<true | false>} default: false
```

Alias de `print-sym=false`, incluido por consistencia con el resto de los comandos de geometría plana.

El comando `\spAfig`

```
\spAfig {<punto, punto, punto, ...>}
\spAfig(<punto, punto, punto, ...>)
\spAfig(<número | letra | letra_{número}>)
\spAfig[<key = val>](<punto, punto, punto, ...>)
```

El comando `\spAfig` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis*, el cual es *completamente opcional*, representando el área de una figura poligonal en la impresión aplicando `\mathcal{A}`.

El argumento admite tres formas: una *lista de vértices* `(<punto, punto, punto, ...>)` (al menos tres); un *número o letra sueltos* `(<1>)`, `(<n>)`; o una *letra de una tabla fija* (F, P, B, L) con subíndice numérico opcional, `(<F>)`, `(<F_{1}>)`.

Si no se proporciona el *argumento opcional* `[<key = val>]` ni el argumento delimitado por paréntesis, `\spAfig` imprimirá ' $\mathcal{A}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área». Con vértices, `\spAfig(A, B, C)` imprimirá ' $\mathcal{A}_{\triangle ABC}$ ' y verbalizará «área del triángulo A B C». Con un número o letra sueltos, `\spAfig(1)` imprimirá ' $\mathcal{A}_1$ ' y verbalizará «área uno»; `\spAfig(n)` imprimirá ' $\mathcal{A}_n$ ' y verbalizará «área n». Con una letra de la tabla, `\spAfig(F)` imprimirá ' $\mathcal{A}_F$ ' y verbalizará «área de la figura»; `\spAfig(F_{1})` imprimirá ' $\mathcal{A}_{F_1}$ ' y verbalizará «área de la figura uno».

Llaves para `\spAfig`

```
read-cmd = {<school | mathcat>} default: school
```

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spAfig(A, B, C)` se verbaliza «área del triángulo A B C»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

```
prefix = {<prefijo>} default: no usado
```

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «área». El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

```
print-sym = {<auto | triangle | square | none>} default: auto
```

Establece si se usará `\triangle` o `\mdlgwhsquare` en la impresión. Con vértices, `auto` infiere el símbolo según el número de puntos (tres o cuatro); `none` lo omite y NVDA/MathCAT verbaliza solo los vértices sin nombrar la figura: `\spAfig[print-sym=none](A, B, C)` imprimirá ' $\mathcal{A}_{ABC}$ ' y verbalizará «área A B C».

Forzar `triangle/square` exige que coincida con el número real de vértices; en caso contrario se produce un error. Con un número o letra sueltos, `auto` y `none` se comportan igual (sin símbolo); `triangle/square` anida el argumento bajo el símbolo: `\spAfig[print-sym=triangle](1)` imprimirá ' $\mathcal{A}_{\Delta_1}$ ' y verbalizará «área del triángulo uno». Esta llave no puede combinarse con una letra de la tabla (F, P, B, L).

`read-sub = {\true | false}` default: false

Establece si NVDA/MathCAT antepone la palabra «sub» al verbalizar el número o letra del argumento único. Si se establece en `true`, `\spAfig[read-sub=true](1)` verbaliza «área sub uno»; si se establece en `false`, verbaliza «área uno». No tiene efecto en la notación de vértices ni cuando el argumento es una letra de la tabla sin subíndice.

`silent-cmd = {\true | false}` default: false

Establece si se *silencia por completo* la verbalización en NVDA/MathCAT, sin afectar la impresión. Si se establece en `true`, NVDA/MathCAT NO verbalizará nada; si se establece en `false`, NVDA/MathCAT verbalizará normalmente.

`read-arg = {\nombre}` default: no usado

Sobrescribe el *nombre de la figura* determinado automáticamente por el número de puntos en el argumento, que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'. Por ejemplo: `\spAfig[read-arg={trapezio}](A, B, C, D)` imprimirá ' $\mathcal{A}_{\square ABCD}$ ' en la salida PDF y NVDA/MathCAT verbalizará «área del trapezio A B C D».

El comando `\spPfig`

```
\spPfig \spPfig
\spPfig(\punto, punto, punto, ...)
\spPfig(\número | letra | letra_{número})
\spPfig[key = val](\punto, punto, punto, ...)
```

El comando `\spPfig` trabaja en *modo matemático* y recibe el *argumento delimitado por paréntesis*, el cual es *completamente opcional*, representando el *perímetro* de una figura poligonal en la impresión aplicando `\mathcal{P}`.

El argumento admite las mismas tres formas que `\spAfig` (§7): una *lista de vértices*, un *número o letra sueltos*, o una *letra de la tabla fija* (F, P, B, L) con subíndice numérico opcional.

Si no se proporciona el *argumento opcional* [`key = val`] ni el argumento delimitado por paréntesis, `\spPfig` imprimirá ' $\mathcal{P}$ ' en la salida PDF y NVDA/MathCAT verbalizará «perímetro». Con vértices, `\spPfig(A, B, C)` imprimirá ' $\mathcal{P}_{\Delta ABC}$ ' y verbalizará «perímetro del triángulo A B C». Con un número o letra sueltos, `\spPfig(2)` imprimirá ' $\mathcal{P}_2$ ' y verbalizará «perímetro dos». Con una letra de la tabla, `\spPfig(P_{\{2\}})` imprimirá ' $\mathcal{P}_{P_2}$ ' y verbalizará «perímetro del polígono dos».

Llaves para `\spPfig`

`read-cmd = {\school | mathcat}` default: school

Establece la *forma* en la que NVDA/MathCAT verbalizará al argumento. Si se establece en `school`, NVDA/MathCAT verbaliza las letras del argumento sin el prefijo «mayúscula»: `\spPfig(A, B, C)` se verbaliza «perímetro del triángulo A B C»; si se establece en `mathcat`, NVDA/MathCAT aplica sus propias reglas, que pueden incluir el prefijo «mayúscula» dependiendo de la configuración del usuario.

`prefix = {\prefijo}` default: no usado

Establece el *prefijo* que será verbalizado por NVDA/MathCAT *antes* de verbalizar el término «perímetro». El valor de esta llave *siempre* debe pasarse entre llaves '{ }'.

`print-sym = {\auto | triangle | square | none}` default: auto

Establece si se usará `\triangle` o `\mdlgwhtsquare` en la impresión, con la misma semántica que en `\spAfig` (§7).

`read-sub = {\true | false}` default: false

Establece si NVDA/MathCAT antepone la palabra «sub» al verbalizar el número o letra del argumento único, con la misma semántica que en `\spAfig` (§7).

`silent-cmd = {\true | false}` default: false

Establece si se *silencia por completo* la verbalización en NVDA/MathCAT, sin afectar la impresión. Si se establece en `true`, NVDA/MathCAT NO verbalizará nada; si se establece en `false`, NVDA/MathCAT verbalizará normalmente.

`read-arg = {\nombre}` default: no usado

Sobrescribe el *nombre de la figura* determinado automáticamente por el número de puntos en el argumento, que será verbalizado por NVDA/MathCAT al ejecutar el comando sin modificar la impresión de este. El valor de esta llave *siempre* debe pasarse entre llaves '{ }'. Por ejemplo: `\spPfig[read-arg={trapezio}](A, B, C, D)` imprimirá ' $\mathcal{P}_{\square ABCD}$ ' en la salida PDF y NVDA/MathCAT verbalizará «perímetro del trapezio A B C D».

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9 Change history

v0.99 (ctan), 2026-09-05 – First public release.

10 Índice de la Documentación

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cifras-sep	3	read-cmd	16
notation-sym	4	Keys for \spmoney provide by spintent :	
read-period-sep	3	cifras-sep	7
read-sign	3	Keys for \spmQ provide by spintent :	
Keys for \spnZ provide by spintent :		wrap-parens	17
read-parens	16	Keys for \spmsym provide by spintent :	
wrap-parens	16	read-sym	13
Keys for \spPfig provide by spintent :		Keys for \spm\ provide by spintent :	
prefix	23	cifras-sep	13
print-sym	23	notation-sym	13
read-arg	23	read-period-sep	13
read-cmd	23	read-sign	13
read-sub	23	read-sym	13
silent-cmd	23	set-sym	13
Keys for \spPoint provide by spintent :		Keys for \spnZ provide by spintent :	
prefix	17	cifras-sep	16
read-cmd	17	read-sign	16
silent-cmd	17	Keys for \spPoly provide by spintent :	
Keys for \spPoly provide by spintent :		print-sym	22
prefix	22	Keys for \spside provide by spintent :	
print-sym	22	read-sty	19
read-cmd	22	Keys for \spunit provide by spintent :	
silent-cmd	22	cifras-sep	4
Keys for \sprayo provide by spintent :		notation-sym	4
prefix	18	read-period-sep	4
read-cmd	18	Keys for \spusep provide by spintent :	
read-sty	18	silent	12
silent-cmd	18	size	12
Keys for \sprecta provide by spintent :			
prefix	18		
read-cmd	18		
silent-cmd	18		
Keys for \spshort provide by spintent :			
read-arg	9		
small-caps	9		
Keys for \spside provide by spintent :			
prefix	19		
read-arg	19		
read-cmd	19		
read-sty	19		
silent-cmd	19		
Keys for \spsiglo provide by spintent :			
print-era	11		
Keys for \spunit provide by spintent :			
print-cdot	4		
read-cdot	4		
read-slash	4		
Keys for \spusep provide by spintent :			
silent	12		
size	12		
Keys for \spd\ provide by spintent :			
cifras-sep	13		

M

Lua module provide by **spintent**:

spintent.lua	6, 7, 9, 10, 17
--------------------	-----------------

P

Packages :

babel-spanish	2
babel	2, 7
documentmetadata-support	1
enumext	1
expl3	3
fontspec	2
fourier-otf	2
hyperref	2
l3keys	3
latex-lab-mathintent	1
libertinus-otf	2
luamml	1
scontents	1
sourcecodepro	2
tagpdf	1, 7, 11
unicode-math	1

11 Implementation

The most recent publicly released version of `spintent` is available at CTAN: <https://www.ctan.org/pkg/spintent>. While general feedback via email is welcomed, specific bugs or feature requests should be reported through the issue tracker: <https://github.com/pablgonz/spintent/issues>.

- The documentation presented here is far from professional, it contains a lot of obvious information that to the eye of a TeXpert are superfluous, but, after so many years developing this project is the only way to remember what does what.

11.1 General conventions

Functions of the form `\__spintent_luafun_some:n` or `\__spintent_luafun_some:nn` are defined (created) in the Lua module `spintent.lua` and only their variants are declared in `expl3`.

Similarly, variables of the form `\l__spintent_luaset_some_tl` or `\l__spintent_luaset_some_str` are defined in the Lua module `spintent.lua` and only declared in `expl3`.

- All variables and functions defined in this package are private and are NOT intended to work or be used by another package or module.

11.2 Initial set up

Start the DocStrip guards.

```
1 (*package)
```

Identify the internal prefix (L^AT_EX3 DocStrip convention) for `l3doc` class.

```
2 (@@=spintent)
```

11.3 Declaration of the package

First we will make sure we have a minimum (super updated) release version of L^AT_EX to work correctly, otherwise, we will return an “error”.

```
3 \NeedsTeXFormat{LaTeX2e}[2026-11-01]
4 \IfFormatAtLeastF { 2026-11-01 }
5 {
6   \PackageError { spintent }
7     { LaTeX~kernel~too~old~need~2026-11-01~or~later }
8     {
9       spintent~requires~a~LaTeX2e~release~2026-11-01~or~
10      later.~Update~your~TeX~distribution.
11     }
12 }
```

Now declare the `spintent` package and we will add the only key `mathcal` that can be passed as a package option.

```
13 \ProvidesExplPackage {spintent} {2026-09-05} {0.99} {Spanish parse intents}
14 \keys_define:nn { spintent }
15 {
16   mathcal .usage:n = load,
17   mathcal .bool_set:N = \l__spintent_mathcal_bool,
18   mathcal .initial:n = true,
19   mathcal .value_required:n = true,
20 }
21 \ProcessKeyOptions [ spintent ]
```

We check if the requirements for running the `spintent` package are met; that is: `\DocumentMetadata` must be active, Lua_T_EX must be running, and the `unicode-math` or `lua-unicode-math` package must be loaded. If neither is present, we will load `unicode-math`. If the conditions cannot be met, we will return a “fatal error”.

```
22 \IfDocumentMetadataTF
23 {
24   \sys_if_engine luatex:TF
25   {
26     \msg_new:nnn { spintent } { package-not-load }
27     {
28       The~'#1'~package~will~be~loaded~as~a~dependency.
29     }
30     \IfPackageLoadedF { unicode-math }
31     {
32       \IfPackageLoadedF { lua-unicode-math }
33       {
34         \msg_info:nnn { spintent } { package-not-load } { unicode-math }
35         \RequirePackage{unicode-math}
36       }
37     }
38     \msg_new:nnn { spintent } { env-success }
39     { Everything~looks~correct,~spintent~will~make~the~intent! }
```



```

40     \msg_info:nn { spintent } { env-success }
41   }
42   {
43     \msg_new:nnnn { spintent } { fatal-engine-requires-luatex }
44     { The~spintent~package~requires~LuaLaTeX. }
45     { This~package~relies~on~Lua~module.~Compile~with~lualatex. }
46     \msg_fatal:nn { spintent } { fatal-engine-requires-luatex }
47   }
48 }
49 {
50   \msg_new:nnnn { spintent } { fatal-metadata-missing }
51   { \token_to_str:N \DocumentMetadata \c_space_tl is~missing. }
52   {
53     The~spintent~package~requires~tagged~PDF~active.\\
54     You~must~declare~\token_to_str:N \DocumentMetadata { tagging=on } ~
55   }
56   \msg_fatal:nn { spintent } { fatal-metadata-missing }
57 }

```

11.4 Some kernel variants

```

\str_if_eq:nVTF
\str_if_eq:nVF
\str_uppercase:V
\str_lowercase:V
\seq_set_from_clist:Ne

```

Non-standard kernel variants.

```

58 \cs_generate_variant:Nn \str_if_eq:nnTF { V }
59 \cs_generate_variant:Nn \str_if_eq:nnF { V }
60 \cs_generate_variant:Nn \str_uppercase:n { V }
61 \cs_generate_variant:Nn \str_lowercase:n { V }
62 \cs_generate_variant:Nn \seq_set_from_clist:Nn { Ne }

```

(End of definition for `\str_if_eq:nVTF` and others.)

11.5 Variables for the key `\mathcal` and `\rightanglesqr`

```

\l__spintent_luaset_cal_fam_ok_str
\l__spintent_luaset_sym_fam_ok_str

```

First, we declare the variables `\l__spintent_luaset_cal_fam_ok_str` and `\l__spintent_luaset_sym_fam_ok_str` before loading the module `spintent.lua`. This is necessary because of the implementation of the functions `\__spintent_luafun_classic_mathcal:nn` and `\__spintent_luafun_right_angle_sqr:` within the Lua module; otherwise, `expl3` will complain about duplicate variable declarations.

```

63 \str_new:N \l__spintent_luaset_cal_fam_ok_str
64 \str_new:N \l__spintent_luaset_sym_fam_ok_str

```

(End of definition for `\l__spintent_luaset_cal_fam_ok_str` and `\l__spintent_luaset_sym_fam_ok_str`.)

11.6 Loading the Lua module and functions

```

\__spintent_luafun_spunit_lookup_alias:V
\__spintent_luafun_spmoney_lookup_data:V
\__spintent_luafun_spmQ_scale_int:Vn
\__spintent_luafun_clean_split_and_set:V

```

Now we load the Lua module `spintent.lua` and generate the variants for the functions `\__spintent_luafun_spunit:` used by `\spunit` (§11.15), `\__spintent_luafun_spmoney_lookup_data:V` used by `\spmoney` (§11.17), `\__spintent_luafun_spmQ_scale_int:Vn` used by `\spmQ` (§11.30) and `\__spintent_luafun_clean_split_and_set:V` used by `\spnum` (§11.14) defined in it.

```

65 \lua_load_module:n { spintent.lua }
66 \cs_generate_variant:Nn \__spintent_luafun_spunit_lookup_alias:n { V }
67 \cs_generate_variant:Nn \__spintent_luafun_spmoney_lookup_data:n { V }
68 \cs_generate_variant:Nn \__spintent_luafun_spmQ_scale_int:nn { VV, Vn }
69 \cs_generate_variant:Nn \__spintent_luafun_clean_split_and_set:n { V }

```

(End of definition for `\__spintent_luafun_spunit_lookup_alias:V` and others.)

11.7 Internal implementation for `\mathcal` and `\rightanglesqr`

To simplify user configuration and obtain the expected output style pdf_TTeX for `\mathcal`, we will provide a fallback to the font NewCMMath-Regular.otf, which contains the glyphs for these characters. Using this same font, we will obtain the symbol `\rightanglesqr` for the key `recto`.

```

mathcal-fallback
right-angle-sqr-fallback

```

First, we'll define two error messages in case our fallback fails.

```

70 \hook_gput_code:nnn { begindocument } { spintent }
71   {
72     \__spintent_luafun_init_math_fallbacks:
73   }
74 \msg_new:nnn { spintent } { mathcal-fallback }
75   {
76     Using~classic~\token_to_str:N \mathcal \c_space_tl as~fallback~(font~not~found~or~
77     no~free~math~family~available).~\msg_line_context:
78   }
79 \msg_new:nnn { spintent } { right-angle-sqr-fallback }
80   {
81     Using~\token_to_str:N \rightangle \c_space_tl as~fallback~for~the~right-angle~

```

```

82     glyph.~\msg_line_context:
83 }

```

(End of definition for `mathcal-fallback` and `right-angle-sqr-fallback`.)

`\__spintenc_mathcal:n`

Now we define the internal function `\__spintenc_mathcal:n` for `\mathcal` which checks the state of the variable `\__spintenc_mathcal_bool` set by the package key `mathcal`; if its state is true, we check the string variable `\__spintenc_luaaset_cal_fam_ok_str` and if the latter has the value true, we call the function `\__spintenc_luafun_classic_mathcal:n` and otherwise (we cannot find the source in the system) we return a warning message and directly use `\mathcal`. Otherwise, that is, `mathcal=false` we simply use `\mathcal`.

```

84 \cs_new_protected:Nn \__spintenc_mathcal:n
85 {
86     \bool_if:NTF \__spintenc_mathcal_bool
87     {
88         \str_if_eq:VnTF \__spintenc_luaaset_cal_fam_ok_str { true }
89         { \__spintenc_luafun_classic_mathcal:n {#1} }
90         {
91             \msg_warning:nn { spintenc } { mathcal-fallback }
92             \mathcal {#1}
93         }
94     }
95     { \mathcal {#1} }
96 }

```

(End of definition for `\__spintenc_mathcal:n`.)

`\__spintenc_right_angle_sqr:`

The internal function `\__spintenc_right_angle_sqr:` establishes the internal copy of the command `\rightanglesqr`. First, we check the string variable `\__spintenc_luaaset_sym_fam_ok_str`. If it is true, we call the function `\__spintenc__luafun_right_angle_sqr:`; otherwise (we do not find the font in the system), we return a warning message and use `\rightangle` instead.

```

97 \cs_new_protected:Nn \__spintenc_right_angle_sqr:
98 {
99     \str_if_eq:VnTF \__spintenc_luaaset_sym_fam_ok_str { true }
100     { \__spintenc__luafun_right_angle_sqr: }
101     {
102         \msg_warning:nn { spintenc } { right-angle-sqr-fallback }
103         \rightangle
104     }
105 }

```

(End of definition for `\__spintenc_right_angle_sqr:`.)

11.8 Internal copy of the unicode invisible separator

`\__spintenc_invisible_sep:`
`\c__spintenc_invisible_sep_tl`

The function `\__spintenc_invisible_sep:` is an internal copy of the Unicode invisible separator U-2063 that uses `\luamml_annotate:en` from the package `luamml`[2] and let `<mo>...</mo>` in the MathML structure embedded in the PDF output.

```

106 \cs_new_protected:Nn \__spintenc_invisible_sep:
107 {
108     \__spintenc_luafun_invisible_sep:
109 }

```

The token list constant `\c__spintenc_invisible_sep_tl` is an internal copy of the Unicode invisible separator U-2063 using the primitive `\Umathchardef` from LuaTeX which leaves `<mi>...</mi>` in the structure MathML embedded in the output PDF.

```

110 \hook_gput_code:nnn { begindocument } { spintenc }
111 {
112     \Umathchardef \c__spintenc_invisible_sep_tl = "0 "0 "2063
113 }

```

- Both copies are used for different purposes, the function `\__spintenc_invisible_sep:` will be used to force the start of a `<mrow>` in the structure MathML, the token list constant `\c__spintenc_invisible_sep_tl` will be used to inject text into the structure MathML when we use `\MathMLintent{<text>}`.

(End of definition for `\__spintenc_invisible_sep:` and `\c__spintenc_invisible_sep_tl`.)

11.9 Normalize argument and affixes

We normalize the $\{\langle argument \rangle\}$ to support affixes (prefixes/suffixes) by trimming leading and trailing spaces, converting internal spaces to hyphens, collapsing consecutive hyphens, and removing leading or trailing hyphens.

`\__spintent_collapse_hyphens:N` A recursive helper to collapse multiple hyphens into a single one.

```

114 \cs_new_protected:Nn \__spintent_collapse_hyphens:N
115   {
116     \tl_if_in:NnT #1 { -- }
117     {
118       \tl_replace_all:Nnn #1 { -- } { - }
119       \__spintent_collapse_hyphens:N #1
120     }
121   }

```

(End of definition for `\__spintent_collapse_hyphens:N`.)

`\__spintent_sanitize_affix:Nn` The function `\__spintent_sanitize_affix:Nn` checks if the $\{\langle argument \rangle\}$ passed in ‘#2’ is *blank*. If so, it clears the target token list variable passed in ‘#1’. Otherwise, trimming bounds, converting spaces ‘`\_`’ to hyphens ‘`-`’, collapsing multiple hyphens, and safely stripping leading/trailing hyphens. For example, if you pass a target variable and raw text:

```
\__spintent_sanitize_affix:Nn \l__spintent_spsomesym_prefix_tl{ foo bar baz }
```

The function will “store” the normalized string `foo-bar-baz` inside `\l__spintent_spsomesym_prefix_tl`, ready to be conditionally appended.

```

122 \cs_new_protected:Nn \__spintent_sanitize_affix:Nn
123   {
124     \tl_if_blank:nTF {#2}
125     { \tl_clear:N #1 }
126     {
127       \tl_set:Ne #1 {#2}
128       \tl_trim_spaces:N #1
129       \tl_replace_all:Nnn #1 { ~ } { - }
130       \__spintent_collapse_hyphens:N #1
131       \tl_put_left:Nn #1 { \q_mark }
132       \tl_put_right:Nn #1 { \q_stop }
133       \tl_replace_all:Nnn #1 { \q_mark - } { \q_mark }
134       \tl_replace_all:Nnn #1 { - \q_stop } { \q_stop }
135       \tl_remove_all:Nn #1 { \q_mark }
136       \tl_remove_all:Nn #1 { \q_stop }
137     }
138   }
139 \cs_generate_variant:Nn \__spintent_sanitize_affix:Nn { cn }

```

(End of definition for `\__spintent_sanitize_affix:Nn`.)

11.10 The command `\setspintent`

The `\setspintent` command is used to configure $\langle keys \rangle$ “globally” for the utilities provided by `spintent`.

11.10.1 Internal variables and error messages for `\setspintent`

`\l__spintent_setspintent_input_arg_cmd_tl` The variable `\l__spintent_setspintent_input_arg_cmd_tl` will store the command or command name passed to the “first” $\{\langle argument \rangle\}$ ‘#1’ of the command `\setspintent`.

```
140 \tl_new:N \l__spintent_setspintent_input_arg_cmd_tl
```

Error message if the $\{\langle argument \rangle\}$ ‘#1’ passed to `\setspintent` is a “non-existent” command.

```

141 \msg_new:nnnn { spintent } { unknown-command-setup }
142   {
143     Cannot~configure~unknown~command~'\exp_not:c{#1}'.
144   }
145   {
146     You~attempted~to~use~\token_to_str:N \setspintent{#1}{...},~but~the~
147     command~\exp_not:c{#1}~is~not~defined.~Please~check~for~typos.
148   }

```

(End of definition for `\l__spintent_setspintent_input_arg_cmd_tl` and `unknown-command-setup`.)

```

__spintent_setspintent_set_keys:nn
__spintent_setspintent_set_keys:en

```

The function `__spintent_setspintent_set_keys:nn` will check if the command passed as `{⟨argument⟩}` exists in ‘#1’ and set the `⟨keys⟩` passed as `{⟨argument⟩}` in ‘#2’ to the correct path; otherwise, it will return an error.

```

149 \cs_new_protected:Nn __spintent_setspintent_set_keys:nn
150 {
151   \cs_if_exist:cTF { #1 }
152   {
153     \keys_set:nn { spintent / #1 } { #2 }
154   }
155   {
156     \msg_error:nnn { spintent } { unknown-command-setup } { #1 }
157   }
158 }
159 \cs_generate_variant:Nn __spintent_setspintent_set_keys:nn { en }

```

(End of definition for `__spintent_setspintent_set_keys:nn`.)

```

__spintent_setspintent_parse_args:nn
__spintent_setspintent_parse_args:Vn

```

The function `__spintent_setspintent_parse_args:nn` will check if the argument passed in #1 begins with ‘\’ or is just a *command name*. If it begins with ‘\’, it will remove it using `\cs_to_str:N` and then execute `__spintent_setspintent_set_keys:en`. Otherwise, i.e., if ‘\’ was not passed, it will directly execute `__spintent_setspintent_set_keys:nn`.

```

160 \cs_new_protected:Nn __spintent_setspintent_parse_args:nn
161 {
162   \tl_if_single:nTF { #1 }
163   {
164     \token_if_cs:NTF #1
165     {
166       __spintent_setspintent_set_keys:en { \cs_to_str:N #1 } { #2 }
167     }
168     {
169       __spintent_setspintent_set_keys:nn { #1 } { #2 }
170     }
171   }
172   {
173     __spintent_setspintent_set_keys:nn { #1 } { #2 }
174   }
175 }
176 \cs_generate_variant:Nn __spintent_setspintent_parse_args:nn { Vn }

```

(End of definition for `__spintent_setspintent_parse_args:nn`.)

`\setspintent`

Now we define the user command `\setspintent`.

```

177 \NewDocumentCommand \setspintent { m m }
178 {
179   \tl_if_blank:nTF { #1 }
180   {
181     \msg_error:nnn { spintent } { unknown-command-setup } { <empty> }
182   }
183   {
184     \tl_set:Nn \l__spintent_setspintent_input_arg_cmd_tl { #1 }
185     \tl_trim_spaces:N \l__spintent_setspintent_input_arg_cmd_tl
186     __spintent_setspintent_parse_args:Vn \l__spintent_setspintent_input_arg_cmd_tl { #2 }
187   }
188 }

```

(End of definition for `\setspintent`. This function is documented on page 3.)

11.11 Handling unknown keys

```

\l__spintent_command_name_str
unknown-choice
cmd-key-unknown
cmd-key-value-unknown

```

We define the variable `\l__spintent_command_name_str` which will store the “name” of each command within the implementation along with the messages `unknown-choice` used by `⟨keys⟩` that use `.choice:n` in their implementation and the messages `cmd-key-unknown`, `cmd-key-value-unknown` used by the `⟨key⟩` `unknown` in commands that use `[⟨key = val⟩]`.

```

189 \str_new:N \l__spintent_command_name_str
190 \msg_new:nnn { spintent } { unknown-choice }
191 {
192   The~value~'~#3'~for~'~#1'~key~is~invalid~use~('~#2').
193 }
194 \msg_new:nnnn { spintent } { cmd-key-unknown }
195 {
196   The~key~'~#1'~is~unknown~by~command~

```

```

197     \c_backslash_str \l__spintent_command_name_str \c_space_tl
198     and~is~being~ignored~\msg_line_context:.
199   }
200   {
201     The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
202     does~not~have~a~key~called~'#1'.\\
203     Check~that~you~have~spelled~the~key~name~correctly.
204   }
205 \msg_new:nnnn { spintent } { cmd-key-value-unknown }
206   {
207     The~key~'#1=#2'~is~unknown~by~command~
208     \c_backslash_str \l__spintent_command_name_str \c_space_tl
209     and~is~being~ignored~\msg_line_context:.
210   }
211   {
212     The~command~\c_backslash_str \l__spintent_command_name_str \c_space_tl
213     does~not~have~a~key~called~'#1'.\\
214     Check~that~you~have~spelled~the~key~name~correctly.
215   }

```

(End of definition for `\l__spintent_command_name_str` and others.)

```

\__spintent_unknown_keys:n
\__spintent_unknown_keys:nn

```

The internal functions `\__spintent_unknown_keys:n` and `\__spintent_unknown_keys:nn` used by the *<key>* unknown.

```

216 \cs_new_protected:Nn \__spintent_unknown_keys_cmd:n
217   {
218     \exp_args:NV \__spintent_unknown_keys_cmd:nn \l_keys_key_str {#1}
219   }
220 \cs_new_protected:Nn \__spintent_unknown_keys_cmd:nn
221   {
222     \tl_if_blank:nTF {#2}
223     { \msg_error:nnn { spintent } { cmd-key-unknown } {#1} }
224     { \msg_error:nnnn { spintent } { cmd-key-value-unknown } {#1} {#2} }
225   }

```

(End of definition for `\__spintent_unknown_keys:n` and `\__spintent_unknown_keys:nn`.)

11.12 Definition of core base variables

```

\__spintent_luafun_clean_split_and_set:n
\__spintent_luafun_clean_split_and_set:V

```

The function `\__spintent_luafun_clean_split_and_set:n` defined in the Lua module `spintent.lua`, is common to several commands in this implementation. It analyzes, processes, cleans and splits the *<argument>* passed to commands using internal functions defined in the Lua module `spintent.lua`. It isolates signs, integer components, decimal separators, decimal parte, decimal periodic part, exponents and final physical “units”, assigning them directly to `expl3` variables.

This function assigns the states of the variables ‘_str’ or ‘_tl’ described below to `true`, `false`, `error` or *stores* information according to the role it fulfills within the implementation.

(End of definition for `\__spintent_luafun_clean_split_and_set:n`.)

```

\l__spintent_luaset_sign_tl
\l__spintent_luaset_part_int_tl
\l__spintent_luaset_dec_sep_tl
\l__spintent_luaset_part_dec_tl
\l__spintent_luaset_part_period_tl
\l__spintent_luaset_has_integer_str
\l__spintent_luaset_has_natural_str

```

The variable `\l__spintent_luaset_sign_tl` stores the explicit ‘+’ or ‘−’ sign if present. The variable `\l__spintent_luaset_part_int_tl` stores the raw digit sequence of the “integer part”. The variable `\l__spintent_luaset_dec_sep_tl` stores the matched ‘,’ or ‘.’ used as the decimal separator. The variable `\l__spintent_luaset_part_dec_tl` stores the raw digits of the “finite decimal part” that follow the separator, and the variable `\l__spintent_luaset_part_period_tl` stores the digits of the “infinite periodic decimal part” that follow a semicolon ‘;’.

The variable `\l__spintent_luaset_has_integer_str` store the state `true` when it is *only* an “integer”. The variable `\l__spintent_luaset_has_natural_str` store the state `true` when it is *only* an “natural number”.

```

226 \tl_new:N \l__spintent_luaset_sign_tl
227 \tl_new:N \l__spintent_luaset_part_int_tl
228 \tl_new:N \l__spintent_luaset_dec_sep_tl
229 \tl_new:N \l__spintent_luaset_part_dec_tl
230 \tl_new:N \l__spintent_luaset_part_period_tl
231 \str_new:N \l__spintent_luaset_has_integer_str
232 \str_new:N \l__spintent_luaset_has_natural_str

```

(End of definition for `\l__spintent_luaset_sign_tl` and others.)


```

\l__spintent_luaset_only_part_int_str
\l__spintent_luaset_only_part_dec_str
\l__spintent_luaset_only_part_period_str
\l__spintent_luaset_format_part_int_str
\l__spintent_luaset_format_part_dec_str

```

The literal entries of the numbers that we will pass to the *intent function* :number of `\MathMLintent` are *stored* in `\l__spintent_luaset_only_part_int_str` for the “*integer part*”, in `\l__spintent_luaset_only_part_dec_str` for the “*finite decimal part*”, and in `\l__spintent_luaset_only_part_period_str` for the “*infinite periodic decimal part*”.

Formatted versions (`cifras-sep=true`) are *stored* in `\l__spintent_luaset_format_part_int_str` for the “*integer part*” and `\l__spintent_luaset_format_part_dec_str` for the “*finite decimal part*”.

```

233 \str_new:N \l__spintent_luaset_only_part_int_str
234 \str_new:N \l__spintent_luaset_only_part_dec_str
235 \str_new:N \l__spintent_luaset_only_part_period_str
236 \str_new:N \l__spintent_luaset_format_part_int_str
237 \str_new:N \l__spintent_luaset_format_part_dec_str

```

(End of definition for `\l__spintent_luaset_only_part_int_str` and others.)

```

\l__spintent_luaset_millions_str
\l__spintent_luaset_decimal_period_str
\l__spintent_luaset_not_decimal_period_str
\l__spintent_luaset_not_decimal_not_period_str
\l__spintent_luaset_dec_is_all_zeros_str
\l__spintent_luaset_has_exponent_str
\l__spintent_luaset_exponent_tl

```

The flag variable `\l__spintent_luaset_millions_str` indicates whether the integer part translates exactly to a multiple of one million when passed to `\MathMLintent`. To properly structure the reading and writing of numbers in MathML, we use the flag variables `\l__spintent_luaset_decimal_period_str`, `\l__spintent_luaset_not_decimal_period_str`, and `\l__spintent_luaset_not_decimal_not_period_str`. The flag variable `\l__spintent_luaset_dec_is_all_zeros_str` monitors whether the “*finite decimal part*” is completely zero (useful for “*pure periodic decimals*”).

The flag variable `\l__spintent_luaset_has_exponent_str` indicates whether active “*scientific notation*” was detected at the end of the entered number, passed with ‘e’ or ‘E’. If true, it will store the digits of the exponent in `\l__spintent_luaset_exponent_tl`.

```

238 \str_new:N \l__spintent_luaset_millions_str
239 \str_new:N \l__spintent_luaset_decimal_period_str
240 \str_new:N \l__spintent_luaset_not_decimal_period_str
241 \str_new:N \l__spintent_luaset_not_decimal_not_period_str
242 \str_new:N \l__spintent_luaset_dec_is_all_zeros_str
243 \str_new:N \l__spintent_luaset_has_exponent_str
244 \tl_new:N \l__spintent_luaset_exponent_tl

```

(End of definition for `\l__spintent_luaset_millions_str` and others.)

```

\l__spintent_luaset_has_units_str
\l__spintent_luaset_status_str
\l__spintent_luaset_arg_above_tl
\l__spintent_luaset_arg_below_tl
\l__spintent_luaset_denom_has_number_str

```

Since the function `\__spintent_luafun_clean_split_and_set:n` is common to `\spnum` (§11.14) and `\spunit` (§11.15) (as well as others), it determines the following unit-related variables after processing the `{⟨arguments⟩}` via the Lua module `spintent.lua`.

First, it checks if there is any residual text *after* the processed number that is a candidate for physical “*units*”. It sets the flag variable `\l__spintent_luaset_has_units_str` accordingly. Following this, it verifies if this text contains more than one “/”. It then sets the variable `\l__spintent_luaset_status_str` to `valid` or `multislash`. If the structure is `valid`, it splits the unit into `\l__spintent_luaset_arg_above_tl` and `\l__spintent_luaset_arg_below_tl`. Finally, it analyzes `\l__spintent_luaset_arg_below_tl`; if the denominator starts with a number, it sets `\l__spintent_luaset_denom_has_number_str` to `true`, otherwise to `false`.

```

245 \str_new:N \l__spintent_luaset_has_units_str
246 \str_new:N \l__spintent_luaset_status_str
247 \tl_new:N \l__spintent_luaset_arg_above_tl
248 \tl_new:N \l__spintent_luaset_arg_below_tl
249 \str_new:N \l__spintent_luaset_denom_has_number_str

```

(End of definition for `\l__spintent_luaset_has_units_str` and others.)

11.13 Common messages for text mode and math mode

Messages shared by the different commands for text mode or math mode.

```

250 \msg_new:nnnn { spintent } { need-math-mode }
251 {
252   Math~mode~required~for~'#1'~\msg_line_context:.
253 }
254 {
255   Use~inside~$...$~or~in~math~environment.
256 }
257 \msg_new:nnnn { spintent } { need-text-mode }
258 {
259   Text~mode~required~for~'#1'~\msg_line_context:.
260 }
261 {
262   Use~outside~of~$...$~or~math~environment.
263 }

```

11.14 The command `\spnum`

The user command `\spnum` is *first* of “the big four” provided by the `spintent` package and is fundamental to the implementation of other commands. It handles the processing of numerical input, scientific notation, and units of measurement, using the Lua module `spintent.lua`, which is responsible for separating and assigning variables defined in `expl3`.

From the `expl3` side, we handle the printing logic and the *intent functions* passed to `\MathMLintent` to achieve a valid MathML structure that is accepted by MathCAT.

11.14.1 Definition of keys and variables for `\spnum`

The variable `\l__spintent_spnum_read_terse_bool` handled by the key `read-sign`, controls the *intent* passed to `\MathMLintent` to spoken the ‘+’ or ‘-’. If set to `clear`, it will say “*positivo*” or “*negativo*” after reading the number; if set to `terse`, it will say “*más*” or “*menos*” before reading the number.

The variable `\l__spintent_spnum_read_period_sep_str` handled by the key `read-period-sep`, controls how the decimal separator ‘;’ (periodic part) is printed and spoken when the `{(argument)}` is a *pure decimal periodic* and is passed the *intent* to `\MathMLintent`. If set to `coma`, it will say “*coma*” and print ‘,’; if set to `punto` it will say “*punto*” and print ‘.’.

The variable `\l__spintent_spnum__notation_sym_tl`, handled by the key `notation-sym`, determines whether ‘x’ will be printed when set to `times` or ‘·’ when set to `cdot`.

The variable `\l__spintent_spnum_intent_read_sign_str` will *store* the spoken translation used in the *intent* function passed to `\MathMLintent` for the ‘+’ or ‘-’ according to the value passed to the key `read-sign`.

The variable `\l__spintent_spnum_intent_read_dec_sep_str` *stores* the spoken translation that will be used in the *intent* function passed to `\MathMLintent` for reading the decimal separation as a “*coma*” or a “*punto*” according to the value found in the variable `\l__spintent_luaset_dec_sep_tl`.

```
264 \bool_new:N \l__spintent_spnum_read_terse_bool
265 \str_new:N \l__spintent_spnum_read_period_sep_str
266 \tl_new:N \l__spintent_spnum_notation_sym_tl
267 \str_new:N \l__spintent_spnum_intent_read_sign_str
268 \str_new:N \l__spintent_spnum_intent_read_dec_sep_str
```

(End of definition for `\l__spintent_spnum_read_terse_bool` and others.)

```
cifras-sep      Now we add the keys cifras-sep, read-sign, read-period-sep and notation-sym.
read-sign
read-period-sep
notation-sym

269 \keys_define:nn { spintent/spnum }
270 {
271   cifras-sep          .bool_set:N = \l__spintent_spnum_cifras_sep_bool,
272   cifras-sep          .initial:n = true,
273   cifras-sep          .value_required:n = true,
274   read-sign           .choices:nn =
275     { terse , clear }
276     {
277       \int_case:nn { \l_keys_choice_int }
278       {
279         {1} { \bool_set_true:N \l__spintent_spnum_read_terse_bool }
280         {2} { \bool_set_false:N \l__spintent_spnum_read_terse_bool }
281       }
282     },
283   read-sign           .initial:n = terse,
284   read-sign           .value_required:n = true,
285   read-sign / unknown .code:n =
286     \msg_error:nneee { spintent } { unknown-choice }
287     { read-sign } { terse , ~clear } { \exp_not:n {#1} },
288   read-period-sep     .choices:nn =
289     { coma , punto }
290     {
291       \int_case:nn { \l_keys_choice_int }
292       {
293         {1} { \str_set:Nn \l__spintent_spnum_read_period_sep_str { coma } }
294         {2} { \str_set:Nn \l__spintent_spnum_read_period_sep_str { punto } }
295       }
296     },
297   read-period-sep     .initial:n = coma,
298   read-period-sep     .value_required:n = true,
299   read-period-sep / unknown .code:n =
300     \msg_error:nneee { spintent } { unknown-choice }
301     { read-period-sep } { coma , ~punto } { \exp_not:n {#1} },
302   notation-sym        .choices:nn =
303     { times , cdot }
```

```

304     {
305       \int_case:nn { \l_keys_choice_int }
306       {
307         {1} { \tl_set:Nn \__spintent_snum_notation_sym_tl { \times } }
308         {2} { \tl_set:Nn \__spintent_snum_notation_sym_tl { \cdot } }
309       }
310     },
311     notation-sym .initial:n = cdot,
312     notation-sym .value_required:n = true,
313     notation-sym / unknown .code:n =
314       \msg_error:nneee { spintent } { unknown-choice }
315       { notation-sym } { times , ~ cdot } { \exp_not:n {#1} },
316     unknown .code:n =
317       {
318         \str_set:Nn \__spintent_command_name_str { snum }
319         \__spintent_unknown_keys_cmd:n {#1}
320       },
321   }

```

(End of definition for *cifras-sep* and others.)

11.14.2 Internal functions for \snum

```

\__spintent_snum_render_part:n
\__spintent_snum_render_int:
\__spintent_snum_render_dec:

```

The function `\__spintent_snum_render_part:n` processes the $\langle argument \rangle$ passed in ‘#1’ used by the functions `\__spintent_snum_render_int:` and `\__spintent_snum_render_dec:` to define the *intent function* :number for the integer and decimal part passed to `\MathMLintenc`. It evaluates whether the key *cifras-sep* is active or not.

```

322 \cs_new_protected:Nn \__spintent_snum_render_part:n
323 {
324   \MathMLintenc { \str_use:c { l__spintenc_luaset_only_part_#1_str } :number }
325   {
326     \bool_if:NTF \__spintenc_snum_cifras_sep_bool
327     { \str_use:c { l__spintenc_luaset_format_part_#1_str } }
328     { \tl_use:c { l__spintenc_luaset_part_#1_tl } }
329   }
330 }
331 \cs_new_protected:Nn \__spintenc_snum_render_int:
332 {
333   \__spintenc_snum_render_part:n { int }
334 }
335 \cs_new_protected:Nn \__spintenc_snum_render_dec:
336 {
337   \__spintenc_snum_render_part:n { dec }
338 }

```

(End of definition for `\__spintenc_snum_render_part:n`, `\__spintenc_snum_render_int:`, and `\__spintenc_snum_render_dec:`.)

```

\__spintenc_snum_render_only_period:
\__spintenc_snum_print_period_bar:
\__spintenc_snum_render_period_sep:

```

The function `\__spintenc_snum_render_only_period:` isolates the “infinite periodic decimal part” from the “finite decimal part”, assigning the *intent function* :number passed to `\MathMLintenc` which will be used by `\MathMLarg` in the function `\__spintenc_snum_print_period_bar:`.

```

339 \cs_new_protected:Nn \__spintenc_snum_render_only_period:
340 {
341   \MathMLintenc { \l__spintenc_luaset_only_part_period_str :number }
342   {
343     \l__spintenc_luaset_part_period_tl
344   }
345 }

```

The function `\__spintenc_snum_print_period_bar:` inserts the function `\__spintenc_snum_render_only_period:` as an argument within the *intent function* `_periódico:postfix($dp)` passed to `\MathMLintenc`, generating the correct spoken for the “infinite periodic decimal part”.

```

346 \cs_new_protected:Nn \__spintenc_snum_print_period_bar:
347 {
348   \MathMLintenc { _periódico:postfix($dp) }
349   {
350     {
351       \overline
352       {
353         \MathMLarg { dp } { \__spintenc_snum_render_only_period: }
354       }
355     }
356 }

```

```
357 }
```

The function `\__spintent_snum_render_period_sep:` sets how the decimal separator handled by the key `read-period-sep` will be printed and spoken when a “*pure periodic decimal*” is passed.

```
358 \cs_new_protected:Nn \__spintent_snum_render_period_sep:
359 {
360   \str_if_eq:VnTF \l__spintent_snum_read_period_sep_str { coma }
361   {
362     \MathMLintent { _coma:prefix($pp) }
363     {
364       \mathord{,}
365       \MathMLarg { pp } { \__spintent_snum_print_period_bar: }
366     }
367   }
368   {
369     \MathMLintent { _punto:prefix($pp) }
370     {
371       \mathord{.}
372       \MathMLarg { pp } { \__spintent_snum_print_period_bar: }
373     }
374   }
375 }
```

(End of definition for `\__spintent_snum_render_only_period:`, `\__spintent_snum_print_period_bar:`, and `\__spintent_snum_render_period_sep:`.)

```
\__spintent_snum_render_decimal_sep:
```

The function `\__spintent_snum_render_decimal_sep:` parses the value of the variable `\l__spintent_luaset` and sets the variable `\l__spintent_snum_intent_read_dec_sep_str` with the function `_coma:prefix($dp)` for reading a “*coma*” or `_punto:prefix($dp)` for reading a “*punto*” to `\MathMLintent`. We print the decimal separator stored in `\l__spintent_luaset_dec_sep_tl` using `\mathord` to remove spaces around it and then use the command `\MathMLarg` where we insert `\__spintent_invisible_sep:` for compatibility with MathML and finally call the function `\__spintent_snum_render_dec:` to print the finite decimal part.

```
376 \cs_new_protected:Nn \__spintent_snum_render_decimal_sep:
377 {
378   \str_if_eq:VnTF \l__spintent_luaset_dec_sep_tl { , }
379   { \str_set:Nn \l__spintent_snum_intent_read_dec_sep_str { _coma } }
380   { \str_set:Nn \l__spintent_snum_intent_read_dec_sep_str { _punto } }
381   \MathMLintent { \l__spintent_snum_intent_read_dec_sep_str :prefix($dp) }
382   {
383     \mathord { \l__spintent_luaset_dec_sep_tl }
384     \MathMLarg { dp }
385     {
386       \__spintent_invisible_sep:
387       \__spintent_snum_render_dec:
388     }
389   }
390   % period part
391   \str_if_eq:VnTF \l__spintent_luaset_decimal_period_str { true }
392   {
393     \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
394     {
395       \__spintent_invisible_sep:
396       \__spintent_snum_print_period_bar:
397     }
398     {
399       \MathMLintent { _con:prefix($pnum) }
400       {
401         \__spintent_invisible_sep:
402         \MathMLarg { pnum }
403         {
404           \__spintent_snum_print_period_bar:
405         }
406       }
407     }
408   }
409 }
```

(End of definition for `\__spintent_snum_render_decimal_sep:`.)

```
\__spintent_snum_print_part_dec:
\__spintent_snum_print_part_period:
```

The function `\__spintent_snum_print_part_dec:` checks if a decimal part exists. If it does, it determines whether the decimal separator is a comma or a period to set the correct speech intent (“*coma*” or “*punto*”).

It then prints the separator, renders the decimal digits, and if a periodic sequence follows, it decides how to attach the periodic bar.

The function `\__spintent_snum_print_part_period:` handles cases where a periodic part exists. It evaluates if it should render the periodic separator directly, particularly when the number has a periodic part but lacks a standard decimal part.

```

410 \cs_new_protected:Nn \__spintent_snum_print_part_dec:
411 {
412   \tl_if_empty:NF \__spintent_luaset_part_dec_tl
413   {
414     \__spintent_snum_render_decimal_sep:
415   }
416 }
417 \cs_new_protected:Nn \__spintent_snum_print_part_period:
418 {
419   \tl_if_empty:NF \__spintent_luaset_part_period_tl
420   {
421     \str_if_eq:VnTF \__spintent_luaset_not_decimal_period_str { true }
422     { \__spintent_snum_render_period_sep: }
423     {
424       \tl_if_empty:NT \__spintent_luaset_part_dec_tl
425       {
426         \__spintent_snum_render_period_sep:
427       }
428     }
429   }
430 }

```

(End of definition for `\__spintent_snum_print_part_dec:` and `\__spintent_snum_print_part_period:`.)

`\__spintent_snum_print_integer:`
`\__spintent_snum_print_num_start:`
`\__spintent_snum_print_number_core:`

The function `\__spintent_snum_print_integer:` checks if the number consists only of an integer. If so, it simply renders it. If a decimal or periodic part exists, it sets up a `MathML` intent to link the integer part (respecting digit separators if enabled) to the subsequent decimal or periodic functions.

The function `\__spintent_snum_print_num_start:` checks if an explicit sign is present. If it finds one, it evaluates the terse reading flag to decide whether the sign should be read as a prefix (“*más*” or “*menos*”) or as a postfix (“*positivo*” or “*negativo*”), and wraps the intent around the rest of the number.

The function `\__spintent_snum_print_number_core:` acts as the main wrapper. It triggers the sign and base number evaluation, and then checks if a scientific exponent is present. If it is, it appends the correct base-10 notation symbol and exponent to complete the sequence.

📌 Note for `\__spintent_invisible_sep:`.

```

431 \cs_new_protected:Nn \__spintent_snum_print_integer:
432 {
433   \str_if_eq:VnTF \__spintent_luaset_not_decimal_not_period_str { true }
434   { \__spintent_snum_render_int: }
435   {
436     \MathMLintent { \__spintent_luaset_only_part_int_str :number:prefix($next) }
437     {
438       \bool_if:NTF \__spintent_snum_cifras_sep_bool
439       { \__spintent_luaset_format_part_int_str }
440       { \__spintent_luaset_part_int_tl }
441       \MathMLarg { next }
442       {
443         \tl_if_empty:NTF \__spintent_luaset_part_dec_tl
444         { \__spintent_snum_print_part_period: }
445         { \__spintent_snum_print_part_dec: }
446       }
447     }
448   }
449 }
450 \cs_new_protected:Nn \__spintent_snum_print_num_start:
451 {
452   \tl_if_empty:NTF \__spintent_luaset_sign_tl
453   {
454     \__spintent_invisible_sep: % need
455     \__spintent_snum_print_integer:
456   }
457   {
458     \bool_if:NTF \__spintent_snum_read_terse_bool
459     {
460       \str_if_eq:VnTF \__spintent_luaset_sign_tl { + }

```

```

461         { \str_set:Nn \__spintent_snum_intent_read_sign_str { _más:prefix($n)} }
462         { \str_set:Nn \__spintent_snum_intent_read_sign_str { _menos:prefix($n)} }
463     }
464     {
465         \str_if_eq:VnTF \__spintent_luaset_sign_tl { + }
466         { \str_set:Nn \__spintent_snum_intent_read_sign_str { _positivo:postfix($n) } }
467         { \str_set:Nn \__spintent_snum_intent_read_sign_str { _negativo:postfix($n) } }
468     }
469     \__spintent_invisible_sep: % need
470     \MathMLintent { \__spintent_snum_intent_read_sign_str }
471     {
472         \__spintent_luaset_sign_tl
473         \MathMLarg { n } { \__spintent_snum_print_integer: }
474     }
475 }
476 }
477 \cs_new_protected:Nn \__spintent_snum_print_number_core:
478 {
479     \__spintent_snum_print_num_start:
480     \str_if_eq:VnTF \__spintent_luaset_has_exponent_str { true }
481     {
482         \MathMLintent { _por } { \__spintent_snum_notation_sym_tl }
483         10^{\__spintent_luaset_exponent_tl }
484     }
485 }

```

(End of definition for `\__spintent_snum_print_integer:`, `\__spintent_snum_print_num_start:`, and `\__spintent_snum_print_number_core:`.)

11.14.3 Definition of error messages

spnum-has-units
spnum-no-digits

We define two error messages for the `\spnum` command. The first “error” triggers if the user includes “units” in the $\langle argument \rangle$ passed in ‘#1’, advising them to use the `\spunit` command instead. The second “error” is raised if the provided $\langle argument \rangle$ contains no numerical digits.

```

486 \msg_new:nnnn { spintent } { spnum-has-units }
487 {
488     The~\token_to_str:N \spnum \c_space_tl command~does~not~accept~units~(''#1'').
489 }
490 {
491     Use~\token_to_str:N \spunit \c_space_tl if~you~need~to~print~numbers~with~units.
492 }
493 \msg_new:nnnn { spintent } { spnum-no-digits }
494 {
495     The~\token_to_str:N \spnum \c_space_tl command~requires~a~valid~numerical~value.
496 }
497 {
498     You~provided~''#1',~which~contains~no~digits.
499 }

```

(End of definition for `spnum-has-units` and `spnum-no-digits`.)

11.14.4 Implementation of the `\spum` command

`\spnum` The public user command `\spnum` accepts an optional $[\langle key = val \rangle]$ argument in ‘#1’ and a mandatory $\langle rational\ number \rangle$ in ‘#2’. It first checks if it is executed inside math mode, raising an error if it is not. If correct, it opens a local group to apply the options and passes the input to Lua for parsing via `\__spintent_luafun_clean_split_and_set:n`.

Before printing, it validates the extracted data. It throws an error if it finds trailing units in the input or if the integer part is completely empty (no valid digits). If all checks pass, it calls `\__spintent_snum_print_number_core:` to render the final number, and finally closes the group.

```

500 \NewDocumentCommand \spnum { O{} m }
501 {
502     \mode_if_math:TF
503     {
504         \group_begin:
505         \keys_set:nn { spintent/spnum } { #1 }
506         \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }
507         \str_if_eq:VnTF \__spintent_luaset_has_units_str { true }
508         { \msg_error:nnn { spintent } { spnum-has-units } { #2 } }
509         {
510             \tl_if_empty:NTF \__spintent_luaset_part_int_tl
511             { \msg_error:nnn { spintent } { spnum-no-digits } { #2 } }
512             { \__spintent_snum_print_number_core: }

```



```

513     }
514     \group_end:
515 }
516 { \msg_error:nnn { spintent } { need-math-mode } { \spnum } }
517 }

```

(End of definition for `\spnum`. This function is documented on page 3.)

11.15 The command `\spunit`

The user command `\spunit` is the *second* of “the big four” provided by the `spintent` package and is used to parse and print physical units and angular systems. It sends the $\langle argument \rangle$ to Lua module `spintent.lua` via `__spintent_luafun_clean_split_and_set:n` to separate and clean the $\langle argument \rangle$.

This ensures that all “units” (such as SI fractions, products, or sexagesimal angles) are processed consistently by the `spintent.lua`.

11.15.1 Definition of variables

```

\__spintent_spunit_luaset_canonical_str
\__spintent_spunit_luaset_lookup_status_str
\__spintent_spunit_luaset_read_str
\__spintent_spunit_luaset_is_compact_str
\__spintent_spunit_luaset_compact_exp_str
\__spintent_spunit_luaset_is_sexagesimal_str
\__spintent_spunit_luaset_status_str

```

These string variables store the data returned by the Lua module `spintent.lua`. They hold the canonical “unit symbol”, the dictionary lookup status, the correct speech text for NVDA/MathCAT, and specific flags that indicate if the “unit” is compact or represents a sexagesimal angle.

```

518 \str_new:N \__spintent_spunit_luaset_canonical_str
519 \str_new:N \__spintent_spunit_luaset_lookup_status_str
520 \str_new:N \__spintent_spunit_luaset_read_str
521 \str_new:N \__spintent_spunit_luaset_is_compact_str
522 \str_new:N \__spintent_spunit_luaset_compact_exp_str
523 \str_new:N \__spintent_spunit_luaset_is_sexagesimal_str
524 \str_new:N \__spintent_spunit_luaset_status_str

```

(End of definition for `\__spintent_spunit_luaset_canonical_str` and others.)

```

\__spintent_spunit_exponent_tl
\__spintent_spunit_unit_name_tl
\__spintent_spunit_read_slash_str
\__spintent_spunit_exp_seq
\__spintent_spunit_mult_seq
\__spintent_spunit_is_mult_bool
\__spintent_spunit_read_cdot_bool

```

These variables are used to process and format the “unit”. The sequences `\__spintent_spunit_mult_seq` and `\__spintent_spunit_exp_seq` split compound “units” at multiplication ‘`*`’ or exponent ‘`^`’ boundaries.

The boolean variables control specific rendering options, such as whether the multiplication dot ‘`\cdot`’ is explicitly read aloud or silently skipped.

```

525 \tl_new:N \__spintent_spunit_exponent_tl
526 \tl_new:N \__spintent_spunit_unit_name_tl
527 \str_new:N \__spintent_spunit_read_slash_str
528 \seq_new:N \__spintent_spunit_exp_seq
529 \seq_new:N \__spintent_spunit_mult_seq
530 \bool_new:N \__spintent_spunit_is_mult_bool
531 \bool_new:N \__spintent_spunit_read_cdot_bool

```

(End of definition for `\__spintent_spunit_exponent_tl` and others.)

11.15.2 Definition of error messages

```

spunit-invalid-unit
spunit-no-units
inline-numeric-denominator

```

We define three error messages for the `\spunit` command. The first “error” triggers if the “unit expression” provided in ‘`#1`’ is unknown or malformed. The second “error” occurs if no “units” are found in the $\langle argument \rangle$, advising the user to use `\spnum` for pure numbers. The third “error” is raised if a numeric coefficient is illegally placed in the denominator of an inline fraction (after a slash).

```

532 \msg_new:nnnn { spintent } { spunit-invalid-unit }
533 {
534   The~expression~'#1'~contains~an~unknown~or~malformed~unit~structure.
535 }
536 {
537   Check~spelling~or~verify~the~unit~is~defined~in~the~dictionary.
538 }
539 \msg_new:nnnn { spintent } { spunit-no-units }
540 {
541   The~\token_to_str:N \spunit \c_space_tl command~requires~at~least~one~unit~('#1').
542 }
543 {
544   Use~\token_to_str:N \spnum \c_space_tl if~you~only~need~to~format~a~pure~number.
545 }
546 \msg_new:nnnn { spintent } { inline-numeric-denominator }
547 {
548   Illegal~numeric~coefficient~in~inline~denominator~'#1'.
549 }
550 {
551   The~inline~mode~of~\token_to_str:N \spunit \c_space_tl
552   does~not~allow~numbers~after~the~slash~(/).
553 }

```

(End of definition for `spunit-invalid-unit`, `spunit-no-units`, and `inline-numeric-denominator`.)

11.15.3 Definition of keys for \spunit

Now we add the keys `print-cdot`, `read-cdot`, `read-slash`, `cifras-sep`, `read-period-sep` and `notation-sym`.

```

554 \keys_define:nn { spintent/spunit }
555 {
556   print-cdot      .bool_set:N = \l__spintent_spunit_print_cdot_bool,
557   print-cdot      .initial:n = false,
558   print-cdot      .value_required:n = true,
559   read-cdot       .bool_set:N = \l__spintent_spunit_read_cdot_bool,
560   read-cdot       .initial:n = false,
561   read-cdot       .value_required:n = true,
562   read-slash      .choices:nn =
563     { por , partido-por }
564   {
565     \int_case:nn { \l_keys_choice_int }
566     {
567       {1} { \str_set:Nn \l__spintent_spunit_read_slash_str { por } }
568       {2} { \str_set:Nn \l__spintent_spunit_read_slash_str { partido-por } }
569     }
570   },
571   read-slash      .initial:n = por,
572   read-slash      .value_required:n = true,
573   read-slash / unknown .code:n =
574     \msg_error:nneee { spintent } { unknown-choice }
575     { read-slash } { por, ~partido-por } { \exp_not:n {#1} },
576   cifras-sep      .meta:nn = { spintent/spnum } { cifras-sep = {#1} },
577   cifras-sep      .value_required:n = true,
578   read-period-sep .meta:nn = { spintent/spnum } { read-period-sep = {#1} },
579   read-period-sep .value_required:n = true,
580   notation-sym    .meta:nn = { spintent/spnum } { notation-sym = {#1} },
581   notation-sym    .value_required:n = true,
582   unknown         .code:n =
583     {
584       \str_set:Nn \l__spintent_command_name_str { spunit }
585       \__spintent_unknown_keys_cmd:n {#1}
586     },
587 }

```

(End of definition for `read-cdot` and others.)

11.15.4 Internal functions for \spunit

The function `\__spintent_spunit_format_canonical:` handles the visual formatting of the “unit symbol”. It specifically checks for the *liter symbol* ‘*ℓ*’ to print it as `\ell`, and formats all other valid “units” in standard upright text using `\mathrm`.

The function `\__spintent_spunit_process_base_exp:n` evaluates a “single unit” along with its exponent. It first looks up the unit ‘*#1*’ in the dictionary. If the “unit” is not found, it flags it as invalid. If found, it handles the spacing and multiplication dot ‘`\cdot`’ intents, and then renders the base unit and its exponent (either using a pre-defined compact exponent from Lua or by manually splitting the string at the ‘`^`’ symbol).

```

588 \cs_new_protected:Nn \__spintent_spunit_format_canonical:
589 {
590   \str_case:VnF \l__spintent_spunit_luaset_canonical_str
591   {
592     { ℓ } { \ell }
593   }
594   { \mathrm { \l__spintent_spunit_luaset_canonical_str } }
595 }
596 \cs_new_protected:Nn \__spintent_spunit_process_base_exp:n
597 {
598   \tl_clear:N \l__spintent_spunit_exponent_tl
599   \__spintent_luafun_spunit_lookup_alias:n { #1 }
600   \str_if_eq:VnTF \l__spintent_spunit_luaset_lookup_status_str { notfound }
601   {
602     \str_set:Nn \l__spintent_luaset_status_str { invalid }
603   }
604   {
605     \bool_if:NF \l__spintent_spunit_is_mult_bool
606     {
607       \bool_if:NTF \l__spintent_spunit_print_cdot_bool
608       {
609         \bool_if:NTF \l__spintent_spunit_read_cdot_bool

```

```

610         { \MathMLintent { por } { \cdot } }
611         { \MathMLintent { :silent } { \cdot } }
612     }
613     {
614         \bool_if:NTF \l__spintent_spunit_read_cdott_bool
615         { \MathMLintent { por } { \, } }
616         { \, }
617     }
618 }
619 \str_if_eq:VnTF \l__spintent_spunit_luaset_is_compact_str { true }
620 {
621     {
622         \MathMLintent { \l__spintent_spunit_luaset_read_str }
623         {
624             \__spintent_spunit_format_canonical:
625         }
626     }
627     ^ { \l__spintent_spunit_luaset_compact_exp_str }
628 }
629 {
630     \seq_set_split:Nnn \l__spintent_spunit_exp_seq { ^ } { #1 }
631     \seq_pop_left:NN \l__spintent_spunit_exp_seq \l__spintent_spunit_unit_name_tl
632     \seq_if_empty:NF \l__spintent_spunit_exp_seq
633     {
634         \seq_pop_left:NN \l__spintent_spunit_exp_seq \l__spintent_spunit_exponent_tl
635     }
636     \MathMLintent { \l__spintent_spunit_luaset_read_str }
637     {
638         \__spintent_spunit_format_canonical:
639     }
640     \tl_if_empty:NF \l__spintent_spunit_exponent_tl
641     {
642         ^ { \l__spintent_spunit_exponent_tl }
643     }
644 }
645 \bool_set_false:N \l__spintent_spunit_is_mult_bool
646 }
647 }

```

(End of definition for `\__spintent_spunit_format_canonical:` and `\__spintent_spunit_process_base_exp:n`.)

`\__spintent_spunit_process_mult:n`
`\__spintent_spunit_process_mult:V`

The function `\__spintent_spunit_process_mult:n` handles compound “units” joined by an explicit multiplication star ‘*’. It splits the `{\argument}` at each “star symbol” and applies `\__spintent_spunit_process_base_exp:n` to evaluate and render each resulting unit individually.

```

648 \cs_new_protected:Nn \__spintent_spunit_process_mult:n
649 {
650     \bool_set_true:N \l__spintent_spunit_is_mult_bool
651     \seq_set_split:Nnn \l__spintent_spunit_mult_seq { * } { #1 }
652     \seq_map_function:NN \l__spintent_spunit_mult_seq \__spintent_spunit_process_base_exp:n
653 }
654 \cs_generate_variant:Nn \__spintent_spunit_process_mult:n { V }

```

(End of definition for `\__spintent_spunit_process_mult:n`.)

`\__spintent_print_spunit_hierarchy:`

The function `\__spintent_print_spunit_hierarchy:` handles the layout of inline “unit” fractions. It first processes the “units” before the slash (the numerator). If a denominator exists, it prints the slash / ‘/’ with its corresponding speech intent, and then processes the remaining “units”.

```

655 \cs_new_protected:Nn \__spintent_print_spunit_hierarchy:
656 {
657     \tl_if_empty:NF \l__spintent_luaset_arg_above_tl
658     {
659         \__spintent_spunit_process_mult:V \l__spintent_luaset_arg_above_tl
660         \tl_if_empty:NF \l__spintent_luaset_arg_below_tl
661         {
662             \MathMLintent { \l__spintent_spunit_read_slash_str } { / }
663             \__spintent_spunit_process_mult:V \l__spintent_luaset_arg_below_tl
664         }
665     }
666 }

```

(End of definition for `\__spintent_print_spunit_hierarchy:`.)

`\__spintent_spunit_safe_exec:n`

The function `\__spintent_spunit_safe_exec:n` performs the main validation and layout for “units”. It first checks if the `{⟨argument⟩}` has units, is a valid expression, and does not contain illegal numbers in the denominator, throwing the appropriate “errors” if any check fails.

If the unit is valid, it checks if it is a sexagesimal angle (degrees, minutes, or seconds). For sexagesimal “units”, it prints the number and the symbol “*without any space in between*”, applying a slight negative space ‘\!’ for arcseconds. For standard units, it prints the number followed by a thin space ‘\,’ and then processes the unit sequence.

```

667 \cs_new_protected:Nn \__spintent_spunit_safe_exec:n
668 {
669   \str_if_eq:VnTF \l__spintent_luaset_has_units_str { false }
670   { \msg_error:nnn { spintent } { spunit-no-units } { #1 } }
671   {
672     \str_if_eq:VnTF \l__spintent_luaset_status_str { valid }
673     {
674       \str_if_eq:VnTF
675       \l__spintent_luaset_denom_has_number_str { true }
676       {
677         \msg_error:nnn { spintent }
678         { inline-numeric-denominator } { #1 }
679       }
680       {
681         \__spintent_luafun_spunit_lookup_alias:V \l__spintent_luaset_arg_above_tl
682         \str_if_eq:VnTF \l__spintent_spunit_luaset_is_sexagesimal_str { true }
683         {
684           \tl_if_empty:NF \l__spintent_luaset_part_int_tl
685           {
686             \__spintent_spnum_print_number_core:
687           }
688           \MathMLintent { \l__spintent_spunit_luaset_read_str }
689           {
690             \mathord
691             {
692               \str_case:Vn \l__spintent_spunit_luaset_canonical_str
693               {
694                 { " } { \prime \! \prime }
695                 { ' } { \prime }
696                 { ° } { ° }
697               }
698             }
699           }
700         }
701         {
702           \tl_if_empty:NF \l__spintent_luaset_part_int_tl
703           {
704             \__spintent_spnum_print_number_core: \,
705           }
706           \__spintent_print_spunit_hierarchy:
707         }
708       }
709     }
710     { \msg_error:nnn { spintent } { spunit-invalid-unit } { #1 } }
711   }
712 }

```

(End of definition for `\__spintent_spunit_safe_exec:n`.)

11.15.5 Implementation of the `\spunit` command

`\spunit`

The public user command `\spunit` accepts an optional `[⟨key = val⟩]` argument in ‘#1’ and a mandatory `{⟨unit expression⟩}` in ‘#2’. It first checks if it is executed inside *math mode*, raising an “error” if it is not. If correct, it opens a local group to apply the options and passes the `{⟨argument⟩}` to Lua module for parsing via `\__spintent_luafun_clean_split_and_set:n` function.

Finally, it delegates the execution to the function `\__spintent_spunit_safe_exec:n` to perform the safety checks and render the parsed data, before closing the group.

```

713 \NewDocumentCommand \spunit { O{} m }
714 {
715   \mode_if_math:TF
716   {
717     \group_begin:
718     \keys_set:nn { spintent/spunit } { #1 }
719     \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #2 } }

```

```

720         \__spintent_spunit_safe_exec:n { #2 }
721     \group_end:
722 }
723 { \msg_error:nnn { spintent } { need-math-mode } { \spunit } }
724 }

```

(End of definition for `\spunit`. This function is documented on page 4.)

11.16 The commands `\spnewunit` and `\spunitalias`

The user commands `\spnewunit` and `\spunitalias` allow users to add new custom “units” or “aliases” to the package’s dictionary. Similar to `\spunit`, these commands delegate the processing and validation to the `spintent.lua` via `\__spintent_luafun_define_custom_unit:nn` and `\__spintent_luafun_define_spunit`.

During the definition, the Lua module checks if the new “unit” or “alias” is valid and ensures it does not conflict with existing “units”. It returns the result of this operation to TeX using the `\__spintent_spunit_luaset_status_str` variable. If a collision is detected or the `{⟨argument⟩}` is invalid, the package throws the corresponding “error” message.

11.16.1 Definition of error messages

spnewunit-duplicate
spunitalias-duplicate
spunitalias-invalid-expr

We define three error messages for the creation of “units” and “aliases”. The first two “errors” trigger if the user attempts to define a “new unit” or alias passed in ‘#1’ that already exists in the dictionary. The third “error” is raised by `\spunitalias` if the target expression provided in ‘#2’ is invalid (for example, if it contains unknown “units” or incorrect syntax).

```

725 \msg_new:nnnn { spintent } { spnewunit-duplicate }
726 {
727     The~unit~symbol~or~alias~'#1'~is~already~defined!
728 }
729 {
730     You~cannot~redefine~with~\token_to_str:N \spnewunit.
731 }
732 \msg_new:nnnn { spintent } { spunitalias-duplicate }
733 {
734     The~alias~'#1'~is~already~defined~or~reserved!
735 }
736 {
737     You~cannot~redefine~an~existing~unit~or~another~alias.
738 }
739 \msg_new:nnnn { spintent } { spunitalias-invalid-expr }
740 {
741     The~expression~'#2'~for~alias~'#1'~is~invalid.
742 }
743 {
744     It~must~contain~only~defined~units,~exponents~and~one~'/'~.
745 }

```

(End of definition for `spnewunit-duplicate`, `spunitalias-duplicate`, and `spunitalias-invalid-expr`.)

`\spnewunit`

The public command `\spnewunit` defines a new custom unit. It passes the new unit symbol ‘#1’ and its speech text ‘#2’ to Lua via `\__spintent_luafun_define_custom_unit:nn`. If the module reports that the unit already exists, it throws the corresponding duplicate error.

```

746 \NewDocumentCommand \spnewunit { m m }
747 {
748     \mode_if_math:TF
749     {
750         \msg_error:nnn { spintent } { need-text-mode } { \spnewunit }
751     }
752     {
753         \__spintent_luafun_define_custom_unit:nn { #1 } { #2 }
754         \str_if_eq:VnT \__spintent_spunit_luaset_status_str { duplicate }
755         {
756             \msg_error:nnn { spintent } { spnewunit-duplicate } { #1 }
757         }
758     }
759 }

```

(End of definition for `\spnewunit`. This function is documented on page 6.)

`\spunitalias`

The public command `\spunitalias` creates a shortcut for a complex unit expression. It passes the new alias name ‘#1’ and the target expression ‘#2’ to Lua via `\__spintent_luafun_define_spunit_alias:nn`. It throws an error if the alias already exists, or if the provided target expression contains invalid syntax or unknown units.

```

760 \NewDocumentCommand \spunitalias { m m }

```

```

761 {
762   \mode_if_math:TF
763   {
764     \msg_error:nnn { spintent } { need-text-mode } { \spunitalias }
765   }
766   {
767     \__spintent_luafun_define_spunit_alias:nn { #1 } { #2 }
768     \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { duplicate }
769     {
770       \msg_error:nnn { spintent } { spunitalias-duplicate } { #1 }
771     }
772     \str_if_eq:VnT \l__spintent_spunit_luaset_status_str { invalid-expr }
773     {
774       \msg_error:nnnn { spintent } { spunitalias-invalid-expr } { #1 } { #2 }
775     }
776   }
777 }

```

(End of definition for `\spunitalias`. This function is documented on page 6.)

11.17 The command `\spmoney`

The user command `\spmoney` is the *third* of “the big four” provided by the `spintent` package, designed to format monetary values and currencies. It delegates `{\argument}` parsing to the Lua module `spintent.lua` via `\__spintent_luafun_clean_split_and_set:n`. The module then retrieves the correct “currency symbol”, pluralization rules, and layout structures from the database.

The command accepts *optional arguments* to override the default “currency symbol” or enable “subunit” rendering. Finally, it constructs the MathML intent for MathCAT.

These string variables store the currency data returned by the Lua module `spintent.lua`. Upon a successful database lookup, the module populates these variables with the “currency symbol”, its layout position (pre or post), the appropriate singular and plural names for both the main unit and subunits, and the execution status flags. TeX then uses this information to render the visual layout and construct the correct MathML speech intents.

```

778 \str_new:N \l__spintent_spmoney_luaset_symbol_str
779 \str_new:N \l__spintent_spmoney_luaset_position_str
780 \str_new:N \l__spintent_spmoney_luaset_sing_str
781 \str_new:N \l__spintent_spmoney_luaset_plur_str
782 \str_new:N \l__spintent_spmoney_luaset_conde_str
783 \str_new:N \l__spintent_spmoney_luaset_subsing_str
784 \str_new:N \l__spintent_spmoney_luaset_subplur_str
785 \str_new:N \l__spintent_spmoney_luaset_print_iso_str
786 \str_new:N \l__spintent_spmoney_luaset_currency_arg_str
787 \str_new:N \l__spintent_spmoney_luaset_status_str

```

(End of definition for `\l__spintent_spmoney_luaset_symbol_str` and others.)

The boolean variable tracks whether subunit rendering is enabled. The string variables assemble the specific speech text (intents) for NVDA/MathCAT, including the sign, the main currency, and the subunits. Finally, the remaining variables temporarily store the numeric value and currency code extracted from the user’s input.

```

788 \bool_new:N \l__spintent_spmoney_sub_units_bool
789 \str_new:N \l__spintent_spmoney_intent_money_str
790 \str_new:N \l__spintent_spmoney_intent_sign_str
791 \str_new:N \l__spintent_spmoney_intent_sub_unit_str
792 \str_new:N \l__spintent_spmoney_res_main_voice_str
793 \int_new:N \l__spintent_spmoney_subnum_int
794 \str_new:N \l__spintent_spmoney_extracted_curr_str
795 \tl_new:N \l__spintent_spmoney_extracted_num_tl

```

(End of definition for `\l__spintent_spmoney_sub_units_bool` and others.)

11.17.1 Definition of error messages

We define two messages to handle invalid currency configurations. The first “error” warns the user if subunit rendering is requested for a currency that does not support fractional divisions (such as JPY or CLP), safely ignoring the flag. The second “error” is triggered if the provided currency identifier is not registered in the Lua database.

```

796 \msg_new:nnnn { spintent } { currency-no-subunits }
797 {
798   The~currency~'#1'~does~not~support~subunits.~
799   The~option~'sub-units=true'~is~ignored.
800 }

```



```

801 {
802     Monies~like~the~Japanese~Yen~(JPY)~or~Chilean~Peso~(CLP)~
803     do~not~have~fractional~subunits~registered~in~Lua~module.~
804 }
805 \msg_new:nnnn { spintent } { invalid-currency }
806 {
807     Invalid~currency~identifier~'#1'~in~\token_to_str:N \spmoney .
808 }
809 {
810     The~provided~symbol,~alias,~or~ISO~code~does~not~exist~in~
811     the~Lua~module.
812 }

```

(End of definition for `currency-no-subunits` and `invalid-currency`.)

11.17.2 Auxiliary functions for \spmoney

```

__spintent_spmoney_set_currency:n
__spintent_spmoney_normalize_key:n
__spintent_spmoney_normalize_key:V

```

The function `\__spintent_spmoney_set_currency:n` validates the currency identifier provided in ‘#1’ against the database. It delegates the lookup to the helper function `\__spintent_spmoney_normalize_key:n`, which interfaces with the Lua module `spintent.lua`. If the module returns a not found status, an “error” is triggered. Otherwise, it stores the valid currency identifier in `\__spintent_spmoney_luaset_currency_arg_str` for subsequent processing.

```

813 \cs_new_protected:Nn \__spintent_spmoney_set_currency:n
814 {
815     \__spintent_spmoney_normalize_key:n {#1}
816     \str_if_eq:VnTF \__spintent_spmoney_luaset_status_str { notfound }
817     {
818         \msg_error:nnn { spintent } { invalid-currency } {#1}
819     }
820     { \str_set:Nn \__spintent_spmoney_luaset_currency_arg_str {#1} }
821 }
822 \cs_new_protected:Nn \__spintent_spmoney_normalize_key:n
823 {
824     \__spintent_luafun_spmoney_normalize_key:n { #1 }
825 }
826 \cs_generate_variant:Nn \__spintent_spmoney_normalize_key:n { V }

```

(End of definition for `\__spintent_spmoney_set_currency:n` and `\__spintent_spmoney_normalize_key:n`.)

11.17.3 Definition of keys for \spmoney

```

currency
sub-units
dolar-math
cifras-sep

```

Now we add the keys `currency`, `sub-units`, `dolar-math` and `cifras-sep`.

```

827 \keys_define:nn { spintent / spmoney }
828 {
829     currency .code:n = { \__spintent_spmoney_set_currency:n {#1} },
830     currency .initial:n = { pesos },
831     currency .value_required:n = true,
832     sub-units .bool_set:N = \__spintent_spmoney_sub_units_bool,
833     sub-units .value_required:n = true,
834     dolar-math .bool_set:N = \__spintent_spmoney_dolar_math_bool,
835     dolar-math .initial:n = { true },
836     dolar-math .value_required:n = true,
837     cifras-sep .meta:nn = { spintent/spnum } { cifras-sep = {#1} },
838     cifras-sep .value_required:n = true,
839     unknown .code:n =
840     {
841         \str_set:Nn \__spintent_command_name_str { spmoney }
842         \__spintent_unknown_keys_cmd:n {#1}
843     },
844 }

```

(End of definition for `currency` and others.)

11.17.4 Internal functions for \spmoney

```
__spintent_spmoney_validate_money:
```

The function `\__spintent_spmoney_validate_money:` verifies if the parsed input contains an embedded currency identifier by checking `\__spintent_luaset_has_units_str`. If a currency is detected, it extracts and normalizes the key from `\__spintent_luaset_arg_above_tl`. It throws an “error” if the currency is not found in the database. If valid, it updates `\__spintent_spmoney_luaset_currency_arg_str` and retrieves the full currency metadata from the `spintent.lua` via `\__spintent_luafun_spmoney_lookup_data:V`.

```

845 \cs_new_protected:Nn \__spintent_spmoney_validate_money:
846 {
847     \str_if_eq:VnT \__spintent_luaset_has_units_str { true }
848     {

```

```

849     \__spintent_spmoney_normalize_key:V \__spintent_luaset_arg_above_tl
850     \str_if_eq:VnTF \__spintent_spmoney_luaset_status_str { notfound }
851     { \msg_error:nne { spintent } { invalid-currency } { \__spintent_luaset_arg_above_tl } }
852     {
853         \str_set:NV
854         \__spintent_spmoney_luaset_currency_arg_str
855         \__spintent_luaset_arg_above_tl
856     }
857 }
858 \str_set:Nc \__spintent_spmoney_luaset_status_str { found }
859 \__spintent_luafun_spmoney_lookup_data:V \__spintent_spmoney_luaset_currency_arg_str
860 }

```

(End of definition for `\__spintent_spmoney_validate_money:`.)

`\__spintent_spmoney_print_symbol:`

The function `\__spintent_spmoney_print_symbol:` renders the “*currency symbol*”. If `\__spintent_spmoney_luaset_print_iso_str` is `true`, it outputs the ISO code using `\mathrm`. Otherwise, it prints the “*standard symbol*”. For dollar signs ‘\$’, it checks `\__spintent_spmoney_dolar_math_bool` to determine whether to render it directly as a math character ‘\$’ or wrap it inside `\text`.

```

861 \cs_new_protected:Nn \__spintent_spmoney_print_symbol:
862 {
863     \str_if_eq:VnTF \__spintent_spmoney_luaset_print_iso_str { true }
864     { \mathrm { \__spintent_spmoney_luaset_currency_arg_str } }
865     {
866         \str_if_eq:VnTF \__spintent_spmoney_luaset_symbol_str { $ }
867         {
868             \bool_if:NTF \__spintent_spmoney_dolar_math_bool
869             { \$ } { \text { \$ } }
870         }
871         { \__spintent_spmoney_luaset_symbol_str }
872     }
873 }

```

(End of definition for `\__spintent_spmoney_print_symbol:`.)

`\__spintent_spmoney_render_layout:`

The function `\__spintent_spmoney_render_layout:` determines the layout order of the “*currency symbol*” relative to the numeric value. It evaluates `\__spintent_spmoney_luaset_position_str`; if set to `pre`, it renders the “*currency symbol*” *before* the number node. Otherwise, it places the number first, followed by the “*currency symbol*”.

```

874 \cs_new_protected:Nn \__spintent_spmoney_render_layout:
875 {
876     \str_if_eq:VnTF \__spintent_spmoney_luaset_position_str { pre }
877     {
878         \__spintent_spmoney_print_symbol:
879         \MathMLarg { n } { \__spintent_spmoney_render_number_node: }
880     }
881     {
882         \MathMLarg { n } { \__spintent_spmoney_render_number_node: }
883         \__spintent_spmoney_print_symbol:
884     }
885 }

```

(End of definition for `\__spintent_spmoney_render_layout:`.)

`\__spintent_spmoney_render_number_node:`
`\__spintent_spmoney_render_only_integer_node:`

The function `\__spintent_spmoney_render_number_node:` renders the full numeric value, encompassing both the integer and decimal components. Conversely, the function `\__spintent_spmoney_render_only_integer_node:` outputs exclusively the integer portion, omitting any decimal data.

```

886 \cs_new_protected:Nn \__spintent_spmoney_render_number_node:
887 {
888     \__spintent_spnum_render_int:
889     \__spintent_spnum_print_part_dec:
890 }
891 \cs_new_protected:Nn \__spintent_spmoney_render_only_integer_node:
892 {
893     \__spintent_spnum_render_int:
894 }

```

(End of definition for `\__spintent_spmoney_render_number_node:` and `\__spintent_spmoney_render_only_integer_node:`.)

`\__spintent_spmoney_print_int:`
`\__spintent_spmoney_intent_int:`

The function `\__spintent_spmoney_print_int:` evaluates the presence of an explicit sign in `\l__spintent_luaset_sign_tl`. If a sign is detected, it assigns a prefix speech intent ‘más’ or ‘menos’ and embeds the sign and subsequent layout inside a `\MathMLintent`. If no sign is present, it routes execution based on whether subunits are enabled.

The function `\__spintent_spmoney_intent_int:` determines the correct grammatical inflection for the currency string (singular, plural, or the prepositional ‘de’ variant for exact millions). It evaluates the integer component, decimal boundaries, and zero-fills to select the appropriate string register, applying it as a postfix `MathML` intent over the layout rendered by `\__spintent_spmoney_render_layout:`.

```

895 \cs_new_protected:Nn \__spintent_spmoney_print_int:
896 {
897   \tl_if_empty:NTF \l__spintent_luaset_sign_tl
898   {
899     \bool_if:NTF \l__spintent_spmoney_sub_units_bool
900     { \__spintent_spmoney_set_and_print_sub_units: }
901     { \__spintent_spmoney_intent_int: }
902   }
903   {
904     \str_if_eq:VnTF \l__spintent_luaset_sign_tl { + }
905     {
906       \str_set:Nn \l__spintent_spmoney_intent_sign_str { _más:prefix ($n) }
907     }
908     {
909       \str_set:Nn \l__spintent_spmoney_intent_sign_str { _menos:prefix ($n) }
910     }
911     \__spintent_invisible_sep:
912     \MathMLintent { \l__spintent_spmoney_intent_sign_str }
913     {
914       \l__spintent_luaset_sign_tl
915       \MathMLarg { n }
916       {
917         \bool_if:NTF \l__spintent_spmoney_sub_units_bool
918         { \__spintent_spmoney_set_and_print_sub_units: }
919         { \__spintent_spmoney_intent_int: }
920       }
921     }
922   }
923 }
924 \cs_new_protected:Nn \__spintent_spmoney_intent_int:
925 {
926   \tl_if_empty:NTF \l__spintent_luaset_dec_sep_tl
927   {
928     \str_if_eq:VnTF \l__spintent_luaset_only_part_int_str { 1 }
929     {
930       \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_sing_str
931     }
932     {
933       \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_str
934     }
935     \str_if_eq:VnT \l__spintent_luaset_millions_str { true }
936     {
937       \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_conde_s
938     }
939   }
940   {
941     \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur_str
942     \str_if_eq:VnT \l__spintent_luaset_millions_str { true }
943     {
944       \str_if_eq:VnTF \l__spintent_luaset_dec_is_all_zeros_str { true }
945       {
946         \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_conde
947       }
948       {
949         \str_set:NV \l__spintent_spmoney_intent_money_str \l__spintent_spmoney_luaset_plur
950       }
951     }
952   }
953   \__spintent_invisible_sep: % need for MathCAT
954   \MathMLintent { \l__spintent_spmoney_intent_money_str :postfix($n) }
955   {
956     \__spintent_spmoney_render_layout:

```

```

957     }
958 }

```

(End of definition for `\__spintent_spmoney_print_int:` and `\__spintent_spmoney_intent_int:`)

```

\__spintent_spmoney_check_sub_units:
spintent_spmoney_set_and_print_sub_units:

```

The function `\__spintent_spmoney_check_sub_units:` verifies whether the selected currency supports “*subunits*”. If the database returns empty subunit data, it triggers a warning and automatically disables the “*subunit*” rendering flag.

The function `\__spintent_spmoney_set_and_print_sub_units:` processes and formats the speech intents when “*subunits*” are active. It normalizes the decimal component (for instance, padding a single-digit decimal like ‘5’ to ‘50’), evaluates the correct singular or plural inflections for both the main “*currency*” and the “*subunits*”, and delegates the final MathML tree assembly to `\__spintent_spmoney_build_tree:`.

```

959 \cs_new_protected:Nn \__spintent_spmoney_check_sub_units:
960 {
961   \str_if_empty:NT \__spintent_spmoney_luaset_subsing_str
962   {
963     \msg_warning:nne { spintent } { currency-no-subunits }
964     { \__spintent_spmoney_luaset_currency_arg_str }
965     \bool_set_false:N \__spintent_spmoney_sub_units_bool
966   }
967 }
968 \cs_new_protected:Nn \__spintent_spmoney_set_and_print_sub_units:
969 {
970   \int_compare:nTF { \str_count:N \__spintent_luaset_only_part_dec_str == 1 }
971   {
972     \int_set:Nn \__spintent_spmoney_subnum_int
973     { \__spintent_luaset_only_part_dec_str * 10 }
974     \str_set:Ne \__spintent_luaset_only_part_dec_str
975     { \__spintent_luaset_only_part_dec_str 0 }
976   }
977   {
978     \int_set:Nn \__spintent_spmoney_subnum_int
979     { \__spintent_luaset_only_part_dec_str }
980   }
981   \int_compare:nTF { \__spintent_spmoney_subnum_int == 1 }
982   {
983     \str_set:Ne \__spintent_spmoney_intent_sub_unit_str
984     { \__spintent_spmoney_luaset_subsing_str :postfix($sub) }
985   }
986   {
987     \str_set:Ne \__spintent_spmoney_intent_sub_unit_str
988     { \__spintent_spmoney_luaset_subplur_str :postfix($sub) }
989   }
990   \str_if_eq:VnTF \__spintent_luaset_only_part_int_str { 1 }
991   {
992     \str_set:NV \__spintent_spmoney_res_main_voice_str
993     \__spintent_spmoney_luaset_sing_str
994   }
995   {
996     \str_set:NV \__spintent_spmoney_res_main_voice_str
997     \__spintent_spmoney_luaset_plur_str
998   }
999   \str_if_eq:VnT \__spintent_luaset_millions_str { true }
1000   {
1001     \str_set:NV \__spintent_spmoney_res_main_voice_str
1002     \__spintent_spmoney_luaset_conde_str
1003   }
1004   \__spintent_spmoney_build_tree:
1005 }

```

(End of definition for `\__spintent_spmoney_check_sub_units:` and `\__spintent_spmoney_set_and_print_sub_units:`)

```

\__spintent_spmoney_build_tree:
spintent_spmoney_print_integer_continue:
\__spintent_spmoney_print_sep_con:
\__spintent_spmoney_print_dec_sub_unit:

```

The function `\__spintent_spmoney_build_tree:` structures the final MathML tree assembly for fractional “*currencies*”. It arranges the integer component, the decimal separator, and the “*subunit*” component based on the layout position (pre or post) of the “*currency symbol*”.

The function `\__spintent_spmoney_print_integer_continue:` isolates the rendering of the integer node with its associated main “*currency*” intent. Furthermore, `\__spintent_spmoney_print_sep_con:` applies the specific speech intent ‘con’ to the decimal separator, while `\__spintent_spmoney_print_dec_sub_unit:` binds the computed “*subunit*” intent to the decimal digits, ensuring accurate accessibility output.

```

1006 \cs_new_protected:Nn \__spintent_spmoney_build_tree:

```

```

1007 {
1008   \str_if_eq:VnTF \l__spintent_spmoney_luaset_position_str { pre }
1009   {
1010     \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
1011     {
1012       \__spintent_spmoney_print_symbol:
1013       \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
1014     }
1015     \__spintent_spmoney_print_sep_con:
1016     \__spintent_spmoney_print_dec_sub_unit:
1017   }
1018   {
1019     \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
1020     {
1021       \__spintent_invisible_sep:
1022       \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
1023     }
1024     \__spintent_spmoney_print_sep_con:
1025     \__spintent_spmoney_print_dec_sub_unit:
1026     \MathMLintent {:silent} { \__spintent_spmoney_print_symbol: }
1027   }
1028 }
1029 \cs_new_protected:Nn \__spintent_spmoney_print_integer_continue:
1030 {
1031   \MathMLintent { \l__spintent_spmoney_res_main_voice_str :postfix($n) }
1032   {
1033     \MathMLarg { n } { \__spintent_spmoney_render_only_integer_node: }
1034   }
1035 }
1036 \cs_new_protected:Nn \__spintent_spmoney_print_sep_con:
1037 {
1038   \MathMLintent { con }
1039   {
1040     \mathord { \l__spintent_luaset_dec_sep_tl }
1041   }
1042 }
1043 \cs_new_protected:Nn \__spintent_spmoney_print_dec_sub_unit:
1044 {
1045   \MathMLintent { \l__spintent_spmoney_intent_sub_unit_str }
1046   {
1047     \__spintent_invisible_sep:
1048     \MathMLarg { sub }
1049     {
1050       \__spintent_spmoney_render_dec:
1051     }
1052   }
1053 }

```

(End of definition for `\__spintent_spmoney_build_tree:` and others.)

`\__spintent_spmoney_parse_and_route:n`

The internal function `\__spintent_spmoney_parse_and_route:n` processes the $\{(\textit{argument})\}$ passed in ‘#1’ using Lua module to extract the numeric value and any “currency” identifier. If a “currency” is found (`\l__spintent_spmoney_extracted_curr_str`), it sets the corresponding variables. It then calls `\__spintent_spmoney_validate_money:` to verify the “currency”. If “subunit” rendering is enabled (`\l__spintent_spmoney_sub_units_bool`), it calls `\__spintent_spmoney_check_sub_units:`. Finally, it calls `\__spintent_spmoney_print_int:` to format the output.

```

1054 \cs_new_protected:Nn \__spintent_spmoney_parse_and_route:n
1055 {
1056   \__spintent_luafun_spmoney_prepare_input:n { #1 }
1057   \__spintent_luafun_clean_split_and_set:V \l__spintent_spmoney_extracted_num_tl
1058   \str_if_empty:NF \l__spintent_spmoney_extracted_curr_str
1059   {
1060     \str_set:Nx \l__spintent_luaset_has_units_str { true }
1061     \str_set:Nx \l__spintent_luaset_arg_above_tl \l__spintent_spmoney_extracted_curr_str
1062   }
1063   \__spintent_spmoney_validate_money:
1064   \bool_if:NT \l__spintent_spmoney_sub_units_bool
1065   { \__spintent_spmoney_check_sub_units: }
1066   \__spintent_spmoney_print_int:
1067 }

```

(End of definition for `\__spintent_spmoney_parse_and_route:n`.)

`\spmoney` The public document command `\spmoney` formats monetary values with accessible speech intents. It accepts an optional argument ‘#1’ for local key-value configurations and a mandatory argument ‘#2’ containing the numeric value and optional embedded currency identifier. It first evaluates the typesetting context, triggering a fatal error if executed outside of math mode. Upon validation, it establishes a local $\TeX$ group to safely isolate the key assignments, delegates the raw input string to `\__spintent_spmoney_parse_and_route:n` for parsing and rendering, and subsequently closes the group to prevent scope leakage.

```

1068 \NewDocumentCommand \spmoney { 0{ } m }
1069 {
1070   \mode_if_math:TF
1071   {
1072     \group_begin:
1073     \keys_set:nn { spintent / spmoney } { #1 }
1074     \__spintent_spmoney_parse_and_route:n { #2 }
1075     \group_end:
1076   }
1077   { \msg_error:nnn { spintent } { need-math-mode } { \spmoney } }
1078 }

```

(End of definition for `\spmoney`. This function is documented on page 6.)

11.18 The command `\spshort`

The user command `\spshort` is the *fourth* and *final* of “the big four”. It manages the semantic injection and typographical formatting of *text mode* abbreviations, ordinals, and acronyms. Unlike *math mode* commands, which rely on MathML attributes, this command uses the PDF *tagging tools* provided by the `tagpdf`[29] package. Specifically, it uses the `E` (expanded) key to ensure that NVDA interpret the full form of the abbreviation, ordinal, or acronym in *text mode*.

These internal string variables serve as the primary data interface with the Lua module `spintent.lua`. During processing, the Lua module “stores” within these registers the execution status, the targeted typographical layout parameters, the phonetic replacement text for the `E` key intended for NVDA, the final sanitized output string, and the isolated structural components for ordinals (base and suffix).

```

1079 \str_new:N \__spintent_spshort_luaset_status_str
1080 \str_new:N \__spintent_spshort_luaset_layout_str
1081 \str_new:N \__spintent_spshort_luaset_expanded_str
1082 \str_new:N \__spintent_spshort_luaset_output_str
1083 \str_new:N \__spintent_spshort_luaset_base_str
1084 \str_new:N \__spintent_spshort_luaset_suffix_str

```

(End of definition for `\__spintent_spshort_luaset_status_str` and others.)

11.18.1 Definition of error messages for `\spshort`

unknown-abbreviation The internal “error” message `unknown-abbreviation` is triggered when the command `\spshort` encounters an invalid lookup state from the Lua module. It acts as a strict fallback exception when the provided `{\argument}` cannot be mapped to any registered database entry in `spintent.lua` nor successfully resolved through the grammatical ordinal expansion rules.

```

1085 \msg_new:nnnn { spintent } { unknown-abbreviation }
1086 { The~abbreviation~or~ordinal~'~#1~'~is~invalid~or~not~recognized. }
1087 { Verify~the~spelling~or~check~the~grammar~rules~for~ordinals. }

```

(End of definition for `unknown-abbreviation`.)

11.18.2 Accessibility interface

`\__spintent_spshort_tag_span:nn` The internal function `\__spintent_spshort_tag_span:nn` processes the `{\argument}` passed in ‘#2’ using the tools provided by `tagpdf`. By utilizing the `tag` and `E` keys, it creates a native PDF /E (#1) structure and expands the `{\argument}` passed in ‘#1’ for proper reading with NVDA.

```

1088 \cs_new_protected:Nn \__spintent_spshort_tag_span:nn
1089 {
1090   \tag_mc_end_push:
1091   \tag_struct_begin:n { tag=Span, E={ #1 } }
1092   \tag_mc_begin:n { tag=Span }
1093   \tag_suspend:n { \spshort }
1094   #2
1095   \tag_resume:n { \spshort }
1096   \tag_mc_end:
1097   \tag_struct_end:
1098   \tag_mc_begin_pop:n {}
1099 }
1100 \cs_generate_variant:Nn \__spintent_spshort_tag_span:nn { en }

```

(End of definition for `\__spintent_spshort_tag_span:nn`.)

11.18.3 Definition of keys for \spshort

small-caps Now we add the keys `small-caps` and `read-arg`.

```
read-arg 1101 \keys_define:nn { spintent / spshort }
1102 {
1103     small-caps .bool_set:N = \l__spintent_spshort_smallcaps_bool,
1104     small-caps .initial:n = false,
1105     small-caps .value_required:n = true,
1106     read-arg .str_set:N = \l__spintent_spshort_read_arg_str,
1107     read-arg .initial:n = {},
1108     read-arg .value_required:n = true,
1109     unknown .code:n =
1110     {
1111         \str_set:Nn \l__spintent_command_name_str { spshort }
1112         \__spintent_unknown_keys_cmd:n {#1}
1113     },
1114 }
```

(End of definition for `small-caps` and `read-arg`.)

11.18.4 Internal functions for \spshort

The internal function `\__spintent_spshort_print:nn` acts as the primary formatter for text abbreviations. It clears the string variable `\l__spintent_spshort_read_arg_str` and evaluates the optional `<keys>` passed in ‘#1’. It then checks if a manual reading override was provided via those keys. If that string variable remains empty, it calls `\__spintent_spshort_process_and_render:n` to process the `{<argument>}` passed in ‘#2’. Otherwise, if an override is detected, it directly delegates the execution to `\__spintent_spshort_render_bypass:n`.

```
1115 \cs_new_protected:Nn \__spintent_spshort_print:nn
1116 {
1117     \str_clear:N \l__spintent_spshort_read_arg_str
1118     \keys_set:nn { spintent / spshort } {#1}
1119     \str_if_empty:NTF \l__spintent_spshort_read_arg_str
1120     {
1121         \__spintent_spshort_process_and_render:n {#2}
1122     }
1123     {
1124         \__spintent_spshort_render_bypass:n {#2}
1125     }
1126 }
```

(End of definition for `\__spintent_spshort_print:nn`.)

The internal function `\__spintent_spshort_process_and_render:n` interfaces with the Lua module `spintent.lua` function `\__spintent_luafun_spshort_process:n` to resolve the abbreviation or ordinal `{<argument>}` passed in ‘#1’.

It subsequently evaluates the execution state stored within the string variable `\l__spintent_spshort_luaset_status` and dynamically routes execution to the corresponding rendering subroutine success, fallback or `error`. Any unrecognized status automatically defaults to the “error” handler.

```
1127 \cs_new_protected:Nn \__spintent_spshort_process_and_render:n
1128 {
1129     \__spintent_luafun_spshort_process:n {#1}
1130     \str_case:NnF \l__spintent_spshort_luaset_status_str
1131     {
1132         { success } { \__spintent_spshort_render_success: }
1133         { fallback } { \__spintent_spshort_render_fallback: }
1134         { error } { \__spintent_spshort_render_error:n {#1} }
1135     }
1136     { \__spintent_spshort_render_error:n {#1} }
1137 }
```

(End of definition for `\__spintent_spshort_process_and_render:n`.)

The function `\__spintent_spshort_render_success:` injects the key `E` and executes the targeted typographical layout. The `\__spintent_spshort_render_fallback:` function handles default cases, applying optional formatting such as `SMALL CAPS` to unrecognized or partially resolved terms.

Conversely, the function `\__spintent_spshort_render_bypass:n` directly maps the user-defined “speech” to the `{<argument>}`, bypassing the automated lookup Lua module. Finally, the function `\__spintent_spshort_rende` triggers a structural exception while yielding the unformatted `{<argument>}` to prevent fatal compilation.

```
1138 \cs_new_protected:Nn \__spintent_spshort_render_success:
1139 {
```

```

1140     \__spintent_spshort_tag_span:en
1141     { \__spintent_spshort_luaset_expanded_str }
1142     { \__spintent_spshort_execute_layout: }
1143   }
1144   \cs_new_protected:Nn \__spintent_spshort_render_fallback:
1145   {
1146     \__spintent_spshort_tag_span:en
1147     { \__spintent_spshort_luaset_expanded_str }
1148     {
1149       \bool_if:NTF \__spintent_spshort_smallcaps_bool
1150       { \textsc{ \__spintent_spshort_luaset_output_str } }
1151       { \__spintent_spshort_luaset_output_str }
1152     }
1153   }
1154   \cs_new_protected:Nn \__spintent_spshort_render_bypass:n
1155   {
1156     \__spintent_spshort_tag_span:en { \__spintent_spshort_read_arg_str } { #1 }
1157   }
1158   \cs_new_protected:Nn \__spintent_spshort_render_error:n
1159   {
1160     \msg_error:nnn { spintent } { unknown-abbreviation } {#1}
1161   }

```

(End of definition for `\__spintent_spshort_render_success:` and others.)

`\__spintent_spshort_execute_layout:`

The internal function `\__spintent_spshort_execute_layout:` executes the typographical layout. It evaluates the descriptor “*stored*” within the string variable `\__spintent_spshort_luaset_layout_str` (such as `linear_regular`, `superscript`, or `small_caps_pure`) and maps it to the corresponding $\text{\TeX}$ formatting. Furthermore, it manages complex nested configurations, dynamically applying combinations such as superscripts integrated with conditional SMALL CAPS based on the active boolean variables.

```

1162   \cs_new_protected:Nn \__spintent_spshort_execute_layout:
1163   {
1164     \str_case:VnF \__spintent_spshort_luaset_layout_str
1165     {
1166       { linear_regular }
1167       {
1168         \bool_if:NTF \__spintent_spshort_smallcaps_bool
1169         { \textsc{ \__spintent_spshort_luaset_output_str } }
1170         { \__spintent_spshort_luaset_output_str }
1171       }
1172       { linear_caps }
1173       { \__spintent_spshort_luaset_output_str }
1174       { linear_mixed }
1175       { \__spintent_spshort_luaset_output_str }
1176       { superscript }
1177       {
1178         \__spintent_spshort_luaset_base_str
1179         \bool_if:NTF \__spintent_spshort_smallcaps_bool
1180         {
1181           \textsuperscript{
1182             \textsc{
1183               \str_lowercase:V \__spintent_spshort_luaset_suffix_str
1184             }
1185           }
1186         }
1187         {
1188           \textsuperscript{ \__spintent_spshort_luaset_suffix_str }
1189         }
1190       }
1191       { small_caps_pure }
1192       {
1193         \bool_if:NTF \__spintent_spshort_smallcaps_bool
1194         {
1195           \textsc{
1196             \str_lowercase:V \__spintent_spshort_luaset_output_str
1197           }
1198         }
1199         { \__spintent_spshort_luaset_output_str }
1200       }
1201       { acronym_long }
1202       { \__spintent_spshort_luaset_output_str }

```

```

1203     }
1204     { \l__spintent_spsshort_luaset_output_str }
1205 }

```

(End of definition for \l__spintent_spsshort_execute_layout:.)

\spsshort The public document command **\spsshort** manages the accessible formatting of *text mode* abbreviations, acronyms, and ordinals. It first triggers an error if invoked within a *math mode*, upon validation, puts **\mode_leave_vertical:** and establishes a local group to call the function **__spintent_spsshort_print:nn** for processing and printing.

```

1206 \NewDocumentCommand \spsshort { 0{} m }
1207 {
1208     \mode_if_math:TF
1209     {
1210         \msg_error:nnn { spintent } { need-text-mode } { \spsshort }
1211     }
1212     {
1213         \mode_leave_vertical:
1214         \group_begin:
1215         \__spintent_spsshort_print:nn {#1} {#2}
1216         \group_end:
1217     }
1218 }

```

(End of definition for \spsshort. This function is documented on page 7.)

11.19 The command \spsiglo

The public document command **\spsiglo** provides a semantic interface for typesetting century designations (e.g., siglo XXI). It accepts both Arabic and Roman numerals as $\langle \textit{argument} \rangle$, interfacing with the Lua module **spintent.lua** to validate, parse, and normalize chronological values.

This command work in dual-mode context awareness, operating uniformly across both *text mode* and *math mode*. When executed within *math mode*, it constructs an accessible **MathML** intent expression tree. Conversely, when invoked in standard *text mode*, it encapsulates the rendered output within a native PDF tagging structure (such as /Alt (#1)) containing an **alt** key.

\l__spintent_spsiglo_luaset_error_str These internal string variables handle data transfer and synchronization with the Lua module **spintent.lua**.
 \l__spintent_spsiglo_luaset_arabic_str They isolate the exception evaluation states, the Arabic numeral required for speech synthesis, the lowercase
 \l__spintent_spsiglo_luaset_roman_min_str Roman numeral designated for small caps typesetting via **\textsc**, and the localized chronological era modifier.
 \l__spintent_spsiglo_print_era_str

```

1219 \str_new:N \l__spintent_spsiglo_luaset_error_str
1220 \str_new:N \l__spintent_spsiglo_luaset_arabic_str
1221 \str_new:N \l__spintent_spsiglo_luaset_roman_min_str
1222 \str_new:N \l__spintent_spsiglo_print_era_str

```

(End of definition for \l__spintent_spsiglo_luaset_error_str and others.)

11.19.1 Definition of error messages for \spsiglo

invalid-century-format This internal error message is issued when the provided century identifier fails both Arabic and Roman numeral validation. It is triggered when the Lua module **spintent.lua** cannot map the $\langle \textit{argument} \rangle$ to an integer within the valid range of 1 to 4000, preventing incorrect speech synthesis.

```

1223 \msg_new:nnnn { spintent } { invalid-century-format }
1224 {
1225     Invalid~century~format~in~'#1'.
1226 }
1227 {
1228     The~argument~'#2'~is~not~a~valid~Arabic~or~Roman~numeral.~
1229     Ensure~the~value~is~between~1~and~4000.
1230 }

```

(End of definition for invalid-century-format.)

11.19.2 Definition of keys for \spsiglo

print-era Now we add the key **print-era**. User configuration determining if an era designation is appended, available values are **none**, **ac** (antes de cristo) and **dc** (después de cristo).

```

1231 \keys_define:nn { spintent / spsiglo }
1232 {
1233     print-era .choices:nn =
1234     { none , ac , dc }
1235     {
1236         \int_case:nn { \l_keys_choice_int }
1237         {

```

```

1238         {1} { \str_set:Nn \l__spintent_spsiglo_print_era_str { none } }
1239         {2} { \str_set:Nn \l__spintent_spsiglo_print_era_str { ac } }
1240         {3} { \str_set:Nn \l__spintent_spsiglo_print_era_str { dc } }
1241     }
1242 },
1243 print-era .initial:n = none,
1244 print-era .value_required:n = true,
1245 print-era / unknown .code:n =
1246     \msg_error:nneee { spintent } { unknown-choice }
1247     { print-era } { none, ~ac, ~dc } { \exp_not:n {#1} },
1248 unknown .code:n =
1249     {
1250         \str_set:Nn \l__spintent_command_name_str { spsiglo }
1251         \__spintent_unknown_keys_cmd:n {#1}
1252     },
1253 }

```

(End of definition for `print-era`.)

11.19.3 Internal functions for `\spsiglo`

The functions `\__spintent_spsiglo_render_simple:nn` and `\__spintent_spsiglo_render_with_era:nnn` generate MathML structures when century designations are evaluated within *math mode*. They construct MathML intent structures, mapping the Arabic numerals to their phonetic equivalents. Furthermore, `\__spintent_spsiglo_render_with_era:nnn` integrates the era modifiers `ac` or `dc`. It adds invisible spacing and postfix intents to ensure correct NVDA/MathCAT pronunciation.

```

1254 \cs_new_protected:Nn \__spintent_spsiglo_render_simple:nn
1255 {
1256     \MathMLintent { #1:number } { \text { \textsc {#2} } } }
1257 }
1258 \cs_generate_variant:Nn \__spintent_spsiglo_render_simple:nn { VV }
1259 \cs_new_protected:Nn \__spintent_spsiglo_render_with_era:nnn
1260 {
1261     \str_if_eq:nnTF { #1 } { ac }
1262     {
1263         \MathMLintent { antes-de-cristo:postfix($z) }
1264         {
1265             \__spintent_invisible_sep:
1266             \MathMLarg { z }
1267             {
1268                 \MathMLintent { #2:number }
1269                 {
1270                     \text { \textsc {#3}~a.~C. }
1271                 }
1272             }
1273         }
1274     }
1275     {
1276         \MathMLintent { después-de-cristo:postfix($z) }
1277         {
1278             \__spintent_invisible_sep:
1279             \MathMLarg { z }
1280             {
1281                 \MathMLintent { #2:number }
1282                 {
1283                     \text { \textsc {#3}~d.~C. }
1284                 }
1285             }
1286         }
1287     }
1288 }
1289 \cs_generate_variant:Nn \__spintent_spsiglo_render_with_era:nnn { VVV }

```

(End of definition for `\__spintent_spsiglo_render_simple:nn` and `\__spintent_spsiglo_render_with_era:nnn`.)

The function `\__spintent_spsiglo_tag_span:nn` handles century formatting within *text mode*. It utilizes the `tagpdf` package to enclose the content inside a PDF /Alt (`#1`) structure. By passing the `{⟨argument⟩}` to the `alt` key, it embeds the chronological value directly into the PDF, ensuring correct NVDA/MathCAT pronunciation without relying on MathML intents.

```

1290 \cs_new_protected:Nn \__spintent_spsiglo_tag_span:nn
1291 {
1292     \tag_mc_end_push:

```

```

1293     \tag_struct_begin:n { tag=Span, alt={ #1 } }
1294     \tag_mc_begin:n { tag=Span }
1295     \tag_suspend:n { \spsiglo }
1296     #2
1297     \tag_resume:n { \spsiglo }
1298     \tag_mc_end:
1299     \tag_struct_end:
1300     \tag_mc_begin_pop:n {}
1301   }
1302   \cs_generate_variant:Nn \__spintent_spsiglo_tag_span:nn { en }

(End of definition for \__spintent_spsiglo_tag_span:nn.)

```

```

\__spintent_spsiglo_print_math:n
\__spintent_spsiglo_print_text:n

```

The functions `\__spintent_spsiglo_print_math:n` and `\__spintent_spsiglo_print_text:n` format century designations based on the active mode. Both first check the “error” status in the variable `\l__spintent_spsiglo_luaset_error_str` returned by the Lua module `spintent.lua` and issue an “error” if validation fails.

If the `{⟨argument⟩}` is valid, the function `\__spintent_spsiglo_print_math:n` operates in *math mode*, applying MathML structures depending on whether an era modifier is present. Conversely, the function `\__spintent_spsiglo_print_text:n` operates in *text mode*, utilizing the `tag` and `alt` keys to expand the value of the `print-era` option into full text (e.g., “antes de cristo”) for NVDA/MathCAT.

```

1303   \cs_new_protected:Nn \__spintent_spsiglo_print_math:n
1304   {
1305     \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
1306     {
1307       \msg_error:nnnn { spintent } { invalid-century-format }
1308       { \spsiglo } { #1 }
1309     }
1310     {
1311       \str_if_eq:VnTF \l__spintent_spsiglo_print_era_str { none }
1312       {
1313         \__spintent_spsiglo_render_simple:VV
1314         \l__spintent_spsiglo_luaset_arabic_str
1315         \l__spintent_spsiglo_luaset_roman_min_str
1316       }
1317       {
1318         \__spintent_spsiglo_render_with_era:VVV
1319         \l__spintent_spsiglo_print_era_str
1320         \l__spintent_spsiglo_luaset_arabic_str
1321         \l__spintent_spsiglo_luaset_roman_min_str
1322       }
1323     }
1324   }
1325   \cs_new_protected:Nn \__spintent_spsiglo_print_text:n
1326   {
1327     \str_if_eq:VnTF \l__spintent_spsiglo_luaset_error_str { true }
1328     {
1329       \msg_error:nnnn { spintent } { invalid-century-format }
1330       { \spsiglo } { #1 }
1331     }
1332     {
1333       \str_case:Vn \l__spintent_spsiglo_print_era_str
1334       {
1335         { none }
1336         {
1337           \__spintent_spsiglo_tag_span:en
1338           { \l__spintent_spsiglo_luaset_arabic_str }
1339           { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } }
1340         }
1341         { ac }
1342         {
1343           \__spintent_spsiglo_tag_span:en
1344           { \l__spintent_spsiglo_luaset_arabic_str ~ antes ~ de ~ cristo }
1345           { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ a.~C. }
1346         }
1347         { dc }
1348         {
1349           \__spintent_spsiglo_tag_span:en
1350           { \l__spintent_spsiglo_luaset_arabic_str ~ después ~ de ~ cristo }
1351           { \textsc { \l__spintent_spsiglo_luaset_roman_min_str } ~ d.~C. }
1352         }
1353       }
1354     }
1355   }

```

```

1353     }
1354   }
1355 }

```

(End of definition for `\__spinttent_spsiglo_print_math:n` and `\__spinttent_spsiglo_print_text:n`)

`\spsiglo` The public command `\spsiglo` is the main interface for century typesetting. It opens a local group and evaluates any optional *(keys)* passed in ‘#1’, then executes the function `\__spinttent_luafun_spsiglo_parse:n` defined in the Lua module `spinttent.lua` passed in ‘#2’ and formats the output for *math mode* or *text mode*.

```

1356 \NewDocumentCommand \spsiglo { 0{} m }
1357 {
1358   \group_begin:
1359     \keys_set:nn { spinttent / spsiglo } { #1 }
1360     \__spinttent_luafun_spsiglo_parse:n { #2 }
1361     \mode_if_math:TF
1362       { \__spinttent_spsiglo_print_math:n { #2 } }
1363       {
1364         \mode_leave_vertical:
1365         \__spinttent_spsiglo_print_text:n { #2 }
1366       }
1367   \group_end:
1368 }

```

(End of definition for `\spsiglo`. This function is documented on page 11.)

11.20 The command `\sptime`

The command `\sptime` is the main public interface for typesetting and tagging time expressions. It parses time string formats (such as HH:MM or HH:MM:SS) using the Lua function `\__spinttent_luafun_sptime_parse:n`. This separates the hours, minutes, seconds, and fractions to construct a MathML *intent* tree for NVDA/MathCAT.

```

\__spinttent_sptime_luaset_seconds_str
\__spinttent_sptime_luaset_has_seconds_str
\__spinttent_sptime_luaset_error_str
\__spinttent_sptime_luaset_is_fraction_str
\__spinttent_sptime_luaset_base_str

```

These internal string variables store the values returned from the Lua module `spinttent.lua`. They hold the extracted time units, error flags, and boolean indicators (such as the presence of seconds or decimal fractions) used during formatting.

```

1369 \str_new:N \__spinttent_sptime_luaset_seconds_str
1370 \str_new:N \__spinttent_sptime_luaset_has_seconds_str
1371 \str_new:N \__spinttent_sptime_luaset_error_str
1372 \str_new:N \__spinttent_sptime_luaset_is_fraction_str
1373 \str_new:N \__spinttent_sptime_luaset_base_str

```

(End of definition for `\__spinttent_sptime_luaset_seconds_str` and others.)

11.20.1 Definition of error messages

`invalid-time-format`

The error message `invalid-time-format` defines the text displayed when an invalid time input is detected. It is triggered when an argument fails the format checks or contains values outside normal ranges, such as hours exceeding 23 or minutes and seconds exceeding 59.

```

1374 \msg_new:nnnn { spinttent } { invalid-time-format }
1375 {
1376   Invalid~time~format~in~#1. \\\
1377   The~argument~'~#2'~does~not~match~any~supported~time~format.
1378 }
1379 {
1380   Allowed~formats:~HH:MM,~HH:MM:SS,~or~HH:MM:SS.ss. \\\
1381   Please~verify~that~hours~are~0-23~and~minutes/seconds~are~0-59.
1382 }

```

(End of definition for `invalid-time-format`.)

11.20.2 Internal functions for `\sptime`

`\__spinttent_sptime_build_seconds:nn`

The internal function `\__spinttent_sptime_build_seconds:nn` builds the MathML structure for the seconds component of a time expression. It wraps the numeric value ‘#2’ in speech intent tags, applying a connector prefix (passed in ‘#1’) and the “segundos” postfix so it is read correctly by NVDA/MathCAT.

```

1383 \cs_new_protected:Nn \__spinttent_sptime_build_seconds:nn
1384 {
1385   \MathMLintent { #1:prefix($z) }
1386   {
1387     \text { : }
1388     \MathMLarg { z }
1389     {
1390       \MathMLintent { segundos:postfix($m) }
1391       {

```



```

1392         \\\spintent_invisible_sep:
1393         \MathMLLarg { m }
1394         {
1395             \text { #2 }
1396         }
1397     }
1398 }
1399 }
1400 }

```

(End of definition for \\\spintent_sptime_build_seconds:nn.)

`\\spintent_sptime_build_tree:` The internal function `\\spintent_sptime_build_tree:` builds the complete MathML structure for the time expression. It checks the string variables `\\l__spintent_sptime_luaset_has_seconds_str` and `\\l__spintent_sptime_luaset_is_fraction_str` to determine if the time includes seconds or decimal fractions. Based on these values, it inserts the base time string and either calls `\\spintent_sptime_build_seconds:` with the appropriate prefix ('con' or 'minutos-con'), or simply appends the 'minutos' intent tag if no seconds are present.

```

1401 \cs_new_protected:Nn \\\spintent_sptime_build_tree:
1402 {
1403     \str_if_eq:VnTF \l__spintent_sptime_luaset_has_seconds_str { true }
1404     {
1405         \MathMLintent { :time }
1406         {
1407             \text { \l__spintent_sptime_luaset_base_str }
1408         }
1409         \str_if_eq:VnTF \l__spintent_sptime_luaset_is_fraction_str { true }
1410         {
1411             \\\spintent_sptime_build_seconds:nn { con }
1412             { \l__spintent_sptime_luaset_seconds_str }
1413         }
1414         {
1415             \\\spintent_sptime_build_seconds:nn { minutos-con }
1416             { \l__spintent_sptime_luaset_seconds_str }
1417         }
1418     }
1419     {
1420         \MathMLintent { :time }
1421         {
1422             \text { \l__spintent_sptime_luaset_base_str }
1423         }
1424         \str_if_eq:VnT \l__spintent_sptime_luaset_is_fraction_str { false }
1425         {
1426             \MathMLintent { minutos }
1427             {
1428                 \\\spintent_invisible_sep:
1429             }
1430         }
1431     }
1432 }

```

(End of definition for \\\spintent_sptime_build_tree:.)

`\sptime` The command `\sptime` is the public interface for formatting time expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it parses the input using `\\spintent_luafun_sptime_parse:n` and checks the error status in `\\l__spintent_sptime_luaset_error_str`. If an invalid format is detected, it issues the `invalid-time-format` error; otherwise, it calls `\\spintent_sptime_build_tree:` to assemble the MathML output.

```

1433 \NewDocumentCommand \sptime { m }
1434 {
1435     \mode_if_math:TF
1436     {
1437         \group_begin:
1438         \\\spintent_luafun_sptime_parse:n { #1 }
1439         \str_if_eq:VnTF \l__spintent_sptime_luaset_error_str { true }
1440         {
1441             \msg_error:nnnn { spintent } { invalid-time-format }
1442             { \sptime } { #1 }
1443         }
1444         {
1445             \\\spintent_sptime_build_tree:

```

```

1446         }
1447     \group_end:
1448 }
1449 { \msg_error:nnn { spintent } { need-math-mode } { \sptime } }
1450 }

```

(End of definition for `\sptime`. This function is documented on page 11.)

11.21 The command `\spdate`

The command `\spdate` is the main public command for typesetting and tagging date expressions. It parses date string formats (such as DD/MM/YYYY or YYYY-MM-DD) using the Lua function `\__spintent_luafun_spdate_parse:n`. This verifies the input and constructs a MathML `intent` tree for NVDA/MathCAT.

```

\__spintent_spdate_luaset_output_str
\__spintent_spdate_luaset_error_str

```

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the formatted date string and the error status used during formatting.

```

1451 \str_new:N \__spintent_spdate_luaset_output_str
1452 \str_new:N \__spintent_spdate_luaset_error_str

```

(End of definition for `\__spintent_spdate_luaset_output_str` and `\__spintent_spdate_luaset_error_str`.)

11.21.1 Definition of error messages

```
invalid-date-format
```

The error message `invalid-date-format` defines the text displayed when an invalid date input is detected. It is triggered when an argument fails the format checks or represents an invalid calendar date, such as `31/02/2024` or using an unsupported pattern like `YYYY/MM/DD`.

```

1453 \msg_new:nnnn { spintent } { invalid-date-format }
1454 {
1455     Invalid~date~or~unsupported~format~in~#1. \\\
1456     The~argument~'~#2'~does~not~match~a~supported~format.
1457 }
1458 {
1459     Allowed~formats:~DD\slash MM\slash YYYY,~DD-MM-YYYY,~or~YYYY-MM-DD. \\\
1460     Ensure~the~day~and~month~are~valid~(e.g.,~avoid~31\slash 02\slash 2024).
1461 }

```

(End of definition for `invalid-date-format`.)

11.21.2 Internal functions for `\spdate`

```
\__spintent_spdate_process:n
```

The internal function `\__spintent_spdate_process:n` parses the input using `\__spintent_luafun_spdate_parse:n` and checks the error status in `\__spintent_spdate_luaset_error_str`. If an invalid format is detected, it issues the `invalid-date-format` error. Otherwise, it builds the MathML `intent` structure using the formatted date stored in `\__spintent_spdate_luaset_output_str`.

```

1462 \cs_new_protected:Nn \__spintent_spdate_process:n
1463 {
1464     \__spintent_luafun_spdate_parse:n { #1 }
1465     \str_if_eq:VnTF \__spintent_spdate_luaset_error_str { true }
1466     {
1467         \msg_error:nnnn { spintent } { invalid-date-format }
1468         { \spdate } { #1 }
1469     }
1470     {
1471         \MathMLintent { :date }
1472         {
1473             \text { \__spintent_spdate_luaset_output_str }
1474         }
1475     }
1476 }

```

(End of definition for `\__spintent_spdate_process:n`.)

```
\spdate
```

The command `\spdate` is the public interface for formatting date expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\__spintent_spdate_process:n` to process the input.

```

1477 \NewDocumentCommand \spdate { m }
1478 {
1479     \mode_if_math:TF
1480     {
1481         \group_begin:
1482         \__spintent_spdate_process:n { #1 }
1483         \group_end:
1484     }
1485     { \msg_error:nnn { spintent } { need-math-mode } { \spdate } }
1486 }

```

(End of definition for `\spdate`. This function is documented on page 11.)

11.22 Semantic Operators

This section provides purely semantic operators. Unlike their formatting counterparts (which parse and format numbers), these commands do not process mandatory arguments. Instead, they solely inject a mathematical glyph wrapped in its corresponding regional `MathCAT` intent. This grants teachers absolute control when building complex algebraic expressions, mixed numbers, or fraction divisions.

11.23 The command `\spusep`

For the use of parentheses, we will provide the command `\spusep` with only two *(keys)* `size` and `silent`. Most of the time we can use `\left` and `\right` or `\Uleft` and `\Uright`, but both forms create a group, and this hinders the code for *tagged* PDF when we want to interrupt something between them.

`spusep-empty-arg`
`spusep-invalid-arg`

The first “error” message `spusep-empty-arg` is raised when `\spusep` receives an empty or blank *{(argument)}*, the second this is when an input is invalid.

```
1487 \msg_new:nnnn { spintent } { spusep-empty-arg }
1488 { Empty~argument~in~\token_to_str:N \spusep. }
1489 { \token_to_str:N \spusep~requires~a~non-empty~argument. }
1490 \msg_new:nnn { spintent } { spusep-invalid-arg }
1491 {
1492   Invalid~argument~'#1'~for~\token_to_str:N \spusep.
1493 }
```

(End of definition for `spusep-empty-arg` and `spusep-invalid-arg`.)

`\l__spintent_spusep_arg_tl`
`\l__spintent_spusep_left_tl`
`\l__spintent_spusep_right_tl`

The variable `\l__spintent_spusep_arg_tl` stores the *{(argument)}*, the variable `\l__spintent_spusep_left_tl` and `\l__spintent_spusep_right_tl` stores the parentheses.

```
1494 \tl_new:N \l__spintent_spusep_arg_tl
1495 \tl_new:N \l__spintent_spusep_left_tl
1496 \tl_new:N \l__spintent_spusep_right_tl
```

(End of definition for `\l__spintent_spusep_arg_tl`, `\l__spintent_spusep_left_tl`, and `\l__spintent_spusep_right_tl`.)

`size`
`silent`
`unknown`

We define the *(keys)* `size` and `silent`.

```
1497 \keys_define:nn { spintent / spusep }
1498 {
1499   size .choices:nn =
1500     { none , big , Big , bigg , Bigg }
1501     {
1502       \int_case:nn { \l_keys_choice_int }
1503       {
1504         { 1 }
1505         {
1506           \tl_clear:N \l__spintent_spusep_left_tl
1507           \tl_clear:N \l__spintent_spusep_right_tl
1508         }
1509         { 2 }
1510         {
1511           \tl_set:Nn \l__spintent_spusep_left_tl { \bigl }
1512           \tl_set:Nn \l__spintent_spusep_right_tl { \bigr }
1513         }
1514         { 3 }
1515         {
1516           \tl_set:Nn \l__spintent_spusep_left_tl { \Bigl }
1517           \tl_set:Nn \l__spintent_spusep_right_tl { \Bigr }
1518         }
1519         { 4 }
1520         {
1521           \tl_set:Nn \l__spintent_spusep_left_tl { \biggl }
1522           \tl_set:Nn \l__spintent_spusep_right_tl { \biggr }
1523         }
1524         { 5 }
1525         {
1526           \tl_set:Nn \l__spintent_spusep_left_tl { \Biggl }
1527           \tl_set:Nn \l__spintent_spusep_right_tl { \Biggr }
1528         }
1529       }
1530     },
1531   size .initial:n = none,
1532   size .value_required:n = true,
1533   size / unknown .code:n =
```

```

1534     \msg_error:nneee { spintent } { unknown-choice }
1535     { size } { none,~big,~Big,~bigg,~Bigg } { \exp_not:n {#1} },
1536     silent      .bool_set:N = \__spintent_spusep_silent_bool,
1537     silent      .initial:n = false,
1538     silent      .value_required:n = true,
1539     unknown     .code:n =
1540     {
1541         \str_set:Nn \__spintent_command_name_str { spusep }
1542         \__spintent_unknown_keys_cmd:n {#1}
1543     },
1544 }

```

(End of definition for `size`, `silent`, and `unknown`.)

`\__spintent_spusep_set_parens:nn`

The function `\__spintent_spusep_set_parens:nn` will place the argument passed in ‘#2’ to the right of the variables `\__spintent_spusep_left_tl` and `\__spintent_spusep_right_tl` and then check the state of the variable `\__spintent_spusep_silent_bool` handled by the key `silent` to print.

```

1545 \cs_new_protected:Nn \__spintent_spusep_set_parens:nn
1546 {
1547     \tl_put_right:cn { \__spintent_spusep_ #1 _tl } { #2 }
1548     \bool_if:NTF \__spintent_spusep_silent_bool
1549     {
1550         \MathMLintent { :silent } { \tl_use:c { \__spintent_spusep_ #1 _tl } }
1551     }
1552     {
1553         \tl_use:c { \__spintent_spusep_ #1 _tl }
1554     }
1555 }

```

(End of definition for `\__spintent_spusep_set_parens:nn`.)

`\__spintent_spusep_convert:n`

The function `\__spintent_spusep_convert:n` captures and stores the $\langle argument \rangle$ passed in ‘#1’ in the variable `\__spintent_spusep_arg_tl`, validates the cases by converting them to the macros provided by `unicode-math`, and passes them to the function `\__spintent_spusep_set_parens:nn` for printing. If the input is not supported or is empty, we return an “error”.

```

1556 \cs_new_protected:Nn \__spintent_spusep_convert:n
1557 {
1558     \tl_set:Ne \__spintent_spusep_arg_tl { \tl_trim_spaces:n {#1} }
1559     \tl_if_empty:NT \__spintent_spusep_arg_tl
1560     {
1561         \msg_error:nnn { spintent } { spusep-invalid-arg } {#1}
1562     }
1563     \str_case:VnF \__spintent_spusep_arg_tl
1564     {
1565         { ( }
1566         {
1567             \__spintent_spusep_set_parens:nn { left } { \lparen }
1568         }
1569         { ) }
1570         {
1571             \__spintent_spusep_set_parens:nn { right } { \rparen }
1572         }
1573         { [ }
1574         {
1575             \__spintent_spusep_set_parens:nn { left } { \lbrack }
1576         }
1577         { ] }
1578         {
1579             \__spintent_spusep_set_parens:nn { right } { \rbrack }
1580         }
1581         { \{ }
1582         {
1583             \__spintent_spusep_set_parens:nn { left } { \lbrace }
1584         }
1585         { \} }
1586         {
1587             \__spintent_spusep_set_parens:nn { right } { \rbrace }
1588         }
1589     }
1590     {
1591         \msg_error:nnn { spintent } { spusep-invalid-arg } {#1}

```

```

1592     }
1593 }

```

(End of definition for `\__spintent_spusep_convert:n`.)

\spusep The command `\spusep` checks that it is running in math mode, opens a local group, sets the keys passed in ‘#1’ and processes the $\langle argument \rangle$ passed in ‘#2’ by means of the function `\__spintent_spusep_convert:n`.

```

1594 \NewDocumentCommand \spusep { O{} m }
1595 {
1596   \mode_if_math:TF
1597   {
1598     \group_begin:
1599     \keys_set:nn { spintent / spusep } {#1}
1600     \__spintent_spusep_convert:n {#2}
1601     \group_end:
1602   }
1603   { \msg_error:nnn { spintent } { need-math-mode } { \spusep } }
1604 }

```

(End of definition for `\spusep`. This function is documented on page 12.)

11.23.1 The command `\spread`

The command `\spread` provides a semantic read for MathCAT of mathematical symbols for situations in which it is necessary to specify gender or plurality in Spanish.

spread-empty-args The error message `spread-empty-args` is raised when `\spread` receives an empty or blank $\langle argument \rangle$.

```

1605 \msg_new:nnnn { spintent } { spread-empty-args }
1606 { Empty~argument~in~\token_to_str:N \spread. }
1607 { \token_to_str:N \spread~requires~two~non-empty~arguments. }

```

(End of definition for `spread-empty-args`.)

__spintent_spread_intent_tl The variable `\__spintent_spread_intent_tl` stores the sanitized semantic reading $\langle argument \rangle$ for symbol to be injected into the `\MathMLintent`.

```

1608 \tl_new:N \__spintent_spread_intent_tl

```

(End of definition for `\__spintent_spread_intent_tl`.)

__spintent_spread_exec:nn The function `\__spintent_spread_exec:nn` implements the core logic for `\spread`. It verifies that neither $\langle argument \rangle$ is empty or blank. If valid, it evaluates the reading $\langle argument \rangle$ ‘#1’ using `\__spintent_sanitize_affix:Nn` and embeds it via `\MathMLintent` into the *math* symbol ‘#2’.

```

1609 \cs_new_protected:Nn \__spintent_spread_exec:nn
1610 {
1611   \bool_lazy_or:nnTF { \tl_if_blank_p:n { #1 } } { \tl_if_blank_p:n { #2 } }
1612   { \msg_error:nn { spintent } { spread-empty-args } }
1613   {
1614     \__spintent_sanitize_affix:Nn \__spintent_spread_intent_tl { #1 }
1615     \MathMLintent { \__spintent_spread_intent_tl } { #2 }
1616   }
1617 }

```

(End of definition for `\__spintent_spread_exec:nn`.)

\spread The command `\spread` assigns a custom semantic *reading* ‘#1’ to any mathematical *symbol* ‘#2’. It enforces math mode and passes execution to `\__spintent_spread_exec:nn`.

```

1618 \NewDocumentCommand \spread { m m }
1619 {
1620   \mode_if_math:TF
1621   {
1622     \group_begin:
1623     \__spintent_spread_exec:nn { #1 } { #2 }
1624     \group_end:
1625   }
1626   { \msg_error:nnn { spintent } { need-math-mode } { \spread } }
1627 }

```

(End of definition for `\spread`. This function is documented on page 11.)

11.23.2 The command `\spdsym`

Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for `\spdsym`.

```
\l__spintent_spdsym_read_sym_tl
\l__spintent_spdsym_symbol_tl
\l__spintent_spdsym_prefix_tl
\l__spintent_spdsym_suffix_tl
```

(End of definition for `\l__spintent_spdsym_read_sym_tl` and others.)

```
prefix We define the (keys) prefix, suffix, set-sym and read-sym for \spdsym.
suffix
set-sym
read-sym
unknown
```

```
1632 \keys_define:nn { spintent / spdsym }
1633 {
1634   prefix .code:n =
1635     { \l__spintent_sanitize_affix:Nn \l__spintent_spdsym_prefix_tl {#1} },
1636   prefix .initial:n = ,
1637   prefix .value_required:n = true,
1638   suffix .code:n =
1639     { \l__spintent_sanitize_affix:Nn \l__spintent_spdsym_suffix_tl {#1} },
1640   suffix .initial:n = ,
1641   suffix .value_required:n = true,
1642   set-sym .choices:nn =
1643     { old-style , two-dots , slash }
1644   {
1645     \int_case:nn { \l_keys_choice_int }
1646     {
1647       {1} { \tl_set:Nn \l__spintent_spdsym_symbol_tl { \div } }
1648       {2} { \tl_set:Nn \l__spintent_spdsym_symbol_tl { : } }
1649       {3} { \tl_set:Nn \l__spintent_spdsym_symbol_tl { / } }
1650     }
1651   },
1652   set-sym .initial:n = two-dots,
1653   set-sym .value_required:n = true,
1654   set-sym / unknown .code:n =
1655     \msg_error:nneee { spintent } { unknown-choice }
1656     { set-sym } { old-style, ~two-dots, ~slash } { \exp_not:n {#1} },
1657   read-sym .choices:nn =
1658     { dividido , dividido-por , dividido-entre }
1659   {
1660     \int_case:nn { \l_keys_choice_int }
1661     {
1662       {1} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido } }
1663       {2} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-por } }
1664       {3} { \tl_set:Nn \l__spintent_spdsym_read_sym_tl { dividido-entre } }
1665     }
1666   },
1667   read-sym .initial:n = dividido,
1668   read-sym .value_required:n = true,
1669   read-sym / unknown .code:n =
1670     \msg_error:nneee { spintent } { unknown-choice }
1671     { read-sym } { dividido, ~dividido-por, ~dividido-entre }
1672     { \exp_not:n {#1} },
1673   unknown .code:n =
1674   {
1675     \str_set:Nn \l__spintent_command_name_str { spdsym }
1676     \l__spintent_unknown_keys_cmd:n {#1}
1677   },
1678 }
```

(End of definition for `prefix` and others.)

`\spdsym` Defines the `\spdsym` operator. It takes a single optional argument for local setup and outputs the configured intent structure, safely attaching prefixes and suffixes if present.

```
1679 \NewDocumentCommand \spdsym { 0{} }
1680 {
1681   \mode_if_math:TF
1682   {
1683     \group_begin:
1684     \keys_set:nn { spintent / spdsym } { #1 }
1685     \MathMLintent
1686     {
```



```

1687         \tl_if_empty:NF \l__spintent_spdsym_prefix_tl
1688         { \l__spintent_spdsym_prefix_tl - }
1689         \l__spintent_spdsym_read_sym_tl
1690         \tl_if_empty:NF \l__spintent_spdsym_suffix_tl
1691         { - \l__spintent_spdsym_suffix_tl }
1692     }
1693     { \l__spintent_spdsym_symbol_tl }
1694 \group_end:
1695 }
1696 { \msg_error:nnn { spintent } { need-math-mode } { \spdsym } }
1697 }

```

(End of definition for `\spdsym`. This function is documented on page 12.)

11.23.3 The command `\spmsym`

Variables to store the regional reading string, the visual operator, and the optional semantic affixes specifically for `\spmsym`.

```

\l__spintent_spmsym_read_sym_tl
\l__spintent_spmsym_symbol_tl
\l__spintent_spmsym_prefix_tl
\l__spintent_spmsym_suffix_tl
1698 \tl_new:N \l__spintent_spmsym_read_sym_tl
1699 \tl_new:N \l__spintent_spmsym_symbol_tl
1700 \tl_new:N \l__spintent_spmsym_prefix_tl
1701 \tl_new:N \l__spintent_spmsym_suffix_tl

```

(End of definition for `\l__spintent_spmsym_read_sym_tl` and others.)

```

prefix    We define the (keys) prefix, suffix, set-sym, read-sym and not-print for \spmsym.
suffix
set-sym   1702 \keys_define:nn { spintent / spmsym }
          {
read-sym  1703   {
          1704     prefix                .code:n =
          1705     { \__spintent_sanitize_affix:Nn \l__spintent_spmsym_prefix_tl {#1} },
not-print 1706     prefix                .initial:n = ,
          1707     prefix                .value_required:n = true,
          1708     suffix                .code:n =
          1709     { \__spintent_sanitize_affix:Nn \l__spintent_spmsym_suffix_tl {#1} },
          1710     suffix                .initial:n = ,
          1711     suffix                .value_required:n = true,
          1712     set-sym              .choices:nn =
          1713     { cross , dot , asterisk }
          1714     {
          1715       \int_case:nn { \l_keys_choice_int }
          1716       {
          1717         {1} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \times } }
          1718         {2} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \cdot } }
          1719         {3} { \tl_set:Nn \l__spintent_spmsym_symbol_tl { \ast } }
          1720       }
          1721     },
          1722     set-sym                .initial:n = dot,
          1723     set-sym                .value_required:n = true,
          1724     set-sym / unknown     .code:n =
          1725     \msg_error:nneee { spintent } { unknown-choice }
          1726     { set-sym } { cross, ~dot, ~asterisk } { \exp_not:n {#1} },
read-sym  1727     read-sym              .choices:nn =
          1728     { por , multiplicado-por , veces }
          1729     {
          1730       \int_case:nn { \l_keys_choice_int }
          1731       {
          1732         {1} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { por } }
          1733         {2} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { multiplicado-por } }
          1734         {3} { \tl_set:Nn \l__spintent_spmsym_read_sym_tl { veces } }
          1735       }
          1736     },
          1737     read-sym              .initial:n = por,
          1738     read-sym              .value_required:n = true,
          1739     read-sym / unknown     .code:n =
          1740     \msg_error:nneee { spintent } { unknown-choice }
          1741     { read-sym } { por, ~multiplicado-por, ~veces } { \exp_not:n {#1} },
not-print 1742     not-print              .bool_set:N = \l__spintent_spmsym_not_print_bool,
not-print 1743     not-print              .initial:n = false,
not-print 1744     not-print              .value_required:n = true,
          1745     unknown              .code:n =
          1746     {
          1747       \str_set:Nn \l__spintent_command_name_str { spmsym }
          1748       \__spintent_unknown_keys_cmd:n {#1}

```

```

1749     },
1750 }

```

(End of definition for *prefix* and others.)

\spmsym Defines the **\spmsym** operator. Similar to **\spdsym**, it outputs the correct mathematical glyph based on the local *⟨keys⟩*, safely attaching prefixes and suffixes if present.

```

1751 \NewDocumentCommand \spmsym { 0{} }
1752 {
1753   \mode_if_math:TF
1754   {
1755     \group_begin:
1756     \keys_set:nn { spintent / spmsym } { #1 }
1757     \MathMLintent
1758     {
1759       \tl_if_empty:NF \l__spintent_spmsym_prefix_tl
1760       { \l__spintent_spmsym_prefix_tl - }
1761       \l__spintent_spmsym_read_sym_tl
1762       \tl_if_empty:NF \l__spintent_spmsym_suffix_tl
1763       { - \l__spintent_spmsym_suffix_tl }
1764     }
1765     {
1766       \bool_if:NTF \l__spintent_spmsym_not_print_bool
1767       {
1768         \invisibletimes
1769       }
1770       { \l__spintent_spmsym_symbol_tl }
1771     }
1772     \group_end:
1773   }
1774   { \msg_error:nnn { spintent } { need-math-mode } { \spmsym } }
1775 }

```

(End of definition for *\spmsym*. This function is documented on page 12.)

11.24 The command **\spdiv**

The user-level command **\spdiv** formats division representations. It takes two mandatory arguments: the *⟨dividend⟩* and the *⟨divisor⟩*. Both *⟨arguments⟩* are internally processed by the **\spnum** command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the *⟨dividend⟩* nor the *⟨divisor⟩* is *blank*.

11.24.1 Definition of keys for **\spdiv**

We define the *⟨keys⟩* **wrap-parens** and **read-parens** for **\spdiv** command. Number-related *⟨keys⟩* **cifras-sep**, **read-sign**, **read-period-sep** and **notation-sym** are forwarded to **\spnum** command using **.meta:nn**, while semantic operator *⟨keys⟩* **set-sym** and **read-sym** are forwarded to the **\spdsym** command using **.meta:nn**.

```

1776 \keys_define:nn { spintent / spdiv }
1777 {
1778   cifras-sep      .meta:nn = { spintent / spnum } { cifras-sep      = {#1} },
1779   read-sign       .meta:nn = { spintent / spnum } { read-sign       = {#1} },
1780   read-period-sep .meta:nn = { spintent / spnum } { read-period-sep = {#1} },
1781   notation-sym    .meta:nn = { spintent / spnum } { notation-sym    = {#1} },
1782   set-sym         .meta:nn = { spintent / spdsym } { set-sym         = {#1} },
1783   read-sym        .meta:nn = { spintent / spdsym } { read-sym        = {#1} },
1784   wrap-parens     .bool_set:N = \l__spintent_spdiv_wrap_parens_bool,
1785   wrap-parens     .initial:n  = false,
1786   wrap-parens     .value_required:n = true,
1787   read-parens     .bool_set:N = \l__spintent_spdiv_read_parens_bool,
1788   read-parens     .initial:n  = true,
1789   read-parens     .value_required:n = true,
1790   unknown         .code:n    =
1791   {
1792     \str_set:Nn \l__spintent_command_name_str { spdiv }
1793     \l__spintent_unknown_keys_cmd:n {#1}
1794   },
1795 }

```

(End of definition for *cifras-sep* and others.)

11.24.2 Definition of error messages

`spdiv-empty-argument` Raised when either the $\langle\textit{dividend}\rangle$ or the $\langle\textit{divisor}\rangle$ is missing or *blank*.

```

1796 \msg_new:nnnn { spintenz } { spdiv-empty-argument }
1797 { The~arguments~for~\token_to_str:N \spdiv~cannot~be~empty. }
1798 {
1799   You~must~provide~both~a~dividend~and~a~divisor.\\
1800 }

```

(End of definition for `spdiv-empty-argument`.)

11.24.3 Internal functions for `\spdiv`

`\__spintenz_spdiv_render_op:nn` The function `\__spintenz_spdiv_render_op:nn` parses both $\langle\textit{arguments}\rangle$ through the `\spnum` command and embeds the semantic operator from `\spdsym`.

```

1801 \cs_new_protected:Nn \__spintenz_spdiv_render_op:nn
1802 {
1803   \spnum {#1}
1804   \MathMLintenz { \__spintenz_spdsym_read_sym_tl }
1805   {
1806     \__spintenz_spdsym_symbol_tl
1807   }
1808   \spnum {#2}
1809 }

```

(End of definition for `\__spintenz_spdiv_render_op:nn`.)

`\__spintenz_spdiv_render_parens:nn` The function `\__spintenz_spdiv_render_parens:nn` handles the structural parentheses around the division.

```

1810 \cs_new_protected:Nn \__spintenz_spdiv_render_parens:nn
1811 {
1812   \bool_if:NTF \__spintenz_spdiv_read_parens_bool
1813   {
1814     ( \__spintenz_spdiv_render_op:nn {#1} {#2} )
1815   }
1816   {
1817     \MathMLintenz { :silent } { ( }
1818     \__spintenz_spdiv_render_op:nn {#1} {#2}
1819     \MathMLintenz { :silent } { ) }
1820   }
1821 }

```

(End of definition for `\__spintenz_spdiv_render_parens:nn`.)

`\__spintenz_spdiv_process:nnn` The function `\__spintenz_spdiv_process:nnn` sets $\langle\textit{keys}\rangle$, validates $\langle\textit{arguments}\rangle$, and decides on parenthesis wrapping.

```

1822 \cs_new_protected:Nn \__spintenz_spdiv_process:nnn
1823 {
1824   \keys_set:nn { spintenz / spdiv } { #1 }
1825   \tl_if_blank:nTF { #2 }
1826   { \msg_error:nnn { spintenz } { spdiv-empty-argument } }
1827   {
1828     \tl_if_blank:nTF { #3 }
1829     { \msg_error:nnn { spintenz } { spdiv-empty-argument } }
1830     {
1831       \bool_if:NTF \__spintenz_spdiv_wrap_parens_bool
1832       { \__spintenz_spdiv_render_parens:nn {#2} {#3} }
1833       { \__spintenz_spdiv_render_op:nn {#2} {#3} }
1834     }
1835   }
1836 }

```

(End of definition for `\__spintenz_spdiv_process:nnn`.)

`\spdiv` Defines the user command `\spdiv`. It strictly relies on *math mode*.

```

1837 \NewDocumentCommand \spdiv { 0{ } m m }
1838 {
1839   \mode_if_math:TF
1840   {
1841     \group_begin:
1842     \__spintenz_spdiv_process:nnn { #1 } { #2 } { #3 }
1843     \group_end:
1844   }
1845   { \msg_error:nnn { spintenz } { need-math-mode } { \spdiv } }
1846 }

```

(End of definition for `\spdiv`. This function is documented on page 13.)

11.25 The command `\spmul`

The user-level command `\spmul` formats multiplication representations. It takes two mandatory arguments: the `{\langle first factor \rangle}` and the `{\langle second factor \rangle}`. Both `{\langle arguments \rangle}` are internally processed by the `\spnum` command to ensure proper digit separation. This command enforces strict validation, ensuring that neither the `{\langle first factor \rangle}` nor the `{\langle second factor \rangle}` is *blank*.

11.25.1 Definition of keys for `\spmul`

We define the `\langle keys \rangle` `wrap-parens` and `read-parens` for `\spmul` command. Number-related `\langle keys \rangle` `cifras-sep`, `read-sign`, `read-period-sep` and `notation-sym` are forwarded to `\spnum` command using `.meta:nn`, while semantic operator `\langle keys \rangle` `set-sym` and `read-sym` are forwarded to the `\spmsym` command using `.meta:nn`.

```

1847 \keys_define:nn { spintent / spmul }
1848 {
1849   cifras-sep      .meta:nn = { spintent / spnum } { cifras-sep      = {#1} },
1850   read-sign       .meta:nn = { spintent / spnum } { read-sign       = {#1} },
1851   read-period-sep .meta:nn = { spintent / spnum } { read-period-sep = {#1} },
1852   notation-sym    .meta:nn = { spintent / spnum } { notation-sym    = {#1} },
1853   set-sym         .meta:nn = { spintent / spmsym } { set-sym         = {#1} },
1854   read-sym        .meta:nn = { spintent / spmsym } { read-sym        = {#1} },
1855   wrap-parens     .bool_set:N = \l__spintent_spmul_wrap_parens_bool,
1856   wrap-parens     .initial:n  = false,
1857   wrap-parens     .value_required:n = true,
1858   read-parens     .bool_set:N = \l__spintent_spmul_read_parens_bool,
1859   read-parens     .initial:n  = true,
1860   read-parens     .value_required:n = true,
1861   unknown         .code:n =
1862     {
1863       \str_set:Nn \l__spintent_command_name_str { spmul }
1864       \__spintent_unknown_keys_cmd:n {#1}
1865     },
1866 }

```

(End of definition for `cifras-sep` and others.)

11.25.2 Definition of error messages

`spmul-empty-argument` Raised when either the `{\langle first factor \rangle}` or the `{\langle second factor \rangle}` is missing or *blank*.

```

1867 \msg_new:nnnn { spintent } { spmul-empty-argument }
1868 { The~arguments~for~\token_to_str:N \spmul~cannot~be~empty. }
1869 {
1870   You~must~provide~both~first~factor~and~second~factor.\\
1871 }

```

(End of definition for `spmul-empty-argument`.)

11.25.3 Internal functions for `\spmul`

`\__spintent_spmul_render_op:nn` The function `\__spintent_spmul_render_op:nn` parses both `{\langle arguments \rangle}` through the `\spnum` command and embeds the semantic operator from `\spmsym`.

```

1872 \cs_new_protected:Nn \__spintent_spmul_render_op:nn
1873 {
1874   \spnum {#1}
1875   \MathMLintent { \l__spintent_spmsym_read_sym_tl }
1876   {
1877     \l__spintent_spmsym_symbol_tl
1878   }
1879   \spnum {#2}
1880 }

```

(End of definition for `\__spintent_spmul_render_op:nn`.)

`\__spintent_spmul_render_parens:nn` The function `\__spintent_spmul_render_parens:nn` handles the structural parentheses around the multiplication.

```

1881 \cs_new_protected:Nn \__spintent_spmul_render_parens:nn
1882 {
1883   \bool_if:NTF \l__spintent_spmul_read_parens_bool
1884   {
1885     ( \__spintent_spmul_render_op:nn {#1} {#2} )
1886   }
1887   {
1888     \MathMLintent { :silent } { ( ) }

```

```

1889     \__spintent_spmul_render_op:nn {#1} {#2}
1890     \MathMLintent { :silent } { ) }
1891   }
1892 }

```

(End of definition for __spintent_spmul_render_parens:nn.)

__spintent_spmul_process:nnn

The function `\__spintent_spmul_process:nnn` sets $\langle keys \rangle$, validates $\langle arguments \rangle$, and decides on parenthesis wrapping.

```

1893 \cs_new_protected:Nn \__spintent_spmul_process:nnn
1894 {
1895   \keys_set:nn { spintent / spmul } { #1 }
1896   \tl_if_blank:nTF { #2 }
1897   { \msg_error:nnn { spintent } { spmul-empty-argument } }
1898   {
1899     \tl_if_blank:nTF { #3 }
1900     { \msg_error:nnn { spintent } { spmul-empty-argument } }
1901     {
1902       \bool_if:NTF \__spintent_spmul_wrap_parens_bool
1903       { \__spintent_spmul_render_parens:nn {#2} {#3} }
1904       { \__spintent_spmul_render_op:nn {#2} {#3} }
1905     }
1906   }
1907 }

```

(End of definition for __spintent_spmul_process:nnn.)

\spmul

Defines the user command `\spmul`. It strictly relies on *math mode*.

```

1908 \NewDocumentCommand \spmul { O{} m m }
1909 {
1910   \mode_if_math:TF
1911   {
1912     \group_begin:
1913     \__spintent_spmul_process:nnn { #1 } { #2 } { #3 }
1914     \group_end:
1915   }
1916   { \msg_error:nnn { spintent } { need-math-mode } { \spmul } }
1917 }

```

(End of definition for `\spmul`. This function is documented on page 13.)

11.26 The commands `\spgcd` and `\spmcn`

The user-level commands `\spgcd` and `\spmcn` provide a semantic interface for the Greatest Common Divisor (MCD in Spanish) and the Least Common Multiple (MCM in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid `MathCAT` intent.

11.26.1 Variables for `\spgcd` and `\spmcn`

Internal variables generated dynamically via a comma-separated list to avoid code duplication for both symmetric commands.

```

\__spintent_spmcm_spmcd_luaset_error_str
\__spintent_mc_args_seq
\__spintent_spmcd_print_result_bool
\__spintent_spmcm_print_result_bool
\__spintent_spmcd_read_terse_bool
\__spintent_spmcm_read_terse_bool
\__spintent_spmcd_print_upper_bool
\__spintent_spmcm_print_upper_bool
\__spintent_spmcd_prefix_tl
\__spintent_spmcm_prefix_tl
\__spintent_spmcd_suffix_tl
\__spintent_spmcm_suffix_tl

```

```

1918 \str_new:N \__spintent_spmcm_spmcd_luaset_error_str
1919 \seq_new:N \__spintent_mc_args_seq
1920 \clist_map_inline:nn { mcd , mcm }
1921 {
1922   \tl_new:c { \__spintent_sp #1 _luaset_ #1 _value_tl }
1923   \bool_new:c { \__spintent_sp #1 _print_result_bool }
1924   \bool_new:c { \__spintent_sp #1 _read_terse_bool }
1925   \bool_new:c { \__spintent_sp #1 _print_upper_bool }
1926   \tl_new:c { \__spintent_sp #1 _prefix_tl }
1927   \tl_new:c { \__spintent_sp #1 _suffix_tl }
1928 }

```

(End of definition for `\__spintent_spmcm_spmcd_luaset_error_str` and others.)

11.26.2 Definition of error messages

mc-invalid-arguments

Exception triggered when the comma-separated sequence contains negative integers, floats, or alphabetical characters.

```

1929 \msg_new:nnnn { spintent } { mc-invalid-arguments }
1930 {
1931   Invalid~arguments~provided~to~#1.
1932 }
1933 {
1934   The~arguments~must~be~a~comma-separated~list. \\\
1935 }

```

(End of definition for `mc-invalid-arguments`.)

11.26.3 Keys for `\spmc` and `\spmcm`

We define the shared *(keys)* `result`, `prefix`, `read-cmd`, `upper-case`, `sample-arg` and `silent-cmd`.

```

1936 \clist_map_inline:nn { mcd , mcm }
1937 {
1938   \keys_define:nn { spintent / sp #1 }
1939   {
1940     result      .bool_set:c = { l__spintent_sp #1 _print_result_bool },
1941     result      .initial:n = false,
1942     result      .value_required:n = true,
1943     prefix      .code:n =
1944       { \__spintent_sanitise_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
1945     prefix      .initial:n = ,
1946     prefix      .value_required:n = true,
1947     read-cmd    .choices:nn =
1948       { terse , clear }
1949     {
1950       \int_case:nn { \l_keys_choice_int }
1951       {
1952         {1} { \bool_set_true:c { l__spintent_sp #1 _read_terse_bool } }
1953         {2} { \bool_set_false:c { l__spintent_sp #1 _read_terse_bool } }
1954       }
1955     },
1956     read-cmd    .initial:n = clear,
1957     read-cmd    .value_required:n = true,
1958     read-cmd / unknown .code:n =
1959       \msg_error:nneee { spintent } { unknown-choice }
1960       { read-cmd } { terse , ~clear } { \exp_not:n {##1} },
1961     upper-case  .bool_set:c = { l__spintent_sp #1 _print_upper_bool },
1962     upper-case  .initial:n = false,
1963     upper-case  .value_required:n = true,
1964     sample-arg  .bool_set:c = { l__spintent_sp #1 _sample_arg_bool },
1965     sample-arg  .initial:n = false,
1966     sample-arg  .value_required:n = true,
1967     silent-cmd  .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
1968     silent-cmd  .initial:n = false,
1969     silent-cmd  .value_required:n = true,
1970     unknown     .code:n =
1971       {
1972         \str_set:Nn \l__spintent_command_name_str { sp #1 }
1973         \__spintent_unknown_keys_cmd:n {##1}
1974       },
1975   }
1976 }

```

(End of definition for `result` and others.)

11.26.4 Internal functions for `\spmc` and `\spmcm`

`\__spintent_mc_process_args:nn` The function `\__spintent_mc_process_args:nn` parses the operation ‘#1’ and its arguments ‘#2’, injecting MathML intents.

```

1977 \cs_new_protected:Nn \__spintent_mc_process_args:nn
1978 {
1979   \seq_set_from_clist:Nn \l__spintent_mc_args_seq {#2}
1980   \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
1981   {
1982     \seq_set_map:NnNn \l__spintent_mc_args_seq \l__spintent_mc_args_seq
1983     { \MathMLintent { :silent } { ##1 } }
1984     \MathMLintent { entre:silent } { ( }
1985     \seq_use:Nnnn \l__spintent_mc_args_seq
1986     { \MathMLintent { _y:silent } { , } }
1987     { \MathMLintent { :silent } { , } }
1988     { \MathMLintent { _y:silent } { , } }
1989   }
1990   {
1991     \MathMLintent { entre } { ( }
1992     \seq_use:Nnnn \l__spintent_mc_args_seq
1993     { \MathMLintent { _y } { , } }
1994     { \MathMLintent { :silent } { , } }
1995     { \MathMLintent { _y } { , } }
1996   }

```



```

1997   \MathMLintent { :silent } { } }
1998 }

```

(End of definition for `\__spintent_mc_process_args:nn`.)

`\__spintent_spmc_process:nnnn`

The internal function `\__spintent_spmc_process:nnnn` serves as a shared central handler for both the `\spmc`d and `\spmc`m commands. It takes the command identifier ‘#1’, its full reading string ‘#2’, the local *(keys)* ‘#3’, and the optional values ‘#4’. It dynamically resolves variables using the identifier to build the `MathML` `intent` string (including any prefixes or suffixes) and prints the math operator. If values are provided in ‘#4’, it calculates the result using the Lua backend. If an error occurs, it throws the `mc-invalid-arguments` error; otherwise, it processes the arguments and conditionally prints the calculated result.

```

1999 \cs_new_protected:Nn \__spintent_spmc_process:nnnn
2000 {
2001   \keys_set:nn { spintent / sp #1 } { #3 }
2002   \MathMLintent
2003   {
2004     \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
2005     { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
2006     \bool_if:cTF { \__spintent_sp #1 _read_terse_bool }
2007     { #1 } { #2 }
2008     \bool_if:cT { \__spintent_sp #1 _silent_cmd_bool }
2009     {
2010       \tl_if_novalue:nTF {#4}
2011       { :silent }
2012       { \bool_if:cT { \__spintent_sp #1 _sample_arg_bool } { :silent } }
2013     }
2014   }
2015   {
2016     \operatorname
2017     {
2018       \bool_if:cTF { \__spintent_sp #1 _print_upper_bool }
2019       { \text_uppercase:n {#1} } { #1 }
2020     }
2021   }
2022   \tl_if_novalue:nF { #4 }
2023   {
2024     \bool_if:cTF { \__spintent_sp #1 _sample_arg_bool }
2025     {
2026       \__spintent_mc_process_args:nn {#1} {#4}
2027     }
2028     {
2029       \use:c { __spintent_luafun_calculate_ #1 :n } { #4 }
2030       \str_if_eq:VnTF \__spintent_spmcm_spmcd_luaset_error_str { true }
2031       {
2032         \msg_error:nne { spintent } { mc-invalid-arguments } { \exp_not:c { sp #1 } }
2033       }
2034       {
2035         \__spintent_mc_process_args:nn {#1} {#4}
2036         \bool_if:cT { \__spintent_sp #1 _print_result_bool }
2037         {
2038           \MathMLintent { es-igual-a } { = }
2039           \tl_use:c { \__spintent_sp #1 _luaset_ #1 _value_tl }
2040         }
2041       }
2042     }
2043   }
2044 }

```

(End of definition for `\__spintent_spmc_process:nnnn`.)

`\spmc`d The commands `\spmc`d and `\spmc`m act as the public interface wrappers for typesetting the greatest common divisor and least common multiple. They enforce math mode, open a local group, and defer execution to the main processing macro `\__spintent_spmc_process:nnnn`, passing along the appropriate reading strings and arguments.

```

2045 \NewDocumentCommand \spmc { 0{} } d() {
2046 {
2047   \mode_if_math:TF
2048   {
2049     \group_begin:
2050     \__spintent_spmc_process:nnnn { mcd } { máximo-común-divisor } {#1} {#2}
2051     \group_end:

```

```

2052     }
2053     { \msg_error:nnn { spintent } { need-math-mode } { \spmc } }
2054 }
2055 \NewDocumentCommand \spmc { 0{} d() }
2056 {
2057     \mode_if_math:TF
2058     {
2059         \group_begin:
2060         __spintent_spmc_process:nnnn { mcm } { mínimo-común-múltiplo } {#1} {#2}
2061         \group_end:
2062     }
2063     { \msg_error:nnn { spintent } { need-math-mode } { \spmc } }
2064 }

```

(End of definition for `\spmc` and `\spmc`. These functions are documented on page 15.)

11.27 The commands `\spnM` and `\spnD`

The user-level commands `\spnM` and `\spnD` provide a semantic interface for the Multiples of a number (“*múltiplos*” in Spanish) and the Divisors of a number (“*divisores*” in Spanish). They delegate argument validation and calculation to the Lua module `spintent.lua`, constructing a valid MathCAT intent.

11.27.1 Variables for `\spnM` and `\spnD`

Internal variables generated dynamically for multiples and divisors.

```

\__spintent_md_out_seq
\__spintent_spnM_luaset_result_tl
\__spintent_spnD_luaset_result_tl
\__spintent_spnM_luaset_is_truncated_str
\__spintent_spnD_luaset_is_truncated_str
\__spintent_spnM_print_result_bool
\__spintent_spnD_print_result_bool
\__spintent_spnM_silent_cmd_bool
\__spintent_spnD_silent_cmd_bool
\__spintent_spnM_prefix_tl
\__spintent_spnD_prefix_tl
\__spintent_spnM_print_result_num_tl
\__spintent_spnD_print_result_num_tl
2065 \seq_new:N \__spintent_md_out_seq
2066 \clist_map_inline:nn { nM , nD }
2067 {
2068     \tl_new:c { \__spintent_sp #1 _luaset_result_tl }
2069     \str_new:c { \__spintent_sp #1 _luaset_is_truncated_str }
2070     \bool_new:c { \__spintent_sp #1 _print_result_bool }
2071     \bool_new:c { \__spintent_sp #1 _silent_cmd_bool }
2072     \tl_new:c { \__spintent_sp #1 _prefix_tl }
2073     \tl_new:c { \__spintent_sp #1 _print_result_num_tl }
2074 }

```

(End of definition for `\__spintent_md_out_seq` and others.)

11.27.2 Keys for `\spnM` and `\spnD`

We define the shared (*keys*) `result`, `silent-cmd`, `result-num` and `prefix`.

```

result
silent-cmd
result-num
prefix
sample-arg
2075 \clist_map_inline:nn { nM , nD }
2076 {
2077     \keys_define:nn { spintent / sp #1 }
2078     {
2079         result .bool_set:c = { \__spintent_sp #1 _print_result_bool },
2080         result .initial:n = false,
2081         result .value_required:n = true,
2082         silent-cmd .bool_set:c = { \__spintent_sp #1 _silent_cmd_bool },
2083         silent-cmd .initial:n = false,
2084         silent-cmd .value_required:n = true,
2085         result-num .tl_set:c = { \__spintent_sp #1 _print_result_num_tl },
2086         result-num .initial:n = ,
2087         prefix .code:n =
2088             { \__spintent_sanitize_affix:cn { \__spintent_sp #1 _prefix_tl } {##1} },
2089         prefix .initial:n = ,
2090         prefix .value_required:n = true,
2091         sample-arg .bool_set:c = { \__spintent_sp #1 _sample_arg_bool },
2092         sample-arg .initial:n = false,
2093         sample-arg .value_required:n = true,
2094         unknown .code:n =
2095             {
2096                 \str_set:Nn \__spintent_command_name_str { sp #1 }
2097                 \__spintent_unknown_keys_cmd:n {##1}
2098             },
2099     }
2100 }

```

(End of definition for `result` and others.)

11.27.3 Definition of error messages for \spnM and \spnD

md-invalid-argument Exception triggered when the argument for multiples or divisors is invalid.

```
2101 \msg_new:nnnn { spintenc } { md-invalid-argument }
2102 { Invalid~argument~provided~to~#1. }
2103 {
2104   The~argument~must~be~a~single~positive~integer. \
2105 }
```

(End of definition for md-invalid-argument.)

11.27.4 Internal functions for \spnM and \spnD

__spintenc_spnM_spnD_process:nnnn

The internal function `\__spintenc_spnM_spnD_process:nnnn` acts as the primary entry point for the multiples and divisors commands. It takes the command identifier ‘#1’, its printed mathematical symbol ‘#2’, its base reading string ‘#3’, the local key-value options ‘#4’, and the optional argument ‘#5’. Its sole purpose is to evaluate and set the provided keys within the current scope before delegating the core branching and typesetting logic to `\__spintenc_spnM_and_D_exec:nnnn`.

```
2106 \cs_new_protected:Nn \__spintenc_spnM_spnD_process:nnnn
2107 {
2108   \keys_set:nn { spintenc / sp #1 } { #4 }
2109   \__spintenc_spnM_spnD_exec:nnnn { #1 } { #2 } { #3 } { #5 }
2110 }
```

(End of definition for __spintenc_spnM_spnD_process:nnnn.)

__spintenc_spnM_spnD_intent:nn

The internal function `\__spintenc_spnM_spnD_intent:nn` resolves the main reading intent string. It takes the command identifier ‘#1’ (such as “nM” or “nD”) and the base reading string ‘#2’. It automatically prepends the localized prefix if defined and handles specific string normalizations, such as ensuring the correct accentuation for “múltiplos”.

```
2111 \cs_new:Nn \__spintenc_spnM_spnD_intent:nn
2112 {
2113   \tl_if_empty:cF { l__spintenc_sp #1 _prefix_tl }
2114   { \tl_use:c { l__spintenc_sp #1 _prefix_tl } - }
2115   \str_case:nnF { #2 }
2116   { { múltiplos } { múltiplos } }
2117   { #2 }
2118 }
```

(End of definition for __spintenc_spnM_spnD_intent:nn.)

__spintenc_spnM_spnD_silent:nn

The internal function `\__spintenc_spnM_and_D_silent:nn` prints the atomized notation structure in full silent mode. It takes the mathematical symbol ‘#1’ (such as “D” or “M”) and the variable or explicit argument ‘#2’. Every component is wrapped in a silent intent, and an invisible separator is injected between the parentheses to prevent assistive technologies from applying default contextual reading rules like “de”.

```
2119 \cs_new_protected:Nn \__spintenc_spnM_spnD_silent:nn
2120 {
2121   \MathMLintent { :silent } { \operatorname { #1 } }
2122   \MathMLintent { :silent } { ( }
2123   \MathMLintent { :silent } { \__spintenc_invisible_sep: }
2124   \MathMLintent { :silent } { #2 }
2125   \MathMLintent { :silent } { ) }
2126 }
```

(End of definition for __spintenc_spnM_spnD_silent:nn.)

__spintenc_spnM_spnD_result:n

The internal function `\__spintenc_spnM_and_D_result:n` handles the calculation and accessible formatting of the computed mathematical set. It takes the command identifier ‘#1’. It invokes the corresponding Lua backend calculation, maps the resulting clist into a sequence, and formats the final printed output utilizing custom accessibility intents for both separators and conjunctions. It also safely manages truncation indicators.

```
2127 \cs_new_protected:Nn \__spintenc_spnM_spnD_result:n
2128 {
2129   \use:c { __spintenc_luafun_calcular_ #1 :nn }
2130   { \l__spintenc_luaseta_part_int_tl }
2131   { \tl_use:c { l__spintenc_sp #1 _print_result_num_tl } }
2132   \MathMLintent { son } { = \{ }
2133   \seq_set_from_clist:Ne
2134   \l__spintenc_md_out_seq { \tl_use:c { l__spintenc_sp #1 _luaseta_result_tl } }
2135   \seq_use:Nnnn \l__spintenc_md_out_seq
2136   { \MathMLintent { _y } { , } }
2137   { \MathMLintent { :silent } { , } }
2138   { \MathMLintent { _y } { , } }
```

```

2139 \str_if_eq:vnT
2140 { \__spintrent_sp #1 \luaset_is_truncated_str } { true }
2141 {
2142   \MathMLintent { :silent } { , } ~ \dots
2143 }
2144 \MathMLintent { :silent } { \} }
2145 }

```

(End of definition for `\__spintrent_spnM_spnD_result:n`.)

`\__spintrent_spnM_spnD_exec:nnnn`

The function `\__spintrent_spnM_and_D_exec:nnnn` implements the multiples and divisors commands. It takes the `{\langle argument \rangle}` ‘#1’ (the command name), the `{\langle argument \rangle}` ‘#2’ (the mathematical symbol), the base reading `{\langle argument \rangle}` ‘#3’, and the optional `{\langle argument \rangle}` ‘#4’. If no `{\langle argument \rangle}` is provided, it evaluates the key `silent-cmd` using the default token ‘n’. If an `{\langle argument \rangle}` is given, it evaluates the key `sample-arg`. If true, it processes the `{\langle argument \rangle}` directly and evaluates the key `silent-cmd`. If false, it bypasses the *silent mode*, validates the `{\langle argument \rangle}` as a natural number, and evaluates the key `result` optionally calculate and append the resulting set.

```

2146 \cs_new_protected:Nn \__spintrent_spnM_spnD_exec:nnnn
2147 {
2148   \tl_if_novalue:nTF { #4 }
2149   {
2150     \bool_if:cTF { \__spintrent_sp #1 _silent_cmd_bool }
2151     { \__spintrent_spnM_spnD_silent:nn { #2 } { n } }
2152     {
2153       \MathMLintent
2154       { \__spintrent_spnM_spnD_intent:nn { #1 } { #3 } }
2155       { \operatorname { #2 } ( n ) }
2156     }
2157   }
2158   {
2159     \bool_if:cTF { \__spintrent_sp #1 _sample_arg_bool }
2160     {
2161       \bool_if:cTF { \__spintrent_sp #1 _silent_cmd_bool }
2162       { \__spintrent_spnM_spnD_silent:nn { #2 } { #4 } }
2163       {
2164         \MathMLintent
2165         { \__spintrent_spnM_spnD_intent:nn { #1 } { #3 } }
2166         { \operatorname { #2 } } ( #4 )
2167       }
2168     }
2169     {
2170       \__spintrent_luafun_clean_split_and_set:n { \exp_not:n { #4 } }
2171       \str_if_eq:VnTF \__spintrent_luaset_has_natural_str { true }
2172       {
2173         \MathMLintent
2174         { \__spintrent_spnM_spnD_intent:nn { #1 } { #3 } }
2175         { \operatorname { #2 } } ( #4 )
2176         \bool_if:cT { \__spintrent_sp #1 _print_result_bool }
2177         { \__spintrent_spnM_spnD_result:n { #1 } }
2178       }
2179       {
2180         \msg_error:nne { spintrent } { md-invalid-argument } { \exp_not:c { sp #1 } }
2181       }
2182     }
2183   }
2184 }

```

(End of definition for `\__spintrent_spnM_spnD_exec:nnnn`.)

`\spnM` The commands `\spnM` and `\spnD` act as the public interface wrappers for typesetting multiples and divisors, respectively. They enforce math mode, open a local group, and defer execution to the function `\__spintrent_spnM_spnD_process:nnnnn`, passing along the optional `\keys` ‘#1’ and the numeric `{\langle argument \rangle}` ‘#2’.

```

2185 \NewDocumentCommand \spnM { O{} d() }
2186 {
2187   \mode_if_math:TF
2188   {
2189     \group_begin:
2190     \__spintrent_spnM_spnD_process:nnnnn { nM } { M } { multiplos } { #1 } { #2 }
2191     \group_end:
2192   }

```

```

2193     { \msg_error:nnn { spintent } { need-math-mode } { \spnM } }
2194   }
2195   \NewDocumentCommand \spnD { 0{} d() }
2196   {
2197     \mode_if_math:TF
2198     {
2199       \group_begin:
2200       \__spintent_spnM_spnD_process:nnnnn { nD } { D } { divisores } {#1} {#2}
2201       \group_end:
2202     }
2203     { \msg_error:nnn { spintent } { need-math-mode } { \spnD } }
2204   }

```

(End of definition for `\spnM` and `\spnD`. These functions are documented on page 14.)

11.28 The command `\spang`

The command `\spang` is the main public command for typesetting and tagging sexagesimal angle expressions (degrees, minutes, and seconds). It parses the input using the Lua function `\__spintent_luafun_spang_parse:n`. This separates the degrees, minutes, and seconds to construct a MathML `intent` tree with the correct postfixes for NVDA/MathCAT.

```

\l__spintent_spang_luaset_grado_str
\l__spintent_spang_luaset_minuto_str
\l__spintent_spang_luaset_segundo_str
\l__spintent_spang_luaset_error_str

```

These internal string variables store the values returned from the Lua module `spintent.lua`. They hold the extracted numerical values for degrees, minutes, and seconds, as well as the error status used during formatting.

```

2205 \str_new:N \l__spintent_spang_luaset_grado_str
2206 \str_new:N \l__spintent_spang_luaset_minuto_str
2207 \str_new:N \l__spintent_spang_luaset_segundo_str
2208 \str_new:N \l__spintent_spang_luaset_error_str

```

(End of definition for `\l__spintent_spang_luaset_grado_str` and others.)

11.28.1 Definition of error messages for `\spang`

invalid-angle-format

The error message `invalid-angle-format` defines the text displayed when an invalid angle input is detected. It is triggered when an argument fails the format checks, does not match the expected sexagesimal pattern, or contains invalid measurement symbols.

```

2209 \msg_new:nnnn { spintent } { invalid-angle-format }
2210 { Invalid~angle~format~in~#1. }
2211 {
2212   The~argument~'#2'~could~not~be~parsed. \
2213 }

```

(End of definition for `invalid-angle-format`.)

11.28.2 Internal functions for `\spang`

__spintent_spang_process:n

The internal function `\__spintent_spang_process:n` parses the angle input using `\__spintent_luafun_spang_parse:n` after resetting the error flag in `\l__spintent_spang_luaset_error_str`. If an invalid format is detected, it issues the `invalid-angle-format` error. Otherwise, it checks the extracted string variables `\l__spintent_spang_luaset_grado_str`, `\l__spintent_spang_luaset_minuto_str`, and `\l__spintent_spang_luaset_segundo_str`. For each non-empty variable, it constructs the corresponding MathML `intent` structure with the correct measurement symbols and postfixes for NVDA/MathCAT.

```

2214 \cs_new_protected:Nn \__spintent_spang_process:n
2215 {
2216   \str_set:Nn \l__spintent_spang_luaset_error_str { false }
2217   \__spintent_luafun_spang_parse:n { #1 }
2218   \str_if_eq:VnTF \l__spintent_spang_luaset_error_str { true }
2219   {
2220     \msg_error:nnnn { spintent } { invalid-angle-format } { \spang } { #1 }
2221   }
2222   {
2223     \str_if_empty:NF \l__spintent_spang_luaset_grado_str
2224     {
2225       \spunit{ \l__spintent_spang_luaset_grado_str °}
2226     }
2227     \str_if_empty:NF \l__spintent_spang_luaset_minuto_str
2228     {
2229       \MathMLintent { minutos : postfix($m) }
2230       {
2231         \MathMLarg { m }
2232         {
2233           \spunit{ \l__spintent_spang_luaset_minuto_str ' }

```

```

2234         }
2235     }
2236 }
2237 \str_if_empty:NF \l__spintent_spang_luaset_segundo_str
2238 {
2239     \MathMLintent { segundos : postfix($s) }
2240     {
2241         \MathMLarg { s }
2242         {
2243             \spunit { \l__spintent_spang_luaset_segundo_str " }
2244         }
2245     }
2246 }
2247 }
2248 }

```

(End of definition for `\l__spintent_spang_process:n`.)

\spang The command `\spang` is the public interface for formatting sexagesimal angle expressions. It first checks if it is being used in math mode, issuing an error if not. Inside a local group, it calls `\l__spintent_spang_process:n` to process the $\langle argument \rangle$.

```

2249 \NewDocumentCommand \spang { m }
2250 {
2251     \mode_if_math:TF
2252     {
2253         \group_begin:
2254         \l__spintent_spang_process:n { #1 }
2255         \group_end:
2256     }
2257     { \msg_error:nnn { spintent } { need-math-mode } { \spang } }
2258 }

```

(End of definition for `\spang`. This function is documented on page 21.)

11.29 The command `\spnZ`

The user-level command `\spnZ` formats integers (the set $\mathbb{Z}$). It includes specific options to automatically enclose numbers (particularly negative ones) in visual parentheses and allows you to control whether these parentheses are read by MathCAT.

This command verifies that the $\langle argument \rangle$ is indeed an *integer*, checking for the complete absence of decimal separators or attached units of measurement.

11.29.1 Definition of keys for `\spnZ`

Now we define the $\langle keys \rangle$ `cifras-sep` and `read-sign`. These are `.meta:nn` $\langle keys \rangle$ passed directly to the `\spnum` command. We also introduce `wrap-parens`, which determines whether the number is enclosed in parentheses, and `read-parens`, which controls if these parentheses are read by MathCAT.

```

2259 \keys_define:nn { spintent / spnZ }
2260 {
2261     cifras-sep .meta:nn = { spintent / spnum } { cifras-sep = {#1} },
2262     read-sign .meta:nn = { spintent / spnum } { read-sign = {#1} },
2263     wrap-parens .bool_set:N = \l__spintent_spnz_wrap_parens_bool,
2264     wrap-parens .initial:n = false,
2265     wrap-parens .value_required:n = true,
2266     read-parens .bool_set:N = \l__spintent_spnz_read_parens_bool,
2267     read-parens .initial:n = true,
2268     read-parens .value_required:n = true,
2269     unknown .code:n =
2270     {
2271         \str_set:Nn \l__spintent_command_name_str { spnZ }
2272         \l__spintent_unknown_keys_cmd:n {#1}
2273     },
2274 }

```

(End of definition for `cifras-sep` and others.)

11.29.2 Definition of error messages

spnz-not-integer Raised when the $\langle argument \rangle$ contains decimal separators or attached units, which conflict with the constraints of a pure integer representation.

```

2275 \msg_new:nnnn { spintent } { spnz-not-integer }
2276 { The~value~'~#1'~is~not~a~valid~integer. }
2277 {

```



```

2278     Command~\token_to_str:N \spnZ~only~accepts~integers.\{
2279 }

```

(End of definition for `spnz-not-integer`.)

11.29.3 Internal functions for `\spnZ`

`\__spintent_spnz_render_parens:` The function `\__spintent_spnz_render_parens:` first checks the state of the boolean variable `\__spintent_spnz` (controlled by the `read-parens` key). If it is `true`, it prints the number surrounded by parentheses. Otherwise, it applies the `:silent function intent` to the parentheses while printing the number.

```

2280 \cs_new_protected:Nn \__spintent_spnz_render_parens:
2281 {
2282   \bool_if:NTF \__spintent_spnz_read_parens_bool
2283   {
2284     ( \__spintent_spnz_print_num_start: )
2285   }
2286   {
2287     \MathMLintent { :silent } { ( }
2288     \__spintent_spnz_print_num_start:
2289     \MathMLintent { :silent } { ) }
2290   }
2291 }

```

(End of definition for `\__spintent_spnz_render_parens:`.)

`\__spintent_spnz_process:nn`

The function `\__spintent_spnz_process:nn` first processes the `<keys>` passed in ‘#1’ and then calls `\__spintent_luafun_clean_split_and_set:n` on ‘#2’. It verifies that the input is a valid integer by checking if the variable `\__spintent_luaset_has_integer_str` is `true`. If not `true`, an “error” is raised.

If it is `true`, it checks whether `\__spintent_luaset_part_int_tl` is *empty* and raises an “error” if so. Otherwise, it checks `\__spintent_spnz_wrap_parens_bool` set by `wrap-parens` and calls either the function `\__spintent_spnz_render_parens:` or `\__spintent_spnz_print_num_start:`.

```

2292 \cs_new_protected:Nn \__spintent_spnz_process:nn
2293 {
2294   \keys_set:nn { spintent / spnZ } {#1}
2295   \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#2} }
2296   \str_if_eq:VnTF \__spintent_luaset_has_integer_str { true }
2297   {
2298     \tl_if_empty:NTF \__spintent_luaset_part_int_tl
2299     { \msg_error:nnn { spintent } { spnum-no-digits } {#2} }
2300     {
2301       \bool_if:NTF \__spintent_spnz_wrap_parens_bool
2302       { \__spintent_spnz_render_parens: }
2303       { \__spintent_spnz_print_num_start: }
2304     }
2305   }
2306   { \msg_error:nnn { spintent } { spnz-not-integer } {#2} }
2307 }

```

(End of definition for `\__spintent_spnz_process:nn`.)

`\spnZ` Defines the user command `\spnZ`, which is executed within a *group* and strictly enforces *math mode*.

```

2308 \NewDocumentCommand \spnZ { O{} m }
2309 {
2310   \mode_if_math:TF
2311   {
2312     \group_begin:
2313     \__spintent_spnz_process:nn { #1 } { #2 }
2314     \group_end:
2315   }
2316   { \msg_error:nnn { spintent } { need-math-mode } { \spnZ } }
2317 }

```

(End of definition for `\spnZ`. This function is documented on page 16.)

11.30 The command `\spmQ`

The command `\spmQ` typesets “mixed numbers” within Spanish mathematical contexts (e.g., $3\frac{1}{2}$). It delegates the scaling of the integer part to the Lua module `spintent.lua` via `\__spintent_luafun_spmQ_scale_int:Vn`, which measures both the integer glyph and the fraction box at the node level and computes a scaling factor to align their heights visually. Finally, it constructs the correct MathML intent for MathCAT.

11.30.1 Definition of variables for \spmQ

The variable `\l__spintrent_spmQ_join_word_str` stores the value set by the key `join-word` which will be verbalized by MathCAT as a grammatical connector between the integer and fractional parts.

The variable `\l__spintrent_spmQ_math_style_tl` stores the current mathematical style used by the function `\__spintrent_spmQ_set_wrap_parens_size:`.

The variable `\l__spintrent_spmQ_intent_read_sign_str` stores the `:intent` passed to `\MathMLintent` when the signs ‘+’ or ‘-’ are present in the integer part, used by the function `\__spintrent_spmQ_print_scale_int:n`.

The variables `\l__spintrent_spmQ_wrap_parens_l_tl` and `\l__spintrent_spmQ_wrap_parens_r_tl` store the sized parentheses set by the function `\__spintrent_spmQ_set_wrap_parens_size:` and used by the key `wrap-parens`.

The integer variable `\l__spintrent_saved_luamml_flag_int` temporarily stores the current value of `\l__luamml_flag_int` before disabling it during the injection of the scaled integer node, and restores it afterwards. This prevents `luamml` from generating a spurious `<math>` element around the `sub_box` processed in isolation, before `\frac` has expanded.

```

2318 \str_new:N \l__spintrent_spmQ_join_word_str
2319 \tl_new:N \l__spintrent_spmQ_math_style_tl
2320 \str_new:N \l__spintrent_spmQ_intent_read_sign_str
2321 \tl_new:N \l__spintrent_spmQ_wrap_parens_l_tl
2322 \tl_new:N \l__spintrent_spmQ_wrap_parens_r_tl
2323 \int_new:N \l__spintrent_saved_luamml_flag_int

```

(End of definition for `\l__spintrent_spmQ_join_word_str` and others.)

11.30.2 Definition of error messages for \spmQ

The error message `spmQ-zero-denominator` is raised when a zero denominator is passed to `\spmQ`. The second error `spmQ-not-integer` is returned when the integer part passed in `{\argument}` is not actually an integer.

```

2324 \msg_new:nnnn { spintrent } { spmQ-zero-denominator }
2325 { Zero~denominator~in~\token_to_str:N \spmQ. }
2326 {
2327   The~denominator~of~a~fraction~cannot~be~zero.\l
2328 }
2329 \msg_new:nnnn { spintrent } { spmQ-not-integer }
2330 { The~value~'#1'~is~not~a~valid~integer. }
2331 {
2332   Command~\token_to_str:N \spmQ \c_space_tl only~accepts~integers.\l
2333 }

```

(End of definition for `spmQ-zero-denominator` and `spmQ-not-integer`.)

11.30.3 Definition of keys for \spmQ

Now we add the keys `join-word`, `use-spnZ`, `wrap-parens`, `read-parens` and `print-dfrac`.

```

join-word
use-spnZ
wrap-parens
read-parens
print-dfrac
unknown
2334 \keys_define:nn { spintrent / spmQ }
2335 {
2336   join-word .choices:nn =
2337   { entero , enteros , y }
2338   {
2339     \int_case:nn { \l_keys_choice_int }
2340     {
2341       {1} { \str_set:Nn \l__spintrent_spmQ_join_word_str { entero } }
2342       {2} { \str_set:Nn \l__spintrent_spmQ_join_word_str { enteros } }
2343       {3} { \str_set:Nn \l__spintrent_spmQ_join_word_str { y } }
2344     }
2345   },
2346   join-word .initial:n = y,
2347   join-word .value_required:n = true,
2348   join-word/unknown .code:n =
2349   \msg_error:nneee { spintrent } { unknown-choice }
2350   { join-word } { entero , ~enteros , ~y } { \exp_not:n {#1} },
2351   use-spnZ .bool_set:N = \l__spintrent_spmQ_use_spnZ_bool,
2352   use-spnZ .initial:n = true,
2353   use-spnZ .value_required:n = true,
2354   print-dfrac .bool_set:N = \l__spintrent_spmQ_print_dfrac_bool,
2355   print-dfrac .initial:n = false,
2356   print-dfrac .value_required:n = true,
2357   wrap-parens .bool_set:N = \l__spintrent_spmQ_wrap_parens_bool,
2358   wrap-parens .initial:n = false,
2359   wrap-parens .value_required:n = true,

```

```

2360     read-parens      .bool_set:N = \__spintent_spmQ_read_parens_bool,
2361     read-parens      .initial:n = true,
2362     read-parens      .value_required:n = true,
2363     unknown          .code:n =
2364     {
2365         \str_set:Nn \__spintent_command_name_str { spmQ }
2366         \__spintent_unknown_keys_cmd:n {#1}
2367     },
2368 }

```

(End of definition for join-word and others.)

11.30.4 Internal functions for \spmQ

The function `\__spintent_spmQ_set_wrap_parens_size:` captures the current mathematical style using the primitive `\mathstyle` from LaTeX and sets the variables `\__spintent_spmQ_wrap_parens_l_tl` and `\__spintent_spmQ_wrap_parens_r_tl` with the appropriate size for the parentheses. Display styles (0 and 1) receive `\Bigl\lparen` and `\Bigr\rparen`; text styles (2 and 3) receive `\bigl\lparen` and `\bigr\rparen`; any other style falls back to `\lparen` and `\rparen`.

If the key `print-dfrac` is active, the size is always forced to `\Bigl\lparen`/`\Bigr\rparen` regardless of the current mathematical style, since `\dfrac` always produces a display-height fraction.

```

2369 \cs_new_protected:Nn \__spintent_spmQ_set_wrap_parens_size:
2370 {
2371     \int_case:nnF { \mathstyle }
2372     {
2373         { 0 }
2374         {
2375             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2376             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2377         }
2378         { 1 }
2379         {
2380             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2381             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2382         }
2383         { 2 }
2384         {
2385             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \bgl \lparen }
2386             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \bgr \rparen }
2387         }
2388         { 3 }
2389         {
2390             \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \bgl \lparen }
2391             \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \bgr \rparen }
2392         }
2393     }
2394     {
2395         \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \lparen }
2396         \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \rparen }
2397     }

```

If the key `print-dfrac` is active, we must reset the variables `\__spintent_spmQ_wrap_parens_l_tl` and `\__spintent_spmQ_wrap_parens_r_tl` to `\Bigl\lparen`/`\Bigr\rparen` unconditionally.

```

2398     \bool_if:NT \__spintent_spmQ_print_dfrac_bool
2399     {
2400         \tl_set:Nn \__spintent_spmQ_wrap_parens_l_tl { \Bigl \lparen }
2401         \tl_set:Nn \__spintent_spmQ_wrap_parens_r_tl { \Bigr \rparen }
2402     }
2403 }

```

(End of definition for `\__spintent_spmQ_set_wrap_parens_size:`)

The functions `\__spintent_spmQ_start_wrap_parens:` and `\__spintent_spmQ_stop_wrap_parens:` open and close the parentheses around the “mixed number” when the key `wrap-parens` is active. They first call `\__spintent_spmQ_set_wrap_parens_size:` to determine the correct size for the current mathematical style, then print the appropriate parenthesis stored in `\__spintent_spmQ_wrap_parens_l_tl` or `\__spintent_spmQ_wrap_parens_r_tl`.

If the variable `\__spintent_spmQ_read_parens_bool`, controlled by the key `read-parens`, is *false*, each parenthesis is wrapped inside `\MathMLintent{:silent}` so that NVDA/MathCAT does not verbalize them.

```

2404 \cs_new_protected:Nn \__spintent_spmQ_start_wrap_parens:
2405 {

```

```

2406 \bool_if:NT \l__spintent_spmQ_wrap_parens_bool
2407 {
2408   \__spintent_spmQ_set_wrap_parens_size:
2409   \bool_if:NTF \l__spintent_spmQ_read_parens_bool
2410     { \l__spintent_spmQ_wrap_parens_l_tl }
2411     { \MathMLintent { :silent } { \l__spintent_spmQ_wrap_parens_l_tl } }
2412 }
2413 }
2414 \cs_new_protected:Nn \__spintent_spmQ_stop_wrap_parens:
2415 {
2416   \bool_if:NT \l__spintent_spmQ_wrap_parens_bool
2417   {
2418     \__spintent_spmQ_set_wrap_parens_size:
2419     \bool_if:NTF \l__spintent_spmQ_read_parens_bool
2420       { \l__spintent_spmQ_wrap_parens_r_tl }
2421       { \MathMLintent { :silent } { \l__spintent_spmQ_wrap_parens_r_tl } }
2422   }
2423 }

```

(End of definition for `\__spintent_spmQ_start_wrap_parens:` and `\__spintent_spmQ_stop_wrap_parens:`.)

`\__spintent_spmQ_set_join_word:`

The function `\__spintent_spmQ_set_join_word:` injects the grammatical connector between the integer part and the fraction by dispatching on the value stored in `\l__spintent_spmQ_join_word_str`, set by the key `join-word`. In all three cases the visual output is the Unicode invisible separator U+2063 via `\c__spintent_invisible_sep_tl`, which produces a `<mi>` node in the MathML structure; what differs is the `:intent` text attached to it via `\MathMLintent`, which NVDA/MathCAT will verbalize as «entero», «enteros» or «y» respectively.

```

2424 \cs_new_protected:Nn \__spintent_spmQ_set_join_word:
2425 {
2426   \str_case:Nn \l__spintent_spmQ_join_word_str
2427   {
2428     { entero } { \MathMLintent { _entero } { \c__spintent_invisible_sep_tl } }
2429     { enteros } { \MathMLintent { _enteros } { \c__spintent_invisible_sep_tl } }
2430     { y } { \MathMLintent { _y- } { \c__spintent_invisible_sep_tl } }
2431   }
2432 }

```

(End of definition for `\__spintent_spmQ_set_join_word:`.)

`\__spintent_spmQ_print_scale_int:`

The function `\__spintent_spmQ_print_scale_int:` prints the integer part of the “mixed number” scaled to match the visual height of the accompanying fraction. Before calling the Lua function `\__spintent_luafun_spmQ_scale_int` it temporarily disables `luamml` by saving the current value of `\l__luamml_flag_int` into `\l__spintent_saved_luamml_flag_int` and setting it to `0` (skip completely). This prevents `luamml` from generating a spurious `<math>` wrapper around the `sub_box` node that holds the scaled integer, which is processed in isolation before `\frac` has expanded. Once the Lua function returns, the original flag value is restored via `\l__spintent_saved_luamml_flag_int`.

The Lua function receives the integer digits from `\l__spintent_luaset_part_int_tl` and the fraction command name (`\frac` or `\dfrac`, determined by the key `print-dfrac`) as its two arguments, and inserts the appropriately scaled `sub_box` node directly into the current horizontal list.

```

2433 \cs_new_protected:Nn \__spintent_spmQ_print_scale_int:
2434 {
2435   \int_set_eq:Nc \l__spintent_saved_luamml_flag_int { \l__luamml_flag_int }
2436   \int_set:Nn \l__luamml_flag_int { 0 }
2437   \bool_if:NTF \l__spintent_spmQ_print_dfrac_bool
2438     { \__spintent_luafun_spmQ_scale_int:Nn \l__spintent_luaset_part_int_tl { dfraction } }
2439     { \__spintent_luafun_spmQ_scale_int:Nn \l__spintent_luaset_part_int_tl { fraction } }
2440   \int_set_eq:Nc \l__luamml_flag_int { \l__spintent_saved_luamml_flag_int }
2441 }

```

(End of definition for `\__spintent_spmQ_print_scale_int:`.)

`\__spintent_spmQ_print_scale_int:n`

The function `\__spintent_spmQ_print_scale_int:n` takes the integer part of the “mixed number” passed in ‘#1’. It first calls `\__spintent_luafun_clean_split_and_set:n` on ‘#1’ to parse and populate the shared Lua variables, then checks the status of `\l__spintent_luaset_has_integer_str`.

If the status is `true`, it inspects `\l__spintent_luaset_sign_tl`. When the sign is *empty*, it delegates directly to `\__spintent_spmQ_print_scale_int:` to render the scaled integer. When a sign is present, it sets `\l__spintent_spmQ_intent_read_sign_str` to either `_más:prefix($e)` or `_menos:prefix($e)` and wraps the sign glyph together with a MathML argument (`$e`) containing the scaled integer inside a `\MathMLintent`, so that NVDA/MathCAT verbalizes the sign as a prefix before the integer value. An

`\__spintent_invisible_sep`: call between the sign and the `MathMLarg` forces the opening of a new `<mrow>` in the MathML structure, which is required for the intent tree to be well-formed.

If the status is `false`, the `spmQ-not-integer` “error” is raised immediately.

```

2442 \cs_new_protected:Nn \__spintent_spmQ_print_scale_int:n
2443 {
2444   \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
2445   \str_if_eq:VnTF \__spintent_luaset_has_integer_str { true }
2446   {
2447     \tl_if_empty:NTF \__spintent_luaset_sign_tl
2448     {
2449       \__spintent_spmQ_print_scale_int:
2450     }
2451     {
2452       \str_if_eq:VnTF \__spintent_luaset_sign_tl { + }
2453       {
2454         \str_set:Nn
2455           \__spintent_spmQ_intent_read_sign_str { _más:prefix($e) }
2456       }
2457       {
2458         \str_set:Nn
2459           \__spintent_spmQ_intent_read_sign_str { _menos:prefix($e) }
2460       }
2461       \MathMLintent { \__spintent_spmQ_intent_read_sign_str }
2462       {
2463         \__spintent_luaset_sign_tl \__spintent_invisible_sep:
2464         \MathMLarg { e } { \__spintent_spmQ_print_scale_int: }
2465       }
2466     }
2467   }
2468   { \msg_error:nn { spintent } { spmQ-not-integer } }
2469 }

```

(End of definition for `\__spintent_spmQ_print_scale_int:n`.)

`\__spintent_spmQ_print_spmZ_dfrac:nn`

The function `\__spintent_spmQ_print_spmZ_dfrac:nn` typesets the fractional part of the “mixed number”. It takes the numerator in ‘#1’ and the denominator in ‘#2’. It first calls `\__spintent_luafun_clean_split_and_set:` on ‘#1’ to validate that the numerator is an integer; if `\__spintent_luaset_has_integer_str` is `false`, the `spmQ-not-integer` “error” is raised immediately.

If the validation passes, the function dispatches on the two boolean variables `\__spintent_spmQ_use_spmZ_bool` and `\__spintent_spmQ_print_dfrac_bool`, controlled respectively by the keys `use-spmZ` and `print-dfrac`. When `use-spmZ` is `true`, both the numerator and denominator are passed through `\spnZ` to gain MathML integer semantics; otherwise, the raw token lists ‘#1’ and ‘#2’ are used directly. In both branches, the fraction command is selected dynamically via `\use:c`: if `print-dfrac` is `true`, `dfrac` is assembled; otherwise, `frac` is used.

```

2470 \cs_new_protected:Nn \__spintent_spmQ_print_spmZ_dfrac:nn
2471 {
2472   \__spintent_luafun_clean_split_and_set:n { \exp_not:n {#1} }
2473   \str_if_eq:VnTF \__spintent_luaset_has_integer_str { false }
2474   { \msg_error:nn { spintent } { spmQ-not-integer } }
2475   \bool_if:NTF \__spintent_spmQ_use_spmZ_bool
2476   {
2477     \use:c { \bool_if:NT \__spintent_spmQ_print_dfrac_bool { d } frac }
2478     { \spnZ { #1 } } { \spnZ { #2 } }
2479   }
2480   {
2481     \use:c { \bool_if:NT \__spintent_spmQ_print_dfrac_bool { d } frac }
2482     { #1 } { #2 }
2483   }
2484 }

```

(End of definition for `\__spintent_spmQ_print_spmZ_dfrac:nn`.)

`\__spintent_spmQ_print:nnn`

The function `\__spintent_spmQ_print:nnn` is the main rendering orchestrator for `\spmQ`. It takes three arguments: the integer part ‘#1’, the numerator ‘#2’, and the denominator ‘#3’. It opens the optional parentheses via `\__spintent_spmQ_start_wrap_parens:`, then calls `\__spintent_spmQ_print_scale_int:n` to scale and render the integer part. Next, it injects the grammatical connector between the integer and the fraction via `\__spintent_spmQ_set_join_word:`. Then it calls `\__spintent_spmQ_print_spmZ_dfrac:nn` to typeset the fraction using the keys `print-dfrac` and `use-spmZ`. Finally, it closes the optional parentheses via `\__spintent_spmQ_stop_wrap_parens:`.

```

2485 \cs_new_protected:Nn \__spintent_spmQ_print:nnn

```

```

2486 {
2487   \__spintent_spmQ_start_wrap_parens:
2488   \__spintent_spmQ_print_scale_int:n { #1 }
2489   \__spintent_spmQ_set_join_word:
2490   \__spintent_spmQ_print_spnZ_dfrac:nn { #2 } { #3 }
2491   \__spintent_spmQ_stop_wrap_parens:
2492 }

```

(End of definition for __spintent_spmQ_print:nnn.)

__spintent_spmQ_exec:nnnn

The internal function __spintent_spmQ_exec:nnnn takes four arguments: the optional *(keys)* ‘#1’, the integer ‘#2’, the numerator ‘#3’, and the denominator ‘#4’. First sets the optional *(keys)* from ‘#1’, then processes the denominator input in ‘#4’ using __spintent_luafun_clean_split_and_set:n to validate it is a non-zero integer.

It checks the status of __spintent_luaset_has_integer_str. If **true** and the integer part is exactly zero, it issues the spmQ-zero-denominator “error”. Otherwise, whether the status is **false** or the integer part is *non-zero*, it calls __spintent_spmQ_print:nnn to format the “mixed number”.

```

2493 \cs_new_protected:Nn \__spintent_spmQ_exec:nnnn
2494 {
2495   \keys_set:nn { spintent / spmQ } { #1 }
2496   \__spintent_luafun_clean_split_and_set:n { \exp_not:n { #4 } }
2497   \str_if_eq:VnTF \__spintent_luaset_has_integer_str { true }
2498   {
2499     \int_compare:nNnTF { \__spintent_luaset_part_int_tl } = { 0 }
2500     { \msg_error:nn { spintent } { spmQ-zero-denominator } }
2501     { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
2502   }
2503   { \__spintent_spmQ_print:nnn { #2 } { #3 } { #4 } }
2504 }

```

(End of definition for __spintent_spmQ_exec:nnnn.)

\spmQ

The command \spmQ is the public interface for formatting “mixed numbers”. It first checks if it is being used in *math mode*, issuing an “error” if not. Inside a local group, it calls __spintent_spmQ_exec:nnnn to process the optional *(keys)* and the *{(arguments)}*.

```

2505 \NewDocumentCommand \spmQ { O{} m m m }
2506 {
2507   \mode_if_math:TF
2508   {
2509     \group_begin:
2510     \__spintent_spmQ_exec:nnnn { #1 } { #2 } { #3 } { #4 }
2511     \group_end:
2512   }
2513   { \msg_error:nnn { spintent } { need-math-mode } { \spmQ } }
2514 }

```

(End of definition for \spmQ. This function is documented on page 16.)

11.31 The commands of the geo family

The commands described in this section provide a semantic interface for geometric objects in Spanish mathematical contexts. They all delegate parsing to **spintent.lua**, sharing the variables __spintent_geo_luaset_error_str, __spintent_geo_luaset_intent_str and __spintent_geo_luaset_print_tl, constructing a valid MathCAT intent.

11.31.1 Definition of shared variables

Internal variables shared across all commands of the family to avoid repeated calls into Lua.

```

\__spintent_geo_luaset_error_str
\__spintent_geo_luaset_intent_str
\__spintent_geo_luaset_print_tl
\__spintent_geo_luaset_is_name_str
\__spintent_geo_luaset_poly_sym_str
\__spintent_geo_luaset_intent_pts_str
\__spintent_geo_luaset_angulo_case_str
\__spintent_geo_intent_tl
\__spintent_geo_fig_op_tl
\__spintent_geo_luaset_concept_str
\__spintent_geo_luaset_letra_word_str
\__spintent_geo_luaset_sub_spoken_str
2515 \str_new:N \__spintent_geo_luaset_error_str
2516 \str_new:N \__spintent_geo_luaset_intent_str
2517 \tl_new:N \__spintent_geo_luaset_print_tl
2518 \str_new:N \__spintent_geo_luaset_points_str
2519 \str_new:N \__spintent_geo_luaset_is_name_str
2520 \str_new:N \__spintent_geo_luaset_poly_sym_str
2521 \str_new:N \__spintent_geo_luaset_intent_pts_str
2522 \str_new:N \__spintent_geo_luaset_angulo_case_str
2523 \tl_new:N \__spintent_geo_intent_tl
2524 \tl_new:N \__spintent_geo_fig_op_tl
2525 \str_new:N \__spintent_geo_luaset_concept_str
2526 \str_new:N \__spintent_geo_luaset_letra_word_str
2527 \str_new:N \__spintent_geo_luaset_sub_spoken_str

```

(End of definition for __spintent_geo_luaset_error_str and others.)

11.32 The command `\spPoint`

The user-level command `\spPoint` provides a semantic interface for a single geometric point. It delegates argument validation to the Lua module `spintent.lua`, constructing a valid MathCAT intent.

11.32.1 Definition of variables for `\spPoint`

Internal variables holding the values set by `read-cmd` and `prefix`.

```
\__spintent_spPoint_mode_str
\__spintent_spPoint_prefix_tl
\__spintent_spPoint_intent_arg_tl
\__spintent_spPoint_kv_tl
2528 \str_new:N \__spintent_spPoint_mode_str
2529 \tl_new:N \__spintent_spPoint_prefix_tl
2530 \tl_new:N \__spintent_spPoint_intent_arg_tl
2531 \tl_new:N \__spintent_spPoint_kv_tl
```

(End of definition for `\__spintent_spPoint_mode_str` and others.)

11.32.2 Definition of error messages for `\spPoint`

The “error” `spPoint-invalid-point` is raised when the argument passed in ‘`#1`’ is not a single uppercase letter with an optional subscript or prime modifier.

```
2532 \msg_new:nnnn { spintent } { spPoint-invalid-point }
2533 {
2534   Invalid~point~format~in~\token_to_str:N \spPoint::~'#1'.
2535 }
2536 {
2537   The~argument~must~be~a~single~uppercase~letter~with~an~optional~
2538   subscript~(\_{...})~or~prime~(').\
2539 }
```

(End of definition for `spPoint-invalid-point`.)

11.32.3 Definition of keys for `\spPoint`

We define the *(keys)* `read-cmd`, `prefix`, `silent-cmd` and `concept`.

```
read-cmd
prefix
silent-cmd
unknown
2540 \keys_define:nn { spintent / spPoint }
2541 {
2542   read-cmd .choices:nn = { school , mathcat }
2543   {
2544     \int_case:nn { \l_keys_choice_int }
2545     {
2546       {1} { \str_set:Nn \__spintent_spPoint_mode_str { school } }
2547       {2} { \str_set:Nn \__spintent_spPoint_mode_str { mathcat } }
2548     }
2549   },
2550   read-cmd .initial:n = school,
2551   read-cmd .value_required:n = true,
2552   read-cmd / unknown .code:n =
2553     \msg_error:nneee { spintent } { unknown-choice }
2554     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
2555   prefix .code:n =
2556     { \__spintent_sanitizize_affix:Nn \__spintent_spPoint_prefix_tl {#1} },
2557   prefix .initial:n = ,
2558   prefix .value_required:n = true,
2559   silent-cmd .bool_set:N = \__spintent_spPoint_silent_cmd_bool,
2560   silent-cmd .initial:n = false,
2561   silent-cmd .value_required:n = true,
2562   concept .bool_set:N = \__spintent_spPoint_concept_bool,
2563   concept .initial:n = true,
2564   concept .value_required:n = true,
2565   unknown .code:n =
2566     {
2567       \str_set:Nn \__spintent_command_name_str { spPoint }
2568       \__spintent_unknown_keys_cmd:n {#1}
2569     },
2570 }
```

(End of definition for `read-cmd` and others.)

11.32.4 Internal functions for `\spPoint`

The internal function `\__spintent_spPoint_print:n` process the *{(argument)}* passed in ‘`#1`’. First, call the function `\__spintent_luafun_geo_parse_and_set:n` to process the *{(argument)}*, returning an “error” if the input is invalid. Then, if `silent-cmd` is `true` or `concept` is `false`, set `\__spintent_spPoint_intent_arg` to `:silent($pt)`; otherwise set it to `punto:prefix($pt)`, prefixed with `prefix` when given. Finally, call `\__spintent_spPoint_mathml:`.

```
2571 \cs_new_protected:Nn \__spintent_spPoint_print:n
```

```

2572 {
2573   \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#1} }
2574   \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
2575   {
2576     \msg_error:nne { spintent } { spPoint-invalid-point } { \exp_not:n {#1} }
2577   }
2578   \bool_if:nTF
2579   {
2580     \bool_lazy_or_p:nn
2581     { \l__spintent_spPoint_silent_cmd_bool }
2582     { ! \l__spintent_spPoint_concept_bool }
2583   }
2584   {
2585     \tl_set:Nn \l__spintent_spPoint_intent_arg_tl { :silent($pt) }
2586   }
2587   {
2588     \tl_set:Ne \l__spintent_spPoint_intent_arg_tl
2589     {
2590       \tl_if_empty:NF \l__spintent_spPoint_prefix_tl
2591       {
2592         \tl_use:N \l__spintent_spPoint_prefix_tl -
2593       }
2594       punto:prefix($pt)
2595     }
2596   }
2597   \__spintent_spPoint_mathml:
2598 }

```

(End of definition for __spintent_spPoint_print:n.)

__spintent_spPoint_mathml: The function __spintent_spPoint_mathml: generates the <mrow> and prints \l__spintent_geo_luaset_print __spintent_invisible_sep: is called first (luamml issue #26).

The kv string is arg="pt", intent="-" when silent-cmd is true; otherwise, for read-cmd=school it is arg="pt", intent="_body-" (literal spelling); for read-cmd=mathcat it is arg="pt" alone. A single call to __spintent_luafun_wrap_mrow_intent_arg:nn applies it, with \l__spintent_spPont_intent_arg_tl as the concept intent.

```

2599 \cs_new_protected:Nn \__spintent_spPoint_mathml:
2600 {
2601   \__spintent_invisible_sep:
2602   \bool_if:NTF \l__spintent_spPoint_silent_cmd_bool
2603   {
2604     \tl_set:Nn \l__spintent_spPoint_kv_tl { arg = "pt", intent = "-" }
2605   }
2606   {
2607     \str_if_eq:VnTF \l__spintent_spPoint_mode_str { school }
2608     {
2609       \tl_set:Ne \l__spintent_spPoint_kv_tl
2610       { arg = "pt", intent = "_ \l__spintent_geo_luaset_intent_str -" }
2611     }
2612     {
2613       \tl_set:Nn \l__spintent_spPoint_kv_tl { arg = "pt" }
2614     }
2615   }
2616   \__spintent_luafun_wrap_mrow_intent_arg:nn
2617   { \l__spintent_spPoint_intent_arg_tl }
2618   { \l__spintent_spPoint_kv_tl }
2619   { \l__spintent_geo_luaset_print_tl }
2620 }

```

(End of definition for __spintent_spPoint_mathml:.)

__spintent_spPoint_exec:nn The function __spintent_spPoint_exec:nn receives the optional <keys> passed in ‘#1’ and the {<argument>} passed in ‘#2’ delegating the rendering of the latter to __spintent_spPoint_print:n.

```

2621 \cs_new_protected:Nn \__spintent_spPoint_exec:nn
2622 {
2623   \keys_set:nn { spintent / spPoint } { #1 }
2624   \__spintent_spPoint_print:n { #2 }
2625 }

```

(End of definition for __spintent_spPoint_exec:nn.)

`\spPoint` The public command `\spPoint` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spPoint_exec:nn`.

```

2626 \NewDocumentCommand \spPoint { 0{} m }
2627 {
2628   \mode_if_math:TF
2629   {
2630     \group_begin:
2631     \__spintent_spPoint_exec:nn { #1 } { #2 }
2632     \group_end:
2633   }
2634   { \msg_error:nnn { spintent } { need-math-mode } { \spPoint } }
2635 }

```

(End of definition for `\spPoint`. This function is documented on page 17.)

`\__spintent_mathmlarg_point:nnn` Annotates one point ([#3](#)) with `arg` and, when applicable, a literal `intent`, without wrapping it in its own `<mrow>`. [#1](#) is the reference name (`arg="#1"`); [#2](#) is the calling command's name, used to look up its own `silent-cmd/read-cmd` variables. Same kv logic as `\__spintent_spPoint_mathml:`. Concept `intent` is passed empty on purpose: the caller applies it once, externally, via a native `\MathMLintent`.

```

2636 \tl_new:N \__spintent_mathmlarg_point_ref_tl
2637 \str_new:N \__spintent_mathmlarg_point_mode_str
2638 \tl_new:N \__spintent_mathmlarg_point_kv_tl
2639 \bool_new:N \__spintent_mathmlarg_point_silent_bool
2640 \cs_new_protected:Nn \__spintent_mathmlarg_point:nnn
2641 {
2642   % referencia del intent ($ref)
2643   \tl_set:Nn \__spintent_mathmlarg_point_ref_tl {#1}
2644   \__spintent_luafun_geo_point_parse_and_set:n { \exp_not:n {#3} }
2645   % Ajustamos la lectura acorde al comando
2646   \bool_set_eq:Nc \__spintent_mathmlarg_point_silent_bool { \__spintent_#2_silent_cmd_bool }
2647   \str_set_eq:Nc \__spintent_mathmlarg_point_mode_str { \__spintent_#2_mode_str }
2648   \__spintent_invisible_sep:
2649   \bool_if:NTF \__spintent_mathmlarg_point_silent_bool
2650   {
2651     \tl_set:Ne \__spintent_mathmlarg_point_kv_tl
2652     { arg = "\__spintent_mathmlarg_point_ref_tl", intent = "-" }
2653   }
2654   {
2655     \str_if_eq:NnTF \__spintent_mathmlarg_point_mode_str { school }
2656     {
2657       \tl_set:Ne \__spintent_mathmlarg_point_kv_tl
2658       { arg = "\__spintent_mathmlarg_point_ref_tl", intent = "_ \__spintent_geo_luaaset_" }
2659     }
2660     {
2661       \tl_set:Ne \__spintent_mathmlarg_point_kv_tl { arg = "\__spintent_mathmlarg_point_ref_tl" }
2662     }
2663   }
2664   \__spintent_luafun_wrap_mrow_intent_arg:nn
2665   { } % debe ir vacío
2666   { \__spintent_mathmlarg_point_kv_tl }
2667   { \__spintent_geo_luaaset_print_tl }
2668 }

```

(End of definition for `\__spintent_mathmlarg_point:nnn`.)

11.33 The command `\sprecta`

The user command `\sprecta` supports two types of notation for the input `{<argument>}`: when the argument contains two points (separated by an optional comma), `\overleftarrow{\rightarrow}` is applied; when the argument is a single point (without comma) or a lowercase letter, it is formatted without any visual decorator. Both notations are automatically detected by the Lua module.

11.33.1 Definition of variables for `\sprecta`

The variable `\__spintent_sprecta_mode_str` store the values set by the key `read-cmd` and the variable `\__spintent_sprayo_prefix_tl` store the value set by the key `prefix`.

```

2669 \str_new:N \__spintent_sprecta_mode_str
2670 \tl_new:N \__spintent_sprecta_prefix_tl
2671 \seq_new:N \__spintent_sprecta_pt_seq
2672 \tl_new:N \__spintent_sprecta_refs_tl
2673 \tl_new:N \__spintent_sprecta_intent_arg_tl
2674 \int_new:N \__spintent_sprecta_pt_idx_int

```

(End of definition for `\__spintent_sprecta_mode_str` and others.)

11.33.2 Definition of error messages for \sprecta

sprecta-invalid-points

The “error” sprecta-invalid-points is raised when the argument matches neither the two-point notation nor the name notation.

```
2675 \msg_new:nnnn { spintent } { sprecta-invalid-points }
2676 {
2677   Invalid~format~in~\token_to_str:N \sprecta:~'#1'~\msg_line_context:.
2678 }
2679 {
2680   See~the~documentation~of~\token_to_str:N \sprecta~for~the~accepted~argument~forms.
2681 }
```

(End of definition for sprecta-invalid-points.)

11.33.3 Definition of keys for \sprecta

read-cmd

We define the *(keys)* read-cmd, prefix and silent-cmd.

prefix

```
2682 \keys_define:nn { spintent / sprecta }
```

silent-cmd

```
2683 {
```

unknown

```
2684   read-cmd               .choices:n = { school , mathcat }
2685   {
2686     \int_case:nn { \l_keys_choice_int }
2687     {
2688       {1} { \str_set:Nn \l__spintent_sprecta_mode_str { school } }
2689       {2} { \str_set:Nn \l__spintent_sprecta_mode_str { mathcat } }
2690     }
2691   },
2692   read-cmd               .initial:n = school,
2693   read-cmd               .value_required:n = true,
2694   read-cmd / unknown     .code:n =
2695     \msg_error:nneee { spintent } { unknown-choice }
2696     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
2697   prefix                  .code:n =
2698     { \__spintent_sanitize_affix:Nn \l__spintent_sprecta_prefix_tl {#1} },
2699   prefix                  .initial:n = ,
2700   prefix                  .value_required:n = true,
2701   silent-cmd              .bool_set:N = \l__spintent_sprecta_silent_cmd_bool,
2702   silent-cmd              .initial:n = false,
2703   silent-cmd              .value_required:n = true,
2704   concept                 .bool_set:N = \l__spintent_sprecta_concept_bool,
2705   concept                 .initial:n = true,
2706   concept                 .value_required:n = true,
2707   unknown                 .code:n =
2708     {
2709       \str_set:Nn \l__spintent_command_name_str { sprecta }
2710       \__spintent_unknown_keys_cmd:n {#1}
2711     },
2712 }
```

(End of definition for read-cmd and others.)

11.33.4 Internal functions for \sprecta

__spintent_sprecta_print:n

The internal function __spintent_sprecta_print:n process the *{(argument)}* passed in ‘#1’. It calls __spintent_luafun_geo_parse_and_set:n to process the *{(argument)}*, returning an “error” if the input is invalid, then calls __spintent_sprecta_mathml:. Unlike \spPoint, no *intent* is built here: both branches of __spintent_sprecta_mathml: build their own, since a proper name (m, l, ...) and a list of points need different amounts of it (a single \$pt versus \$pt1, \$pt2, \ldots).

```
2713 \cs_new_protected:Nn \__spintent_sprecta_print:n
2714 {
2715   \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
2716   \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
2717   {
2718     \msg_error:nne { spintent } { sprecta-invalid-points } { \exp_not:n {#1} }
2719   }
2720   \__spintent_sprecta_mathml:
2721 }
```

(End of definition for __spintent_sprecta_print:n.)

__spintent_sprecta_mathml:

The function __spintent_sprecta_mathml: branches on \l__spintent_geo_luaset_is_name_str. For a proper name (m, l, ...), the logic mirrors __spintent_spPoint_mathml: exactly: the concept *intent* is :silent(\$pt) when silent-cmd is true or concept is false; otherwise it is recta:prefix(\$pt),

prefixed with `prefix` when given. The kv string, however, depends *only* on `silent-cmd` — `concept=false` alone must not silence the letter itself: `arg="pt", intent="-"` when silent; otherwise `arg="pt"` plus a literal `intent` for `read-cmd=school`, or `arg="pt"` alone for `read-cmd=mathcat`. A single call to `\__spintent_luafun_wrap_mrow_intent_arg:nn` applies both.

For a list of points, `\__spintent_geo_luaset_points_str` (set by Lua, one raw point per element, unmerged) is split into `\__spintent_sprecta_pt_seq` and walked once to build `\__spintent_sprecta_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each point by position. The concept `intent` is `_:nofix(refs)` when `silent-cmd` is `true` or `concept` is `false`; otherwise `recta:prefix(refs)`, prefixed as above. It is applied with a native `\MathMLintent` wrapping `\overleftrightharrow`, preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, luamml issue #26). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptn` reference name and `silent-cmd/read-cmd` looked up under `sprecta`: it alone decides `arg/literal intent` per point, so `concept` plays no role at that level.

```

2722 \cs_new_protected:Nn \__spintent_sprecta_mathml:
2723 {
2724   \str_if_eq:VnTF \__spintent_geo_luaset_is_name_str { true }
2725   {
2726     \__spintent_invisible_sep:
2727     \bool_if:nTF
2728     {
2729       \bool_lazy_or_p:nn
2730       { \__spintent_sprecta_silent_cmd_bool }
2731       { ! \__spintent_sprecta_concept_bool }
2732     }
2733     {
2734       \tl_set:Nn \__spintent_sprecta_intent_arg_tl { :silent($pt) }
2735     }
2736     {
2737       \tl_set:Ne \__spintent_sprecta_intent_arg_tl
2738       {
2739         \tl_if_empty:NF \__spintent_sprecta_prefix_tl
2740         { \tl_use:N \__spintent_sprecta_prefix_tl - }
2741         recta:prefix($pt)
2742       }
2743     }
2744     \bool_if:NTF \__spintent_sprecta_silent_cmd_bool
2745     {
2746       \tl_set:Nn \__spintent_sprecta_kv_tl { arg = "pt", intent = "-" }
2747     }
2748     {
2749       \str_if_eq:VnTF \__spintent_sprecta_mode_str { school }
2750       {
2751         \tl_set:Ne \__spintent_sprecta_kv_tl
2752         { arg = "pt", intent = "_ \__spintent_geo_luaset_intent_str -" }
2753       }
2754       {
2755         \tl_set:Nn \__spintent_sprecta_kv_tl { arg = "pt" }
2756       }
2757     }
2758     \__spintent_luafun_wrap_mrow_intent_arg:nn
2759     { \__spintent_sprecta_intent_arg_tl }
2760     { \__spintent_sprecta_kv_tl }
2761     { \__spintent_geo_luaset_print_tl }
2762   }
2763   {
2764     \seq_set_from_clist:NN \__spintent_sprecta_pt_seq \__spintent_geo_luaset_points_str
2765     \tl_clear:N \__spintent_sprecta_refs_tl
2766     \int_zero:N \__spintent_sprecta_pt_idx_int
2767     \seq_map_inline:Nn \__spintent_sprecta_pt_seq
2768     {
2769       \int_incr:N \__spintent_sprecta_pt_idx_int
2770       \tl_if_empty:NF \__spintent_sprecta_refs_tl
2771       { \tl_put_right:Nn \__spintent_sprecta_refs_tl { , } }
2772       \tl_put_right:Ne \__spintent_sprecta_refs_tl { $ pt \int_use:N \__spintent_sprecta_
2773     }
2774     \bool_if:nTF
2775     {
2776       \bool_lazy_or_p:nn
2777       { \__spintent_sprecta_silent_cmd_bool }
2778       { ! \__spintent_sprecta_concept_bool }

```

```

2779     }
2780     {
2781       \tl_set:Nx \l__spintent_sprecta_intent_arg_tl { _-:nofix( \l__spintent_sprecta_refs_tl ) }
2782     }
2783     {
2784       \tl_set:Nx \l__spintent_sprecta_intent_arg_tl
2785       {
2786         \tl_if_empty:NF \l__spintent_sprecta_prefix_tl
2787         { \tl_use:N \l__spintent_sprecta_prefix_tl - }
2788         recta:prefix( \l__spintent_sprecta_refs_tl )
2789       }
2790     }
2791     \__spintent_invisible_sep:
2792     \MathMLintent { \l__spintent_sprecta_intent_arg_tl }
2793     {
2794       \__spintent_luafun_inner_sep:
2795       \overleftrightharrow
2796       {
2797         \int_zero:N \l__spintent_sprecta_pt_idx_int
2798         \seq_map_inline:Nn \l__spintent_sprecta_pt_seq
2799         {
2800           \int_incr:N \l__spintent_sprecta_pt_idx_int
2801           \__spintent_mathmlarg_point:nnn
2802           { pt \int_use:N \l__spintent_sprecta_pt_idx_int }
2803           { sprecta }
2804           { ##1 }
2805         }
2806       }
2807     }
2808   }
2809 }

```

(End of definition for `\__spintent_sprecta_mathml:`)

`\__spintent_sprecta_exec:nn`

The function `\__spintent_sprecta_exec:nn` receives the optional *(keys)* passed in ‘#1’ and the *(argument)* passed in ‘#2’ delegating the rendering of the latter to `\__spintent_sprecta_print:n`.

```

2810 \cs_new_protected:Nn \__spintent_sprecta_exec:nn
2811 {
2812   \keys_set:nn { spintent / sprecta } { #1 }
2813   \__spintent_sprecta_print:n { #2 }
2814 }

```

(End of definition for `\__spintent_sprecta_exec:nn`)

`\sprecta`

The public command `\sprecta` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sprecta_exec:nn`.

```

2815 \NewDocumentCommand \sprecta { 0{} m }
2816 {
2817   \mode_if_math:TF
2818   {
2819     \group_begin:
2820     \__spintent_sprecta_exec:nn { #1 } { #2 }
2821     \group_end:
2822   }
2823   { \msg_error:nnn { spintent } { need-math-mode } { \sprecta } }
2824 }

```

(End of definition for `\sprecta`. This function is documented on page 18.)

11.34 The command `\sprayo`

The user command `\sprayo` provides a semantic interface for a ray. The key `read-sty` selects the verbalized term, *rayo* or *semirrecta*, both rendered identically with `\overrightarrow`.

11.34.1 Definition of variables for `\sprayo`

`\l__spintent_sprayo_mode_str`
`\l__spintent_sprayo_prefix_tl`
`\l__spintent_sprayo_read_sty_str`

The variable `\l__spintent_sprayo_mode_str` store the values set by the key `read-cmd`; the variable `\l__spintent_sprayo_prefix_tl` store the value set by the key `prefix` and the variable `\l__spintent_sprayo_read_sty_str` stores the value set by the key `read-sty`.

🌱 The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```

2825 \str_new:N \l__spintent_sprayo_mode_str
2826 \tl_new:N \l__spintent_sprayo_prefix_tl
2827 \str_new:N \l__spintent_sprayo_read_sty_str

```


(End of definition for `\l__spintent_sprayo_mode_str`, `\l__spintent_sprayo_prefix_tl`, and `\l__spintent_sprayo_read_sty_str`.)

11.34.2 Definition of error messages for `\sprayo`

sprayo-invalid-points

Raised when the argument is not a comma-separated pair of uppercase letters.

```
2828 \msg_new:nnnn { spintent } { sprayo-invalid-points }
2829 {
2830   Invalid~point~format~in~\token_to_str:N \sprayo:~'#1'~\msg_line_context:.
2831 }
2832 {
2833   See~the~documentation~of~\token_to_str:N \sprayo~for~the~accepted~point~formats.
2834 }
```

(End of definition for `sprayo-invalid-points`.)

11.34.3 Definition of keys for `\sprayo`

read-cmd

We define the `(keys)` `read-cmd`, `prefix`, `silent-cmd`, `concept` and `read-sty`.

```
prefix 2835 \keys_define:nn { spintent / sprayo }
silent-cmd 2836 {
read-sty 2837   read-cmd .choices:nn = { school , mathcat }
unknown 2838   {
2839     \int_case:nn { \l_keys_choice_int }
2840     {
2841       {1} { \str_set:Nn \l__spintent_sprayo_mode_str { school } }
2842       {2} { \str_set:Nn \l__spintent_sprayo_mode_str { mathcat } }
2843     }
2844   },
2845   read-cmd .initial:n = school,
2846   read-cmd .value_required:n = true,
2847   read-cmd / unknown .code:n =
2848     \msg_error:nneee { spintent } { unknown-choice }
2849     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
2850   prefix .code:n =
2851     { \__spintent_sanitize_affix:Nn \l__spintent_sprayo_prefix_tl {#1} },
2852   prefix .initial:n = ,
2853   prefix .value_required:n = true,
2854   silent-cmd .bool_set:N = \l__spintent_sprayo_silent_cmd_bool,
2855   silent-cmd .initial:n = false,
2856   silent-cmd .value_required:n = true,
2857   concept .bool_set:N = \l__spintent_sprayo_concept_bool,
2858   concept .initial:n = true,
2859   concept .value_required:n = true,
2860   read-sty .choices:nn = { rayo , semirrecta }
2861   {
2862     \int_case:nn { \l_keys_choice_int }
2863     {
2864       {1} { \str_set:Nn \l__spintent_sprayo_read_sty_str { rayo } }
2865       {2} { \str_set:Nn \l__spintent_sprayo_read_sty_str { semirrecta } }
2866     }
2867   },
2868   read-sty .initial:n = rayo,
2869   read-sty .value_required:n = true,
2870   read-sty / unknown .code:n =
2871     \msg_error:nneee { spintent } { unknown-choice }
2872     { read-sty } { rayo,~semirrecta } { \exp_not:n {#1} },
2873   unknown .code:n =
2874   {
2875     \str_set:Nn \l__spintent_command_name_str { sprayo }
2876     \__spintent_unknown_keys_cmd:n {#1}
2877   },
2878 }
```

(End of definition for `read-cmd` and others.)

11.34.4 Internal functions for `\sprayo`

`\l__spintent_sprayo_pt_seq`
`\l__spintent_sprayo_refs_tl`
`\l__spintent_sprayo_intent_arg_tl`
`\l__spintent_sprayo_pt_idx_int`

Auxiliary variables used by `\__spintent_sprayo_mathml:` to split `\l__spintent_geo_luaset_points_str` and build the `$pts1`, `$pts2`, ... reference list.

```
2879 \seq_new:N \l__spintent_sprayo_pt_seq
2880 \tl_new:N \l__spintent_sprayo_refs_tl
2881 \tl_new:N \l__spintent_sprayo_intent_arg_tl
2882 \int_new:N \l__spintent_sprayo_pt_idx_int
```

(End of definition for `\l__spintent_sprayo_pt_seq` and others.)

`\__spintent_sprayo_print:n` The internal function `\__spintent_sprayo_print:n` process the $\langle argument \rangle$ passed in ‘#1’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process the $\langle argument \rangle$, returning an “error” if the input is invalid, then calls `\__spintent_sprayo_mathml:`. Unlike `\sprecta`, `\sprayo` has no proper-name argument: it always takes a list of points, so no `intent` is built here.

```
2883 \cs_new_protected:Nn \__spintent_sprayo_print:n
2884 {
2885   \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#1} }
2886   \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
2887   {
2888     \msg_error:nne { spintent } { sprayo-invalid-points } { \exp_not:n {#1} }
2889   }
2890   \__spintent_sprayo_mathml:
2891 }
```

(End of definition for `\__spintent_sprayo_print:n`.)

`\__spintent_sprayo_mathml:` The function `\__spintent_sprayo_mathml:` splits `\l__spintent_geo_luaset_points_str` (set by Lua, one raw point per element) into `\l__spintent_sprayo_pt_seq` and walks it once to build `\l__spintent_sprayo_refs_tl` the comma-separated $\$pts1, \$pts2, \dots$ reference list, naming each point by position. The concept `intent` is `_-:nofix(refs)` when `silent-cmd` is `true` or `concept` is `false`; otherwise it is `\l__spintent_sprayo_read_sty_str:prefix(refs)` — the concept word itself (`rayo` or `semirrecta`) comes from the `read-sty` key, not a literal, since `\sprayo` verbalizes under two names — prefixed with `prefix` when given. It is applied with a native `\MathMLintent` wrapping `\overrightarrow`, preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptsn` reference name and `silent-cmd/read-cmd` looked up under `sprayo`: it alone decides arg/literal `intent` per point, so concept plays no role at that level.

```
2892 \cs_new_protected:Nn \__spintent_sprayo_mathml:
2893 {
2894   \seq_set_from_clist:NN \l__spintent_sprayo_pt_seq \l__spintent_geo_luaset_points_str
2895   \tl_clear:N \l__spintent_sprayo_refs_tl
2896   \int_zero:N \l__spintent_sprayo_pt_idx_int
2897   \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
2898   {
2899     \int_incr:N \l__spintent_sprayo_pt_idx_int
2900     \tl_if_empty:NF \l__spintent_sprayo_refs_tl
2901     { \tl_put_right:Nn \l__spintent_sprayo_refs_tl { , } }
2902     \tl_put_right:Ne \l__spintent_sprayo_refs_tl { $ pts \int_use:N \l__spintent_sprayo_pt_idx_int }
2903   }
2904   \bool_if:nTF
2905   {
2906     \bool_lazy_or:pnn
2907     { \l__spintent_sprayo_silent_cmd_bool }
2908     { ! \l__spintent_sprayo_concept_bool }
2909   }
2910   {
2911     \tl_set:Ne \l__spintent_sprayo_intent_arg_tl { _-:nofix( \l__spintent_sprayo_refs_tl ) }
2912   }
2913   {
2914     \tl_set:Ne \l__spintent_sprayo_intent_arg_tl
2915     {
2916       \tl_if_empty:NF \l__spintent_sprayo_prefix_tl
2917       { \tl_use:N \l__spintent_sprayo_prefix_tl - }
2918       \l__spintent_sprayo_read_sty_str :prefix( \l__spintent_sprayo_refs_tl )
2919     }
2920   }
2921   \__spintent_invisible_sep:
2922   \MathMLintent { \l__spintent_sprayo_intent_arg_tl }
2923   {
2924     \__spintent_luafun_inner_sep:
2925     \overrightarrow
2926     {
2927       \int_zero:N \l__spintent_sprayo_pt_idx_int
2928       \seq_map_inline:Nn \l__spintent_sprayo_pt_seq
2929       {
2930         \int_incr:N \l__spintent_sprayo_pt_idx_int
2931         \__spintent_mathmlarg_point:nnn
2932         { pts \int_use:N \l__spintent_sprayo_pt_idx_int }
```

```

2933         { sprayo }
2934         { ##1 }
2935     }
2936 }
2937 }
2938 }

```

(End of definition for `\__spintent_sprayo_mathml:`.)

`\__spintent_sprayo_exec:nn` The function `\__spintent_sprayo_exec:nn` receives the optional $\langle keys \rangle$ passed in ‘#1’ and the $\{ \langle argument \rangle \}$ passed in ‘#2’ delegating the rendering of the latter to `\__spintent_sprayo_print:n`.

```

2939 \cs_new_protected:Nn \__spintent_sprayo_exec:nn
2940 {
2941     \keys_set:nn { spintent / sprayo } { #1 }
2942     \__spintent_sprayo_print:n { #2 }
2943 }

```

(End of definition for `\__spintent_sprayo_exec:nn`.)

`\sprayo` The public command `\sprayo` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sprayo_exec:nn`.

```

2944 \NewDocumentCommand \sprayo { O{} m }
2945 {
2946     \mode_if_math:TF
2947     {
2948         \group_begin:
2949         \__spintent_sprayo_exec:nn { #1 } { #2 }
2950         \group_end:
2951     }
2952     { \msg_error:nnn { spintent } { need-math-mode } { \sprayo } }
2953 }

```

(End of definition for `\sprayo`. This function is documented on page 18.)

11.35 The commands `\spside` and `\spmside`

The user commands `\spside` and `\spmside` provide a semantic interface for a side, segment or trace determined by two points. The key `read-sty` selects the verbalized term — *lado*, *segmento*, or *trazo*, the latter used in straightedge-and-compass constructions — and the key `read-arg` overrides that term entirely with a free-form string. Additionally, the command `\spmside` provides the key `print-m`, controlling whether `\operatorname{m}` is typeset before the argument.

11.35.1 Definition of variables for `\spside` and `\spmside`

`\__spintent_spside_mode_str` Each pair of variables is declared once for both `\spside` and `\spmside` via `\clist_map_inline:nn`. The variables `\__spintent_spside_mode_str` and `\__spintent_spside_mode_str` store the values set by the key `read-cmd`; the variables `\__spintent_spside_prefix_tl` and `\__spintent_spmside_prefix_tl` store the values set by the key `prefix`.

`\__spintent_spside_read_sty_str` The variables `\__spintent_spside_read_sty_str` and `\__spintent_spmside_read_sty_str` stores the value set by the key `read-sty`.

`\__spintent_spside_read_arg_tl` The variables `\__spintent_spside_read_arg_tl` and `\__spintent_spmside_read_arg_tl` stores the value set by the key `read-arg`.

🔗 The variables are defined here as they are assigned using `.choices:nn` or `.code:n` rather than a direct property within `\keys_define:nn`.

```

2954 \clist_map_inline:nn { side , mside }
2955 {
2956     \str_new:c { \__spintent_sp #1 _mode_str }
2957     \tl_new:c { \__spintent_sp #1 _prefix_tl }
2958     \str_new:c { \__spintent_sp #1 _read_sty_str }
2959     \tl_new:c { \__spintent_sp #1 _read_arg_tl }
2960 }

```

(End of definition for `\__spintent_spside_mode_str` and others.)

11.35.2 Definition of error messages for \spside and \spmside

spside-invalid-points

spmside-invalid-points

We define an error message for the commands `\spside` and `\spmside`. The “error” is triggered if the user includes a invalid “*point*” in the $\langle\textit{argument}\rangle$ passed in ‘#1’ to the command.

```

2961 \clist_map_inline:nn { side , mside }
2962 {
2963   \msg_new:nnnn { spintent } { sp #1 -invalid-points }
2964   {
2965     Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
2966   }
2967   {
2968     See~the~documentation~of~##1~for~the~accepted~point~formats.
2969   }
2970 }
```

(End of definition for *spside-invalid-points* and *spmside-invalid-points*.)

11.35.3 Definition of keys for \spside and \spmside

read-cmd

prefix

silent-cmd

concept

read-sty

read-arg

print-m

unknown

We define the shared $\langle\textit{keys}\rangle$ `read-cmd`, `prefix`, `silent-cmd`, `concept`, `read-sty`, `read-arg` and `print-m`.

```

2971 \clist_map_inline:nn { side , mside }
2972 {
2973   \keys_define:nn { spintent / sp #1 }
2974   {
2975     read-cmd      .choices:nn = { school , mathcat }
2976     {
2977       \int_case:nn { \l_keys_choice_int }
2978       {
2979         {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
2980         {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
2981       }
2982     },
2983     read-cmd      .initial:n = school,
2984     read-cmd      .value_required:n = true,
2985     read-cmd / unknown .code:n =
2986       \msg_error:nneee { spintent } { unknown-choice }
2987       { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
2988     prefix        .code:n =
2989       { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
2990     prefix        .initial:n = ,
2991     prefix        .value_required:n = true,
2992     silent-cmd    .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
2993     silent-cmd    .initial:n = false,
2994     silent-cmd    .value_required:n = true,
2995     concept       .bool_set:c = { l__spintent_sp #1 _concept_bool },
2996     concept       .initial:n = true,
2997     concept       .value_required:n = true,
2998     read-sty      .choices:nn = { lado , segmento , trazo }
2999     {
3000       \int_case:nn { \l_keys_choice_int }
3001       {
3002         {1} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { lado } }
3003         {2} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { segmento } }
3004         {3} { \str_set:cn { l__spintent_sp #1 _read_sty_str } { trazo } }
3005       }
3006     },
3007     read-sty      .initial:n = lado,
3008     read-sty      .value_required:n = true,
3009     read-sty / unknown .code:n =
3010       \msg_error:nneee { spintent } { unknown-choice }
3011       { read-sty } { lado,~segmento,~trazo } { \exp_not:n {##1} },
3012     read-arg      .code:n =
3013       {
3014         \__spintent_sanitize_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1}
3015       },
3016     read-arg      .initial:n = ,
3017     read-arg      .value_required:n = true,
3018     print-m       .bool_set:c = { l__spintent_sp #1 _print_m_bool },
3019     print-m       .initial:n = false,
3020     print-m       .value_required:n = true,
3021     unknown       .code:n =
3022     {
3023       \str_set:Nn \l__spintent_command_name_str { sp #1 }

```

```

3024         \__spintent_unknown_keys_cmd:n {##1}
3025     },
3026 }
3027 }

```

• The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spside`.

(End of definition for `read-cmd` and others.)

`\__spintent_spside_intent_tl` `\__spintent_spside_intent_tl` holds the concept word (`read-arg` or `read-sty`). The rest are scratch variables shared by both commands (the function itself is shared) to split `\__spintent_geo_luaset_points_str` and build the `$pts1`, `$pts2`, ... reference list.

```

\__spintent_spside_intent_tl
\__spintent_spside_pt_seq
\__spintent_spside_refs_tl
\__spintent_spside_intent_arg_tl
\__spintent_spside_pt_idx_int
3028 \tl_new:N \__spintent_spside_intent_tl
3029 \seq_new:N \__spintent_spside_pt_seq
3030 \tl_new:N \__spintent_spside_refs_tl
3031 \tl_new:N \__spintent_spside_intent_arg_tl
3032 \int_new:N \__spintent_spside_pt_idx_int

```

(End of definition for `\__spintent_spside_intent_tl` and others.)

`\__spintent_spside_process:nn`

The internal function `\__spintent_spside_process:nn` receives the short name of the command (side or mside) in ‘#1’ and the `{\argument}` in ‘#2’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process ‘#2’, returning an “error” if invalid, then builds `\__spintent_spside_intent_tl`. When `read-arg` is empty, it is set from `read-sty`, prefixed with ‘medida-del-’ for `\spmside`. When `read-arg` is given, it is used as is, and ‘medida-del-’ is prepended only if `concept` is `true`: with `concept=false`, `read-arg` is the verbatim escape hatch for a full user-written phrase (e.g. `read-arg={medida de la hipotenusa}`), gender and article left to the user. No `intent` is built here: it needs the reference list, which `\__spintent_spside_math` builds after splitting the points.

```

3033 \cs_new_protected:Nn \__spintent_spside_process:nn
3034 {
3035     \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
3036     \str_if_eq:VnT \__spintent_geo_luaset_error_str { true }
3037     {
3038         \msg_error:nnee { spintent } { sp #1 -invalid-points }
3039         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
3040     }
3041     \tl_if_empty:cTF { \__spintent_sp #1 _read_arg_tl }
3042     {
3043         \tl_set:Ne \__spintent_spside_intent_tl { \str_use:c { \__spintent_sp #1 _read_sty_str } }
3044         \str_if_eq:eeT {#1} { mside }
3045         {
3046             \tl_put_left:Nn \__spintent_spside_intent_tl { medida-del- }
3047         }
3048     }
3049     {
3050         \tl_set:Ne \__spintent_spside_intent_tl { \tl_use:c { \__spintent_sp #1 _read_arg_tl } }
3051         \bool_if:cT { \__spintent_sp #1 _concept_bool }
3052         {
3053             \str_if_eq:eeT {#1} { mside }
3054             {
3055                 \tl_put_left:Nn \__spintent_spside_intent_tl { medida-del- }
3056             }
3057         }
3058     }
3059     \__spintent_spside_mathml:n { #1 }
3060 }

```

(End of definition for `\__spintent_spside_process:nn`.)

`\__spintent_spside_mathml:n`

The function `\__spintent_spside_mathml:n` receives the short command name in ‘#1’. It splits `\__spintent_geo_luaset_points_str` into `\__spintent_spside_pt_seq` and walks it once to build `\__spintent_spside_refs_tl`, the comma-separated `$pts1`, `$pts2`, ... reference list, naming each point by position.

The concept `intent` is `_(refs)` when `silent-cmd` is `true`; otherwise, if `read-arg` is empty, it is `_(refs)` when `concept` is `false` or `\__spintent_spside_intent_tl:prefix(refs)` when `true`; if `read-arg` is given, it is always `\__spintent_spside_intent_tl:prefix(refs)` — `concept` never silences a user-given `read-arg` (see `\__spintent_spside_process:nn`). The `:prefix(...)` form is prefixed with `prefix` when given. It is applied with a native `\MathMLintent`, preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26).

Inside, for `#1=side` the points are always wrapped in `\overline`; for `#1=mside`, when `print-m` is `true` a silent `\operatorname{m}` precedes the `\overline`-wrapped points; when `false`, the points are typeset

bare, with no `\overline` at all — the recursive lookup in the shared `mathml_filter` callback finds each point either way, whether nested inside `\overline` or sitting directly in the `\MathMLintent` group. In every case the sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `pts` reference name and `silent-cmd/read-cmd` looked up under `sp#1`: it alone decides `arg/literal intent` per point, so `concept` plays no role at that level.

```

3061 \cs_new_protected:Nn \__spintent_spside_mathml:n
3062 {
3063   \seq_set_from_clist:NN \__spintent_spside_pt_seq \__spintent_geo_luaset_points_str
3064   \tl_clear:N \__spintent_spside_refs_tl
3065   \int_zero:N \__spintent_spside_pt_idx_int
3066   \seq_map_inline:Nn \__spintent_spside_pt_seq
3067   {
3068     \int_incr:N \__spintent_spside_pt_idx_int
3069     \tl_if_empty:NF \__spintent_spside_refs_tl
3070     { \tl_put_right:Nn \__spintent_spside_refs_tl { , } }
3071     \tl_put_right:Ne \__spintent_spside_refs_tl { $ pts \int_use:N \__spintent_spside_pt_idx_int }
3072   }
3073   \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
3074   {
3075     \tl_set:Ne \__spintent_spside_intent_arg_tl { _ ( \__spintent_spside_refs_tl ) }
3076   }
3077   {
3078     \tl_if_empty:cTF { \__spintent_sp #1 _read_arg_tl }
3079     {
3080       \bool_if:cTF { \__spintent_sp #1 _concept_bool }
3081       {
3082         \tl_set:Ne \__spintent_spside_intent_arg_tl
3083         {
3084           \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
3085           { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
3086           \__spintent_spside_intent_tl :prefix( \__spintent_spside_refs_tl )
3087         }
3088       }
3089       {
3090         \tl_set:Ne \__spintent_spside_intent_arg_tl { _ ( \__spintent_spside_refs_tl ) }
3091       }
3092     }
3093     {
3094       \tl_set:Ne \__spintent_spside_intent_arg_tl
3095       {
3096         \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
3097         { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
3098         \__spintent_spside_intent_tl :prefix( \__spintent_spside_refs_tl )
3099       }
3100     }
3101   }
3102   \__spintent_invisible_sep: % (luamml issues #26)
3103   \MathMLintent { \__spintent_spside_intent_arg_tl }
3104   {
3105     \__spintent_luafun_inner_sep:
3106     \str_if_eq:eeTF {#1} { side }
3107     {
3108       \overline { \__spintent_spside_points_loop:n {#1} }
3109     }
3110     {
3111       \bool_if:cTF { \__spintent_sp #1 _print_m_bool }
3112       {
3113         \__spintent_luafun_wrap_mrow_intent_arg:nn
3114         { } { intent = "silent" }
3115         { \operatorname { m } } }
3116       \overline { \__spintent_spside_points_loop:n {#1} }
3117     }
3118     {
3119       \__spintent_spside_points_loop:n {#1}
3120     }
3121   }
3122 }
3123 }

```

(End of definition for `\__spintent_spside_mathml:n`.)

`\__spintent_spside_points_loop:n`

Walks `\__spintent_spside_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once

per point. Factored out of `\__spintent_spside_mathml:n` since it is needed identically in three places there (`side`, `mside` with and without `print-m`); it takes the short command name in ‘`#1`’ again since a separate macro does not inherit its caller’s own `#1`.

```

3124 \cs_new_protected:Nn \__spintent_spside_points_loop:n
3125 {
3126   \int_zero:N \l__spintent_spside_pt_idx_int
3127   \seq_map_inline:Nn \l__spintent_spside_pt_seq
3128   {
3129     \int_incr:N \l__spintent_spside_pt_idx_int
3130     \__spintent_mathmlarg_point:nnn
3131     { pts \int_use:N \l__spintent_spside_pt_idx_int }
3132     { sp #1 }
3133     { ##1 }
3134   }
3135 }

```

(End of definition for `\__spintent_spside_points_loop:n`.)

`\__spintent_spside_exec:nnn`

The function `\__spintent_spside_exec:nnn` receives the *short name* of the command passed in ‘`#1`’, the optional *keys* passed in ‘`#2`’ and the *argument* passed in ‘`#3`’, delegating the rendering to `\__spintent_spside_process:nn`.

```

3136 \cs_new_protected:Nn \__spintent_spside_exec:nnn
3137 {
3138   \keys_set:nn { spintent / sp #1 } { #2 }
3139   \__spintent_spside_process:nn { #1 } { #3 }
3140 }

```

(End of definition for `\__spintent_spside_exec:nnn`.)

`\spside`
`\spmside`

Both public commands `\spside` and `\spmside` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spside_exec:nnn`.

```

3141 \NewDocumentCommand \spside { O{} m }
3142 {
3143   \mode_if_math:TF
3144   {
3145     \group_begin:
3146     \__spintent_spside_exec:nnn { side } { #1 } { #2 }
3147     \group_end:
3148   }
3149   { \msg_error:nnn { spintent } { need-math-mode } { \spside } }
3150 }
3151 \NewDocumentCommand \spmside { O{} m }
3152 {
3153   \mode_if_math:TF
3154   {
3155     \group_begin:
3156     \__spintent_spside_exec:nnn { mside } { #1 } { #2 }
3157     \group_end:
3158   }
3159   { \msg_error:nnn { spintent } { need-math-mode } { \spmside } }
3160 }

```

(End of definition for `\spside` and `\spmside`. These functions are documented on page 18.)

11.36 The commands `\spangle` and `\spmangle`

The user commands `\spangle` and `\spmangle`, provide a semantic interface for angles, following the same generic pattern as `\spside/\spmside`. The key `print-m` chooses between the `\angle/\rightanglesqr` pair, typeset together with an explicit `\operatorname{m}`, and the `\measuredangle/\measuredrightangle` pair, which already conveys the notion of measure on its own; the key `recto` independently chooses the right-angle variant of whichever pair is active.

11.36.1 Definition of variables for `\spangle` and `\spmangle`

`\l__spintent_spangle_mode_str`
`\l__spintent_spmangle_mode_str`
`\l__spintent_spangle_prefix_tl`
`\l__spintent_spmangle_prefix_tl`

Each pair of variables is declared once for both `\spangle` and `\spmangle` via `\clist_map_inline:nn`. The variables `\l__spintent_spangle_mode_str` and `\l__spintent_spmangle_mode_str` store the values set by the key `read-cmd`; the variables `\l__spintent_spangle_prefix_tl` and `\l__spintent_spmangle_prefix_tl` store the values set by the key `prefix`.

🌱 The variables defined here are assigned using `.choices:nn` or `.code:n` within `\keys_define:nn`.

```

3161 \clist_map_inline:nn { angle , mangle }
3162 {
3163   \str_new:c { l__spintent_sp #1 _mode_str }

```



```

3164     \tl_new:c { l__spintent_sp #1 _prefix_tl }
3165 }

```

(End of definition for `l__spintent_spangle_mode_str` and others.)

11.36.2 Definition of error messages for `\spangle` and `\spmangle`

`spangle-invalid-arg` We define an “error” message for the commands `\spangle` and `\spmangle`. The “error” is triggered if the user includes a invalid `{⟨argument⟩}` passed in ‘`#1`’ to the command.

```

3166 \clist_map_inline:nn { angle , mangle }
3167 {
3168     \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
3169     {
3170         Invalid~format~in~##1:~'##2'~\msg_line_context:.
3171     }
3172     {
3173         See~the~documentation~of~##1~for~the~accepted~argument~forms.
3174     }
3175 }

```

(End of definition for `spangle-invalid-arg` and `spmangle-invalid-arg`.)

11.36.3 Definition of keys for `\spangle` and `\spmangle`

`read-cmd` We define the shared `⟨keys⟩` `read-cmd`, `prefix`, `recto`, `print-m`, `silent-cmd` and `concept`,
`prefix`
`recto`
`print-m`
`silent-cmd`
`concept`
`unknown`

```

3176 \clist_map_inline:nn { angle , mangle }
3177 {
3178     \keys_define:nn { spintent / sp #1 }
3179     {
3180         read-cmd .choices:nn = { school , mathcat }
3181         {
3182             \int_case:nn { \l_keys_choice_int }
3183             {
3184                 {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
3185                 {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
3186             }
3187         },
3188         read-cmd .initial:n = school,
3189         read-cmd .value_required:n = true,
3190         read-cmd / unknown .code:n =
3191             \msg_error:nnee { spintent } { unknown-choice }
3192             { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
3193         prefix .code:n =
3194             { \__spintent_sanitize_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
3195         prefix .initial:n = ,
3196         prefix .value_required:n = true,
3197         recto .bool_set:c = { l__spintent_sp #1 _recto_bool },
3198         recto .initial:n = false,
3199         recto .value_required:n = true,
3200         print-m .bool_set:c = { l__spintent_sp #1 _print_m_bool },
3201         print-m .initial:n = false,
3202         print-m .value_required:n = true,
3203         silent-cmd .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
3204         silent-cmd .initial:n = false,
3205         silent-cmd .value_required:n = true,
3206         concept .bool_set:c = { l__spintent_sp #1 _concept_bool },
3207         concept .initial:n = true,
3208         concept .value_required:n = true,
3209         unknown .code:n =
3210             {
3211                 \str_set:Nn \__spintent_command_name_str { sp #1 }
3212                 \__spintent_unknown_keys_cmd:n {##1}
3213             },
3214     }
3215 }

```

🔗 The keys `print-m` and `concept` are defined for both commands for simplicity, but has no effect on `\spangle`.

(End of definition for `read-cmd` and others.)

11.36.4 Internal functions for \spangle and \spmangle

__spintent_spangle_sym:n

The internal function `\__spintent_spangle_sym:n` dispatches on `#1`, `print-m` and `recto` to select one of four symbols: `\angle/\rightanglesqr` when `print-m` is true; `\measuredangle/\measuredrightangle` otherwise. When `#1` is `angle`, only `recto` applies, choosing between `\angle` and `\rightangle`.

```

3216 \cs_new_protected:Nn \__spintent_spangle_sym:n
3217 {
3218   \str_if_eq:eeTF {#1} { mangle }
3219   {
3220     \bool_if:cTF { \__spintent_sp #1 _print_m_bool }
3221     {
3222       \bool_if:cTF { \__spintent_sp #1 _recto_bool }
3223       { \__spintent_right_angle_sqr: } { \angle }
3224     }
3225     {
3226       \bool_if:cTF { \__spintent_sp #1 _recto_bool }
3227       { \measuredrightangle } { \measuredangle }
3228     }
3229   }
3230   {
3231     \bool_if:cTF { \__spintent_sp #1 _recto_bool }
3232     { \__spintent_right_angle_sqr: } { \angle }
3233   }
3234 }

```

(End of definition for `\__spintent_spangle_sym:n`.)

\l__spintent_spangle_intent_tl

`\l__spintent_spangle_intent_tl` holds the concept word (ángulo or medida-del-ángulo, ‘-recto’-suffixed when `recto` is true). The rest are scratch variables shared by both commands to split `\l__spintent_geo_lua` and build the reference list — used only in the tres-puntos and un-punto cases (see `\__spintent_spangle_mathml:n`):

\l__spintent_spangle_pt_seq
\l__spintent_spangle_refs_tl
\l__spintent_spangle_intent_arg_tl
\l__spintent_spangle_pt_idx_int
\l__spintent_spangle_print_m_now_bool

```

3235 \tl_new:N \l__spintent_spangle_intent_tl
3236 \seq_new:N \l__spintent_spangle_pt_seq
3237 \tl_new:N \l__spintent_spangle_refs_tl
3238 \tl_new:N \l__spintent_spangle_intent_arg_tl
3239 \int_new:N \l__spintent_spangle_pt_idx_int
3240 \str_new:N \l__spintent_spangle_kv_tl
3241 \bool_new:N \l__spintent_spangle_print_m_now_bool

```

(End of definition for `\l__spintent_spangle_intent_tl` and others.)

__spintent_spangle_process:nn

The internal function `\__spintent_spangle_process:nn` receives the short name of the command (`angle` or `mangle`) in ‘`#1`’ and the `{\argument}` in ‘`#2`’. It calls `\__spintent_luafun_geo_angulo_parse_and_set:n` to process ‘`#2`’, returning an “error” if invalid, then sets `\l__spintent_spangle_intent_tl` to ángulo (or medida-del-ángulo for `\spmangle`), suffixed with ‘-recto’ when `recto` is true. No `intent` is built here: it depends on `\l__spintent_geo_luaaset_angulo_case_str` (tres-puntos, un-punto or nombre), which `\__spintent_spangle_mathml:n` branches on.

```

3242 \cs_new_protected:Nn \__spintent_spangle_process:nn
3243 {
3244   \__spintent_luafun_geo_angulo_parse_and_set:n { \exp_not:n {#2} }
3245   \str_if_eq:VnT \l__spintent_geo_luaaset_error_str { true }
3246   {
3247     \msg_error:nnee { spintent } { sp #1 -invalid-arg }
3248     { \c_backslash_str sp #1 } { \exp_not:n {#2} }
3249   }
3250   \str_if_eq:eeTF {#1} { mangle }
3251   {
3252     \tl_set:Nn \l__spintent_spangle_intent_tl { medida-del-ángulo }
3253   }
3254   {
3255     \tl_set:Nn \l__spintent_spangle_intent_tl { ángulo }
3256   }
3257   \bool_if:cT { \__spintent_sp #1 _recto_bool }
3258   {
3259     \tl_put_right:Nn \l__spintent_spangle_intent_tl { -recto }
3260   }
3261   \__spintent_spangle_mathml:n {#1}
3262 }

```

(End of definition for `\__spintent_spangle_process:nn`.)

`\__spintent_spangle_mathml:n` For tres-puntos/un-punto, a single native `\MathMLintent` wraps the symbol and the points together, with `\l__spintent_spangle_intent_tl:prefix(refs)` — the same shape as `\__spintent_spside_mathml:n`. The earlier split into two sibling `\MathMLintent` calls worked around a MathCAT bug with several explicit `$`-references in `:prefix(...)` that only affects the demo build; the stable NVDA/updated Acrobat Reader combination reads this shape correctly, so the split is no longer needed and both commands now share the same structure.

For `\spmangle` with `print-m true`, a silent `\operatorname{m}` precedes `\__spintent_spangle_sym:n` inside the same `\MathMLintent`, exactly as `\__spintent_spside_mathml:n` does for its own `m`. `nombre` is unchanged (see below): it never references more than one thing, so it was never affected by the bug this simplifies away from.

```

3263 \cs_new_protected:Nn \__spintent_spangle_mathml:n
3264 {
3265   \__spintent_invisible_sep: % (luamml issues #26)
3266   \str_if_eq:VnTF \__spintent_geo_luaset_angulo_case_str { nombre }
3267   {
3268     \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
3269     {
3270       \tl_set:Nn \__spintent_spangle_intent_arg_tl { _- }
3271     }
3272     {
3273       \bool_if:cTF { \__spintent_sp #1 _concept_bool }
3274       {
3275         \tl_set:Nc \__spintent_spangle_intent_arg_tl
3276         {
3277           \tl_if_empty:cF { \__spintent_sp #1 _prefix_tl }
3278           { \tl_use:c { \__spintent_sp #1 _prefix_tl } - }
3279           \__spintent_spangle_intent_tl :prefix
3280         }
3281       }
3282       {
3283         \tl_set:Nn \__spintent_spangle_intent_arg_tl { _- }
3284       }
3285     }
3286     \tl_set:Nc \__spintent_spangle_kv_tl
3287     { intent = "\__spintent_spangle_intent_arg_tl" }
3288     \__spintent_luafun_wrap_mrow_intent_arg:nn
3289     { } { \__spintent_spangle_kv_tl }
3290     { \__spintent_spangle_sym:n {#1} }
3291     \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
3292     {
3293       \__spintent_luafun_wrap_mrow_intent_arg:nn { } { intent = "_- " }
3294       { \__spintent_geo_luaset_print_tl }
3295     }
3296     {
3297       \str_if_eq:VnTF \__spintent_geo_luaset_nombre_es_num_str { true }
3298       {
3299         \tl_set:Nc \__spintent_spangle_kv_tl
3300         { intent = "_ \__spintent_geo_luaset_intent_str -" }
3301         \__spintent_luafun_wrap_mrow_intent_arg:nn
3302         { } { \__spintent_spangle_kv_tl }
3303         { \__spintent_geo_luaset_print_tl }
3304       }
3305       { \__spintent_geo_luaset_print_tl }
3306     }
3307   }
3308   {
3309     \seq_set_from_clist:NN \__spintent_spangle_pt_seq \__spintent_geo_luaset_points_str
3310     \tl_clear:N \__spintent_spangle_refs_tl
3311     \int_zero:N \__spintent_spangle_pt_idx_int
3312     \seq_map_inline:Nn \__spintent_spangle_pt_seq
3313     {
3314       \int_incr:N \__spintent_spangle_pt_idx_int
3315       \tl_if_empty:NF \__spintent_spangle_refs_tl
3316       { \tl_put_right:Nn \__spintent_spangle_refs_tl { , } }
3317       \tl_put_right:Ne \__spintent_spangle_refs_tl { $ pt \int_use:N \__spintent_spangle_
3318     }
3319     \bool_if:cTF { \__spintent_sp #1 _silent_cmd_bool }
3320     {
3321       \tl_set:Nn \__spintent_spangle_intent_arg_tl { _- }
3322     }

```

```

3323     {
3324       \bool_if:cTF { l__spintent_sp #1 _concept_bool }
3325       {
3326         \tl_set:Nx \l__spintent_spangle_intent_arg_tl
3327         {
3328           \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
3329           { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
3330           \l__spintent_spangle_intent_tl :prefix( \l__spintent_spangle_refs_tl )
3331         }
3332       }
3333       {
3334         \tl_set:Nn \l__spintent_spangle_intent_arg_tl { _- }
3335       }
3336     }
3337     \bool_set_false:N \l__spintent_spangle_print_m_now_bool
3338     \str_if_eq:eeT {#1} { mangle }
3339     {
3340       \bool_if:cT { l__spintent_sp #1 _print_m_bool }
3341       { \bool_set_true:N \l__spintent_spangle_print_m_now_bool }
3342     }
3343     \MathMLIntent { \l__spintent_spangle_intent_arg_tl }
3344     {
3345       \__spintent_luafun_inner_sep:
3346       \bool_if:NT \l__spintent_spangle_print_m_now_bool
3347       {
3348         \__spintent_luafun_wrap_mrow_intent_arg:nn
3349         { } { intent = ":silent" }
3350         { \operatorname { m } }
3351       }
3352       \__spintent_luafun_wrap_mrow_intent_arg:nn
3353       { } { intent = ":silent" }
3354       { \__spintent_spangle_sym:n {#1} }
3355       \int_zero:N \l__spintent_spangle_pt_idx_int
3356       \seq_map_inline:Nn \l__spintent_spangle_pt_seq
3357       {
3358         \int_incr:N \l__spintent_spangle_pt_idx_int
3359         \__spintent_mathmlarg_point:nnn
3360         { pt \int_use:N \l__spintent_spangle_pt_idx_int
3361         { sp#1 }
3362         { ##1 }
3363       }
3364     }
3365   }
3366 }

```

(End of definition for `\__spintent_spangle_mathml:n`.)

`\__spintent_spangle_exec:nnn`

The function `\__spintent_spangle_exec:nnn` receives the *short name* of the command passed in ‘#1’, the optional *keys* passed in ‘#2’ and the *{argument}* passed in ‘#3’, delegating the rendering to `\__spintent_spangle_process:nn`.

```

3367 \cs_new_protected:Nn \__spintent_spangle_exec:nnn
3368 {
3369   \keys_set:nn { spintent / sp #1 } { #2 }
3370   \__spintent_spangle_process:nn { #1 } { #3 }
3371 }

```

(End of definition for `\__spintent_spangle_exec:nnn`.)

`\spangle`
`\spmangle`

Both public commands `\spangle` and `\spmangle` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_spangle_exec:nnn`.

```

3372 \NewDocumentCommand \spangle { 0{} m }
3373 {
3374   \mode_if_math:TF
3375   {
3376     \group_begin:
3377     \__spintent_spangle_exec:nnn { angle } { #1 } { #2 }
3378     \group_end:
3379   }
3380   { \msg_error:nnn { spintent } { need-math-mode } { \spangle } }
3381 }
3382 \NewDocumentCommand \spmangle { 0{} m }

```

```

3383 {
3384   \mode_if_math:TF
3385   {
3386     \group_begin:
3387     \__spintent_spangle_exec:nnn { mangle } { #1 } { #2 }
3388     \group_end:
3389   }
3390   { \msg_error:nnn { spintent } { need-math-mode } { \spmangle } }
3391 }

```

(End of definition for `\spangle` and `\spmangle`. These functions are documented on page 20.)

11.37 The commands `\sparc` and `\spmarc`

The user commands `\sparc` and `\spmarc` provide a semantic interface for arcs. Unlike `\spmside` and `\spmangle`, `\spmarc` has no `print-m` key: `\operatorname{m}` is always present, since $m\widehat{AB}$ is the only standard notation for the measure of an arc.

11.37.1 Definition of variables for `\sparc` and `\spmarc`

Each pair of variables is declared once for both `\sparc` and `\spmarc` via `\clist_map_inline:nn`. The variables `\__spintent_sparc_mode_str` and `\__spintent_spmarc_mode_str` store the values set by the key `read-cmd`; the variables `\__spintent_sparc_prefix_tl` and `\__spintent_spmarc_prefix_tl` store the values set by the key `prefix`.

```

3392 \clist_map_inline:nn { arc , marc }
3393 {
3394   \str_new:c { \__spintent_sp #1 _mode_str }
3395   \tl_new:c { \__spintent_sp #1 _prefix_tl }
3396 }

```

(End of definition for `\__spintent_sparc_mode_str` and others.)

11.37.2 Definition of error messages for `\sparc` and `\spmarc`

We define an “error” message for the commands `\sparc` and `\spmarc`. The “error” is triggered if the user includes a invalid `{⟨argument⟩}` passed in ‘#1’ to the command.

```

3397 \clist_map_inline:nn { arc , marc }
3398 {
3399   \msg_new:nnnn { spintent } { sp #1 -invalid-points }
3400   {
3401     Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
3402   }
3403   {
3404     See~the~documentation~of~##1~for~the~accepted~point~formats.
3405   }
3406 }

```

(End of definition for `sparc-invalid-points` and `spmarc-invalid-points`.)

11.37.3 Definition of keys for `\sparc` and `\spmarc`

We define the shared `⟨keys⟩` `read-cmd`, `prefix`, `silent-cmd` and `concept`.

```

read-cmd 3407 \clist_map_inline:nn { arc , marc }
prefix   3408 {
silent-cmd 3409   \keys_define:nn { spintent / sp #1 }
concept    3410   {
unknown    3411     read-cmd .choices:nn = { school , mathcat }
           3412     {
           3413       \int_case:nn { \l_keys_choice_int }
           3414       {
           3415         {1} { \str_set:cn { \__spintent_sp #1 _mode_str } { school } }
           3416         {2} { \str_set:cn { \__spintent_sp #1 _mode_str } { mathcat } }
           3417       }
           3418     },
           3419     read-cmd .initial:n = school,
           3420     read-cmd .value_required:n = true,
           3421     read-cmd / unknown .code:n =
           3422     \msg_error:nneee { spintent } { unknown-choice }
           3423     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
           3424     prefix .code:n =
           3425     { \__spintent_sanitize_affix:cn { \__spintent_sp #1 _prefix_tl } {##1} },
           3426     prefix .initial:n = ,
           3427     prefix .value_required:n = true,
           3428     silent-cmd .bool_set:c = { \__spintent_sp #1 _silent_cmd_bool },
           3429     silent-cmd .initial:n = false,

```

```

3430         silent-cmd             .value_required:n = true,
3431         concept                 .bool_set:c = { l__spintent_sp #1 _concept_bool },
3432         concept                 .initial:n = true,
3433         concept                 .value_required:n = true,
3434         unknown                 .code:n =
3435         {
3436             \str_set:Nn \l__spintent_command_name_str { sp #1 }
3437             \__spintent_unknown_keys_cmd:n {##1}
3438         },
3439     }
3440 }

```

• The key `print-m` is defined for both commands for simplicity, but has no effect on `\sparc`.

(End of definition for `read-cmd` and others.)

11.37.4 Internal functions for `\sparc` and `\spmarc`

`\__spintent_sparc_wide_arc:n`

The function `\__spintent_sparc_wide_arc:n` acts as a fallback when the mathematical font used in the document does not provide the command `\widearc`. The definition of this function is a direct adaptation of a response given by Clea F. Rees (@cfr) to a query in the chat room of TeX-SX.

```

3441 \cs_new_protected_nopar:Nn \__spintent_sparc_wide_arc:n
3442 {
3443     \Umathaccent 7 \symoperators "023DC \scan_stop: {#1}
3444 }

```

(End of definition for `\__spintent_sparc_wide_arc:n`.)

`\__spintent_sparc_print_arc:n`

The function `\__spintent_sparc_print_arc:n` applies the command `\widearc` to the argument passed to `\sparc` or `\spmarc`. If the mathematical font set in the document does not provide the command `\widearc`, we will use the built-in function `\__spintent_sparc_wide_arc:n`.

```

3445 \cs_new_protected:Nn \__spintent_sparc_print_arc:n
3446 {
3447     \cs_if_exist:NTF \widearc
3448     { \widearc {#1} }
3449     { \__spintent_sparc_wide_arc:n {#1} }
3450 }

```

(End of definition for `\__spintent_sparc_print_arc:n`.)

`\l__spintent_sparc_intent_tl`

`\l__spintent_sparc_intent_tl` holds the concept word (arco or medida-del-arco). The rest are scratch variables to split `\l__spintent_geo_luaset_points_str` and build the `$pt1`, `$pt2`, ... reference list used in the concept `intent`.

```

3451 \tl_new:N \l__spintent_sparc_intent_tl
3452 \seq_new:N \l__spintent_sparc_pt_seq
3453 \tl_new:N \l__spintent_sparc_refs_tl
3454 \tl_new:N \l__spintent_sparc_intent_arg_tl
3455 \int_new:N \l__spintent_sparc_pt_idx_int

```

(End of definition for `\l__spintent_sparc_intent_tl` and others.)

`\__spintent_sparc_process:nn`

The internal function `\__spintent_sparc_process:nn` receives the short name of the command (arc or marc) in ‘#1’ and the `{(argument)}` in ‘#2’. It calls `\__spintent_luafun_geo_parse_and_set:n` to process ‘#2’, returning an “error” if invalid, then sets `\l__spintent_sparc_intent_tl` to arco (or medida-del-arco for `\spmarc`). No `intent` is built here: it needs the reference list, which `\__spintent_sparc_mathl` builds after splitting the points.

```

3456 \cs_new_protected:Nn \__spintent_sparc_process:nn
3457 {
3458     \__spintent_luafun_geo_parse_and_set:n { \exp_not:n {#2} }
3459     \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
3460     {
3461         \msg_error:nnee { spintent } { sp #1 -invalid-points }
3462         { \c_backslash_str sp #1 } { \exp_not:n {#2} }
3463     }
3464     \str_if_eq:eeTF {#1} { marc }
3465     { \tl_set:Nn \l__spintent_sparc_intent_tl { medida-del-arco } }
3466     { \tl_set:Nn \l__spintent_sparc_intent_tl { arco } }
3467     \__spintent_sparc_mathl:n {#1}
3468 }

```

(End of definition for `\__spintent_sparc_process:nn`.)

`\__spintent_sparc_mathml:n`

The function `\__spintent_sparc_mathml:n` receives the short command name in ‘#1’. It splits `\l__spintent_geo_` into `\l__spintent_sparc_pt_seq` and walks it once to build `\l__spintent_sparc_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each point by position. The concept `intent` is `_` when `silent-cmd` is `true` or `concept` is `false`; otherwise `\l__spintent_sparc_intent` prefixed with `prefix` when given. It is applied with a single native `\MathMLintent` wrapping `\__spintent_sparc_print` directly (same shape as `\__spintent_spside_mathml:n`), preceded by `\__spintent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, luamml issue #26). A silent `\operatorname{m}` precedes it for `\spmarc` (always, no `print-m` key here). The sequence is walked a second time, calling `\__spintent_mathmlarg_point:nnn` once per point with its `ptn` reference name, looked up under `sp#1`.

```

3469 \cs_new_protected:Nn \__spintent_sparc_mathml:n
3470 {
3471   \seq_set_from_clist:NN \l__spintent_sparc_pt_seq \l__spintent_geo_lua_set_points_str
3472
3473   \tl_clear:N \l__spintent_sparc_refs_tl
3474   \int_zero:N \l__spintent_sparc_pt_idx_int
3475   \seq_map_inline:Nn \l__spintent_sparc_pt_seq
3476   {
3477     \int_incr:N \l__spintent_sparc_pt_idx_int
3478     \tl_if_empty:NF \l__spintent_sparc_refs_tl
3479     { \tl_put_right:Nn \l__spintent_sparc_refs_tl { , } }
3480     \tl_put_right:Ne \l__spintent_sparc_refs_tl { $ pt \int_use:N \l__spintent_sparc_pt_idx_int }
3481   }
3482
3483   \bool_if:cTF { l__spintent_sp #1 _silent_cmd_bool }
3484   {
3485     \tl_set:Nn \l__spintent_sparc_intent_arg_tl { _- }
3486   }
3487   {
3488     \bool_if:cTF { l__spintent_sp #1 _concept_bool }
3489     {
3490       \tl_set:Ne \l__spintent_sparc_intent_arg_tl
3491       {
3492         \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
3493         { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
3494         \l__spintent_sparc_intent_tl :prefix( \l__spintent_sparc_refs_tl )
3495       }
3496     }
3497     {
3498       \tl_set:Nn \l__spintent_sparc_intent_arg_tl { _- }
3499     }
3500   }
3501
3502   \__spintent_invisible_sep: % need by MathCAT (bug in luamml issues #26)
3503   \MathMLintent { \l__spintent_sparc_intent_arg_tl }
3504   {
3505     \__spintent_luafun_inner_sep:
3506     \str_if_eq:eeT {#1} { marc }
3507     {
3508       \__spintent_luafun_wrap_mrow_intent_arg:nn
3509       { } { intent = "silent" }
3510       { \operatorname { m } } }
3511   }
3512   \__spintent_sparc_print_arc:n
3513   {
3514     \int_zero:N \l__spintent_sparc_pt_idx_int
3515     \seq_map_inline:Nn \l__spintent_sparc_pt_seq
3516     {
3517       \int_incr:N \l__spintent_sparc_pt_idx_int
3518       \__spintent_mathmlarg_point:nnn
3519       { pt \int_use:N \l__spintent_sparc_pt_idx_int }
3520       { sp #1 }
3521       { ##1 }
3522     }
3523   }
3524 }
3525 }
```

(End of definition for `\__spintent_sparc_mathml:n`.)

`\__spintent_sparc_exec:nnn`

The function `\__spintent_sparc_exec:nnn` receives the *short name* of the command passed in ‘#1’,

the optional $\langle keys \rangle$ passed in ‘#2’ and the $\langle argument \rangle$ passed in ‘#3’, delegating the rendering to `\__spintent_sparc_process:nn`.

```

3526 \cs_new_protected:Nn \__spintent_sparc_exec:nnn
3527 {
3528     \keys_set:nn { spintent / sp #1 } { #2 }
3529     \__spintent_sparc_process:nn { #1 } { #3 }
3530 }

```

(End of definition for `\__spintent_sparc_exec:nnn`.)

`\sparc` Both public commands `\sparc` and `\spmarc` enforces *math mode*, opens a local group, and delegates execution to `\__spintent_sparc_exec:nnn`.

```

3531 \NewDocumentCommand \sparc { O{} m }
3532 {
3533     \mode_if_math:TF
3534     {
3535         \group_begin:
3536         \__spintent_sparc_exec:nnn { arc } { #1 } { #2 }
3537         \group_end:
3538     }
3539     { \msg_error:nnn { spintent } { need-math-mode } { \sparc } }
3540 }
3541 \NewDocumentCommand \spmarc { O{} m }
3542 {
3543     \mode_if_math:TF
3544     {
3545         \group_begin:
3546         \__spintent_sparc_exec:nnn { marc } { #1 } { #2 }
3547         \group_end:
3548     }
3549     { \msg_error:nnn { spintent } { need-math-mode } { \spmarc } }
3550 }

```

(End of definition for `\sparc` and `\spmarc`. These functions are documented on page 21.)

11.38 The command `\spPoly`

The user command `\spPoly` typesets a polygon defined by its vertices. The figure type and its visual symbol are inferred from the vertex count: three points produce a triangle, four a square, five or more a generic polygon with no symbol. It never sends a boolean to Lua: `\l__spintent_geo_luaset_intent_str` (the full *intent*, with the figure name) and `\l__spintent_geo_luaset_intent_pts_str` (vertices only) are both returned unconditionally, and `expl3` picks between them according to `print-sym`.

11.38.1 Definition of variables for `\spPoly`

`\l__spintent_spPoly_mode_str` The variable `\l__spintent_spPoly_mode_str` stores the value set by the key `read-cmd`; the variable `\l__spintent_spPoly_prefix_tl` stores the value set by the key `prefix`.

```

3551 \str_new:N \l__spintent_spPoly_mode_str
3552 \tl_new:N \l__spintent_spPoly_prefix_tl

```

(End of definition for `\l__spintent_spPoly_mode_str` and `\l__spintent_spPoly_prefix_tl`.)

11.38.2 Definition of error messages for `\spPoly`

`spPoly-invalid-points` Raised when the argument is not a comma-separated list of at least three uppercase letters.

```

3553 \msg_new:nnnn { spintent } { spPoly-invalid-points }
3554 {
3555     Invalid~point~format~in~\token_to_str:N \spPoly:~'#1'~\msg_line_context:.
3556 }
3557 {
3558     See~the~documentation~of~\token_to_str:N \spPoly~for~the~accepted~point~formats.
3559 }

```

(End of definition for `spPoly-invalid-points`.)

11.38.3 Definition of keys for `\spPoly`

`read-cmd` We define the $\langle keys \rangle$ `read-cmd`, `prefix`, `print-sym`, `silent-cmd` and `concept`.

`prefix`

`print-sym`

`silent-cmd`

`concept`

`unknown`

```

3560 \keys_define:nn { spintent / spPoly }
3561 {
3562     read-cmd .choices:nn = { school , mathcat }
3563     {
3564         \int_case:nn { \l_keys_choice_int }
3565         {
3566             {1} { \str_set:Nn \l__spintent_spPoly_mode_str { school } }

```

```

3567         {2} { \str_set:Nn \l__spintent_spPoly_mode_str { mathcat } }
3568     }
3569 },
3570 read-cmd          .initial:n = school,
3571 read-cmd          .value_required:n = true,
3572 read-cmd / unknown .code:n =
3573     \msg_error:nneee { spintent } { unknown-choice }
3574     { read-cmd } { school,~mathcat } { \exp_not:n {#1} },
3575 prefix           .code:n =
3576     { \__spintent_sanitize_affix:Nn \l__spintent_spPoly_prefix_tl {#1} },
3577 prefix           .initial:n = ,
3578 prefix           .value_required:n = true,
3579 print-sym        .bool_set:N = \l__spintent_spPoly_print_sym_bool,
3580 print-sym        .initial:n = true,
3581 print-sym        .value_required:n = true,
3582 silent-cmd       .bool_set:N = \l__spintent_spPoly_silent_cmd_bool,
3583 silent-cmd       .initial:n = false,
3584 silent-cmd       .value_required:n = true,
3585 concept          .bool_set:N = \l__spintent_spPoly_concept_bool,
3586 concept          .initial:n = true,
3587 concept          .value_required:n = true,
3588 unknown         .code:n =
3589     {
3590         \str_set:Nn \l__spintent_command_name_str { spPoly }
3591         \__spintent_unknown_keys_cmd:n {#1}
3592     },
3593 }

```

(End of definition for `read-cmd` and others.)

11.38.4 Internal functions for `\spPoly`

```

\l__spintent_spPoly_pt_seq
\l__spintent_spPoly_refs_tl
\l__spintent_spPoly_intent_arg_tl
\l__spintent_spPoly_pt_idx_int

```

Scratch variables to split `\l__spintent_geo_luaset_points_str` and build the `$pt1, $pt2, \dots` reference list, and to hold the final concept `intent` once built.

```

3594 \seq_new:N \l__spintent_spPoly_pt_seq
3595 \tl_new:N \l__spintent_spPoly_refs_tl
3596 \tl_new:N \l__spintent_spPoly_intent_arg_tl
3597 \int_new:N \l__spintent_spPoly_pt_idx_int

```

(End of definition for `\l__spintent_spPoly_pt_seq` and others.)

```
\__spintent_spPoly_sym:
```

Unchanged: reads `\l__spintent_geo_luaset_poly_sym_str` and typesets the corresponding symbol.

```

3598 \cs_new_protected:Nn \__spintent_spPoly_sym:
3599 {
3600     \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
3601     {
3602         { triangle } { \triangle }
3603         { square } { \mdlgwhtsquare }
3604     }
3605 }

```

(End of definition for `\__spintent_spPoly_sym:`.)

```
\__spintent_spPoly_build_refs:
```

Splits `\l__spintent_geo_luaset_points_str` into `\l__spintent_spPoly_pt_seq` and walks it once to build `\l__spintent_spPoly_refs_tl`, the comma-separated `$pt1, $pt2, \dots` reference list, naming each vertex by position.

```

3606 \cs_new_protected:Nn \__spintent_spPoly_build_refs:
3607 {
3608     \seq_set_from_clist:NN \l__spintent_spPoly_pt_seq \l__spintent_geo_luaset_points_str
3609     \tl_clear:N \l__spintent_spPoly_refs_tl
3610     \int_zero:N \l__spintent_spPoly_pt_idx_int
3611     \seq_map_inline:Nn \l__spintent_spPoly_pt_seq
3612     {
3613         \int_incr:N \l__spintent_spPoly_pt_idx_int
3614         \tl_if_empty:NF \l__spintent_spPoly_refs_tl
3615         { \tl_put_right:Nn \l__spintent_spPoly_refs_tl { , } }
3616         \tl_put_right:Ne \l__spintent_spPoly_refs_tl { $ pt \int_use:N \l__spintent_spPoly_pt_idx_int }
3617     }
3618 }

```

(End of definition for `\__spintent_spPoly_build_refs:`.)

```

\__spintrent_spPoly_build_concept: Builds \__spintrent_spPoly_intent_arg_tl, independently of print-sym: _(refs) when silent-
cmd is true or concept is false; otherwise \__spintrent_geo_luaset_concept_str:prefix(refs),
prefixed with prefix when given. Must be called after \__spintrent_spPoly_build_refs:.

3619 \cs_new_protected:Nn \__spintrent_spPoly_build_concept:
3620 {
3621   \bool_if:NTF \__spintrent_spPoly_silent_cmd_bool
3622   {
3623     \tl_set:Nc \__spintrent_spPoly_intent_arg_tl { _ ( \__spintrent_spPoly_refs_tl ) }
3624   }
3625   {
3626     \bool_if:NTF \__spintrent_spPoly_concept_bool
3627     {
3628       \tl_set:Nc \__spintrent_spPoly_intent_arg_tl
3629       {
3630         \tl_if_empty:NF \__spintrent_spPoly_prefix_tl
3631         { \tl_use:N \__spintrent_spPoly_prefix_tl - }
3632         \__spintrent_geo_luaset_concept_str :prefix( \__spintrent_spPoly_refs_tl )
3633       }
3634     }
3635     {
3636       \tl_set:Nc \__spintrent_spPoly_intent_arg_tl { _ ( \__spintrent_spPoly_refs_tl ) }
3637     }
3638   }
3639 }

```

(End of definition for `\__spintrent_spPoly_build_concept:`.)

`\__spintrent_spPoly_points_loop:` Walks `\__spintrent_spPoly_pt_seq` a second time, calling `\__spintrent_mathmlarg_point:nnn` once per vertex with its `pt`*n* reference name, looked up under `spPoly`: it alone decides `arg`/literal `intent` per vertex — replacing the single combined `\MathMLarg{pts}{...}` that lost `arg` on any vertex with a subscript or prime (the same `annotate_attribute/sub-sup` limitation already found and worked around for `\spPoint`).

```

3640 \cs_new_protected:Nn \__spintrent_spPoly_points_loop:
3641 {
3642   \int_zero:N \__spintrent_spPoly_pt_idx_int
3643   \seq_map_inline:Nn \__spintrent_spPoly_pt_seq
3644   {
3645     \int_incr:N \__spintrent_spPoly_pt_idx_int
3646     \__spintrent_mathmlarg_point:nnn
3647     { pt \int_use:N \__spintrent_spPoly_pt_idx_int }
3648     { spPoly }
3649     { ##1 }
3650   }
3651 }

```

(End of definition for `\__spintrent_spPoly_points_loop:`.)

`\__spintrent_spPoly_process:n` Calls `\__spintrent_luafun_geo_poly_parse_and_set:n`, reports `spPoly-invalid-points` on failure, then delegates to `\__spintrent_spPoly_mathml:`. No `intent` is built here: it needs the reference list, which `\__spintrent_spPoly_mathml:` builds after splitting the vertices.

```

3652 \cs_new_protected:Nn \__spintrent_spPoly_process:n
3653 {
3654   \__spintrent_luafun_geo_poly_parse_and_set:n { \exp_not:n {#1} }
3655   \str_if_eq:VnT \__spintrent_geo_luaset_error_str { true }
3656   {
3657     \msg_error:nne { spintrent } { spPoly-invalid-points }
3658     { \exp_not:n {#1} }
3659   }
3660   \__spintrent_spPoly_mathml:
3661 }

```

(End of definition for `\__spintrent_spPoly_process:n`.)

`\__spintrent_spPoly_mathml:` Builds the reference list and the concept `intent` (`\__spintrent_spPoly_build_refs:`, `\__spintrent_spPoly_build_concept:`) then wraps everything in a single native `\MathMLintent`, preceded by `\__spintrent_luafun_inner_sep:` (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26). `print-sym` and `concept/silent-cmd` are independent: the former only decides whether the glyph (`\__spintrent_spPoly_sym:`, silenced with `\__spintrent_luafun_wrap_mrow_intent_arg:nn`) is attempted — it is skipped anyway when `\__spintrent_geo_luaset_poly_sym_str` is empty (no symbol defined for that vertex count, e.g. pentagon and above) — and never affects whether the concept word itself is spoken.

```

3662 \cs_new_protected:Nn \__spintrent_spPoly_mathml:

```

```

3663 {
3664   \__spintent_invisible_sep: % (luamml issues #26)
3665   \__spintent_spPoly_build_refs:
3666   \__spintent_spPoly_build_concept:
3667   \MathMLintent { \__spintent_spPoly_intent_arg_tl }
3668   {
3669     \__spintent_luafun_inner_sep:
3670     \bool_if:NT \__spintent_spPoly_print_sym_bool
3671     {
3672       \str_if_empty:NF \__spintent_geo_luaset_poly_sym_str
3673       {
3674         \__spintent_luafun_wrap_mrow_intent_arg:nn
3675         { } { intent = ":silent" }
3676         { \__spintent_spPoly_sym: }
3677       }
3678     }
3679     \__spintent_spPoly_points_loop:
3680   }
3681 }

```

(End of definition for __spintent_spPoly_mathml:.)

__spintent_spPoly_exec:nn The function __spintent_spPoly_exec:nn receives the optional keys in #1 and the argument in #2, delegating the rendering to __spintent_spPoly_process:n.

```

3682 \cs_new_protected:Nn \__spintent_spPoly_exec:nn
3683 {
3684   \keys_set:nn { spintent / spPoly } { #1 }
3685   \__spintent_spPoly_process:n { #2 }
3686 }

```

(End of definition for __spintent_spPoly_exec:nn.)

\spPoly The public command \spPoly enforces *math mode*, opens a local group, and delegates execution to __spintent_spPoly_exec:nn.

```

3687 \NewDocumentCommand \spPoly { 0{ } m }
3688 {
3689   \mode_if_math:TF
3690   {
3691     \group_begin:
3692     \__spintent_spPoly_exec:nn { #1 } { #2 }
3693     \group_end:
3694   }
3695   { \msg_error:nnn { spintent } { need-math-mode } { \spPoly } }
3696 }

```

(End of definition for \spPoly. This function is documented on page 22.)

11.39 The commands \spAfig and \spPfig

The user commands \spAfig and \spPfig typeset the area and the perimeter of a polygonal figure, using the delimited-optional argument syntax d() already established by \spcmd/\spmcm. The key print-sym is never sent to Lua: the symbol type is returned unconditionally in __spintent_geo_luaset_poly_sym_str, and expl3 decides whether to wrap it in \MathMLintent{<:silent>} — the same mechanism already used by \spPoly.

11.39.1 Definition of variables for \spAfig and \spPfig

Scratch variables used by __spintent_spfig_process:nn to assemble the visual and the intent. __spintent_geo_sym_word_tl holds the spoken symbol word (triángulo/cuadrado, or empty). __spintent_geo_final_print_tl holds the final visual placed under \mathcal's subscript. __spintent_geo and __spintent_geo_mathcat_concept_str hold, respectively, the final literal intent and the concept forwarded when read-cmd is mathcat.

```

3697 \tl_new:N \__spintent_geo_sym_word_tl
3698 \tl_new:N \__spintent_geo_final_print_tl
3699 \str_new:N \__spintent_geo_final_intent_str
3700 \str_new:N \__spintent_geo_mathcat_concept_str

```

(End of definition for __spintent_geo_sym_word_tl and others.)

These variables store the value set by each corresponding key: read-cmd, print-sym, prefix and read-arg, respectively. The value set by read-arg overrides the figure name inferred from the vertex count, independently of print-sym.

```

3701 \clist_map_inline:nn { Afig , Pfig }

```

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```

3702 {
3703     \str_new:c { l__spintent_sp #1 _mode_str      }
3704     \str_new:c { l__spintent_sp #1 _print_sym_str }
3705     \tl_new:c  { l__spintent_sp #1 _prefix_tl     }
3706     \tl_new:c  { l__spintent_sp #1 _read_arg_tl   }
3707 }

```

(End of definition for `l__spintent_spAfig_mode_str` and others.)

11.39.2 Definition of error messages for `\spAfig` and `\spPfig`

spAfig-invalid-points All eight messages are generated by the same map, using the same `##1/##2` doubling convention as
 spPfig-invalid-points `\spside/\spmside`. `sp #1-invalid-points` is currently unused by `l__spintent_spfig_process:nn`,
 spAfig-invalid-arg which raises only `sp #1-invalid-arg` for any parse failure, regardless of whether a comma was present.
 spPfig-invalid-arg `sp #1-sym-lettra-conflict` and `sp #1-sym-cardinality-conflict` are raised when `print-sym` is
 spAfig-sym-lettra-conflict forced to a value incompatible with the actual argument.
 spPfig-sym-lettra-conflict

```

3708 \clist_map_inline:nn { Afig , Pfig }
3709 {
3710     \msg_new:nnnn { spintent } { sp #1 -invalid-points }
3711     {
3712         Invalid~point~format~in~##1:~'##2'~\msg_line_context:.
3713     }
3714     {
3715         See~the~documentation~of~##1~for~the~accepted~point~formats.
3716     }
3717     \msg_new:nnnn { spintent } { sp #1 -invalid-arg }
3718     {
3719         Invalid~argument~format~in~##1:~'##2'~\msg_line_context:.
3720     }
3721     {
3722         See~the~documentation~of~##1~for~the~accepted~argument~forms.
3723     }
3724     \msg_new:nnnn { spintent } { sp #1 -sym-lettra-conflict }
3725     {
3726         print-sym~cannot~be~combined~with~a~lettered~argument~in~'##1'~\msg_line_context:.
3727     }
3728     {
3729         See~the~documentation~of~##1~for~details~on~print-sym.
3730     }
3731     \msg_new:nnnn { spintent } { sp #1 -sym-cardinality-conflict }
3732     {
3733         print-sym~does~not~match~the~vertex~count~in~'##1'~\msg_line_context:.
3734     }
3735     {
3736         See~the~documentation~of~##1~for~details~on~print-sym.
3737     }
3738 }

```

(End of definition for `spAfig-invalid-points` and others.)

11.39.3 Definition of keys for `\spAfig` and `\spPfig`

read-cmd The key `print-sym` chooses the visual symbol: `auto` (default) infers it from the vertex count (three points
 prefix give `\triangle`, four give `\mdlgwhtsquare`, five or more give none); `triangle/square` force one, requiring
 print-sym the vertex count to match; `none` omits it, even with vertices. With a bare number/letter argument, `auto` and
 read-arg `none` behave the same (no symbol); `triangle/square` nest the argument under the forced symbol instead,
 read-sub and cannot be combined with a table letter (F/P/B/L). The key `read-arg` overrides the spoken figure name
 silent-cmd for real vertices, independently of `print-sym`. The key `read-sub` controls whether the bare number/letter
 concept argument is read with the word «sub» or as a plain number.
 unknown

```

3739 \clist_map_inline:nn { Afig , Pfig }
3740 {
3741     \keys_define:nn { spintent / sp #1 }
3742     {
3743         read-cmd .choices:nn = { school , mathcat }
3744         {
3745             \int_case:nn { \l_keys_choice_int }
3746             {
3747                 {1} { \str_set:cn { l__spintent_sp #1 _mode_str } { school } }
3748                 {2} { \str_set:cn { l__spintent_sp #1 _mode_str } { mathcat } }
3749             }
3750         },
3751         read-cmd .initial:n = school,
3752         read-cmd .value_required:n = true,

```

```

3753     read-cmd / unknown .code:n =
3754         \msg_error:nneee { spintent } { unknown-choice }
3755         { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
3756     prefix .code:n =
3757         { \__spintent_sanitizé_affix:cn { l__spintent_sp #1 _prefix_tl } {##1} },
3758     prefix .initial:n = ,
3759     prefix .value_required:n = true,
3760     print-sym .choices:nn = { auto , triangle , square , none }
3761     {
3762         \int_case:nn { \l_keys_choice_int }
3763         {
3764             {1} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { auto } }
3765             {2} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { triangle } }
3766             {3} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { square } }
3767             {4} { \str_set:cn { l__spintent_sp #1 _print_sym_str } { none } }
3768         }
3769     },
3770     print-sym .initial:n = auto,
3771     print-sym .value_required:n = true,
3772     print-sym / unknown .code:n =
3773         \msg_error:nneee { spintent } { unknown-choice }
3774         { print-sym } { auto,~triangle,~square,~none } { \exp_not:n {##1} },
3775     read-arg .code:n =
3776         { \__spintent_sanitizé_affix:cn { l__spintent_sp #1 _read_arg_tl } {##1} },
3777     read-arg .initial:n = ,
3778     read-arg .value_required:n = true,
3779     read-sub .bool_set:c = { l__spintent_sp #1 _read_sub_bool },
3780     read-sub .initial:n = false,
3781     read-sub .value_required:n = true,
3782     silent-cmd .bool_set:c = { l__spintent_sp #1 _silent_cmd_bool },
3783     silent-cmd .initial:n = false,
3784     silent-cmd .value_required:n = true,
3785     concept .bool_set:c = { l__spintent_sp #1 _concept_bool },
3786     concept .initial:n = true,
3787     concept .value_required:n = true,
3788     unknown .code:n =
3789     {
3790         \str_set:Nn \l__spintent_command_name_str { sp #1 }
3791         \__spintent_unknown_keys_cmd:n {##1}
3792     },
3793 }
3794 }

```

(End of definition for read-cmd and others.)

11.39.4 Internal functions for \spAfig and \spPfig

__spintent_spfig_sym:

The function __spintent_spfig_sym: reads \l__spintent_geo_print_sym_str and typesets the corresponding symbol: \triangle for the value triangle, \mdlgwhtsquare for the value square.

```

3795 \cs_new_protected:Nn \__spintent_spfig_sym:
3796 {
3797     \str_case:Vn \l__spintent_geo_print_sym_str
3798     {
3799         { triangle } { \triangle }
3800         { square } { \mdlgwhtsquare }
3801     }
3802 }

```

(End of definition for __spintent_spfig_sym:.)

\l__spintent_spfig_op_tl
\l__spintent_spfig_intent_tl
\l__spintent_spfig_symbol_tl
\l__spintent_spfig_pt_seq
\l__spintent_spfig_refs_tl
\l__spintent_spfig_intent_arg_tl
\l__spintent_spfig_pt_idx_int

\l__spintent_spfig_op_tl holds área or perímetro. \l__spintent_spfig_intent_tl holds the concept word for the vertices case (read-arg or the auto-detected figure name). The rest are scratch variables to split \l__spintent_geo_luaset_points_str and build the \$pt1,\$pt2,... reference list.

```

3803 \tl_new:N \l__spintent_spfig_op_tl
3804 \tl_new:N \l__spintent_spfig_intent_tl
3805 \tl_new:N \l__spintent_spfig_desc_tl
3806 \tl_new:N \l__spintent_spfig_symbol_tl
3807 \seq_new:N \l__spintent_spfig_pt_seq
3808 \tl_new:N \l__spintent_spfig_refs_tl
3809 \tl_new:N \l__spintent_spfig_intent_arg_tl
3810 \int_new:N \l__spintent_spfig_pt_idx_int

```

(End of definition for \l__spintent_spfig_op_tl and others.)

`\__spintent_spfing_process:nn` Receives the short command name (Afig or Pfig) in ‘#1’ and the $\{\langle argument \rangle\}$ in ‘#2’. Calls `\__spintent_luafun_geo` or `\__spintent_luafun_geo_pfig_parse_and_set:n`, reports `sp#1-invalid-arg` on failure, sets `\l__spintent_spfing_op_tl` to `área/perímetro`, then delegates to `\__spintent_spfing_mathml:n`.

```

3811 \cs_new_protected:Nn \__spintent_spfing_process:nn
3812 {
3813   \str_if_eq:eeTF {#1} { Afig }
3814   { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
3815   { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
3816   \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
3817   {
3818     \msg_error:nnee { spintent } { sp #1 -invalid-arg }
3819     { \c_backslash_str sp #1 } { \exp_not:n {#2} }
3820   }
3821   \str_if_eq:eeTF {#1} { Afig }
3822   { \tl_set:Nn \l__spintent_spfing_op_tl { área } }
3823   { \tl_set:Nn \l__spintent_spfing_op_tl { perímetro } }
3824   \str_set:Ne \l__spintent_geo_print_sym_str { \str_use:c { \l__spintent_sp #1 _print_sym_str } }
3825   \__spintent_spfing_mathml:n {#1}
3826 }

```

(End of definition for `\__spintent_spfing_process:nn`)

`\__spintent_spfing_build_simple:n` Handles the argument-less, table-letter and bare number/letter cases (no vertices). Builds `\l__spintent_geo_final_print_tl` and `\l__spintent_spfing_desc_tl` (the descriptive part, without `área/perímetro`). `\l__spintent_spfing_intent_tl` is `\l__spintent_spfing_op_tl-\l__spintent_spfing_desc_tl`. `silent-cmd` overrides `\l__spintent_spfing_intent_tl` with `_`; with `silent-cmd=false` and `concept false`, it becomes `\l__spintent_spfing_desc_tl-` (the descriptive part alone, literally, without `área/perímetro`); with `concept true` it is `\l__spintent_spfing_intent_tl`. Preserves the `sym- letra-conflict` check (a table letter cannot combine with a forced `print-sym`). Receives the short command name in ‘#1’.

```

3827 \cs_new_protected:Nn \__spintent_spfing_build_simple:n
3828 {
3829   \tl_if_empty:nTF { \l__spintent_geo_luaset_print_tl }
3830   {
3831     \str_case:VnF \l__spintent_geo_print_sym_str
3832     {
3833       { auto }
3834       {
3835         \tl_clear:N \l__spintent_geo_final_print_tl
3836         \tl_clear:N \l__spintent_spfing_desc_tl
3837       }
3838       { none }
3839       {
3840         \tl_clear:N \l__spintent_geo_final_print_tl
3841         \tl_clear:N \l__spintent_spfing_desc_tl
3842       }
3843     }
3844     {
3845       \tl_set:Nn \l__spintent_geo_final_print_tl { \__spintent_spfing_sym: }
3846       \tl_set:Ne \l__spintent_spfing_desc_tl
3847       {
3848         \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
3849         { triángulo } { cuadrado }
3850       }
3851     }
3852   }
3853   {
3854     \str_if_empty:NTF \l__spintent_geo_luaset_letra_word_str
3855     {
3856       \str_case:VnF \l__spintent_geo_print_sym_str
3857       {
3858         { auto } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_print_tl } }
3859         { none } { \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_print_tl } }
3860       }
3861       {
3862         \tl_set:Nn \l__spintent_geo_final_print_tl
3863         {
3864           \__spintent_spfing_sym:
3865           \c_math_subscript_token { \l__spintent_geo_luaset_print_tl }
3866         }
3867       }
3868       \str_case:VnF \l__spintent_geo_print_sym_str

```



```

3869         {
3870             { auto }
3871             {
3872                 \tl_set:Nc \l__spintent_spfig_desc_tl
3873                 {
3874                     \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
3875                     \l__spintent_geo_luaset_sub_spoken_str
3876                 }
3877             }
3878         { none }
3879         {
3880             \tl_set:Nc \l__spintent_spfig_desc_tl
3881             {
3882                 \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
3883                 \l__spintent_geo_luaset_sub_spoken_str
3884             }
3885         }
3886     }
3887     {
3888         \tl_set:Nc \l__spintent_spfig_desc_tl
3889         {
3890             \str_if_eq:VnTF \l__spintent_geo_print_sym_str { triangle }
3891             { triángulo } { cuadrado }
3892             -
3893             \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
3894             \l__spintent_geo_luaset_sub_spoken_str
3895         }
3896     }
3897 }
3898 {
3899     \str_case:VnF \l__spintent_geo_print_sym_str
3900     {
3901         { auto } { }
3902         { none } { }
3903     }
3904     {
3905         \msg_error:nne { spintent } { sp #1 -sym-letra-conflict }
3906         { \c_backslash_str sp #1 }
3907     }
3908     \tl_set:Nn \l__spintent_geo_final_print_tl { \l__spintent_geo_luaset_print_tl }
3909     \tl_set:Nc \l__spintent_spfig_desc_tl
3910     {
3911         \str_if_empty:NF \l__spintent_geo_luaset_letra_conector_str
3912         { \l__spintent_geo_luaset_letra_conector_str - }
3913         \l__spintent_geo_luaset_letra_word_str
3914         \str_if_empty:NF \l__spintent_geo_luaset_sub_spoken_str
3915         {
3916             -
3917             \bool_if:cT { \l__spintent_sp #1 _read_sub_bool } { sub - }
3918             \l__spintent_geo_luaset_sub_spoken_str
3919         }
3920     }
3921 }
3922 }
3923 \tl_set:Nc \l__spintent_spfig_intent_tl
3924 {
3925     \tl_if_empty:NTF \l__spintent_spfig_desc_tl
3926     { \l__spintent_spfig_op_tl }
3927     { \l__spintent_spfig_op_tl - \l__spintent_spfig_desc_tl }
3928 }
3929 \bool_if:cTF { \l__spintent_sp #1 _silent_cmd_bool }
3930 { \tl_set:Nn \l__spintent_spfig_intent_arg_tl { _- } }
3931 {
3932     \bool_if:cTF { \l__spintent_sp #1 _concept_bool }
3933     { \tl_set:Nc \l__spintent_spfig_intent_arg_tl { \l__spintent_spfig_intent_tl } }
3934     { \tl_set:Nc \l__spintent_spfig_intent_arg_tl { _ \l__spintent_spfig_desc_tl - } }
3935 }
3936 }

```

(End of definition for \l__spintent_spfig_build_simple:n.)

\l__spintent_spfig_build_refs: Splits \l__spintent_geo_luaset_points_str into \l__spintent_spfig_pt_seq and walks it once to

build `\l__spintent_spfig_refs_tl`, the comma-separated `$pt1,$pt2,...` reference list, naming each vertex by position.

```

3937 \cs_new_protected:Nn \l__spintent_spfig_build_refs:
3938 {
3939   \seq_set_from_clist:NN \l__spintent_spfig_pt_seq \l__spintent_geo_luaset_points_str
3940   \tl_clear:N \l__spintent_spfig_refs_tl
3941   \int_zero:N \l__spintent_spfig_pt_idx_int
3942   \seq_map_inline:Nn \l__spintent_spfig_pt_seq
3943   {
3944     \int_incr:N \l__spintent_spfig_pt_idx_int
3945     \tl_if_empty:NF \l__spintent_spfig_refs_tl
3946     { \tl_put_right:Nn \l__spintent_spfig_refs_tl { , } }
3947     \tl_put_right:Ne \l__spintent_spfig_refs_tl { $ pt \int_use:N \l__spintent_spfig_pt_idx_int
3948   }
3949 }
```

(End of definition for `\l__spintent_spfig_build_refs:.`)

`\l__spintent_spfig_build_concept_vertices:n` Handles the vertices case. `read-arg`, when given, overrides the auto-detected figure name; with `concept true` it is still prefixed with `área-del-/perímetro-del-`, with `concept false` it is used verbatim, and never silenced (only `silent-cmd` silences when `read-arg` is given — same convention as `\l__spintent_spside_process:nn`). With `read-arg` empty, `concept` controls silencing as usual. Also determines `\l__spintent_spfig_symbol_tl` (empty when none applies), preserving the sym-cardinality-conflict check (a forced `print-sym` must match the actual vertex count). Receives the short command name in `#1`. Must be called after `\l__spintent_spfig_build_refs:.`

```

3950 \cs_new_protected:Nn \l__spintent_spfig_build_concept_vertices:n
3951 {
3952   \tl_if_empty:cTF { \l__spintent_sp #1_read_arg_tl }
3953   { \tl_set:Ne \l__spintent_spfig_intent_tl { \l__spintent_geo_luaset_concept_str } }
3954   {
3955     \bool_if:cTF { \l__spintent_sp #1_concept_bool }
3956     {
3957       \tl_set:Ne \l__spintent_spfig_intent_tl
3958       {
3959         \l__spintent_spfig_op_tl -del-
3960         \tl_use:c { \l__spintent_sp #1_read_arg_tl }
3961       }
3962     }
3963     {
3964       \tl_set:Ne \l__spintent_spfig_intent_tl { \tl_use:c { \l__spintent_sp #1_read_arg_tl } }
3965     }
3966   }
3967
3968   \str_case:VnF \l__spintent_geo_print_sym_str
3969   {
3970     { auto }
3971     {
3972       \str_case:Vn \l__spintent_geo_luaset_poly_sym_str
3973       {
3974         { triangle } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \triangle } }
3975         { square } { \tl_set:Nn \l__spintent_spfig_symbol_tl { \mdlgwhtsquare } }
3976       }
3977       \str_if_empty:NT \l__spintent_geo_luaset_poly_sym_str
3978       { \tl_clear:N \l__spintent_spfig_symbol_tl }
3979     }
3980     { none }
3981     { \tl_clear:N \l__spintent_spfig_symbol_tl }
3982   }
3983   {
3984     \str_if_eq:VVTF \l__spintent_geo_print_sym_str \l__spintent_geo_luaset_poly_sym_str
3985     { \tl_set:Nn \l__spintent_spfig_symbol_tl { \l__spintent_spfig_sym: } }
3986     {
3987       \msg_error:nne { spintent } { sp #1 -sym-cardinality-conflict }
3988       { \c_backslash_str sp #1 }
3989     }
3990   }
3991
3992   \bool_if:cTF { \l__spintent_sp #1_silent_cmd_bool }
3993   {
3994     \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ ( \l__spintent_spfig_refs_tl ) }
3995   }
```

```

3996 {
3997   \tl_if_empty:cTF { l__spintent_sp #1 _read_arg_tl }
3998   {
3999     \bool_if:cTF { l__spintent_sp #1 _concept_bool }
4000     {
4001       \tl_set:Ne \l__spintent_spfig_intent_arg_tl
4002       {
4003         \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4004         { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4005         \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )
4006       }
4007     }
4008     {
4009       \tl_set:Ne \l__spintent_spfig_intent_arg_tl { _ ( \l__spintent_spfig_refs_tl ) }
4010     }
4011   }
4012   {
4013     \tl_set:Ne \l__spintent_spfig_intent_arg_tl
4014     {
4015       \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4016       { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4017       \l__spintent_spfig_intent_tl :prefix( \l__spintent_spfig_refs_tl )
4018     }
4019   }
4020 }
4021 }

```

(End of definition for `\__spintent_spfig_build_concept_vertices:n`.)

`\__spintent_spfig_points_loop:n` Walks `\__spintent_spfig_pt_seq` a second time, calling `\__spintent_mathmlarg_point:nnn` once per vertex with its `ptn` reference name, looked up under `sp#1`. Receives the short command name in ‘`#1`’.

```

4022 \cs_new_protected:Nn \__spintent_spfig_points_loop:n
4023 {
4024   \int_zero:N \l__spintent_spfig_pt_idx_int
4025   \seq_map_inline:Nn \l__spintent_spfig_pt_seq
4026   {
4027     \int_incr:N \l__spintent_spfig_pt_idx_int
4028     \__spintent_mathmlarg_point:nnn
4029     { pt \int_use:N \l__spintent_spfig_pt_idx_int }
4030     { sp #1 }
4031     { ##1 }
4032   }
4033 }

```

(End of definition for `\__spintent_spfig_points_loop:n`.)

`\__spintent_spfig_mathml:n` Receives the short command name in ‘`#1`’. Branches on whether `\l__spintent_geo_luaset_points_str` is empty.

Without vertices, `\__spintent_spfig_build_simple:n` builds the content; a `\MathMLintent` wraps it only when `concept` is `true` — with `concept=false` the content is typeset directly, with no `intent` at all. With vertices, `\__spintent_spfig_build_refs:` then `\__spintent_spfig_build_concept_vertices:n` build the reference list and the concept `intent`, applied with a single native `\MathMLintent`. Inside, `\__spintent_mathcal:n{⟨A|P⟩}` is silenced with `\__spintent_luafun_wrap_mrow_intent_arg:nn` (it would otherwise be claimed by the queue meant for the symbol/vertices, the same ordering issue already found and fixed for `\spangle`’s own symbol); `\l__spintent_spfig_symbol_tl` is silenced the same way when non-empty; the sequence is then walked by `\__spintent_spfig_points_loop:n`. All of this sits nested inside a subscript (`\c_math_subscript_token`), reached by the shared `mathml_filter` callback’s recursive lookup through a noad’s sub field, not just its nucleus.

`\__spintent_invisible_sep:` precedes both branches (a second sibling so the sole `<mrow>` does not collapse into `<math>`, `luamml` issue #26).

```

4034 \cs_new_protected:Nn \__spintent_spfig_mathml:n
4035 {
4036   \__spintent_invisible_sep: % need by MathCAT (bug in luamml issues #26)
4037   \str_if_empty:NTF \l__spintent_geo_luaset_points_str
4038   {
4039     \__spintent_spfig_build_simple:n {#1}
4040     \MathMLintent { \l__spintent_spfig_intent_arg_tl }
4041     {
4042       \__spintent_luafun_inner_sep:
4043       \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n

```

```

4044         \tl_if_empty:NF \l__spintent_geo_final_print_tl
4045         {
4046             \c_math_subscript_token { \l__spintent_geo_final_print_tl }
4047         }
4048     }
4049 }
4050 {
4051     \__spintent_spfig_build_refs:
4052     \__spintent_spfig_build_concept_vertices:n {#1}
4053     \MathMLintent { \l__spintent_spfig_intent_arg_tl }
4054     {
4055         \__spintent_luafun_inner_sep:
4056         \__spintent_luafun_wrap_mrow_intent_arg:nn
4057         { } { intent = ":silent" }
4058         {
4059             \str_if_eq:eeTF {#1} { Afig } { \__spintent_mathcal:n { A } } { \__spintent_mathcal:n { A } }
4060         }
4061         \c_math_subscript_token
4062         {
4063             \tl_if_empty:NF \l__spintent_spfig_symbol_tl
4064             {
4065                 \__spintent_luafun_wrap_mrow_intent_arg:nn
4066                 { } { intent = ":silent" }
4067                 { \tl_use:N \l__spintent_spfig_symbol_tl }
4068             }
4069             \__spintent_spfig_points_loop:n {#1}
4070         }
4071     }
4072 }
4073 }

```

(End of definition for `\__spintent_spfig_mathml:n`.)

`\__spintent_spfig_exec:nnn`

The function `\__spintent_spfig_exec:nnn` receives the short name of the command in `#1`, the optional keys in `#2`, and the argument in `#3`, delegating the rendering to `\__spintent_spfig_process:nn`.

```

4074 \cs_new_protected:Nn \__spintent_spfig_exec:nnn
4075 {
4076     \keys_set:nn { spintent / sp #1 } { #2 }
4077     \__spintent_spfig_process:nn { #1 } { #3 }
4078 }

```

(End of definition for `\__spintent_spfig_exec:nnn`.)

`\spAfig`

`\spPfig`

Both public commands `\spAfig` and `\spPfig` enforce math mode, open a local group, and delegate execution to `\__spintent_spfig_exec:nnn`.

```

4079 \NewDocumentCommand \spAfig { 0{} d() }
4080 {
4081     \mode_if_math:TF
4082     {
4083         \group_begin:
4084         \tl_if_novalue:nTF {#2}
4085         { \__spintent_spfig_exec:nnn { Afig } { #1 } { } }
4086         { \__spintent_spfig_exec:nnn { Afig } { #1 } { #2 } }
4087         \group_end:
4088     }
4089     { \msg_error:nnn { spintent } { need-math-mode } { \spAfig } }
4090 }
4091 \NewDocumentCommand \spPfig { 0{} d() }
4092 {
4093     \mode_if_math:TF
4094     {
4095         \group_begin:
4096         \tl_if_novalue:nTF {#2}
4097         { \__spintent_spfig_exec:nnn { Pfig } { #1 } { } }
4098         { \__spintent_spfig_exec:nnn { Pfig } { #1 } { #2 } }
4099         \group_end:
4100     }
4101     { \msg_error:nnn { spintent } { need-math-mode } { \spPfig } }
4102 }

```

(End of definition for `\spAfig` and `\spPfig`. These functions are documented on page 22.)

11.40 The command \spcirc

The user command `\spcirc` provides a semantic interface for a circle or a circumference, using the delimited-optional argument syntax `d()` already established by `\spmc`/`\spmcm` and `\spAfig`/`\spPfig`. The argument holds a required center point and an optional radius, separated by a comma. The radius is never required to declare its own kind via a key: its shape alone decides whether it is a pure number, a measure (number + unit), a bare letter, or a two-letter segment — reusing, respectively, `\spnum`, `\spunit`, and `\spside`/`\spmside` directly as sub-commands, rather than reimplementing any of their parsing.

11.40.1 Definition of variables for \spcirc

Shared outputs of `\__spintent_luafun_geo_circ_parse_and_set:n`. `\__spintent_geo_luaset_circ_rad` holds one of `number`, `measure`, `label`, `segment`, or an empty string when no radius was given. `\__spintent_geo_lu` holds the corresponding raw text, ready to be forwarded to the sub-command that renders it.

```
4103 \str_new:N \__spintent_geo_luaset_circ_radio_type_str
4104 \str_new:N \__spintent_geo_luaset_circ_radio_raw_str
```

(End of definition for `\__spintent_geo_luaset_circ_radio_type_str` and `\__spintent_geo_luaset_circ_radio_raw_str`.)

The variable `\__spintent_spcirc_mode_str` stores the value set by the key `read-cmd`; the variable `\__spintent_spcirc_prefix_tl` stores the value set by the key `prefix`. The variable `\__spintent_spcirc_print_sym_str` stores the value set by the key `print-sym` (usual or `odot`). The variable `\__spintent_spcirc_read_sty_str` stores the value set by the key `read-sty` (`circun` or `circle`). The variable `\__spintent_spcirc_radio_sty_str` stores the value set by the key `radio-sty`, only meaningful when the radius is a segment. The variable `\__spintent_spcirc_sub_index_str` stores the value set by the key `sub-index`. The variable `\__spintent_spcirc_articulo_str` is set internally by `\__spintent_spcirc_word_and_article:` to `la` or `el`, matching the grammatical gender of the chosen word.

```
4105 \str_new:N \__spintent_spcirc_mode_str
4106 \tl_new:N \__spintent_spcirc_prefix_tl
4107 \str_new:N \__spintent_spcirc_print_sym_str
4108 \str_new:N \__spintent_spcirc_read_sty_str
4109 \str_new:N \__spintent_spcirc_radio_sty_str
4110 \str_new:N \__spintent_spcirc_sub_index_str
4111 \str_new:N \__spintent_spcirc_articulo_str
4112 \tl_new:N \__spintent_spcirc_read_arg_tl
```

(End of definition for `\__spintent_spcirc_mode_str` and others.)

Scratch variables used by `\__spintent_spcirc_process:n` to assemble the chosen word (`circunferencia/círculo` or the `read-arg` override) and the final literal `intent` string.

```
4113 \tl_new:N \__spintent_geo_circ_word_tl
4114 \str_new:N \__spintent_geo_circ_intent_str
```

(End of definition for `\__spintent_geo_circ_word_tl` and `\__spintent_geo_circ_intent_str`.)

11.40.2 Definition of error messages for \spcirc

`spcirc-invalid-arg` Raised when the argument does not match a single center point, optionally followed by a comma and a valid radius shape.

```
4115 \msg_new:nnnn { spintent } { spcirc-invalid-arg }
4116 {
4117   Invalid~argument~format~in~\token_to_str:N \spcirc:~'#1'~\msg_line_context:.
4118 }
4119 {
4120   See~the~documentation~of~\token_to_str:N \spcirc~for~the~accepted~argument~forms.
4121 }
```

(End of definition for `spcirc-invalid-arg`.)

11.40.3 Definition of keys for \spcirc

The key `print-sym` chooses the visual symbol: `usual` for a plain C, `odot` (default) for `\odot`. The key `read-sty` chooses the verbalized word and its article: `circun` (default) for «*la circunferencia*», `circle` for «*el círculo*». The key `radio-sty` chooses, only when the radius is a two-letter segment, whether it is rendered as the *object* (side, via `\spside`) or its *measure* (mside, default, via `\spmside`). The key `sub-index` sets a subscript on the symbol; the key `read-sub` controls whether it is read with the word «sub» or as a plain number, matching the same convention as `\spAfig`.

```
4122 \keys_define:nn { spintent / spcirc }
4123 {
4124   read-cmd      .choices:nn = { school , mathcat }
4125   spside        {
4126     \int_case:nn { \l_keys_choice_int }
```

```

4127         {
4128             {1} { \str_set:Nn \l__spintent_spcirc_mode_str { school } }
4129             {2} { \str_set:Nn \l__spintent_spcirc_mode_str { mathcat } }
4130         }
4131     },
4132     read-cmd      .initial:n = school,
4133     read-cmd      .value_required:n = true,
4134     prefix        .code:n =
4135     { \__spintent_sanitizize_affix:Nn \l__spintent_spcirc_prefix_tl {#1} },
4136     prefix        .initial:n = ,
4137     prefix        .value_required:n = true,
4138     print-sym     .choices:nn = { usual , odot }
4139     {
4140         \int_case:nn { \l_keys_choice_int }
4141         {
4142             {1} { \str_set:Nn \l__spintent_spcirc_print_sym_str { usual } }
4143             {2} { \str_set:Nn \l__spintent_spcirc_print_sym_str { odot } }
4144         }
4145     },
4146     print-sym     .initial:n = odot,
4147     print-sym     .value_required:n = true,
4148     read-sty      .choices:nn = { circun , circle }
4149     {
4150         \int_case:nn { \l_keys_choice_int }
4151         {
4152             {1} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circun } }
4153             {2} { \str_set:Nn \l__spintent_spcirc_read_sty_str { circle } }
4154         }
4155     },
4156     read-sty      .initial:n = circun,
4157     read-sty      .value_required:n = true,
4158     radio-sty     .choices:nn = { side , mside }
4159     {
4160         \int_case:nn { \l_keys_choice_int }
4161         {
4162             {1} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { side } }
4163             {2} { \str_set:Nn \l__spintent_spcirc_radio_sty_str { mside } }
4164         }
4165     },
4166     radio-sty     .initial:n = mside,
4167     radio-sty     .value_required:n = true,
4168     sub-index     .code:n =
4169     { \str_set:Nn \l__spintent_spcirc_sub_index_str {#1} },
4170     sub-index     .initial:n = ,
4171     sub-index     .value_required:n = true,
4172     read-sub      .bool_set:N = \l__spintent_spcirc_read_sub_bool,
4173     read-sub      .initial:n = false,
4174     read-sub      .value_required:n = true,
4175     read-arg      .code:n =
4176     { \__spintent_sanitizize_affix:Nn \l__spintent_spcirc_read_arg_tl {#1} },
4177     read-arg      .initial:n = ,
4178     read-arg      .value_required:n = true,
4179     silent-cmd    .bool_set:N = \l__spintent_spcirc_silent_cmd_bool,
4180     silent-cmd    .initial:n = false,
4181     silent-cmd    .value_required:n = true,
4182     spside        .meta:nn = { spintent / spside } { #1 },
4183     spside        .value_required:n = true,
4184     spmside       .meta:nn = { spintent / spmside } { #1 },
4185     spmside       .value_required:n = true,
4186     unknown       .code:n =
4187     {
4188         \str_set:Nn \l__spintent_command_name_str { spcirc }
4189         \__spintent_unknown_keys_cmd:n {#1}
4190     },
4191 }

```

(End of definition for read-cmd and others.)

11.40.4 Internal functions for \spcirc

`\__spintent_spcirc_sym:` The function `\__spintent_spcirc_sym:` typesets `\bigodot` or the plain letter C, according to the value set by the key `print-sym`.

4192 `\cs_new_protected:Nn \__spintent_spcirc_sym:`

```

4193 {
4194   \str_if_eq:VnTF \l__spintent_spcirc_print_sym_str { odot }
4195   {
4196     \group_begin:
4197     \textstyle\bigodot
4198     \group_end:
4199   }
4200   { C }
4201 }

```

(End of definition for `\__spintent_spcirc_sym:`)

`\__spintent_spcirc_word_and_article:`

The function `\__spintent_spcirc_word_and_article:` sets `\l__spintent_geo_circ_word_tl` to the chosen word — either the value set by the key `read-arg`, or `circunferencia/círculo` according to the key `read-sty` — and `\l__spintent_spcirc_articulo_str` to the matching article (`la/el`). The article always follows `read-sty`, even when `read-arg` overrides the word itself.

```

4202 \cs_new_protected:Nn \__spintent_spcirc_word_and_article:
4203 {
4204   \tl_if_empty:NTF \l__spintent_spcirc_read_arg_tl
4205   {
4206     \str_if_eq:VnTF \l__spintent_spcirc_read_sty_str { circun }
4207     {
4208       \tl_set:Nn \l__spintent_geo_circ_word_tl { circunferencia }
4209       \str_set:Nn \l__spintent_spcirc_articulo_str { la }
4210     }
4211     {
4212       \tl_set:Nn \l__spintent_geo_circ_word_tl { círculo }
4213       \str_set:Nn \l__spintent_spcirc_articulo_str { el }
4214     }
4215   }
4216   {
4217     \tl_set:Nn \l__spintent_geo_circ_word_tl { \l__spintent_spcirc_read_arg_tl }
4218     \str_if_eq:VnTF \l__spintent_spcirc_read_sty_str { circun }
4219     { \str_set:Nn \l__spintent_spcirc_articulo_str { la } }
4220     { \str_set:Nn \l__spintent_spcirc_articulo_str { el } }
4221   }
4222 }

```

(End of definition for `\__spintent_spcirc_word_and_article:`)

`\__spintent_spcirc_radio_print:`

The function `\__spintent_spcirc_radio_print:` dispatches on `\l__spintent_geo_luaset_circ_radio_type` reusing the actual public sub-commands rather than reimplementing any parsing: `number` and `measure` delegate to `\spnum/\spunit`; `segment` delegates to `\spside/\spmside` according to the key `radio-sty` — both already accept the concatenated two-letter form that Lua hands over here. `label` is typeset as a bare letter, with no special intent wrapping. In every case, `\exp_args:No` is required to expand `\l__spintent_geo_luaset_circ_radio_raw_str` into the sub-command's argument, since a plain brace group does not expand a `str` variable on its own.

```

4223 \cs_new_protected:Nn \__spintent_spcirc_radio_print:
4224 {
4225   \str_case:Vn \l__spintent_geo_luaset_circ_radio_type_str
4226   {
4227     { number } { \exp_args:No \spnum { \l__spintent_geo_luaset_circ_radio_raw_str } }
4228     { measure } { \exp_args:No \spunit { \l__spintent_geo_luaset_circ_radio_raw_str } }
4229     { label } { \l__spintent_geo_luaset_circ_radio_raw_str }
4230     { segment }
4231     {
4232       \str_if_eq:VnTF \l__spintent_spcirc_radio_sty_str { side }
4233       { \exp_args:No \spside { \l__spintent_geo_luaset_circ_radio_raw_str } }
4234       { \exp_args:No \spmside { \l__spintent_geo_luaset_circ_radio_raw_str } }
4235     }
4236   }
4237 }

```

(End of definition for `\__spintent_spcirc_radio_print:`)

`\__spintent_spcirc_process:n`

The function `\__spintent_spcirc_process:n` calls `\__spintent_luafun_geo_circ_parse_and_set:n` to resolve the center and classify the radius, then assembles `\l__spintent_geo_circ_intent_str` — the word, an optional `sub-index` suffix, `de-centro-en-`, the center, and `-y-radius` when a radius is present — and delegates the rendering to `\__spintent_spcirc_mathml:`.

```

4238 \cs_new_protected:Nn \__spintent_spcirc_process:n
4239 {

```



```

4240 \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#1} }
4241 \str_if_eq:VnT \l__spintent_geo_luaset_error_str { true }
4242 {
4243   \msg_error:nne { spintent } { spirc-invalid-arg } { \exp_not:n {#1} }
4244 }
4245 \__spintent_spcirc_word_and_article:
4246 \str_set:Ne \l__spintent_geo_circ_intent_str { \l__spintent_geo_circ_word_tl }
4247 \str_if_empty:NF \l__spintent_spcirc_sub_index_str
4248 {
4249   \str_put_right:Ne \l__spintent_geo_circ_intent_str
4250   {
4251     -
4252     \bool_if:NT \l__spintent_spcirc_read_sub_bool { sub - }
4253     \l__spintent_spcirc_sub_index_str
4254   }
4255 }
4256 \str_put_right:Ne \l__spintent_geo_circ_intent_str
4257 {
4258   -de-centro-en-\l__spintent_geo_luaset_print_tl
4259 }
4260 \str_if_empty:NF \l__spintent_geo_luaset_circ_radio_type_str
4261 {
4262   \str_put_right:Nn \l__spintent_geo_circ_intent_str { -y-radio }
4263 }
4264 \__spintent_spcirc_mathml:
4265 }

```

(End of definition for __spintent_spcirc_process:n.)

__spintent_spcirc_mathml: The function __spintent_spcirc_mathml: reads the state already set by __spintent_spcirc_process:n and renders it. When a radius is present, the visual is split across *two* separate \MathMLintent calls with __spintent_spcirc_radio_print: — which typesets a sub-command carrying its own independent intent — sitting as a *sibling* in between, rather than nested inside either one. Embedding a full sub-command’s own \MathMLintent inside another command’s content argument makes MathCAT read only the outer literal string, the same problem already seen with mover/msub structures; as *siblings*, both are read in sequence as one continuous phrase.

```

4266 \cs_new_protected:Nn \__spintent_spcirc_mathml:
4267 {
4268   \__spintent_invisible_sep: % need by MathCAT (bug in luamml https://github.com/latex3/luamml/issues/26)
4269   \str_if_empty:NTF \l__spintent_geo_luaset_circ_radio_type_str
4270   {
4271     \MathMLintent
4272     {
4273       \tl_if_empty:NF \l__spintent_spcirc_prefix_tl
4274       {
4275         \tl_use:N \l__spintent_spcirc_prefix_tl -
4276       }
4277       \l__spintent_geo_circ_intent_str
4278     }
4279     {
4280       \c__spintent_invisible_sep_tl
4281       \__spintent_spcirc_sym:
4282       \str_if_empty:NF \l__spintent_spcirc_sub_index_str
4283       {
4284         \c_math_subscript_token { \l__spintent_spcirc_sub_index_str }
4285       }
4286       ( \l__spintent_geo_luaset_print_tl )
4287     }
4288   }
4289   {
4290     \MathMLintent
4291     {
4292       \tl_if_empty:NF \l__spintent_spcirc_prefix_tl
4293       {
4294         \tl_use:N \l__spintent_spcirc_prefix_tl -
4295       }
4296       \l__spintent_geo_circ_intent_str
4297     }
4298     {
4299       \c__spintent_invisible_sep_tl
4300       \__spintent_spcirc_sym:

```

```

4301         \str_if_empty:NF \l__spintrent_spcirc_sub_index_str
4302         {
4303             \c_math_subscript_token { \l__spintrent_spcirc_sub_index_str }
4304         }
4305         ( \l__spintrent_geo_luaset_print_tl ,
4306         }
4307         \__spintrent_invisible_sep:
4308         \__spintrent_spcirc_radio_print:
4309         \MathMLIntent { :silent } { ) }
4310     }
4311 }

```

(End of definition for `\__spintrent_spcirc_mathml:`.)

`\__spintrent_spcirc_exec:nn`

The function `\__spintrent_spcirc_exec:nn` receives the optional keys in `#1` and the argument in `#2`, delegating the rendering to `\__spintrent_spcirc_print:n`.

```

4312 \cs_new_protected:Nn \__spintrent_spcirc_exec:nn
4313 {
4314     \keys_set:nn { spintrent / spcirc } { #1 }
4315     \__spintrent_spcirc_process:n { #2 }
4316 }

```

(End of definition for `\__spintrent_spcirc_exec:nn`.)

`\spcirc`

The public command `\spcirc` enforces math mode, opens a local group, and delegates execution to `\__spintrent_spcirc_exec:nn`. It follows the same `d()`-argument pattern as `\spmcdd/\spmcmm` and `\spAfig/\spPfig`; an omitted radius forwards an explicit empty group rather than the `-NoValue-` marker.

```

4317 \NewDocumentCommand \spcirc { 0{} d() }
4318 {
4319     \mode_if_math:TF
4320     {
4321         \group_begin:
4322         \tl_if_novalue:nTF {#2}
4323         { \__spintrent_spcirc_exec:nn { #1 } { } }
4324         { \__spintrent_spcirc_exec:nn { #1 } { #2 } }
4325         \group_end:
4326     }
4327     { \msg_error:nnn { spintrent } { need-math-mode } { \spcirc } }
4328 }

```

(End of definition for `\spcirc`. This function is documented on page ??.)

11.41 The commands `\spAcirc` and `\spPcirc`

The user commands `\spAcirc` and `\spPcirc` typeset the area and the perimeter of a circle or a circumference. The argument follows `\spcirc`'s own point-and-radius grammar for a real center (with an optional radius); when no comma is present and no valid center is found, it falls back to a bare number/letter shortcut, reusing the same Lua entry point as `\spAfig/\spPfig`. A table letter matched by that fallback is rejected, since it has no meaning for a circle.

11.41.1 Definition of variables for `\spAcirc` and `\spPcirc`

These variables store the value set by each corresponding key: `read-cmd`, `print-sym`, `read-sty`, `radio-sty`, `sub-index`, `prefix` and `read-arg`, respectively.

```

\__spintrent_spAcirc_mode_str
\__spintrent_spPcirc_mode_str
\__spintrent_spAcirc_print_sym_str
\__spintrent_spPcirc_print_sym_str
\__spintrent_spAcirc_read_sty_str
\__spintrent_spPcirc_read_sty_str
\__spintrent_spAcirc_radio_sty_str
\__spintrent_spPcirc_radio_sty_str
\__spintrent_spAcirc_sub_index_str
\__spintrent_spPcirc_sub_index_str
\__spintrent_spAcirc_prefix_tl
\__spintrent_spPcirc_prefix_tl
\__spintrent_spAcirc_read_arg_tl
\__spintrent_spPcirc_read_arg_tl
\__spintrent_geo_circfig_mode_str
4329 \clist_map_inline:nn { Acirc , Pcirc }
4330 {
4331     \str_new:c { \__spintrent_sp #1 _mode_str }
4332     \str_new:c { \__spintrent_sp #1 _print_sym_str }
4333     \str_new:c { \__spintrent_sp #1 _read_sty_str }
4334     \str_new:c { \__spintrent_sp #1 _radio_sty_str }
4335     \str_new:c { \__spintrent_sp #1 _sub_index_str }
4336     \tl_new:c { \__spintrent_sp #1 _prefix_tl }
4337     \tl_new:c { \__spintrent_sp #1 _read_arg_tl }
4338 }

```

(End of definition for `\__spintrent_spAcirc_mode_str` and others.)

The variable `\__spintrent_geo_circfig_mode_str` stores whether the argument was resolved as a real circle (`circle`) or as the bare number/letter shortcut (`bare`).

```

4339 \str_new:N \__spintrent_geo_circfig_mode_str

```

(End of definition for `\__spintrent_geo_circfig_mode_str`.)

11.41.2 Definition of error messages for `\spAcirc` and `\spPcirc`

`spAcirc-invalid-arg`
`spPcirc-invalid-arg`

The error message shown when the argument does not match any of the accepted forms: empty, a center point (optionally followed by a comma and a radius), or a bare number/letter.

```

4340 \clist_map_inline:nn { Acirc , Pcirc }
4341 {
4342   \msg_new:nnnn { spintenc } { sp #1 -invalid-arg }
4343   {
4344     Invalid~argument~in~##1:~'##2'~\msg_line_context:.
4345   }
4346   {
4347     See~the~documentation~of~##1~for~the~accepted~argument.
4348   }

```

(End of definition for `spAcirc-invalid-arg` and `spPcirc-invalid-arg`.)

11.41.3 Definition of keys for `\spAcirc` and `\spPcirc`

`read-cmd`
`prefix`
`print-sym`
`read-sty`
`radio-sty`
`sub-index`
`read-sub`
`read-arg`
`silent-cmd`
`spside`
`spmside`
`unknown`

We define the shared `(keys)` `read-cmd`, `prefix`, `print-m`, `read-sty`, `sub-index`, `read-sub`, `read-arg`, and `silent-cmd`.

The keys `spside` and `spmside` are of the type `.meta:nn` passed to the commands `\spside` and `\spmside`

```

4349 \keys_define:nn { spintenc / sp #1 }
4350 {
4351   read-cmd .choices:nn = { school , mathcat }
4352   {
4353     \int_case:nn { \l_keys_choice_int }
4354     {
4355       {1} { \str_set:cn { l__spintenc_sp #1 _mode_str } { school } }
4356       {2} { \str_set:cn { l__spintenc_sp #1 _mode_str } { mathcat } }
4357     }
4358   },
4359   read-cmd .initial:n = school,
4360   read-cmd .value_required:n = true,
4361   read-cmd / unknown .code:n =
4362     \msg_error:nneee { spintenc } { unknown-choice }
4363     { read-cmd } { school,~mathcat } { \exp_not:n {##1} },
4364   prefix .code:n =
4365     { \__spintenc_sanitiz_affix:cn { l__spintenc_sp #1 _prefix_tl } {##1} },
4366   prefix .initial:n = ,
4367   prefix .value_required:n = true,
4368   print-sym .choices:nn = { usual , odot }
4369   {
4370     \int_case:nn { \l_keys_choice_int }
4371     {
4372       {1} { \str_set:cn { l__spintenc_sp #1 _print_sym_str } { usual } }
4373       {2} { \str_set:cn { l__spintenc_sp #1 _print_sym_str } { odot } }
4374     }
4375   },
4376   print-sym .initial:n = odot,
4377   print-sym .value_required:n = true,
4378   print-sym / unknown .code:n =
4379     \msg_error:nneee { spintenc } { unknown-choice }
4380     { print-sym } { usual,~odot } { \exp_not:n {##1} },
4381   read-sty .choices:nn = { circun , circle }
4382   {
4383     \int_case:nn { \l_keys_choice_int }
4384     {
4385       {1} { \str_set:cn { l__spintenc_sp #1 _read_sty_str } { circun } }
4386       {2} { \str_set:cn { l__spintenc_sp #1 _read_sty_str } { circle } }
4387     }
4388   },
4389   read-sty .initial:n = circun,
4390   read-sty .value_required:n = true,
4391   radio-sty .choices:nn = { side , mside }
4392   {
4393     \int_case:nn { \l_keys_choice_int }
4394     {
4395       {1} { \str_set:cn { l__spintenc_sp #1 _radio_sty_str } { side } }
4396       {2} { \str_set:cn { l__spintenc_sp #1 _radio_sty_str } { mside } }
4397     }
4398   },
4399   radio-sty .initial:n = mside,

```

```

4400     radio-sty           .value_required:n = true,
4401     sub-index           .code:n =
4402     { \str_set:cn { \__spintent_sp #1 _sub_index_str } {##1} },
4403     sub-index           .initial:n = ,
4404     sub-index           .value_required:n = true,
4405     read-sub            .bool_set:c = { \__spintent_sp #1 _read_sub_bool },
4406     read-sub            .initial:n = false,
4407     read-sub            .value_required:n = true,
4408     read-arg            .code:n =
4409     { \__spintent_sanitiz_affix:cn { \__spintent_sp #1 _read_arg_tl } {##1} },
4410     read-arg            .initial:n = ,
4411     read-arg            .value_required:n = true,
4412     silent-cmd          .bool_set:c = { \__spintent_sp #1 _silent_cmd_bool },
4413     silent-cmd          .initial:n = false,
4414     silent-cmd          .value_required:n = true,
4415     spside              .meta:nn = { spintent / spside } { ##1 },
4416     spside              .value_required:n = true,
4417     spmside             .meta:nn = { spintent / spmside } { ##1 },
4418     spmside             .value_required:n = true,
4419     unknown             .code:n =
4420     {
4421       \str_set:Nn \__spintent_command_name_str { sp #1 }
4422       \__spintent_unknown_keys_cmd:n {##1}
4423     },
4424   }
4425 }

```

(End of definition for read-cmd and others.)

11.4.1.4 Internal functions for \spAcirc and \spPcirc

__spintent_spcircfig_sym:n

The function `\__spintent_spcircfig_sym:n` typesets `\bigodot` or the plain letter C, according to the value set by the key `print-sym`.

```

4426 \cs_new_protected:Nn \__spintent_spcircfig_sym:n
4427 {
4428   \str_if_eq:vnTF { \__spintent_sp #1 _print_sym_str } { odot }
4429   {
4430     \bigodot
4431   }
4432   { C }
4433 }

```

(End of definition for __spintent_spcircfig_sym:n.)

__spintent_spcircfig_radio_print:n

The function `\__spintent_spcircfig_radio_print:n` typesets the radius according to the type set by `\__spintent_luafun_geo_circ_parse_and_set:n`: `\spnum` for a plain number, `\spunit` for a measure, the raw text for a label, or `\spside/\spmside` for a segment, according to the value set by the key `radio-sty`.

```

4434 \cs_new_protected:Nn \__spintent_spcircfig_radio_print:n
4435 {
4436   \str_case:Vn \__spintent_geo_luaset_circ_radio_type_str
4437   {
4438     { number } { \exp_args:No \spnum { \__spintent_geo_luaset_circ_radio_raw_str } }
4439     { measure } { \exp_args:No \spunit { \__spintent_geo_luaset_circ_radio_raw_str } }
4440     { label } { \__spintent_geo_luaset_circ_radio_raw_str }
4441     { segment }
4442     {
4443       \str_if_eq:vnTF { \__spintent_sp #1 _radio_sty_str } { side }
4444       { \exp_args:No \spside { \__spintent_geo_luaset_circ_radio_raw_str } }
4445       { \exp_args:No \spmside { \__spintent_geo_luaset_circ_radio_raw_str } }
4446     }
4447   }
4448 }

```

(End of definition for __spintent_spcircfig_radio_print:n.)

__spintent_spcircfig_word_and_article:n

The function `\__spintent_spcircfig_word_and_article:n` sets `\__spintent_geo_circ_word_tl` to the chosen word — either the value set by the key `read-arg`, or `circunferencia/círculo` according to the key `read-sty` — and `\__spintent_spcirc_articulo_str` to `la` or `el`, matching the grammatical gender of the chosen word.

```

4449 \cs_new_protected:Nn \__spintent_spcircfig_word_and_article:n
4450 {
4451   \tl_if_empty:cTF { \__spintent_sp #1 _read_arg_tl }

```

```

4452     {
4453         \str_if_eq:vnTF { \__spintent_sp #1 _read_sty_str } { circun }
4454         {
4455             \tl_set:Nn \__spintent_geo_circ_word_tl { circunferencia }
4456             \str_set:Nn \__spintent_spcirc_articulo_str { la }
4457         }
4458         {
4459             \tl_set:Nn \__spintent_geo_circ_word_tl { círculo }
4460             \str_set:Nn \__spintent_spcirc_articulo_str { el }
4461         }
4462     }
4463     {
4464         \tl_set:Ne \__spintent_geo_circ_word_tl { \tl_use:c { \__spintent_sp #1 _read_arg_tl } }
4465         \str_if_eq:vnTF { \__spintent_sp #1 _read_sty_str } { circun }
4466         { \str_set:Nn \__spintent_spcirc_articulo_str { la } }
4467         { \str_set:Nn \__spintent_spcirc_articulo_str { el } }
4468     }
4469 }

```

(End of definition for __spintent_spcircfig_word_and_article:n.)

__spintent_spcircfig_process:nn

The function __spintent_spcircfig_process:nn sets __spintent_geo_fig_op_tl to área or perímetro. When #2 is not empty, it calls __spintent_luafun_geo_circ_parse_and_set:n; if that call reports an error and #2 contains no comma, it falls back to __spintent_luafun_geo_afig_parse_and_set:n / __spintent_luafun_geo_pfig_parse_and_set:n (the same Lua entry point used by \spAfig/\spPfig) to try the bare number/letter shortcut, setting __spintent_geo_circfig_mode_str to bare on success. A table letter match from that fallback is rejected (forced back to an error), since table letters are not meaningful for a circle.

__spintent_spcircfig_word_and_article:n sets the word/article used to build __spintent_geo_circ_int for the bare, empty, and real-center cases, delegating the rendering to __spintent_spcircfig_mathml:nn.

```

4470 \cs_new_protected:Nn \__spintent_spcircfig_process:nn
4471 {
4472     \str_if_eq:eeTF {#1} { Acirc }
4473     { \tl_set:Nn \__spintent_geo_fig_op_tl { área } }
4474     { \tl_set:Nn \__spintent_geo_fig_op_tl { perímetro } }
4475     \str_set:Nn \__spintent_geo_circfig_mode_str { circle }
4476     \tl_if_empty:nF {#2}
4477     {
4478         \__spintent_luafun_geo_circ_parse_and_set:n { \exp_not:n {#2} }
4479         \str_if_eq:VnT \__spintent_geo_luaerror_str { true }
4480         {
4481             \tl_if_in:nnF {#2} { , }
4482             {
4483                 \str_if_eq:eeTF {#1} { Acirc }
4484                 { \__spintent_luafun_geo_afig_parse_and_set:n { \exp_not:n {#2} } }
4485                 { \__spintent_luafun_geo_pfig_parse_and_set:n { \exp_not:n {#2} } }
4486                 \str_if_eq:VnF \__spintent_geo_luaerror_str { true }
4487                 {
4488                     \str_if_empty:NNTF \__spintent_geo_luaerror_letra_word_str
4489                     { \str_set:Nn \__spintent_geo_circfig_mode_str { bare } }
4490                     { \str_set:Nn \__spintent_geo_luaerror_str { true } }
4491                 }
4492             }
4493         }
4494         \str_if_eq:VnF \__spintent_geo_circfig_mode_str { bare }
4495         {
4496             \str_if_eq:VnT \__spintent_geo_luaerror_str { true }
4497             {
4498                 \msg_error:nnee { spintent } { sp #1 -invalid-arg }
4499                 { \c_backslash_str sp #1 } { \exp_not:n {#2} }
4500             }
4501         }
4502     }
4503     \__spintent_spcircfig_word_and_article:n {#1}
4504     \str_if_eq:VnTF \__spintent_geo_circfig_mode_str { bare }
4505     {
4506         \str_set:Ne \__spintent_geo_circ_intent_str
4507         {
4508             \__spintent_geo_fig_op_tl -
4509             \bool_if:cT { \__spintent_sp #1 _read_sub_bool } { sub- }
4510             \__spintent_geo_luaerror_sub_spoken_str

```

```

4511     }
4512     \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
4513     {
4514         \str_set:Ne \l__spintent_geo_circ_intent_str
4515         {
4516             _ \l__spintent_geo_fig_op_tl -
4517             \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
4518             \l__spintent_geo_luaset_sub_spoken_str :prefix($pts)
4519         }
4520     }
4521 }
4522 {
4523     \tl_if_empty:nTF {#2}
4524     {
4525         \str_set:Ne \l__spintent_geo_circ_intent_str
4526         {
4527             \l__spintent_geo_fig_op_tl -de- \l__spintent_spcirc_articulo_str -
4528             \l__spintent_geo_circ_word_tl
4529         }
4530         \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
4531         {
4532             \str_set:Ne \l__spintent_geo_circ_intent_str
4533             {
4534                 _ \l__spintent_geo_fig_op_tl -de-
4535                 \l__spintent_spcirc_articulo_str -
4536                 \l__spintent_geo_circ_word_tl :prefix($pts)
4537             }
4538         }
4539     }
4540     {
4541         \str_set:Ne \l__spintent_geo_circ_intent_str
4542         {
4543             \l__spintent_geo_fig_op_tl -de- \l__spintent_spcirc_articulo_str -
4544             \l__spintent_geo_circ_word_tl
4545         }
4546         \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
4547         {
4548             \str_put_right:Ne \l__spintent_geo_circ_intent_str
4549             {
4550                 -
4551                 \bool_if:cT { l__spintent_sp #1 _read_sub_bool } { sub- }
4552                 \str_use:c { l__spintent_sp #1 _sub_index_str }
4553             }
4554         }
4555         \str_put_right:Ne \l__spintent_geo_circ_intent_str
4556         { -de-centro-en- \l__spintent_geo_luaset_print_tl }
4557         \str_if_empty:NF \l__spintent_geo_luaset_circ_radio_type_str
4558         { \str_put_right:Nn \l__spintent_geo_circ_intent_str { -y-radio } }
4559         \str_if_eq:eeT { \str_use:c { l__spintent_sp #1 _mode_str } } { mathcat }
4560         {
4561             \str_set:Ne \l__spintent_geo_circ_intent_str
4562             {
4563                 _ \l__spintent_geo_fig_op_tl -de-
4564                 \l__spintent_spcirc_articulo_str -
4565                 \l__spintent_geo_circ_word_tl :prefix($pts)
4566             }
4567         }
4568     }
4569 }
4570 \__spintent_spcircfig_mathml:nn { #1 } { #2 }
4571 }

```

(End of definition for `\__spintent_spcircfig_process:nn`.)

`\__spintent_spcircfig_mathml:nn`

The function `\__spintent_spcircfig_mathml:nn` reads the state already set by `\__spintent_spcircfig_process:nn` and renders it. Three cases follow: the bare shortcut prints the symbol with the shortcut value as its own subscript; an empty `#2` prints the symbol alone; and a real center (with an optional radius) prints the symbol, the optional sub- index subscript, and the center between parentheses. When a radius is present, `\mathcal` (`\__spintent_mathcal:n`) gets its own `\MathMLintent` holding the full spoken intent, and the circle's own visual pieces — symbol/center/comma, the radius (via `\__spintent_spcircfig_radio_print:n`, kept audible), and the closing parenthesis — become three sibling `\MathMLintent` calls sharing a single

`\c_math_subscript_token` group, so the whole notation is typeset at subscript size.

```

4572 \cs_new_protected:Nn \__spintent_spcircfig_mathml:nn
4573 {
4574   \__spintent_invisible_sep:
4575   \str_if_eq:VnTF \__spintent_geo_circfig_mode_str { bare }
4576   {
4577     %%\__spintent_invisible_sep:
4578     \MathMLintent
4579     {
4580       \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4581       { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4582       \l__spintent_geo_circ_intent_str
4583     }
4584     {
4585       \c__spintent_invisible_sep_tl
4586       \str_if_eq:eeTF {#1} { Acirc }
4587       { \__spintent_mathcal:n { A } }
4588       { \__spintent_mathcal:n { P } }
4589       \c_math_subscript_token
4590       {
4591         \__spintent_spcircfig_sym:n {#1}
4592         \c_math_subscript_token { \l__spintent_geo_luaset_print_tl }
4593       }
4594     }
4595   }
4596   {
4597     \tl_if_empty:nTF {#2}
4598     {
4599       %%\__spintent_invisible_sep:
4600       \MathMLintent
4601       {
4602         \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4603         { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4604         \l__spintent_geo_circ_intent_str
4605       }
4606       {
4607         \c__spintent_invisible_sep_tl
4608         \str_if_eq:eeTF {#1} { Acirc }
4609         { \__spintent_mathcal:n { A } }
4610         { \__spintent_mathcal:n { P } }
4611         \c_math_subscript_token { \__spintent_spcircfig_sym:n {#1} }
4612       }
4613     }
4614     {
4615       %%\__spintent_invisible_sep:
4616       \str_if_empty:NTF \l__spintent_geo_luaset_circ_radio_type_str
4617       {
4618         \MathMLintent
4619         {
4620           \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4621           { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4622           \l__spintent_geo_circ_intent_str
4623         }
4624         {
4625           \c__spintent_invisible_sep_tl
4626           \str_if_eq:eeTF {#1} { Acirc }
4627           { \__spintent_mathcal:n { A } }
4628           { \__spintent_mathcal:n { P } }
4629           \c_math_subscript_token
4630           {
4631             \__spintent_spcircfig_sym:n {#1}
4632             \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
4633             {
4634               \c_math_subscript_token
4635               { \str_use:c { l__spintent_sp #1 _sub_index_str } }
4636             }
4637             ( \l__spintent_geo_luaset_print_tl )
4638           }
4639         }
4640       }
4641     }

```



```

4642         \MathMLintent
4643         {
4644             \tl_if_empty:cF { l__spintent_sp #1 _prefix_tl }
4645             { \tl_use:c { l__spintent_sp #1 _prefix_tl } - }
4646             \l__spintent_geo_circ_intent_str
4647         }
4648         {
4649             \c__spintent_invisible_sep_tl
4650             \str_if_eq:eeTF {#1} { Acirc }
4651             { \__spintent_mathcal:n { A } }
4652             { \__spintent_mathcal:n { P } }
4653         }
4654     \c_math_subscript_token
4655     {
4656         \MathMLintent
4657         { :silent }
4658         {
4659             \MathMLintent{ :silent } { \__spintent_spcircfig_sym:n {#1} }
4660             \str_if_empty:cF { l__spintent_sp #1 _sub_index_str }
4661             {
4662                 \c_math_subscript_token
4663                 { \str_use:c { l__spintent_sp #1 _sub_index_str } }
4664             }
4665             ( \l__spintent_geo_luaset_print_tl , \,
4666             }
4667             %%\__spintent_invisible_sep:
4668             \__spintent_spcircfig_radio_print:n {#1}
4669             \MathMLintent { :silent } { ) }
4670         }
4671     }
4672 }
4673 }
4674 }

```

(End of definition for __spintent_spcircfig_mathml:nn.)

__spintent_spcircfig_exec:nnn The function __spintent_spcircfig_exec:nnn receives the short name of the command in #1, the optional keys in #2, and the argument in #3, delegating the rendering to __spintent_spcircfig_process:nn.

```

4675 \cs_new_protected:Nn \__spintent_spcircfig_exec:nnn
4676 {
4677     \keys_set:nn { spintent / sp #1 } { #2 }
4678     \__spintent_spcircfig_process:nn { #1 } { #3 }
4679 }

```

(End of definition for __spintent_spcircfig_exec:nnn.)

\spAcirc Both public commands \spAcirc and \spPcirc enforce math mode, open a local group, and delegate execution to __spintent_spcircfig_exec:nnn.

```

4680 \NewDocumentCommand \spAcirc { 0{} d() }
4681 {
4682     \mode_if_math:TF
4683     {
4684         \group_begin:
4685         \tl_if_novalue:nTF {#2}
4686         { \__spintent_spcircfig_exec:nnn { Acirc } { #1 } { } }
4687         { \__spintent_spcircfig_exec:nnn { Acirc } { #1 } { #2 } }
4688         \group_end:
4689     }
4690     { \msg_error:nnn { spintent } { need-math-mode } { \spAcirc } }
4691 }
4692 \NewDocumentCommand \spPcirc { 0{} d() }
4693 {
4694     \mode_if_math:TF
4695     {
4696         \group_begin:
4697         \tl_if_novalue:nTF {#2}
4698         { \__spintent_spcircfig_exec:nnn { Pcirc } { #1 } { } }
4699         { \__spintent_spcircfig_exec:nnn { Pcirc } { #1 } { #2 } }
4700         \group_end:
4701     }
4702     { \msg_error:nnn { spintent } { need-math-mode } { \spPcirc } }
4703 }

```

(End of definition for \spAcirc and \spPcirc. These functions are documented on page ??.)

11.42 Finish package

Finish package implementation.

```
4704 \file_input_stop:
4705 \endpackage
```

12 Índice de la Implementación

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